



**TSMC 0.13 μ m (CL013G) Process
1.2-Volt SAGE-X™
Standard Cell Library
Databook**

January 2004

Release 2.6

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Revision History

This document contains the release history for the TSMC 0.13 μ m (CL013G) Process 1.2V SAGE-X™ Standard Cell Library databook.

Part Number	Release Number	Date of Release	Updates
DB-SX-TSM014-1.0/13	1.0	March 2001	• Initial release
DB-SX-TSM014-2.0/13	2.0	August 2001	• Data recharacterization
DB-SX-TSM014-2.1/13	2.1	October 2001	• Update adder cells
DB-SX-TSM014-2.2/13	2.2	January 2002	• Update cell list
DB-SX-TSM014-2.3/13	2.3	March 2002	• Data recharacterization
DB-SX-TSM014-2.4/13	2.4	April 2002	• Data recharacterization
DB-SX-TSM014-2.5/13	2.5	June 2002	• Footer corrected
DB-SX-TSM014-2.6/13	2.6	January 2004	• Update TLATNTSCA schematic, add ACHCON and ACHCIN datasheets

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Artisan's SAGE-X standard cell library builds upon our SAGE architecture, producing the optimum combination of high-density with high-performance. The cell line-up is derived from extensive customer design, synthesis, and place-and-route benchmark analysis. Library optimization is achieved by carefully matching the library functions and drive strengths to leading synthesis and place-and-route tools, producing superior RTL-to-GDSII results.

How This Book Is Organized

This section is organized into three sections:

- ***Global Parameters*** provides an overview of parameters specific to your SAGE-X library.
- ***Special Cells*** details the types of special cells you will find included in the library.
- ***Reading the Standard Cell Datasheet*** describes what you will find in each datasheet.

Datasheets for each cell in this library are provided after the introduction. The datasheets are included in alphabetical order within the following categories:

- Base Cells
- Advanced Arithmetic Cells
- Register File Cells
- Synthesis Optimized Arithmetic Cells

Global Parameters

This section specifies global parameters for the TSMC 0.13 μ m (CL013G) 1.2V SAGE-X™ Standard Cell Library. It covers physical specifications, electrical specifications, derating factors, propagation delay calculation, timing constraints, power calculation, and power-rail strapping.

Specifications

Table 1 shows the physical design specifications of this library.

Table 1. Physical Specifications

Drawn Gate Length (μm)	0.13
Layers of Metal	4, 5, 6, 7, or 8
Layout Grid (μm)	0.005
Vertical Pin Grid (μm)	0.41
Horizontal Pin Grid (μm)	0.46
Cell Power and Ground Rail Width (μm)	0.5

In this library, all pins are located on the vertical and horizontal pin grids. Most place-and-route tools work more efficiently with all pins on grids, and some tools even require it.

The SAGE-X library also supports designs with eight layers of metal. You may need to change the design rules in the technology file, because the top-level metal has a greater minimum width and greater minimum spacing requirement. See "TSMC 0.13 μm LOGIC 1P8M Salicide 1.0V/2.5V, 1.2V/2.5V, 1.0V/3.3V, 1.2V/3.3 Design Rule" design rule manual. You must define these correctly for the place-and-route tool.

Table 2 describes the electrical specifications for this library.

Table 2. Electrical Specifications

Parameter	Minimum	Maximum
DC Supply Voltage (Vdd)	1.08 V	1.32 V
Junction Temperature	-40°C	125°C

Table 3 shows the derating factors for this SAGE-X Standard Cell Library.

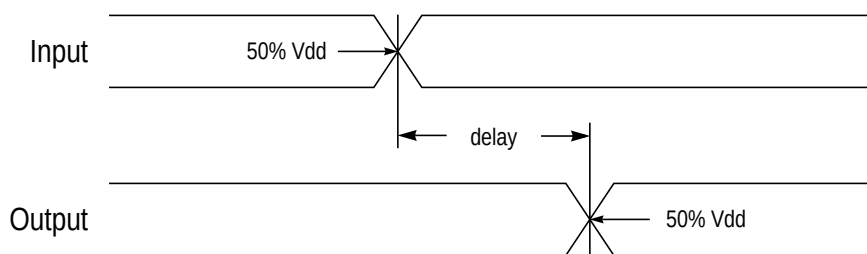
Table 3. Derating Factors

K _{Process} (slow)	1.248
K _{Process} (typical)	1.000 (by definition)
K _{Process} (fast)	0.812
K _{Volt} (1.2V to 1.08V)	-1.512/V
K _{Volt} (1.2V to 1.32V)	-1.009/V
K _{Temp} (25°C to 125°C)	0.00091°C
K _{Temp} (25°C to -40°C)	0.00103°C

Propagation Delay and Transition Time

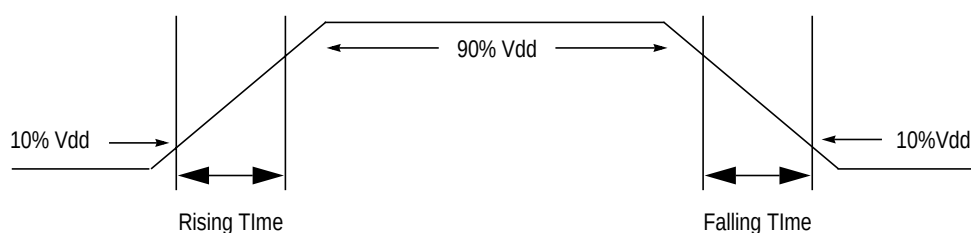
The propagation delay through a cell is the sum of the intrinsic delay, the load-dependent delay, and the input-slew dependent delay. Delays are defined as the time interval between the input stimulus crossing 50% of V_{dd} and the output crossing 50% of V_{dd} . Figure 1 illustrates the propagation delay.

Figure 1. Propagation Delay



The transition times (slews) on input and output pins are defined as the time interval between the signal crossing 10% of V_{dd} and 90% of V_{dd} . Figure 2 illustrates transition time measurements for rising and falling signals. For the functional models, these slews were extrapolated from 30/70% characterization values for added accuracy.

Figure 2. Transition Time



Factors that affect propagation delays and transition time include: temperature, supply voltage, process variations, fanout loading, interconnect loading, input-transition time, input-signal polarity, and timing constraints. The timing models provided with this library include the effects of input-transition time on propagation delays. Also, all timing models use a table lookup method to calculate accurate timing. To simplify calculations, the standard cell datasheets

provide all timing numbers for an input slew of 0.028ns and a linearized load factor, K_{load} , which is not as accurate as the timing models. All cells have been characterized with a fully populated metal2 (0.46 μ m horizontal pitch) and metal3 (0.41 μ m vertical pitch) routing grid across the entire cell layout.

The SAGE-X Standard Cell Library may contain negative propagation delays. Although most third-party verification tools can handle negative propagation delays, some tools will turn negative delays into a zero value.

Derating Factors

Derating factors are coefficients that the typical process characterization data is multiplied by to arrive at timing data that reflects appropriate operating conditions. Table 3 on page 12 provides derating factors for variations in process case, temperature, and voltage.

Derating factors are derived by averaging the performance of many different cells in the library. A particular combination of cells may perform better or worse than indicated by these derating factors.

Delay Calculation

Using the delay data in the datasheets ($t_{intrinsic}$, K_{load} , and C_{load}) and the delay derating factors, the estimated total propagation delay is calculated as such:

$$t_{TPD} = (K_{Process}) \cdot [1 + (K_{V_{olt}} \cdot \Delta V_{dd})] \cdot [1 + (K_{Temp} \cdot \Delta T)] \cdot t_{typical}$$

$$t_{typical} = t_{intrinsic} + (K_{load} \cdot C_{load})$$

where:

t_{TPD} = total propagation delay (ns);

$t_{typical}$ = delay at typical corner—1.2V, 25°C, typical process (ns);

$t_{intrinsic}$ = delay through the cell when there is no output load (ns);

K_{load} = load delay multiplier (ns/pF);

C_{load} = total output load capacitance (pF);

$K_{Process}$ = process derating factor, where process is slow, typical, or fast;

$K_{V_{olt}}$ = voltage derating factor (/V);

ΔV_{dd} = $V_{dd} - 1.2V$;

K_{Temp} = temperature derating factor (/°C);

ΔT = junction temperature - 25°C.

Timing Constraints

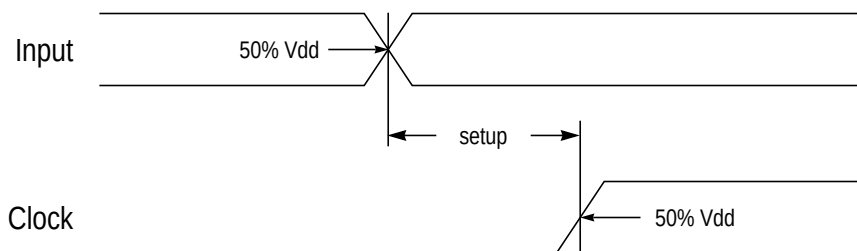
Timing constraints define minimum time intervals during which specific signals must be held steady in order to ensure the correct functioning of any given cell. Timing constraints include: setup time, hold time, recovery time, and minimum pulse width.

The sequential-cell timing models provided with this library include the effects of input-transition time and data-signal and clock-signal polarity on timing constraints. To simplify calculations, the datasheets specify timing constraint values for 0.028ns data slew and 0.028ns clock slew. Other factors that affect timing constraints include temperature, supply voltage, and process case variations. All cells have been characterized with a fully populated metal2 (0.46 μ m horizontal pitch) and metal3 (0.41 μ m vertical pitch) routing grid across the entire cell layout.

Timing constraints can affect propagation delays. The intrinsic delays given in the datasheets are measured with relaxed timing constraints (longer than necessary setup times, hold times, recovery times, and pulse widths). The use of shorter timing constraint intervals may increase delay. Each cell is considered functional as long as the actual delay does not exceed the delay given in the datasheets by more than 10%.

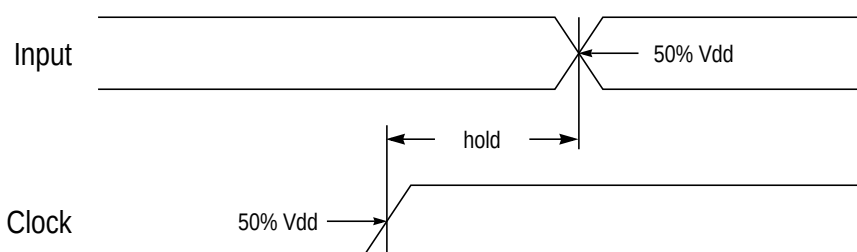
Setup Time

The setup time for a sequential cell is the minimum length of time the data-input signal must remain stable before the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large setup time) by more than 10%. Setup-constraint values are measured as the interval between the data signal crossing 50% of Vdd and the clock signal crossing 50% of Vdd. For the measurement of setup time, the data input signal is kept stable after the active clock edge for an infinite hold time. Figure 3 illustrates setup time for a positive-edge-triggered sequential cell.

Figure 3. Setup Time

Hold Time

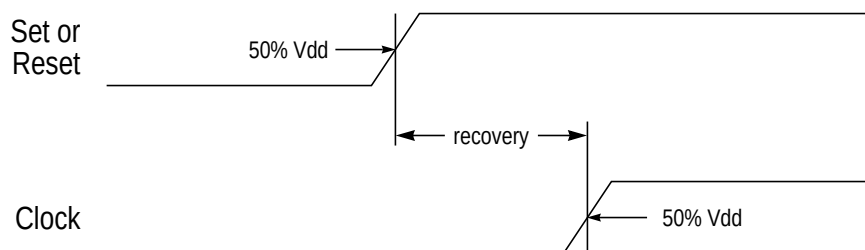
The hold time for a sequential cell is the minimum length of time the data-input signal must remain stable after the active edge of the clock (or other specified signal) to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large hold time) by more than 10%. Hold-constraint values are measured as the interval between the data signal crossing 50% of Vdd and the clock signal crossing 50% of Vdd. For the measurement of hold time, the data input signal is held stable before the active clock edge for an infinite setup time. Figure 4 illustrates hold time for a positive-edge-triggered sequential cell.

Figure 4. Hold Time

Recovery Time

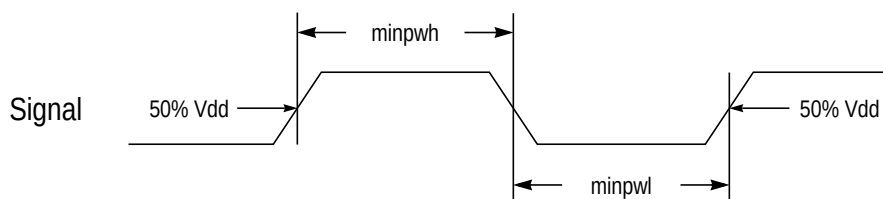
Recovery time for sequential cells is the minimum length of time that the active-low set or reset signal must remain high before the active edge of the clock to ensure correct functioning of the cell. The cell is considered functional as long as the delay for the output reaching its expected value does not exceed the reference delay (measured with a large recovery time) by more than 10%. Recovery constraint values are measured as the interval between the set or reset signal crossing 50% of V_{dd} and the clock signal crossing 50% of V_{dd} . For the measurement of recovery time, the set or reset signal is held stable after the active clock edge for an infinite hold time. Figure 5 illustrates recovery time.

Figure 5. Recovery Time



Minimum Pulse Width

Minimum pulse width is the minimum length of time between the leading and trailing edges of a pulse waveform. Minimum pulse width high (minpwh) is measured as the interval between the rising edge of the signal crossing 50% of V_{dd} and the falling edge of the signal crossing 50% of V_{dd} . Minimum pulse width low (minpwl) is measured as the interval between the falling edge of the signal crossing 50% of V_{dd} and the rising edge of the signal crossing 50% of V_{dd} . Figure 6 illustrates minimum pulse width.

Figure 6. Minimum Pulse Width

Minimum pulse width is defined as 0.501ns for all set/reset pins (SN, RN) and 0.210ns for all clock pins (G, GN, CK, CKN). These are the largest minimum pulse widths measured from all the cells in the library. An input pulse of shorter duration will produce unpredictable results.

Power Dissipation

The SAGE-X Standard Cell Library is designed to dissipate only AC power, except for the small reverse-bias leakage currents which are normally present in all CMOS circuits.

The power dissipation internal to a cell when a given input switches is primarily dependent upon the cell design itself. The power dissipation of a complete design, or part of a design, using cells from the library is primarily a function of the switching frequency of the design's internal nets. These nets include the inputs and outputs of each cell and the capacitive load associated with the outputs of each cell.

The SAGE-X library datasheets contain both an AC power table which documents the internal energy consumption of each cell and a pin capacitance table which gives input-pin capacitance data used to compute output loading. This information, coupled with design-specific information, can be used to estimate the total power dissipation of a cell within a design.

The AC power tables specify the amount of energy consumed within a cell ($\mu\text{W}/\text{MHz}$) when the corresponding pin changes state at 25°C, 1.2V, and typical process. The energy data in the tables were measured for an input slew of 0.028ns and no loading at the outputs.

For combinatorial cells, energy values are provided for only input pins. The energy value for each input pin is the average of energies associated with the input transitions which result in an output transition.

For sequential cells, the energy associated with each input pin is the average energy of those input transitions which *do not* result in an output transition. The energy associated with the output pin of a sequential cell is the average energy of all cases where an output transition is the result of a clock-input transition, minus the energy associated with the clock input pin. In the event that a sequential cell has multiple outputs, all output energy data will be associated with only one output pin.

Power Calculation

Power dissipation is dependent upon the power-supply voltage, frequency of operation, internal capacitance, and output load. The power dissipated by each cell is:

$$P_{avg} = \sum_{n=1}^x (E_{in} \cdot f_{in}) + \sum_{n=1}^y \left(C_{on} \cdot V_{dd}^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

where:

- P_{avg} = average power (μ W);
- x = number of input pins;
- E_{in} = energy associated with the n th input pin (μ W/MHz);
- f_{in} = frequency at which the n th input pin changes state during the normal operation of the design (MHz);
- y = number of output pins;
- C_{on} = external capacitive loading on the n th output pin, including the capacitance of each input pin connected to the output driver, plus the route wire capacitance, actual or estimated (pF);
- V_{dd} = operating voltage = 1.2V;
- f_{on} = frequency at which the n th output pin changes state during the normal operation of the design (MHz);
- E_{os} = energy associated with the output pin for sequential cells only (μ W/MHz).

The switching frequency of inputs and outputs of a particular cell in a design can be obtained from a gate-level logic simulator (e.g. Verilog) by applying typical input stimuli and measuring the activity on each node of interest. The total average power for the design can be computed by adding the average power for each cell.

For example, for a DFFXL cell with clock switching at 133MHz, input and output pins switching at 20MHz, an external capacitive loading on the output pin of 0.02pF, and using the power table in the example DFF datasheet shown on page 27, the power dissipated by the DFFXL is given by the following equation:

$$P_{avg} = \sum_{n=1}^x (E_{in} \cdot f_{in}) + \sum_{n=1}^y \left(C_{on} \cdot Vdd^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

Substituting,

$$x = 2;$$

$$E_{i1} = 0.0060 \mu\text{W/MHz};$$

$$E_{i2} = 0.0077 \mu\text{W/MHz};$$

$$f_{i1} = 20 \text{ MHz};$$

$$f_{i2} = 133 \text{ MHz};$$

$$y = 2;$$

$$C_{o1} = 0.02 \text{ pF};$$

$$C_{o2} = 0.02 \text{ pF};$$

$$Vdd = 1.2\text{V};$$

$$f_{o1} = 20 \text{ MHz};$$

$$f_{o2} = 20 \text{ MHz};$$

$$E_{os} = 0.0085 \mu\text{W/MHz},$$

we have:

$$P_{avg} = \sum_{n=1}^2 (E_{in} \cdot f_{in}) + \sum_{n=1}^2 \left(C_{on} \cdot Vdd^2 \cdot \frac{1}{2} f_{on} \right) + E_{os} \cdot f_{o1}$$

$$\begin{aligned} P_{avg} &= (E_{i1} \cdot f_{i1}) + (E_{i2} \cdot f_{i2}) \\ &\quad + \left(C_{o1} \cdot 1.2^2 \cdot \frac{1}{2} f_{o1} \right) + \left(C_{o2} \cdot 1.2^2 \cdot \frac{1}{2} f_{o2} \right) \\ &\quad + (E_{os} \cdot f_{o1}) \end{aligned}$$

$$\begin{aligned}
P_{\text{avg}} &= (0.0060 \bullet 20) + (0.0077 \bullet 133) \\
&\quad + \left(0.02 \bullet 1.44 \bullet \frac{1}{2}(20)\right) + \left(0.02 \bullet 1.44 \bullet \frac{1}{2}(20)\right) \\
&\quad + (0.0085 \bullet 20) \\
P_{\text{avg}} &= 1.89 \text{ } \mu\text{W}
\end{aligned}$$

Power-Rail Strapping

You must determine the required amount of vertical power-rail strapping to satisfy all requirements imposed by the design methodology for a given design. Power-rail strapping should be sized small enough to optimize standard cell height and maximize router efficiency, yet it must be large enough to provide sufficient power to the cells.

The guidelines below provide a rough estimate with many simplifying assumptions. For a given module design, you can estimate the amount of vertical power-rail strapping that is required to fulfill electromigration requirements.

Given:

- I_{avg} = total average current for the module, calculated from previous section (mA);
- w_{m1} = VSS/VDD metal1 wire width (μm), see Physical Specifications;
- r = number of rows in module;
- d_{m1} = maximum metal1 current density allowed for the process (mA/ μm);
- d_{m2} = maximum metal2 current density allowed for the process (mA/ μm);
- I_{m1} = maximum current that can be supported by all horizontal metal1 wires (mA);
- I_{strap} = total current that must be supported by the vertical metal2 strapping (mA);
- w_{m2} = metal2 wire width required for vertical strapping (μm);
- c = minimum number of metal2 straps;

we have:

$$I_{m1} = w_{m1} \bullet r \bullet 2 \bullet d_{m1},$$

where multiplying by 2 assumes metal1 wires are supplied from both ends;

$$I_{strap} = \frac{(I_{avg} - I_{m1})}{2},$$

where dividing by 2 assumes the metal2 vertical strap wires are supplied from both ends;

$$w_{m2} = \frac{I_{strap}}{d_{m2}},$$

It is recommended that the metal2 wire width, w_{m2} , be divided into c equal portions which are spaced equidistant across the module, where

$$c = \frac{I_{avg}}{I_{m1}}, \text{ rounded up to the next integer.}$$

The same consideration must be given to the number of vias used to connect the metal1 and metal2 straps.

Adding Routing Channels

In the SAGE-X Standard Cell Library, each cell is designed with a uniform cell height of $3.69\mu\text{m}$ (i.e., 9 tracks tall with $0.41\mu\text{m}$ per track). The cell layouts allow neighboring rows of cells to share common power or ground rails when cells abut each other at the top and bottom edges of the cell bounding box. The sea-of-cells layout with no channels between rows will usually yield the minimum area. In case of extremely congested areas, you may want to separate some rows of cells to increase the number of routing channels within a particular layout region. Because geometries must overlap cell boundaries, a particular spacing between the rows may result in DRC violations for layer spacing. It is recommended that you do not use spacings that cause DRC violations. If these spacings must be used, the DRC violations must be fixed manually by filling the void between the rows with the appropriate layer(s).

Table 4 indicates which DRC violations to expect and how to correct them for a separation between rows of cells.

Table 4. Correcting DRC Violations

Row Separation in Number of Grids	Expected DRC Violations	Action to Correct DRC Violations
0 (Rows Abut)	None	None
1	NP/PP space < 0.31	Draw layer between rows to merge NP/PP regions above and below row separation.
2	NWELL space < 0.62 μm	Draw NWELL layer between rows to merge NWELL regions above and below row separation.
3	NWELL space < 0.62 μm	Draw NWELL layer between rows to merge NWELL regions above and below row separation.
4	None	None
5 or more	None	None

Special Cells

This section discusses special cells in the SAGE-X Standard Cell Library.

Antenna-Fix Cell

The library contains an antenna-fix cell which must be inserted manually. However, most place and route tools will indicate which nets require the antenna-fix cell. The TSMC antenna effect prevention guideline, "TSMC 0.13 μm LOGIC 1P8M Salicide 1.0V/2.5V, 1.2V/2.5V, 1.0V/3.3V, 1.2V/3.3V Design Rule," specifies a maximum wire length. During place and route, the router may connect wires to the input gates of cells that are longer than the maximum length allowable by the guideline. The antenna cell can be used in this case to add an optional diode on the net close to the input gates which do not meet the guideline. Pin A on the antenna cell connects to a diode, reverse biased to ground. A diode can be added to either P or N.

Fill Cells

The library contains several FILL cells: FILL1, FILL2, FILL4, FILL8, FILL16, FILL32, FILL64. The number appended to "FILL" in the cell name denotes the width of the cell in tracks.

During place and route, the FILL cells are used to connect power and ground rails across an area containing no cells. The FILL cells are also used to ensure gaps do not occur between well or implant layers which could cause design rule violations. Using wider cells where appropriate reduces the size of the layout database.

Low-Power (XL) Cells

The library contains a wide variety of cells, denoted by an "XL" suffix in the cell name, that are designed specifically for low-power applications. Input capacitance for the XL cells is much lower than that for corresponding X1 (1x drive strength) cells. Because XL cells have been designed for the sole purpose of reducing power consumption, output rise and fall times for these cells may not be equal. Due to the low-drive capability of the XL cells, these cells are not intended for use in critical timing paths, or to drive heavily loaded nets.

TIEHI/LO Cells

The library contains a TIEHI cell and a TIELO cell. The outputs of the TIEHI and TIELO cells are driven through diffusion to provide isolation from the power and ground rails for better ESD protection. The standard cell abstract methodology assumes that the TIEHI and TIELO cells are used to tie off any inputs to power and ground. If these cells are not used and the router is allowed to drop vias on the power rail, DRC errors or shorts may result.

Delay Cells

The library contains delay cells that have the same width. These delay cells allow you to adjust a given delay path with a simple cell substitution after place and route.

Reading the Standard Cell Datasheet

Please refer to the datasheet for DFF on pages 27 and 28 for the arrangement of each of the following datasheet sections.

1. Cell Name

The cell name field contains the cell name. The datasheets are presented alphabetically by cell name.

2. Cell Description

The cell description gives the function of the cell. When applicable, the equation(s) for the output pins are provided.

3. Function Table

The function table gives all possible combinations of input and output signals for the cell. Table 5 defines the symbols used in datasheet function tables.

Table 5. Function Table Key

Symbol	Description
0	Logic Low
1	Logic High
	High to Low Transition
	Low to High Transition
x	Don't Care
IL	Illegal/Undefined
Z	High Impedance

4. Logic Symbol

The logic symbol is a graphical representation of the cell, similar to the view in the schematic editor when the cell is instantiated. The symbol shows the name and location of the input and output pins.

5. Cell Size

This cell size table gives the height and width (μm) for each drive strength of the cell.

6. Functional Schematic

The functional schematic provides a functional representation of the cell.

7. Drive Strength

The drive strength of each cell is indicated by an “X” followed by the unit strength.

8. AC Power

The AC power table shows the amount of energy consumed ($\mu\text{W}/\text{MHz}$) within the cell when the corresponding pin changes state. The energy data for each drive strength of the cell are calculated at 25°C , 1.2 V, typical process, input slew of 0.028ns, and no external load at the output pins.

9. Delay

The delay table shows the intrinsic delay (ns) which is the delay through the cell when there is no load on the output, and the load multiplier for load dependent delay, K_{load} (ns/pF). The delays and load multiplier for each drive strength of the cell are calculated at 25°C , 1.2 V, typical process, and input slew of 0.028ns.

10. Timing Constraints

The timing constraints table shows the timing conditions (ns) required at 25°C , 1.2 V, and typical process to maintain proper functionality. Setup constraint values are measured for 0.028ns data slew and 0.028ns clock slew. Hold constraint values are measured for 0.028ns data slew and 0.028ns clock slew. Minimum pulse width is defined to be 0.501ns for all set/reset pins and 0.210ns for all clock pins. These are the largest minimum pulse widths measured from all the cells in the library.

11. Pin Capacitance

The pin capacitance table shows the typical loading at the input pins of the cell (pF) for each drive strength of the cell.

NOTE: The data shown in the following datasheet is intended for use as a reference only. Refer to the values in the DFF datasheet on pages 81 and 82 for actual data.



DFF

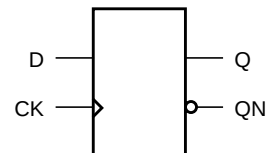
Cell Description

The DFF cell is a positive-edge-triggered, static D-type flip-flop.

Function

D	CK	Q[n+1]	QN[n+1]
0		0	1
1		1	0
x		Q[n]	QN[n]

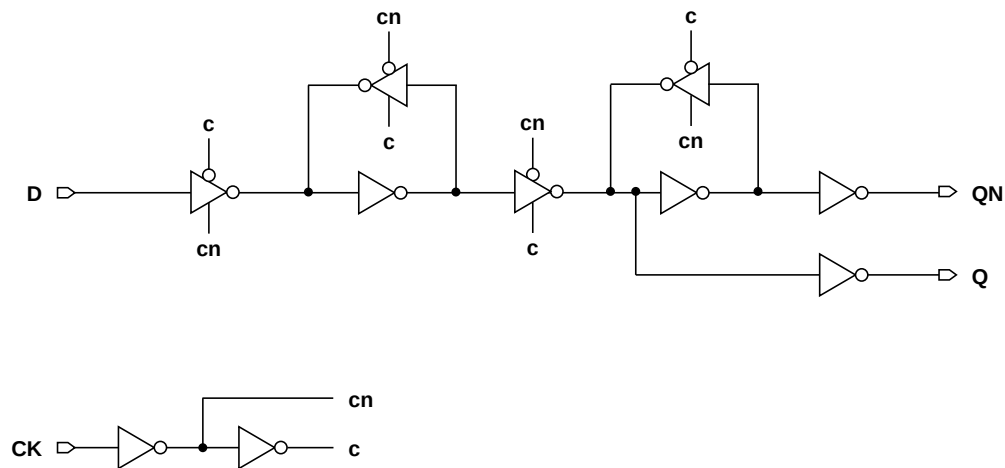
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFXL	3.69	7.36
DFFX1	3.69	7.36
DFFX2	3.69	8.74
DFFX4	3.69	11.50

Functional Schematic





DFF

11

8

AC Power

7

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0060	0.0065	0.0085	0.0125
CK	0.0077	0.0086	0.0108	0.0170
Q	0.0085	0.0111	0.0173	0.0289

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0013	0.0013	0.0015	0.0023
CK	0.0015	0.0019	0.0023	0.0035

9

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2200	0.1975	0.1815	0.1556	10.0234	5.4930	2.3655	1.1817
CK $\rightarrow$ Q $\downarrow$	0.1772	0.1482	0.1325	0.1240	5.4212	3.1463	1.5528	0.7766
CK $\rightarrow$ QN $\uparrow$	0.2282	0.1956	0.1795	0.1746	10.0086	5.4770	2.3481	1.1751
CK $\rightarrow$ QN $\downarrow$	0.2837	0.2679	0.2507	0.2146	4.8742	2.8717	1.4914	0.7376

10

Timing Constraints at 25°C, 1.2V, Typical Process

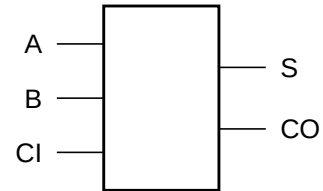
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0547	0.0625	0.0469	0.0469
	setup $\downarrow$ $\rightarrow$ CK	0.1406	0.1641	0.1406	0.1328
	hold $\uparrow$ $\rightarrow$ CK	-0.0391	-0.0391	-0.0312	-0.0312
	hold $\downarrow$ $\rightarrow$ CK	-0.0469	-0.0625	-0.0469	-0.0391
CK	minpwh	0.1127	0.1127	0.0933	0.0835
	minpwl	0.1807	0.1516	0.1418	0.1127

**Cell Description**

The ADDF cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B) with carry in (CI). The two outputs (S, CO) are represented by the logic equations:

$$S = (A \oplus B \oplus CI)$$

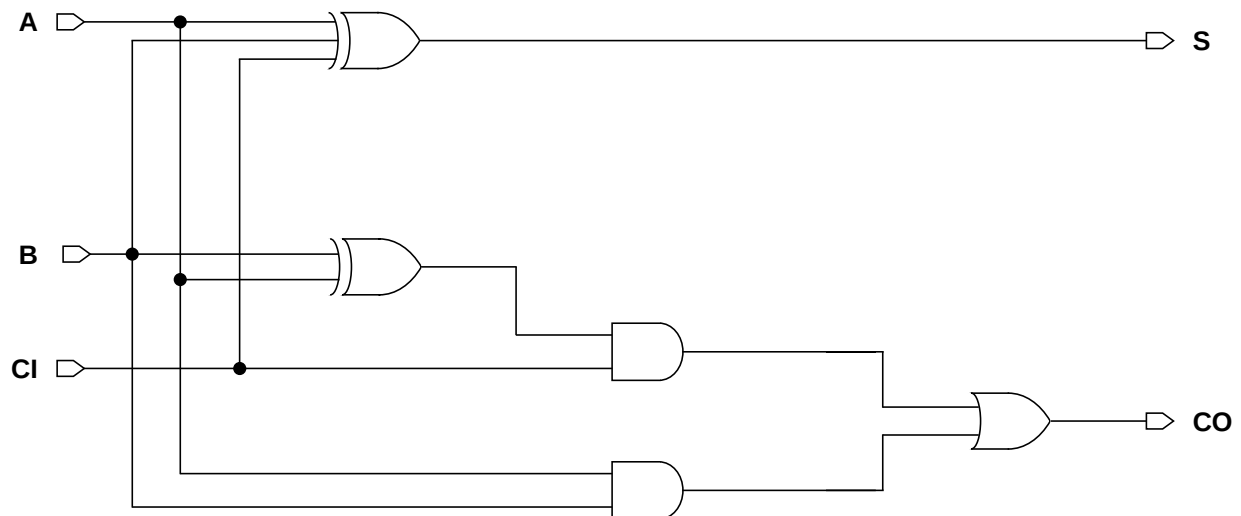
$$CO = (A \oplus B) \bullet CI + (A \bullet B)$$

Logic Symbol**Functions**

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDFXL	3.69	9.20
ADDFX1	3.69	9.66
ADDFX2	3.69	9.66
ADDFX4	3.69	10.58

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0281	0.0341	0.0398	0.0565
B	0.0341	0.0431	0.0517	0.0658
CI	0.0154	0.0185	0.0246	0.0408

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0020	0.0048	0.0048	0.0048
B	0.0022	0.0049	0.0049	0.0049
CI	0.0036	0.0045	0.0045	0.0045

Delays (25°C, 1.2V, Typical Process)

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.2904	0.1741	0.1871	0.2240	9.7159	5.2395	2.3167	1.1824
A $\rightarrow$ S $\downarrow$	0.3248	0.2229	0.2547	0.3020	6.7779	3.2780	1.7937	0.9852
B $\rightarrow$ S $\uparrow$	0.3458	0.2054	0.2166	0.2445	9.7938	5.2391	2.3279	1.1899
B $\rightarrow$ S $\downarrow$	0.3984	0.2550	0.2868	0.3342	6.7735	3.2777	1.7937	0.9852
CI $\rightarrow$ S $\uparrow$	0.1894	0.1468	0.1669	0.2125	9.6132	5.2195	2.3208	1.1916
CI $\rightarrow$ S $\downarrow$	0.1170	0.1167	0.1468	0.1935	6.4252	3.3213	1.8407	1.0235
A $\rightarrow$ CO $\uparrow$	0.3296	0.2141	0.2344	0.2792	9.4853	5.1388	2.2718	1.1549
A $\rightarrow$ CO $\downarrow$	0.3200	0.2070	0.2363	0.2792	5.6300	3.0412	1.6935	0.9420
B $\rightarrow$ CO $\uparrow$	0.3725	0.2458	0.2660	0.3108	9.3724	5.1377	2.2715	1.1547
B $\rightarrow$ CO $\downarrow$	0.3734	0.2264	0.2502	0.2864	5.2715	2.8228	1.5533	0.8427
CI $\rightarrow$ CO $\uparrow$	0.1391	0.1145	0.1332	0.1790	9.5380	5.1899	2.3044	1.1792
CI $\rightarrow$ CO $\downarrow$	0.1877	0.1488	0.1778	0.2221	5.9497	3.1368	1.7370	0.9633



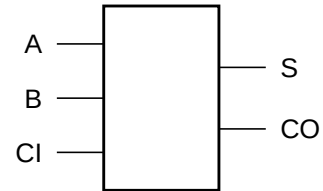
Cell Description

The ADDFH cell is a high-speed cell providing the arithmetic sum (S) and carry out (CO) of two operands (A, B) with carry in (CI). The two outputs (S, CO) are represented by the logic equations:

$$S = (A \oplus B \oplus CI)$$

$$CO = (A \oplus B) \bullet CI + (A \bullet B)$$

Logic Symbol



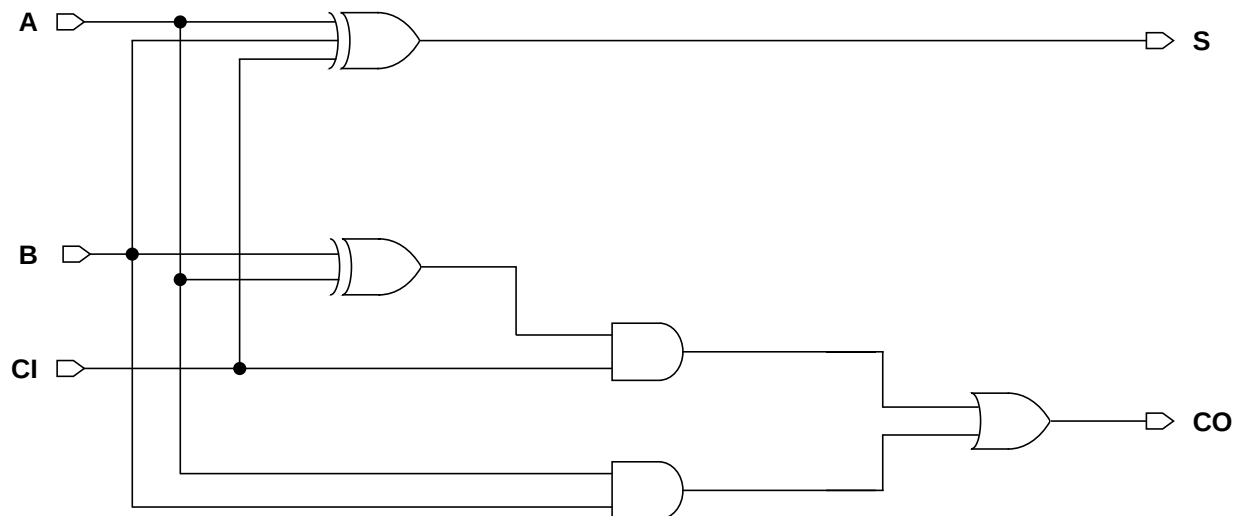
Functions

CI	A	B	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDFHXL	3.69	10.12
ADDFHX1	3.69	10.58
ADDFHX2	3.69	14.72
ADDFHX4	3.69	23.92

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0298	0.0414	0.0748	0.1281
B	0.0258	0.0357	0.0659	0.1199
CI	0.0170	0.0220	0.0362	0.0710

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0025	0.0043	0.0070	0.0132
B	0.0054	0.0090	0.0167	0.0308
CI	0.0016	0.0030	0.0047	0.0091

Delays (25°C, 1.2V, Typical Process)

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.3175	0.2038	0.1939	0.1717	9.4286	5.1538	2.2627	1.1328
A $\rightarrow$ S $\downarrow$	0.2849	0.2245	0.2105	0.1856	5.3829	2.9495	1.4998	0.7547
B $\rightarrow$ S $\uparrow$	0.2410	0.1512	0.1380	0.1226	9.5310	5.1773	2.2692	1.1363
B $\rightarrow$ S $\downarrow$	0.2700	0.1755	0.1594	0.1463	5.4166	2.9482	1.5003	0.7553
CI $\rightarrow$ S $\uparrow$	0.2770	0.1649	0.1414	0.1420	9.4875	5.1657	2.2645	1.1337
CI $\rightarrow$ S $\downarrow$	0.2350	0.1646	0.1436	0.1445	5.5465	2.9760	1.5073	0.7588
A $\rightarrow$ CO $\uparrow$	0.3265	0.2068	0.1935	0.1681	9.4949	5.1657	2.2673	1.1335
A $\rightarrow$ CO $\downarrow$	0.3106	0.2268	0.2090	0.1815	5.6479	2.9551	1.5024	0.7507
B $\rightarrow$ CO $\uparrow$	0.2028	0.1387	0.1243	0.1121	9.5552	5.1727	2.2696	1.1345
B $\rightarrow$ CO $\downarrow$	0.2912	0.1711	0.1497	0.1371	5.6642	2.8597	1.4767	0.7358
CI $\rightarrow$ CO $\uparrow$	0.1321	0.0930	0.0833	0.0791	9.6299	5.1880	2.2704	1.1352
CI $\rightarrow$ CO $\downarrow$	0.1970	0.1306	0.1120	0.1081	6.4724	3.0848	1.5364	0.7678



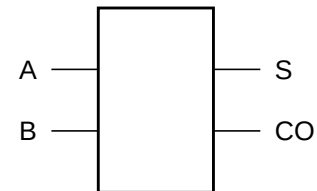
Cell Description

The ADDH cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B). The two outputs (S, CO) are represented by the logic equations:

$$S = (\bar{A} \bullet B) + (A \bullet \bar{B})$$

$$CO = A \bullet B$$

Logic Symbol



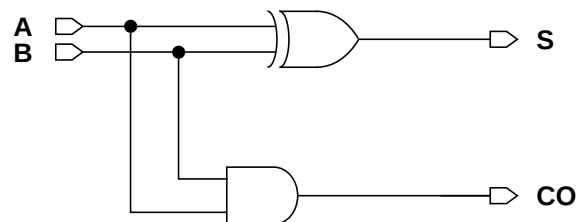
Functions

A	B	S	CO
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ADDHXL	3.69	5.06
ADDHX1	3.69	5.52
ADDHX2	3.69	8.74
ADDHX4	3.69	13.80

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0166	0.0335	0.0621	0.1199
B	0.0129	0.0183	0.0329	0.0614

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0025	0.0073	0.0143	0.0267
B	0.0039	0.0056	0.0087	0.0158

Delays (25°C, 1.2V, Typical Process)

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ S $\uparrow$	0.1196	0.0691	0.0637	0.0644	10.7408	2.9536	1.5130	0.7395
A $\rightarrow$ S $\downarrow$	0.1185	0.0777	0.0704	0.0681	5.6653	1.9402	1.0041	0.4823
B $\rightarrow$ S $\uparrow$	0.0646	0.0440	0.0412	0.0410	10.6347	3.0739	1.5350	0.7394
B $\rightarrow$ S $\downarrow$	0.0787	0.0708	0.0606	0.0588	5.1565	1.7175	0.9031	0.4331
A $\rightarrow$ CO $\uparrow$	0.0645	0.0689	0.0658	0.0643	9.3844	5.1565	2.2595	1.1311
A $\rightarrow$ CO $\downarrow$	0.0845	0.0999	0.0967	0.0844	4.6537	2.9439	1.4653	0.7229
B $\rightarrow$ CO $\uparrow$	0.0627	0.0677	0.0635	0.0615	9.3852	5.1495	2.2597	1.1309
B $\rightarrow$ CO $\downarrow$	0.0765	0.0927	0.0885	0.0767	4.6165	2.9231	1.4591	0.7205

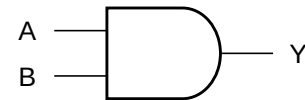


Cell Description

The AND2 cell provides the logical AND of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B)$$

Logic Symbol



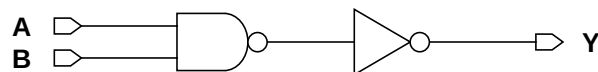
Functions

A	B	Y
0	x	0
x	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND2XL	3.69	1.84
AND2X1	3.69	1.84
AND2X2	3.69	1.84
AND2X4	3.69	2.76
AND2X6	3.69	4.14
AND2X8	3.69	4.60

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0046	0.0058	0.0088	0.0166	0.0243	0.0301
B	0.0052	0.0066	0.0105	0.0186	0.0281	0.0350

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0013	0.0015	0.0021	0.0041	0.0063	0.0080
B	0.0015	0.0017	0.0025	0.0044	0.0068	0.0084

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0732	0.0716	0.0641	0.0601	0.0575	0.0569
A $\rightarrow$ Y $\downarrow$	0.0794	0.0975	0.0932	0.0742	0.0742	0.0710
B $\rightarrow$ Y $\uparrow$	0.0781	0.0769	0.0688	0.0642	0.0616	0.0608
B $\rightarrow$ Y $\downarrow$	0.0891	0.1107	0.1058	0.0834	0.0843	0.0799

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.4143	5.1586	2.2610	1.1305	0.7525	0.5645
A $\rightarrow$ Y $\downarrow$	4.6801	2.7928	1.4684	0.7179	0.4781	0.3787
B $\rightarrow$ Y $\uparrow$	9.4156	5.1590	2.2611	1.1306	0.7526	0.5646
B $\rightarrow$ Y $\downarrow$	4.7133	2.8140	1.4764	0.7210	0.4804	0.3802

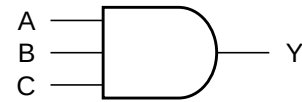


Cell Description

The AND3 cell provides the logical AND of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B \bullet C)$$

Logic Symbol



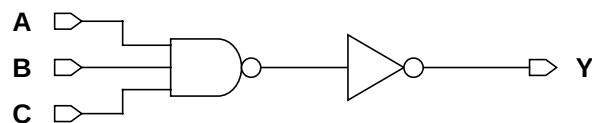
Functions

A	B	C	Y
0	x	x	0
x	0	x	0
x	x	0	0
1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND3XL	3.69	2.30
AND3X1	3.69	2.30
AND3X2	3.69	2.30
AND3X4	3.69	3.22
AND3X6	3.69	5.06
AND3X8	3.69	6.44

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0050	0.0063	0.0096	0.0169	0.0249	0.0329
B	0.0056	0.0073	0.0110	0.0195	0.0298	0.0380
C	0.0061	0.0080	0.0125	0.0219	0.0339	0.0439

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0014	0.0015	0.0024	0.0043	0.0069	0.0087
B	0.0016	0.0016	0.0024	0.0043	0.0073	0.0091
C	0.0014	0.0017	0.0028	0.0043	0.0077	0.0097

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.1093	0.0921	0.0798	0.0831	0.0701	0.0699
A $\rightarrow$ Y $\downarrow$	0.0939	0.1059	0.1052	0.0789	0.0805	0.0760
B $\rightarrow$ Y $\uparrow$	0.1158	0.0978	0.0861	0.0905	0.0773	0.0767
B $\rightarrow$ Y $\downarrow$	0.1022	0.1167	0.1179	0.0876	0.0917	0.0861
C $\rightarrow$ Y $\uparrow$	0.1181	0.1028	0.0915	0.0946	0.0811	0.0811
C $\rightarrow$ Y $\downarrow$	0.1077	0.1287	0.1318	0.0952	0.1015	0.0955

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.5501	5.2038	2.2845	1.1478	0.7633	0.5722
A $\rightarrow$ Y $\downarrow$	4.7990	2.8149	1.4957	0.7235	0.4821	0.3599
B $\rightarrow$ Y $\uparrow$	9.5505	5.2045	2.2846	1.1478	0.7634	0.5721
B $\rightarrow$ Y $\downarrow$	4.8333	2.8340	1.5053	0.7268	0.4848	0.3617
C $\rightarrow$ Y $\uparrow$	9.5445	5.2039	2.2848	1.1479	0.7632	0.5721
C $\rightarrow$ Y $\downarrow$	4.8659	2.8567	1.5167	0.7312	0.4881	0.3641

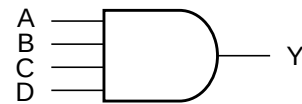


Cell Description

The AND4 cell provides the logical AND of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B \bullet C \bullet D)$$

Logic Symbol



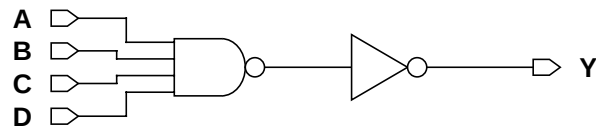
Functions

A	B	C	D	Y
0	x	x	x	0
x	0	x	x	0
x	x	0	x	0
x	x	x	0	0
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AND4XL	3.69	2.76
AND4X1	3.69	2.76
AND4X2	3.69	3.22
AND4X4	3.69	4.60
AND4X6	3.69	5.98
AND4X8	3.69	7.36

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0045	0.0057	0.0098	0.0179	0.0263	0.0345
B	0.0050	0.0067	0.0118	0.0206	0.0320	0.0412
C	0.0056	0.0075	0.0133	0.0232	0.0361	0.0473
D	0.0059	0.0084	0.0145	0.0265	0.0400	0.0524

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0014	0.0017	0.0026	0.0047	0.0073	0.0102
B	0.0013	0.0017	0.0027	0.0049	0.0076	0.0101
C	0.0014	0.0018	0.0027	0.0055	0.0081	0.0102
D	0.0015	0.0018	0.0027	0.0061	0.0086	0.0106

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.1227	0.0969	0.0879	0.0833	0.0826	0.0807
A $\rightarrow$ Y $\downarrow$	0.0892	0.1038	0.0986	0.0852	0.0868	0.0899
B $\rightarrow$ Y $\uparrow$	0.1303	0.1061	0.0978	0.0933	0.0928	0.0901
B $\rightarrow$ Y $\downarrow$	0.0962	0.1168	0.1121	0.0969	0.0993	0.1024
C $\rightarrow$ Y $\uparrow$	0.1376	0.1134	0.1043	0.1001	0.1001	0.0972
C $\rightarrow$ Y $\downarrow$	0.1038	0.1293	0.1238	0.1074	0.1108	0.1144
D $\rightarrow$ Y $\uparrow$	0.1413	0.1175	0.1087	0.1055	0.1048	0.1020
D $\rightarrow$ Y $\downarrow$	0.1097	0.1399	0.1348	0.1177	0.1210	0.1252

**Delays at 25°C, 1.2V, Typical Process**

Description	K _{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A → Y↑	9.6909	5.2488	2.3092	1.1562	0.7692	0.5773
A → Y↓	4.7983	2.8357	1.4806	0.7297	0.4867	0.3858
B → Y↑	9.6911	5.2492	2.3095	1.1562	0.7691	0.5773
B → Y↓	4.8332	2.8600	1.4903	0.7343	0.4896	0.3881
C → Y↑	9.6920	5.2495	2.3096	1.1563	0.7692	0.5773
C → Y↓	4.8748	2.8870	1.5021	0.7395	0.4932	0.3910
D → Y↑	9.6920	5.2498	2.3097	1.1563	0.7692	0.5773
D → Y↓	4.9206	2.9197	1.5169	0.7465	0.4981	0.3948

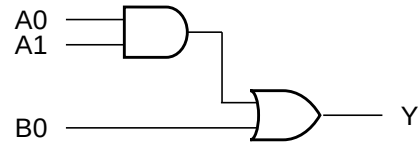
**Cell Description**

The AO21 cell provides the logical OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

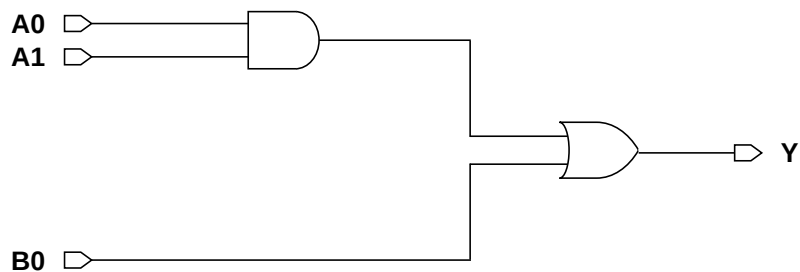
$$Y = (A0 \bullet A1) + B0$$

Functions

A0	A1	B0	Y
0	x	0	0
x	0	0	0
x	x	1	1
1	1	x	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
AO21XL	3.69	2.30
AO21X1	3.69	2.30
AO21X2	3.69	2.76
AO21X4	3.69	3.68

Functional Schematic

**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0054	0.0066	0.0105	0.0185
A1	0.0060	0.0072	0.0124	0.0209
B0	0.0051	0.0064	0.0104	0.0181

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0015	0.0016	0.0025	0.0047
A1	0.0013	0.0014	0.0025	0.0045
B0	0.0015	0.0016	0.0023	0.0042

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0766	0.0808	0.0717	0.0723	9.4497	5.1814	2.2717	1.2181
A0 $\rightarrow$ Y $\downarrow$	0.1315	0.1515	0.1517	0.1099	5.0448	3.0000	1.5790	0.7505
A1 $\rightarrow$ Y $\uparrow$	0.0789	0.0830	0.0754	0.0758	9.4516	5.1820	2.2718	1.2182
A1 $\rightarrow$ Y $\downarrow$	0.1453	0.1650	0.1572	0.1219	5.1107	3.0316	1.5666	0.7549
B0 $\rightarrow$ Y $\uparrow$	0.0619	0.0636	0.0577	0.0550	9.3860	5.1426	2.2532	1.2096
B0 $\rightarrow$ Y $\downarrow$	0.1253	0.1456	0.1386	0.1044	5.1095	3.0317	1.5665	0.7549

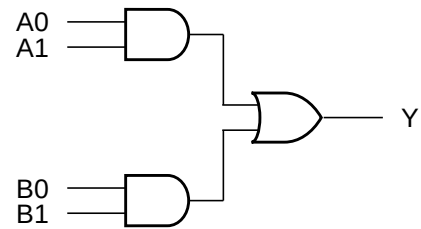
**Cell Description**

The AO22 cell provides the logical OR of two AND groups. The output (Y) is represented by the logic equation:

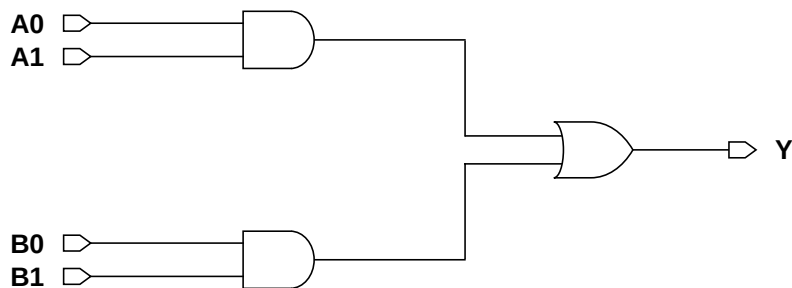
$$Y = (A0 \bullet A1) + (B0 \bullet B1)$$

Functions

A0	A1	B0	B1	Y
0	x	0	x	0
0	x	x	0	0
x	0	0	x	0
x	0	x	0	0
x	x	1	1	1
1	1	x	x	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
AO22XL	3.69	2.76
AO22X1	3.69	2.76
AO22X2	3.69	3.22
AO22X4	3.69	4.14

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0056	0.0067	0.0110	0.0192
A1	0.0062	0.0075	0.0123	0.0218
B0	0.0066	0.0079	0.0129	0.0230
B1	0.0073	0.0086	0.0143	0.0255

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0016	0.0017	0.0027	0.0048
A1	0.0016	0.0016	0.0027	0.0049
B0	0.0015	0.0014	0.0026	0.0047
B1	0.0013	0.0013	0.0024	0.0046

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0772	0.0808	0.0722	0.0705	9.4139	5.1732	2.2729	1.2198
A0 $\rightarrow$ Y $\downarrow$	0.1350	0.1543	0.1550	0.1131	5.2944	3.1122	1.6506	0.7678
A1 $\rightarrow$ Y $\uparrow$	0.0803	0.0837	0.0761	0.0746	9.4061	5.1718	2.2731	1.2198
A1 $\rightarrow$ Y $\downarrow$	0.1499	0.1689	0.1730	0.1258	5.3623	3.1447	1.6636	0.7722
B0 $\rightarrow$ Y $\uparrow$	0.0916	0.0951	0.0847	0.0863	9.4861	5.2072	2.2839	1.2230
B0 $\rightarrow$ Y $\downarrow$	0.1704	0.1878	0.1843	0.1366	5.3027	3.1149	1.6502	0.7680
B1 $\rightarrow$ Y $\uparrow$	0.0936	0.0979	0.0875	0.0898	9.4860	5.2072	2.2840	1.2229
B1 $\rightarrow$ Y $\downarrow$	0.1824	0.2018	0.1995	0.1483	5.3751	3.1490	1.6638	0.7723



Cell Description

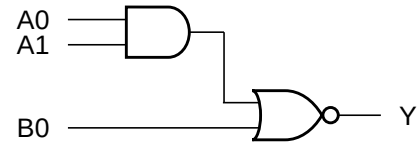
The AOI21 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + B0}$$

Functions

A0	A1	B0	Y
0	x	0	1
x	0	0	1
x	x	1	0
1	1	x	0

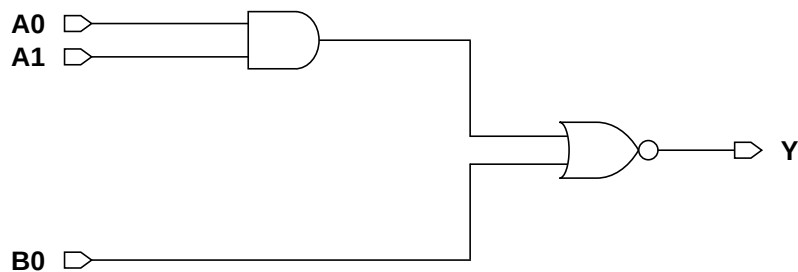
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI21XL	3.69	2.30
AOI21X1	3.69	2.30
AOI21X2	3.69	3.68
AOI21X4	3.69	5.06

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0044	0.0069	0.0143	0.0268
A1	0.0054	0.0085	0.0173	0.0339
B0	0.0043	0.0069	0.0139	0.0266

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0021	0.0033	0.0069	0.0123
A1	0.0020	0.0031	0.0062	0.0122
B0	0.0019	0.0031	0.0058	0.0108

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0623	0.0513	0.0513	0.0473	14.4533	7.2857	3.6455	1.7791
A0 $\rightarrow$ Y $\downarrow$	0.0274	0.0276	0.0274	0.0271	5.3269	3.1897	1.5968	0.8214
A1 $\rightarrow$ Y $\uparrow$	0.0758	0.0636	0.0629	0.0598	14.4052	7.2698	3.6354	1.7762
A1 $\rightarrow$ Y $\downarrow$	0.0301	0.0308	0.0303	0.0307	5.3277	3.1903	1.5970	0.8215
B0 $\rightarrow$ Y $\uparrow$	0.0545	0.0455	0.0438	0.0426	14.4523	7.2869	3.6422	1.7787
B0 $\rightarrow$ Y $\downarrow$	0.0192	0.0186	0.0190	0.0185	4.3677	2.5732	1.3762	0.6876



Cell Description

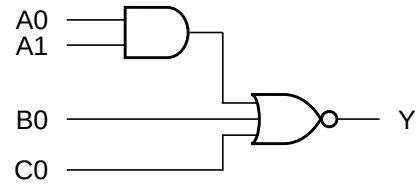
The AOI211 cell provides the logical inverted OR of one AND group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + B0 + C0}$$

Functions

A0	A1	B0	C0	Y
0	x	0	0	1
x	0	0	0	1
x	x	x	1	0
x	x	1	x	0
1	1	x	x	0

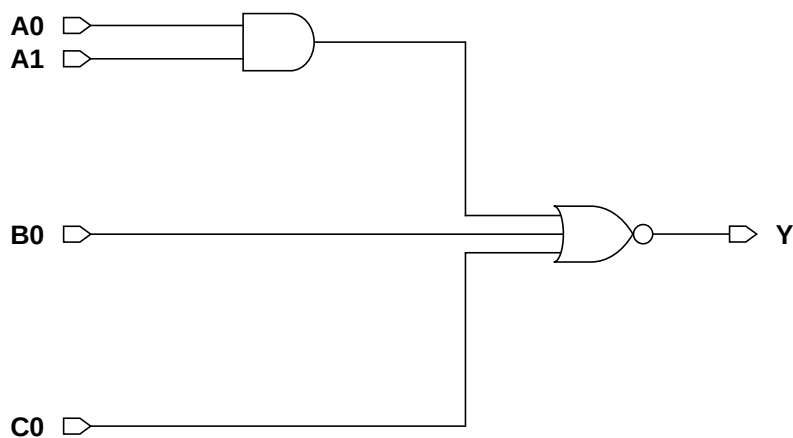
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI211XL	3.69	2.30
AOI211X1	3.69	2.30
AOI211X2	3.69	4.14
AOI211X4	3.69	4.14

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0066	0.0105	0.0237	0.0235
A1	0.0076	0.0123	0.0272	0.0248
B0	0.0053	0.0084	0.0183	0.0213
C0	0.0062	0.0098	0.0218	0.0232

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0023	0.0035	0.0075	0.0024
A1	0.0021	0.0033	0.0071	0.0022
B0	0.0022	0.0036	0.0065	0.0023
C0	0.0021	0.0033	0.0070	0.0022

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1001	0.0815	0.0810	0.2340	17.7614	8.8782	4.0110	1.1265
A0 $\rightarrow$ Y $\downarrow$	0.0353	0.0362	0.0382	0.1405	5.6136	3.3643	1.6366	0.7139
A1 $\rightarrow$ Y $\uparrow$	0.1163	0.0962	0.0942	0.2511	17.7076	8.8582	4.0022	1.1265
A1 $\rightarrow$ Y $\downarrow$	0.0381	0.0397	0.0413	0.1436	5.6109	3.3642	1.6364	0.7140
B0 $\rightarrow$ Y $\uparrow$	0.0732	0.0597	0.0555	0.2082	17.7691	8.8837	4.0081	1.1265
B0 $\rightarrow$ Y $\downarrow$	0.0228	0.0220	0.0236	0.1136	4.3704	2.5741	1.3771	0.7123
C0 $\rightarrow$ Y $\uparrow$	0.0969	0.0790	0.0777	0.2323	17.7317	8.8654	4.0058	1.1266
C0 $\rightarrow$ Y $\downarrow$	0.0276	0.0268	0.0307	0.1208	4.4101	2.5905	1.3867	0.7123



Cell Description

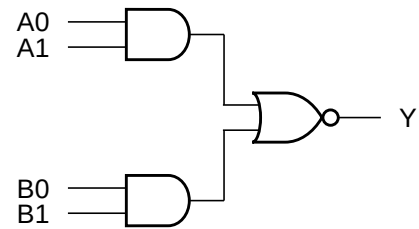
The AOI22 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1)}$$

Functions

A0	A1	B0	B1	Y
0	x	0	x	1
0	x	x	0	1
x	0	0	x	1
x	0	x	0	1
x	x	1	1	0
1	1	x	x	0

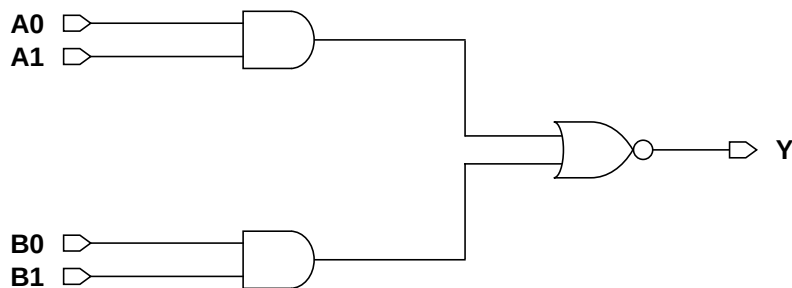
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI22XL	3.69	2.76
AOI22X1	3.69	2.76
AOI22X2	3.69	4.60
AOI22X4	3.69	6.44

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0048	0.0080	0.0155	0.0312
A1	0.0058	0.0097	0.0191	0.0375
B0	0.0064	0.0104	0.0200	0.0386
B1	0.0074	0.0120	0.0235	0.0451

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0020	0.0034	0.0065	0.0130
A1	0.0022	0.0034	0.0070	0.0125
B0	0.0021	0.0033	0.0062	0.0122
B1	0.0020	0.0032	0.0066	0.0121

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0615	0.0527	0.0520	0.0508	14.7399	7.2953	3.6473	1.7558
A0 $\rightarrow$ Y $\downarrow$	0.0243	0.0239	0.0236	0.0242	5.3784	3.1532	1.5762	0.8133
A1 $\rightarrow$ Y $\uparrow$	0.0773	0.0658	0.0655	0.0620	14.7173	7.2834	3.6415	1.7521
A1 $\rightarrow$ Y $\downarrow$	0.0282	0.0277	0.0274	0.0274	5.3850	3.1560	1.5778	0.8137
B0 $\rightarrow$ Y $\uparrow$	0.0937	0.0786	0.0761	0.0720	14.7235	7.2873	3.6435	1.7531
B0 $\rightarrow$ Y $\downarrow$	0.0383	0.0386	0.0376	0.0382	5.4768	3.1951	1.5955	0.8213
B1 $\rightarrow$ Y $\uparrow$	0.1070	0.0907	0.0890	0.0836	14.6856	7.2725	3.6369	1.7508
B1 $\rightarrow$ Y $\downarrow$	0.0411	0.0418	0.0412	0.0416	5.4766	3.1955	1.5961	0.8212



Cell Description

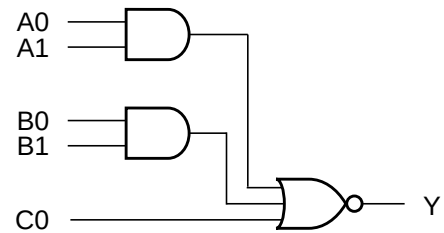
The AOI221 cell provides the logical inverted OR of two AND groups and a third input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + C0}$$

Functions

A0	A1	B0	B1	C0	Y
0	x	0	x	0	1
0	x	x	0	0	1
x	0	0	x	0	1
x	0	x	0	0	1
x	x	x	x	1	0
x	x	1	1	x	0
1	1	x	x	x	0

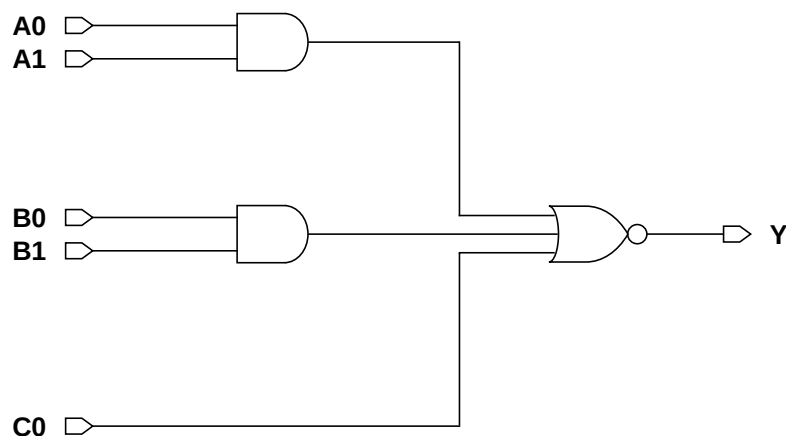
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI221XL	3.69	3.22
AOI221X1	3.69	3.68
AOI221X2	3.69	5.98
AOI221X4	3.69	5.52

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0068	0.0115	0.0225	0.0238
A1	0.0078	0.0131	0.0261	0.0250
B0	0.0087	0.0145	0.0283	0.0259
B1	0.0096	0.0162	0.0320	0.0265
C0	0.0065	0.0108	0.0211	0.0231

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0022	0.0037	0.0070	0.0024
A1	0.0022	0.0036	0.0076	0.0025
B0	0.0023	0.0038	0.0071	0.0024
B1	0.0021	0.0037	0.0075	0.0023
C0	0.0021	0.0035	0.0066	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1206	0.0906	0.0867	0.2538	17.7769	8.0284	4.0149	1.1273
A0 $\rightarrow$ Y $\downarrow$	0.0375	0.0375	0.0357	0.1445	5.5128	3.2084	1.6050	0.7152
A1 $\rightarrow$ Y $\uparrow$	0.1372	0.1042	0.1016	0.2713	17.7304	8.0110	4.0077	1.1276
A1 $\rightarrow$ Y $\downarrow$	0.0406	0.0409	0.0397	0.1478	5.5157	3.2074	1.6052	0.7151
B0 $\rightarrow$ Y $\uparrow$	0.1449	0.1106	0.1063	0.2758	17.7703	8.0268	4.0148	1.1275
B0 $\rightarrow$ Y $\downarrow$	0.0406	0.0438	0.0416	0.1536	5.8003	3.3483	1.6763	0.7162
B1 $\rightarrow$ Y $\uparrow$	0.1608	0.1238	0.1206	0.2926	17.7199	8.0100	4.0069	1.1275
B1 $\rightarrow$ Y $\downarrow$	0.0431	0.0469	0.0452	0.1563	5.7990	3.3480	1.6765	0.7161
C0 $\rightarrow$ Y $\uparrow$	0.0844	0.0671	0.0626	0.2252	17.7700	8.0268	4.0127	1.1276
C0 $\rightarrow$ Y $\downarrow$	0.0238	0.0236	0.0218	0.1198	4.3949	2.5837	1.2934	0.7137



Cell Description

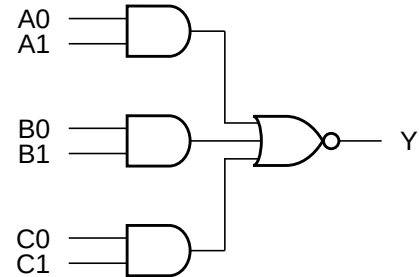
The AOI222 cell provides the logical inverted OR of three AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1) + (B0 \bullet B1) + (C0 \bullet C1)}$$

Functions

A0	A1	B0	B1	C0	C1	Y
0	x	0	x	0	x	1
0	x	0	x	x	0	1
0	x	x	0	0	x	1
0	x	x	0	x	0	1
x	0	0	x	0	x	1
x	0	0	x	x	0	1
x	0	x	0	0	x	1
x	0	x	0	x	0	1
x	x	x	x	1	1	0
x	x	1	1	x	x	0
1	1	x	x	x	x	0

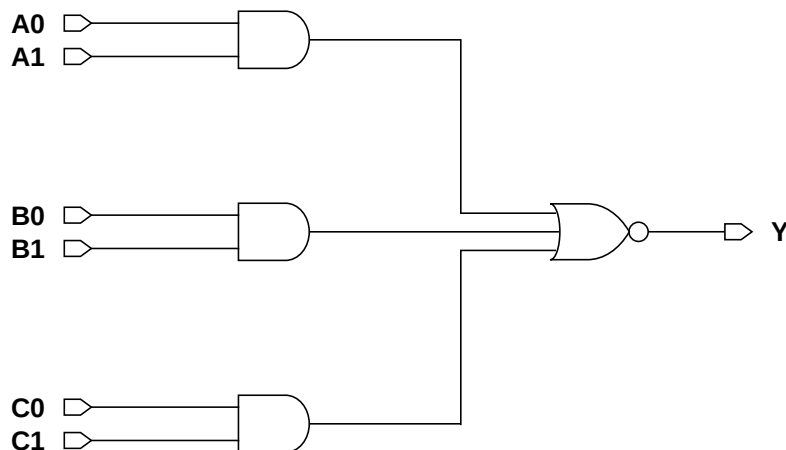
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI222XL	3.69	3.68
AOI222X1	3.69	4.14
AOI222X2	3.69	7.36
AOI222X4	3.69	5.98

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0074	0.0125	0.0230	0.0240
A1	0.0085	0.0143	0.0266	0.0251
B0	0.0091	0.0155	0.0286	0.0259
B1	0.0101	0.0171	0.0322	0.0268
C0	0.0110	0.0185	0.0347	0.0276
C1	0.0119	0.0202	0.0381	0.0291

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0022	0.0038	0.0073	0.0025
A1	0.0024	0.0039	0.0078	0.0026
B0	0.0023	0.0038	0.0071	0.0024
B1	0.0023	0.0037	0.0074	0.0024
C0	0.0022	0.0037	0.0071	0.0023
C1	0.0023	0.0037	0.0073	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0934	0.0763	0.0718	0.2400	17.7713	8.0300	4.0161	1.1280
A0 $\rightarrow$ Y $\downarrow$	0.0315	0.0311	0.0291	0.1414	5.4140	3.1728	1.5868	0.7156
A1 $\rightarrow$ Y $\uparrow$	0.1129	0.0910	0.0864	0.2589	17.7504	8.0199	4.0105	1.1281
A1 $\rightarrow$ Y $\downarrow$	0.0355	0.0349	0.0328	0.1452	5.4204	3.1774	1.5886	0.7156
B0 $\rightarrow$ Y $\uparrow$	0.1663	0.1277	0.1189	0.2985	17.7646	8.0263	4.0136	1.1280
B0 $\rightarrow$ Y $\downarrow$	0.0505	0.0515	0.0481	0.1601	5.4875	3.1946	1.5961	0.7164
B1 $\rightarrow$ Y $\uparrow$	0.1833	0.1408	0.1326	0.3156	17.7264	8.0122	4.0065	1.1282
B1 $\rightarrow$ Y $\downarrow$	0.0538	0.0547	0.0516	0.1632	5.4885	3.1948	1.5962	0.7163
C0 $\rightarrow$ Y $\uparrow$	0.1908	0.1473	0.1383	0.3207	17.7611	8.0251	4.0129	1.1281
C0 $\rightarrow$ Y $\downarrow$	0.0602	0.0638	0.0604	0.1723	5.7027	3.2942	1.6467	0.7162
C1 $\rightarrow$ Y $\uparrow$	0.2076	0.1608	0.1523	0.3380	17.7225	8.0115	4.0066	1.1282
C1 $\rightarrow$ Y $\downarrow$	0.0636	0.0674	0.0640	0.1758	5.7031	3.2941	1.6466	0.7163



Cell Description

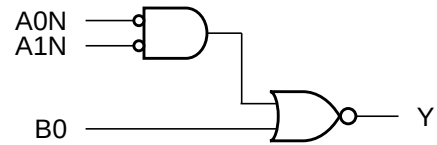
The AOI2BB1 cell provides the logical inverted OR of one AND group of two inverted inputs (A0N, A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A0N} \bullet \overline{A1N}) + B0}$$

Functions

A0N	A1N	B0	Y
1	x	0	1
x	1	0	1
x	x	1	0
0	0	x	0

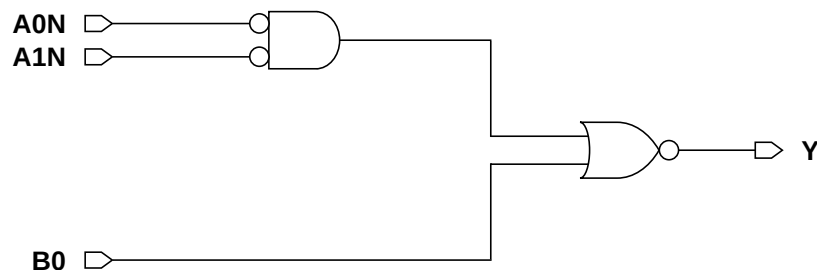
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI2BB1XL	3.69	2.30
AOI2BB1X1	3.69	2.30
AOI2BB1X2	3.69	2.76
AOI2BB1X4	3.69	4.60

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0051	0.0067	0.0101	0.0215
A1N	0.0056	0.0075	0.0115	0.0244
B0	0.0037	0.0059	0.0109	0.0222

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0015	0.0017	0.0026	0.0052
A1N	0.0015	0.0017	0.0029	0.0057
B0	0.0018	0.0030	0.0061	0.0108

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0739	0.0684	0.0616	0.0626	14.7461	7.2934	3.6426	1.7238
A0N $\rightarrow$ Y $\downarrow$	0.1246	0.1366	0.1165	0.1124	5.3612	2.9625	1.5266	0.7539
A1N $\rightarrow$ Y $\uparrow$	0.0785	0.0729	0.0676	0.0685	14.7502	7.2964	3.6438	1.7247
A1N $\rightarrow$ Y $\downarrow$	0.1361	0.1472	0.1282	0.1229	5.3619	2.9633	1.5267	0.7538
B0 $\rightarrow$ Y $\uparrow$	0.0502	0.0423	0.0376	0.0369	14.6797	7.2681	3.6358	1.7197
B0 $\rightarrow$ Y $\downarrow$	0.0220	0.0214	0.0210	0.0214	4.7017	2.5833	1.3812	0.6901



Cell Description

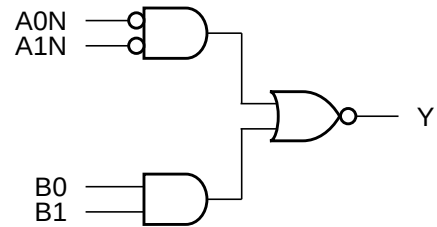
The AOI2BB2 cell provides the logical inverted OR of one AND group of two inverted inputs (A0N, A1N) and one AND group of two non-inverted inputs (B0, B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(A0N \bullet A1N)} + (B0 \bullet B1)$$

Functions

A0N	A1N	B0	B1	Y
1	x	0	x	1
1	x	x	0	1
x	1	0	x	1
x	1	x	0	1
x	x	1	1	0
0	0	x	x	0

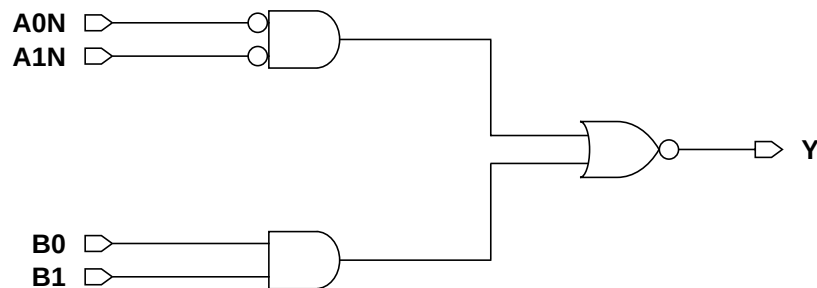
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI2BB2XL	3.69	2.76
AOI2BB2X1	3.69	2.76
AOI2BB2X2	3.69	4.14
AOI2BB2X4	3.69	6.90

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0049	0.0059	0.0103	0.0192
A1N	0.0055	0.0064	0.0116	0.0222
B0	0.0043	0.0064	0.0134	0.0268
B1	0.0052	0.0077	0.0173	0.0327

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0015	0.0017	0.0029	0.0053
A1N	0.0013	0.0014	0.0028	0.0056
B0	0.0022	0.0030	0.0069	0.0121
B1	0.0020	0.0031	0.0073	0.0113

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0652	0.0611	0.0620	0.0573	10.9616	6.2610	2.9303	1.3081
A0N $\rightarrow$ Y $\downarrow$	0.1233	0.1332	0.1167	0.1058	5.0519	2.9487	1.5387	0.7430
A1N $\rightarrow$ Y $\uparrow$	0.0684	0.0641	0.0669	0.0631	10.9673	6.2646	2.9328	1.3094
A1N $\rightarrow$ Y $\downarrow$	0.1308	0.1406	0.1259	0.1161	5.0528	2.9479	1.5390	0.7428
B0 $\rightarrow$ Y $\uparrow$	0.0637	0.0542	0.0522	0.0468	14.7462	8.3903	3.9006	1.7234
B0 $\rightarrow$ Y $\downarrow$	0.0280	0.0276	0.0275	0.0292	5.4599	3.5057	1.6966	0.8951
B1 $\rightarrow$ Y $\uparrow$	0.0766	0.0680	0.0653	0.0610	14.8437	8.5245	3.8926	1.8482
B1 $\rightarrow$ Y $\downarrow$	0.0301	0.0307	0.0311	0.0324	5.4586	3.5060	1.6979	0.8950



Cell Description

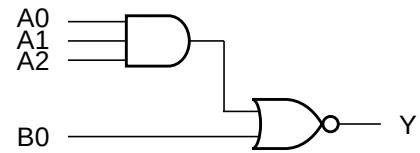
The AOI31 cell provides the logical inverted OR of one AND group and an additional input. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + B0}$$

Functions

A0	A1	A2	B0	Y
0	x	x	0	1
x	0	x	0	1
x	x	0	0	1
x	x	x	1	0
1	1	1	x	0

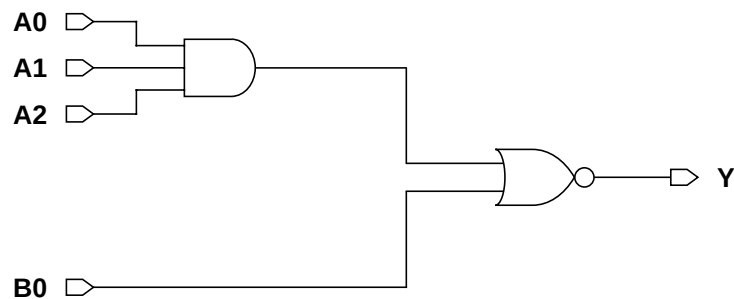
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI31XL	3.69	2.76
AOI31X1	3.69	2.76
AOI31X2	3.69	4.60
AOI31X4	3.69	4.60

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0049	0.0077	0.0162	0.0216
A1	0.0060	0.0097	0.0199	0.0228
A2	0.0070	0.0115	0.0235	0.0243
B0	0.0056	0.0094	0.0188	0.0227

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0021	0.0035	0.0076	0.0022
A1	0.0021	0.0035	0.0071	0.0022
A2	0.0020	0.0034	0.0066	0.0021
B0	0.0019	0.0031	0.0057	0.0021

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0687	0.0576	0.0587	0.1908	14.7527	7.2967	3.6491	1.1262
A0 $\rightarrow$ Y $\downarrow$	0.0335	0.0339	0.0346	0.1403	6.6863	3.8903	1.9773	0.7150
A1 $\rightarrow$ Y $\uparrow$	0.0854	0.0727	0.0726	0.2108	14.7082	7.2802	3.6408	1.1263
A1 $\rightarrow$ Y $\downarrow$	0.0397	0.0405	0.0404	0.1471	6.6864	3.8889	1.9764	0.7150
A2 $\rightarrow$ Y $\uparrow$	0.0992	0.0849	0.0851	0.2279	14.7320	7.2982	3.6500	1.1264
A2 $\rightarrow$ Y $\downarrow$	0.0425	0.0436	0.0440	0.1502	6.6840	3.8892	1.9765	0.7150
B0 $\rightarrow$ Y $\uparrow$	0.0739	0.0630	0.0600	0.2053	14.7819	7.3163	3.6575	1.1263
B0 $\rightarrow$ Y $\downarrow$	0.0195	0.0193	0.0196	0.1141	4.3695	2.5719	1.3765	0.7134



Cell Description

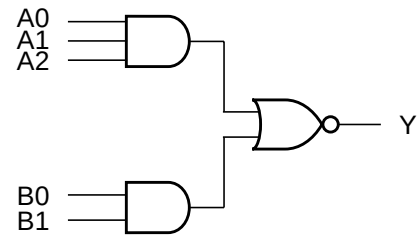
The AOI32 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1)}$$

Functions

A0	A1	A2	B0	B1	Y
0	x	x	0	x	1
0	x	x	x	0	1
x	0	x	0	x	1
x	0	x	x	0	1
x	x	0	0	x	1
x	x	0	x	0	1
x	x	x	1	1	0
1	1	1	x	x	0

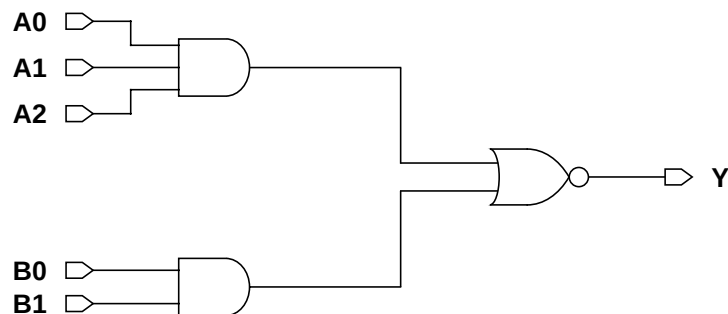
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AOI32XL	3.69	3.22
AOI32X1	3.69	3.22
AOI32X2	3.69	5.06
AOI32X4	3.69	4.60

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0070	0.0113	0.0220	0.0233
A1	0.0081	0.0134	0.0257	0.0249
A2	0.0092	0.0154	0.0293	0.0264
B0	0.0064	0.0109	0.0201	0.0229
B1	0.0074	0.0126	0.0240	0.0244

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0022	0.0035	0.0065	0.0024
A1	0.0022	0.0034	0.0069	0.0023
A2	0.0021	0.0033	0.0071	0.0021
B0	0.0021	0.0034	0.0060	0.0025
B1	0.0022	0.0034	0.0069	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1021	0.0854	0.0827	0.2373	14.7380	7.2941	3.6458	1.1266
A0 $\rightarrow$ Y $\downarrow$	0.0482	0.0479	0.0469	0.1613	6.6824	3.8880	1.9674	0.7162
A1 $\rightarrow$ Y $\uparrow$	0.1181	0.1009	0.0979	0.2545	14.7019	7.2807	3.6398	1.1268
A1 $\rightarrow$ Y $\downarrow$	0.0542	0.0549	0.0540	0.1667	6.6822	3.8867	1.9673	0.7161
A2 $\rightarrow$ Y $\uparrow$	0.1322	0.1141	0.1111	0.2713	14.7181	7.2973	3.6479	1.1269
A2 $\rightarrow$ Y $\downarrow$	0.0570	0.0584	0.0576	0.1697	6.6821	3.8871	1.9675	0.7162
B0 $\rightarrow$ Y $\uparrow$	0.0837	0.0728	0.0687	0.2201	14.7839	7.3229	3.6477	1.1268
B0 $\rightarrow$ Y $\downarrow$	0.0257	0.0249	0.0240	0.1338	5.3868	3.1547	1.5762	0.7156
B1 $\rightarrow$ Y $\uparrow$	0.0995	0.0860	0.0832	0.2344	14.7556	7.3092	3.6525	1.1269
B1 $\rightarrow$ Y $\downarrow$	0.0295	0.0287	0.0282	0.1363	5.3931	3.1575	1.5781	0.7155

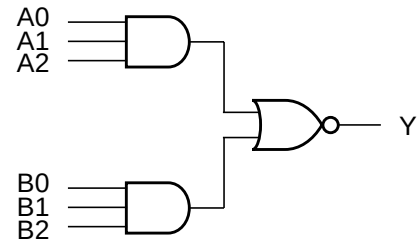
**Cell Description**

The AOI33 cell provides the logical inverted OR of two AND groups. The output (Y) is represented by the logic equation:

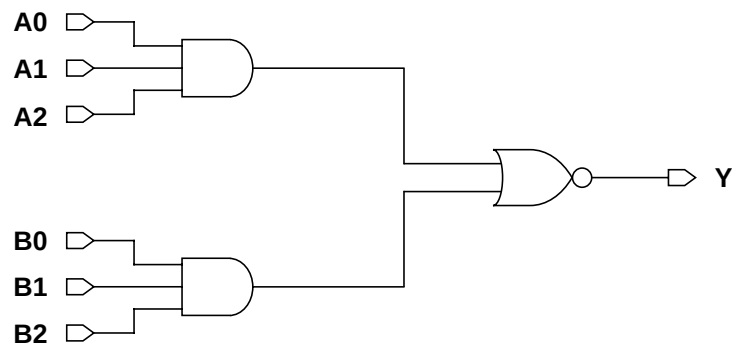
$$Y = \overline{(A0 \bullet A1 \bullet A2) + (B0 \bullet B1 \bullet B2)}$$

Functions

A0	A1	A2	B0	B1	B2	Y
0	x	x	0	x	x	1
0	x	x	x	0	x	1
0	x	x	x	x	0	1
x	0	x	0	x	x	1
x	0	x	x	0	x	1
x	0	x	x	x	0	1
x	x	0	0	x	x	1
x	x	0	x	0	x	1
x	x	0	x	x	0	1
x	x	x	1	1	1	0
1	1	1	x	x	x	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
AOI33XL	3.69	3.68
AOI33X1	3.69	3.68
AOI33X2	3.69	6.44
AOI33X4	3.69	5.06

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0070	0.0118	0.0222	0.0231
A1	0.0081	0.0137	0.0262	0.0253
A2	0.0092	0.0156	0.0299	0.0265
B0	0.0089	0.0149	0.0286	0.0256
B1	0.0099	0.0168	0.0325	0.0269
B2	0.0110	0.0185	0.0361	0.0281

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0021	0.0035	0.0068	0.0024
A1	0.0023	0.0035	0.0073	0.0025
A2	0.0023	0.0036	0.0076	0.0025
B0	0.0021	0.0035	0.0065	0.0024
B1	0.0022	0.0035	0.0071	0.0023
B2	0.0021	0.0034	0.0074	0.0022

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0965	0.0828	0.0765	0.2298	14.7743	7.3139	3.6584	1.1272
A0 $\rightarrow$ Y $\downarrow$	0.0342	0.0331	0.0311	0.1524	6.5872	3.8439	1.9481	0.7176
A1 $\rightarrow$ Y $\uparrow$	0.1138	0.0979	0.0923	0.2498	14.7465	7.3027	3.6528	1.1273
A1 $\rightarrow$ Y $\downarrow$	0.0408	0.0399	0.0383	0.1593	6.5913	3.8434	1.9481	0.7178
A2 $\rightarrow$ Y $\uparrow$	0.1311	0.1125	0.1063	0.2692	14.7625	7.3128	3.6581	1.1274
A2 $\rightarrow$ Y $\downarrow$	0.0452	0.0440	0.0422	0.1637	6.5960	3.8477	1.9494	0.7175
B0 $\rightarrow$ Y $\uparrow$	0.1343	0.1143	0.1101	0.2733	14.7613	7.3110	3.6562	1.1273
B0 $\rightarrow$ Y $\downarrow$	0.0607	0.0612	0.0601	0.1760	6.6565	3.8756	1.9679	0.7167
B1 $\rightarrow$ Y $\uparrow$	0.1511	0.1290	0.1254	0.2920	14.7318	7.2974	3.6504	1.1274
B1 $\rightarrow$ Y $\downarrow$	0.0672	0.0676	0.0670	0.1822	6.6566	3.8755	1.9679	0.7167
B2 $\rightarrow$ Y $\uparrow$	0.1653	0.1418	0.1391	0.3091	14.7418	7.3075	3.6548	1.1274
B2 $\rightarrow$ Y $\downarrow$	0.0699	0.0708	0.0712	0.1855	6.6568	3.8753	1.9680	0.7167

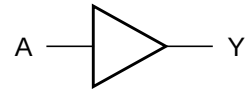
**Cell Description**

The BUF cell provides the logical buffer of a single input (A). The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
BUFX2	3.69	1.84
BUFX3	3.69	2.30
BUFX4	3.69	2.30
BUFX6	3.69	3.22
BUFX8	3.69	3.68
BUFX12	3.69	4.60
BUFX16	3.69	6.44
BUFX20	3.69	7.82

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)							
	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0091	0.0118	0.0155	0.0222	0.0302	0.0422	0.0554	0.0708

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0024	0.0031	0.0039	0.0058	0.0073	0.0095	0.0141	0.0178

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0547	0.0484	0.0486	0.0457	0.0458	0.0485	0.0458	0.0457
A $\rightarrow$ Y $\downarrow$	0.0847	0.0705	0.0678	0.0676	0.0637	0.0659	0.0616	0.0615

Description	K_{load} (ns/pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	2.4899	1.5552	1.1239	0.7487	0.5618	0.3792	0.2980	0.2250
A $\rightarrow$ Y $\downarrow$	1.3727	0.9330	0.7116	0.4739	0.3543	0.2400	0.1915	0.1440



Cell Description

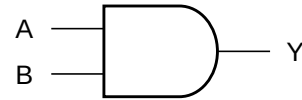
The CLKAND2 cell provides the logical AND of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = A \bullet B$$

Functions

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

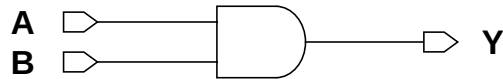
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKAND2X2	3.69	2.30
CLKAND2X3	3.69	2.30
CLKAND2X4	3.69	2.76
CLKAND2X6	3.69	3.22
CLKAND2X8	3.69	4.14
CLKAND2X12	3.69	5.98

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	X2	X3	X4	X6	X8	X12
A	0.0078	0.0098	0.0135	0.0200	0.0256	0.0382
B	0.0086	0.0108	0.0144	0.0211	0.0274	0.0408

Pin Capacitance

Pin	Capacitance (pF)					
	X2	X3	X4	X6	X8	X12
A	0.0020	0.0026	0.0032	0.0043	0.0061	0.0102
B	0.0021	0.0027	0.0035	0.0046	0.0067	0.0100

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X2	X3	X4	X6	X8	X12
A $\rightarrow$ Y $\uparrow$	0.0775	0.0760	0.0777	0.0781	0.0841	0.0772
A $\rightarrow$ Y $\downarrow$	0.0789	0.0699	0.0700	0.0710	0.0633	0.0613
B $\rightarrow$ Y $\uparrow$	0.0804	0.0799	0.0821	0.0827	0.0883	0.0798
B $\rightarrow$ Y $\downarrow$	0.0839	0.0749	0.0750	0.0763	0.0673	0.0645

Delays at 25°C, 1.2V, Typical Process

Description	K_{load} (ns/pF)					
	X2	X3	X4	X6	X8	X12
A $\rightarrow$ Y $\uparrow$	2.2780	1.5760	1.1414	0.7611	0.6177	0.4281
A $\rightarrow$ Y $\downarrow$	2.4499	1.7079	1.3541	0.8541	0.6463	0.4304
B $\rightarrow$ Y $\uparrow$	2.2779	1.5758	1.1413	0.7611	0.6176	0.4280
B $\rightarrow$ Y $\downarrow$	2.4537	1.7103	1.3556	0.8551	0.6468	0.4308



Cell Description

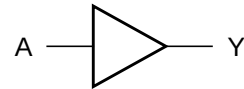
The CLKBUF cell provides the logical buffer of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKBUFX2	3.69	1.84
CLKBUFX3	3.69	1.84
CLKBUFX4	3.69	2.30
CLKBUFX6	3.69	2.76
CLKBUFX8	3.69	3.22
CLKBUFX12	3.69	6.90
CLKBUFX16	3.69	10.58
CLKBUFX20	3.69	13.34

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)							
	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0072	0.0094	0.0121	0.0184	0.0215	0.0595	0.0803	0.1020

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0019	0.0023	0.0027	0.0038	0.0051	0.0130	0.0179	0.0218

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0660	0.0656	0.0646	0.0658	0.0631	0.0624	0.0615	0.0641
A $\rightarrow$ Y $\downarrow$	0.0874	0.0846	0.0744	0.0698	0.0665	0.0616	0.0632	0.0644

Description	K_{load} (ns/pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	2.4509	1.5617	1.1293	0.7530	0.6404	0.2219	0.1610	0.1235
A $\rightarrow$ Y $\downarrow$	2.4636	1.7184	1.2242	0.8422	0.6289	0.2416	0.1760	0.1409



Cell Description

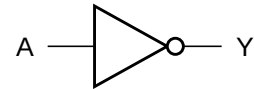
The CLKINV cell provides the logical inversion of a single input (A), with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$$Y = \bar{A}$$

Functions

A	Y
0	1
1	0

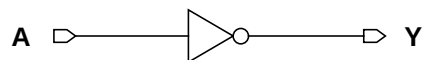
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKINVX1	3.69	0.92
CLKINVX2	3.69	1.38
CLKINVX3	3.69	1.38
CLKINVX4	3.69	1.84
CLKINVX6	3.69	2.30
CLKINVX8	3.69	2.76
CLKINVX12	3.69	5.98
CLKINVX16	3.69	7.36
CLKINVX20	3.69	8.74

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)								
	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0041	0.0056	0.0082	0.0107	0.0164	0.0218	0.0592	0.0802	0.0999

Pin Capacitance

Pin	Capacitance (pF)								
	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0027	0.0038	0.0055	0.0072	0.0111	0.0149	0.0404	0.0542	0.0678

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)								
	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0174	0.0162	0.0157	0.0151	0.0155	0.0154	0.0147	0.0147	0.0146
A $\rightarrow$ Y $\downarrow$	0.0178	0.0177	0.0175	0.0179	0.0169	0.0171	0.0167	0.0169	0.0168

Description	K_{load} (ns/pF)								
	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	3.4339	2.2360	1.5473	1.1178	0.7454	0.5590	0.2026	0.1489	0.1191
A $\rightarrow$ Y $\downarrow$	3.4031	2.4101	1.7107	1.3696	0.8036	0.6219	0.2290	0.1727	0.1374



Cell Description

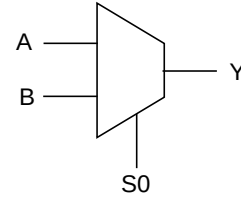
The CLKMUX2 cell is a non-inverting 2 to 1 multiplexer with balanced delays for clock signals. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (S0 \bullet B) + (\overline{S0} \bullet A)$$

Functions

S0	A	B	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

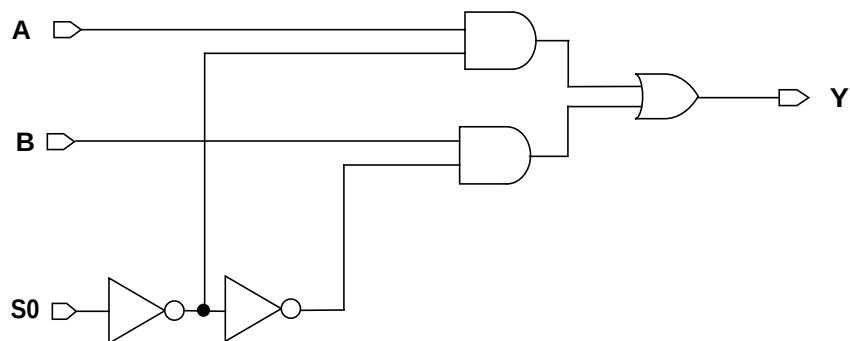
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKMX2X2	3.69	3.68
CLKMX2X3	3.69	4.14
CLKMX2X4	3.69	4.14
CLKMX2X6	3.69	8.28
CLKMX2X8	3.69	6.90
CLKMX2X12	3.69	8.28

Functional Schematic





AC Power

Pin	Power (μ W/MHz)					
	X2	X3	X4	X6	X8	X12
S0	0.0169	0.0230	0.0255	0.0456	0.0380	0.0538
B	0.0140	0.0181	0.0210	0.0392	0.0356	0.0496
A	0.0131	0.0166	0.0188	0.0345	0.0338	0.0487

Pin Capacitance

Pin	Capacitance (pF)					
	X2	X3	X4	X6	X8	X12
S0	0.0052	0.0064	0.0068	0.0126	0.0043	0.0051
B	0.0032	0.0040	0.0044	0.0088	0.0023	0.0031
A	0.0035	0.0043	0.0048	0.0084	0.0026	0.0033

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	X2	X3	X4	X6	X8	X12
S0 $\rightarrow$ Y $\uparrow$	0.1078	0.1093	0.1077	0.1022	0.2015	0.1995
S0 $\rightarrow$ Y $\downarrow$	0.0979	0.1043	0.1128	0.1082	0.1954	0.1915
B $\rightarrow$ Y $\uparrow$	0.0860	0.0821	0.0784	0.0885	0.1834	0.1726
B $\rightarrow$ Y $\downarrow$	0.0944	0.0963	0.1073	0.0949	0.1968	0.1834
A $\rightarrow$ Y $\uparrow$	0.0887	0.0848	0.0796	0.0867	0.1845	0.1744
A $\rightarrow$ Y $\downarrow$	0.0913	0.0923	0.1019	0.0912	0.1935	0.1793



Delays at 25°C, 1.2V, Typical Process

Description	K _{load} (ns/pF)					
	X2	X3	X4	X6	X8	X12
S0 → Y↑	2.2759	1.5742	1.1377	0.7605	0.5635	0.3867
S0 → Y↓	2.4410	1.7171	1.3855	0.8697	0.6757	0.4256
B → Y↑	2.2748	1.5726	1.1369	0.7612	0.5634	0.3866
B → Y↓	2.4408	1.7168	1.3854	0.8694	0.6757	0.4256
A → Y↑	2.2748	1.5739	1.1360	0.7602	0.5635	0.3866
A → Y↓	2.4376	1.7162	1.3829	0.8687	0.6757	0.4256



Cell Description

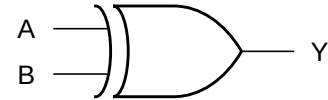
The CLKXOR2 cell provides a logical EXCLUSIVE OR of two inputs (A, B) with balanced delays for clock signals. The output (Y) is represented by the logic equation:

$$Y = (A \bullet \bar{B}) + (\bar{A} \bullet B)$$

Functions

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

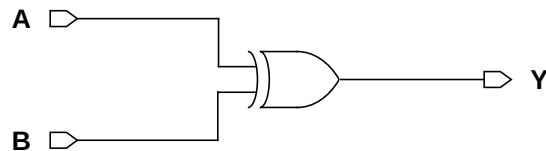
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
CLKXOR2X1	3.69	3.68
CLKXOR2X2	3.69	3.68
CLKXOR2X4	3.69	5.98
CLKXOR2X8	3.69	9.66

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
A	0.0105	0.0158	0.0278	0.0507
B	0.0122	0.0183	0.0349	0.0635

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
A	0.0039	0.0047	0.0070	0.0124
B	0.0023	0.0037	0.0063	0.0116

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
A $\rightarrow$ Y $\uparrow$	0.1073	0.1123	0.1165	0.1061	5.2872	2.2801	1.1420	0.5705
A $\rightarrow$ Y $\downarrow$	0.1084	0.1112	0.1096	0.1103	4.6246	2.5094	1.2488	0.6244
B $\rightarrow$ Y $\uparrow$	0.1497	0.1235	0.1275	0.1203	5.2899	2.2792	1.1443	0.5716
B $\rightarrow$ Y $\downarrow$	0.1557	0.1391	0.1321	0.1271	4.6229	2.5092	1.2487	0.6242



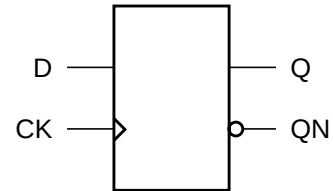
Cell Description

The DFF cell is a positive-edge triggered, static D-type flip-flop.

Function Table

D	CK	Q[n+1]	QN[n+1]
0		0	1
1		1	0
x		Q[n]	QN[n]

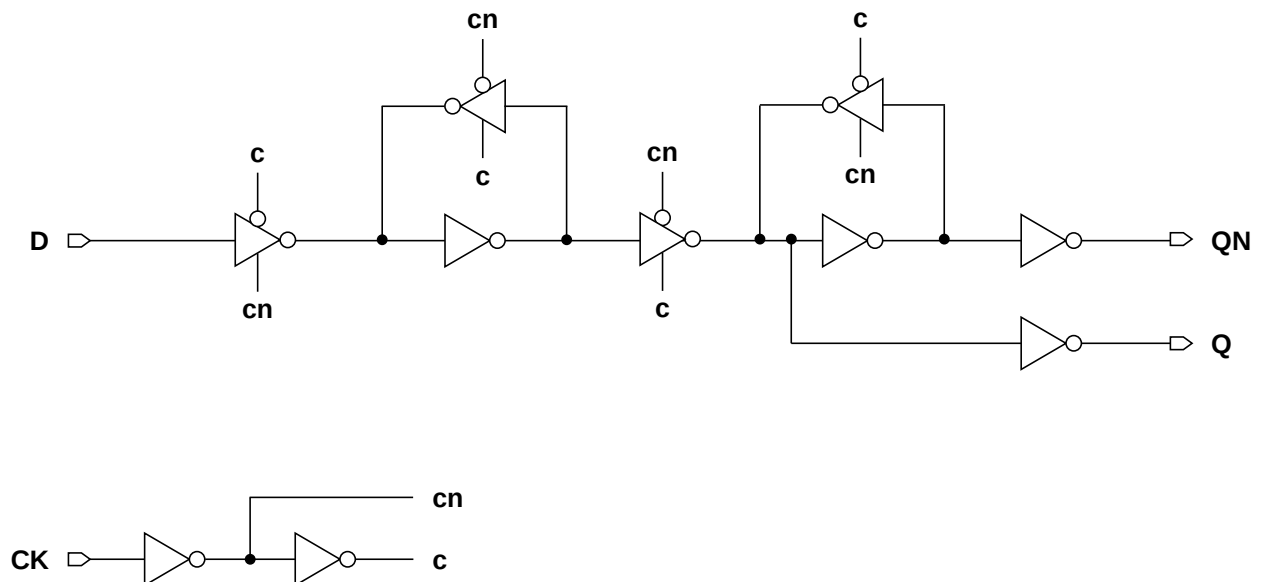
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFXL	3.69	7.36
DFFX1	3.69	7.36
DFFX2	3.69	8.74
DFFX4	3.69	11.50

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0086	0.0094	0.0113	0.0182
CK	0.0098	0.0105	0.0133	0.0189
Q	0.0097	0.0125	0.0195	0.0341

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0014	0.0014	0.0015	0.0024
CK	0.0015	0.0020	0.0023	0.0036

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2217	0.1945	0.1790	0.1502	9.4333	5.1676	2.2686	1.1328
CK $\rightarrow$ Q $\downarrow$	0.1807	0.1470	0.1325	0.1228	5.3750	3.0501	1.5140	0.7569
CK $\rightarrow$ QN $\uparrow$	0.2361	0.1974	0.1778	0.1714	9.4059	5.1477	2.2498	1.1259
CK $\rightarrow$ QN $\downarrow$	0.2858	0.2636	0.2478	0.2083	4.8120	2.7719	1.4499	0.7160

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0547	0.0664	0.0508	0.0508
	setup $\downarrow$ $\rightarrow$ CK	0.1367	0.1602	0.1367	0.1250
	hold $\uparrow$ $\rightarrow$ CK	-0.0391	-0.0469	-0.0352	-0.0312
	hold $\downarrow$ $\rightarrow$ CK	-0.0430	-0.0625	-0.0469	-0.0430
CK	minpwh	0.1176	0.1030	0.0933	0.0787
	minpwl	0.1807	0.1516	0.1418	0.1127



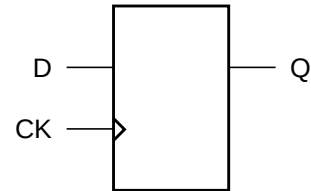
Cell Description

The DFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop. The cell has a single output (Q) and fast clock-to-out path.

Functions

D	CK	Q[n+1]
0		0
1		1
x		Q[n]

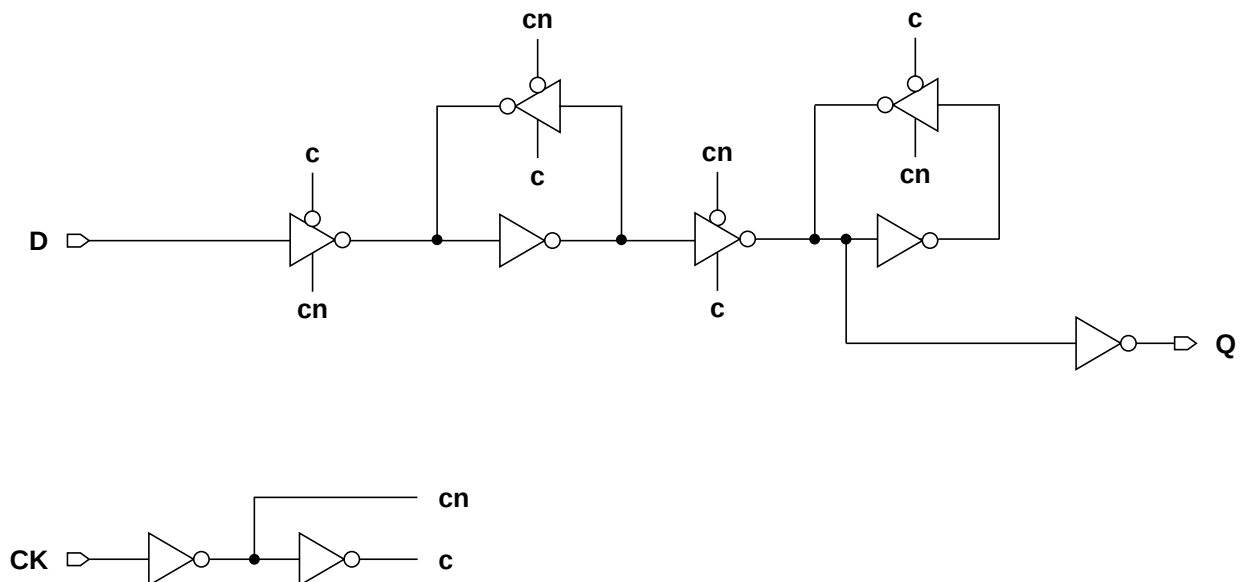
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFHQX1	3.69	9.66
DFFHQX2	3.69	9.66
DFFHQX4	3.69	12.42
DFFHQX8	3.69	13.34

Functional Schematic





AC Power

Pin	Power (μ W/MHz)			
	X1	X2	X4	X8
D	0.0129	0.0182	0.0303	0.0303
CK	0.0174	0.0225	0.0358	0.0359
Q	0.0074	0.0099	0.0148	0.0261

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0014	0.0018	0.0027	0.0028
CK	0.0030	0.0040	0.0060	0.0059

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1420	0.1187	0.1096	0.1234	4.3050	2.2243	1.1115	0.5589
CK $\rightarrow$ Q $\downarrow$	0.1459	0.1284	0.1201	0.1384	2.8631	1.4767	0.7308	0.3765

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup $\uparrow$ $\rightarrow$ CK	0.0898	0.0820	0.0664	0.0664
	setup $\downarrow$ $\rightarrow$ CK	0.1133	0.0977	0.0781	0.0781
	hold $\uparrow$ $\rightarrow$ CK	-0.0234	-0.0234	-0.0117	-0.0078
	hold $\downarrow$ $\rightarrow$ CK	-0.0664	-0.0508	-0.0352	-0.0312
CK	minpwh	0.0641	0.0544	0.0495	0.0544
	minpwl	0.2099	0.1661	0.1418	0.1467



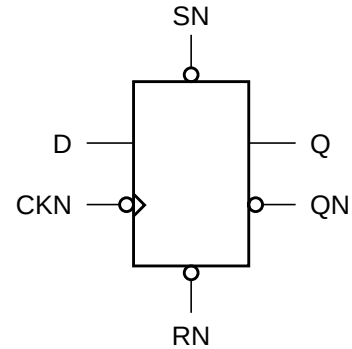
Cell Description

The DFFNSR cell is a negative-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Functions

RN	SN	D	CKN	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0		0	1
1	1	1		1	0
1	1	x		Q[n]	QN[n]

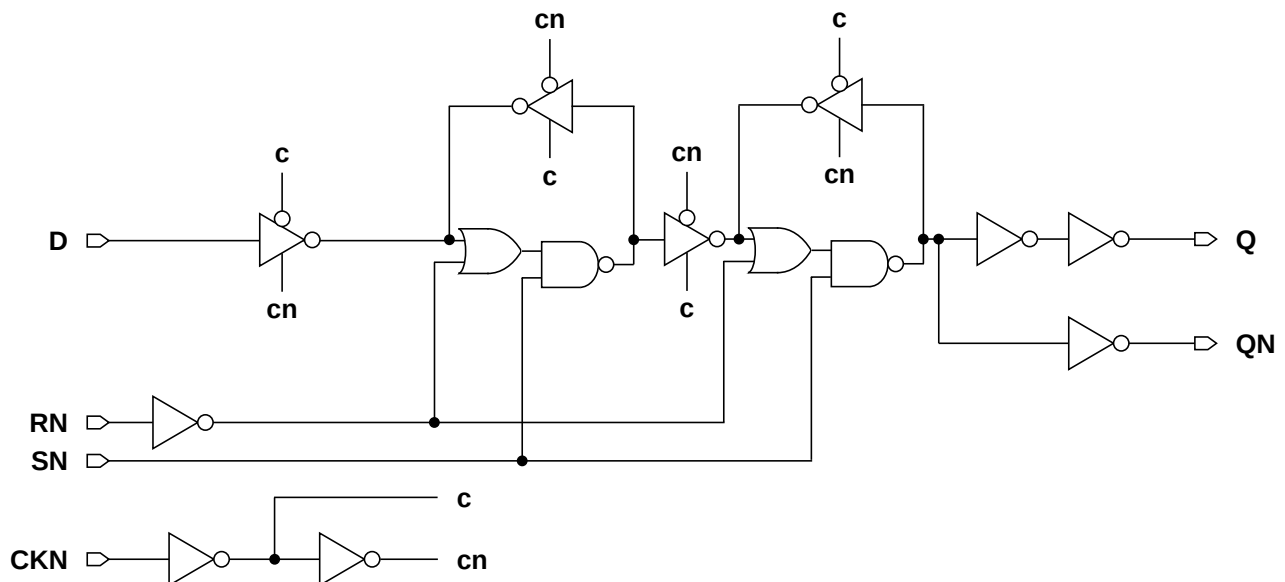
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFNSRXL	3.69	10.12
DFFNSRX1	3.69	10.12
DFFNSRX2	3.69	11.04
DFFNSRX4	3.69	15.18

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0092	0.0110	0.0161	0.0273
CKN	0.0094	0.0100	0.0112	0.0153
SN	0.0018	0.0018	0.0019	0.0024
RN	0.0044	0.0044	0.0047	0.0058
Q	0.0101	0.0130	0.0199	0.0362

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0012	0.0016	0.0024	0.0043
CKN	0.0016	0.0016	0.0016	0.0024
SN	0.0025	0.0029	0.0037	0.0066
RN	0.0015	0.0015	0.0014	0.0022

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3144	0.3172	0.3042	0.3110	9.9177	5.1613	2.2628	1.1330
CKN $\rightarrow$ Q $\downarrow$	0.2685	0.2861	0.2821	0.2606	4.9514	2.8439	1.5044	0.7461
SN $\rightarrow$ Q $\uparrow$	0.2263	0.2600	0.3494	0.5619	9.8999	5.1572	2.2690	1.1421
SN $\rightarrow$ Q $\downarrow$	0.1945	0.2100	0.2373	0.2919	4.9326	2.8396	1.5062	0.7512
RN $\rightarrow$ Q $\downarrow$	0.2410	0.2782	0.3311	0.3767	4.9230	2.8373	1.5053	0.7509
CKN $\rightarrow$ QN $\uparrow$	0.2087	0.2045	0.1903	0.1716	10.1451	5.2394	2.5683	1.1439
CKN $\rightarrow$ QN $\downarrow$	0.2600	0.2635	0.2440	0.2367	6.6743	3.3277	1.6453	0.8713
SN $\rightarrow$ QN $\uparrow$	0.1348	0.1270	0.1385	0.1822	10.0769	5.2529	2.6151	1.2105
SN $\rightarrow$ QN $\downarrow$	0.1748	0.2067	0.2801	0.4534	5.6258	3.2187	1.8520	1.1071
RN $\rightarrow$ QN $\uparrow$	0.1812	0.1952	0.2317	0.2652	10.0228	5.2315	2.6064	1.2100

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup↑ → CKN	0.0469	-0.0156	-0.0352	-0.0234
	setup↓ → CKN	0.1602	0.1289	0.0977	0.0703
	hold↑ → CKN	0.0820	0.0898	0.0820	0.0781
	hold↓ → CKN	-0.0859	-0.0820	-0.0664	-0.0391
CKN	minpwl	0.2050	0.1953	0.1759	0.1807
	minpwh	0.1418	0.1370	0.1370	0.1176
SN	minpwl	0.1661	0.1953	0.2779	0.4722
	recovery	-0.0234	-0.0430	-0.0547	-0.0547
	removal	0.0703	0.0898	0.1055	0.1172
RN	minpwl	0.1467	0.1710	0.2099	0.2584
	recovery	0.0312	-0.0352	-0.0508	-0.0430
	removal	0.0820	0.1250	0.1797	0.2188



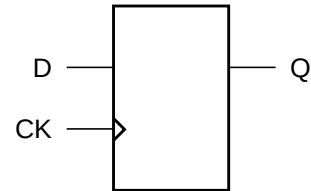
Cell Description

The DFFQ cell is a positive-edge triggered, static D-type flip-flop. The cell has a single output (Q) .

Functions

D	CK	Q[n+1]
0		0
1		1
x		Q[n]

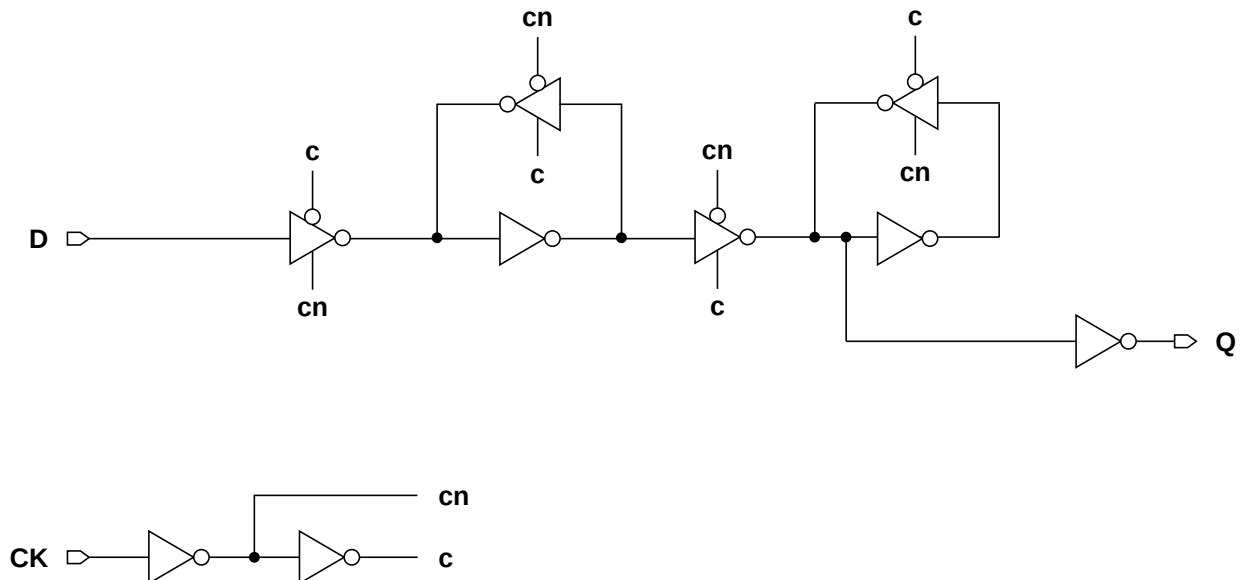
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFQXL	3.69	6.90
DFFQX1	3.69	6.90
DFFQX2	3.69	7.36
DFFQX4	3.69	9.20

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0086	0.0090	0.0107	0.0152
CK	0.0103	0.0111	0.0122	0.0167
Q	0.0069	0.0081	0.0105	0.0153

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0012	0.0012	0.0014	0.0021
CK	0.0015	0.0020	0.0023	0.0030

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2187	0.2019	0.1688	0.1586	9.4612	5.1774	2.2669	1.1333
CK $\rightarrow$ Q $\downarrow$	0.1864	0.1619	0.1358	0.1291	5.3574	2.9736	1.5139	0.7564

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0547	0.0664	0.0625	0.0508
	setup $\downarrow$ $\rightarrow$ CK	0.1250	0.1445	0.1523	0.1367
	hold $\uparrow$ $\rightarrow$ CK	-0.0391	-0.0430	-0.0391	-0.0352
	hold $\downarrow$ $\rightarrow$ CK	-0.0352	-0.0469	-0.0664	-0.0508
CK	minpwh	0.1127	0.1078	0.0884	0.0835
	minpwl	0.1856	0.1564	0.1418	0.1176



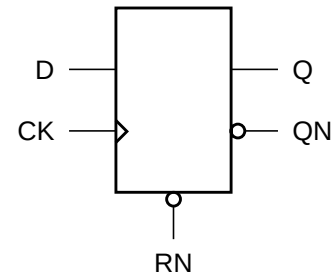
Cell Description

The DFFR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN).

Functions

RN	D	CK	Q[n+1]	QN[n+1]
0	x	x	0	1
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

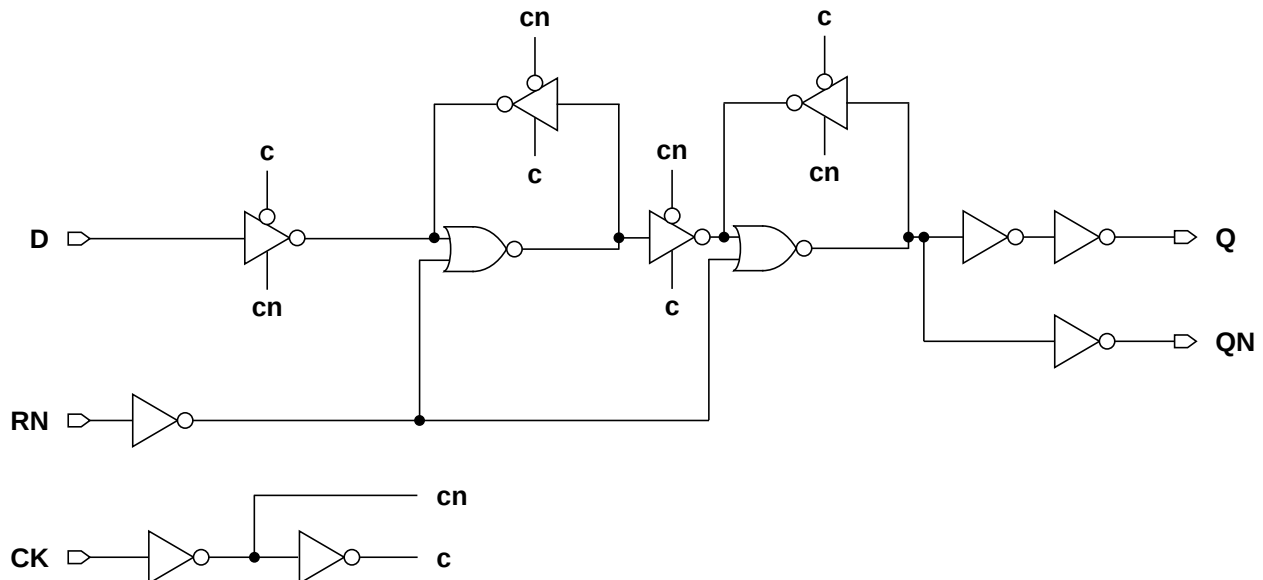
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFRXL	3.69	8.74
DFFRX1	3.69	8.74
DFFRX2	3.69	10.12
DFFRX4	3.69	12.88

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0091	0.0107	0.0144	0.0241
CK	0.0095	0.0099	0.0118	0.0168
RN	0.0018	0.0020	0.0023	0.0033
Q	0.0094	0.0123	0.0196	0.0327

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0011	0.0013	0.0019	0.0031
CK	0.0014	0.0014	0.0017	0.0024
RN	0.0041	0.0045	0.0057	0.0077

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2723	0.2468	0.2255	0.2076	9.9764	5.1832	2.5869	1.1357
CK $\rightarrow$ Q $\downarrow$	0.3099	0.3118	0.3140	0.2926	5.3725	2.9640	1.4658	0.7562
RN $\rightarrow$ Q $\downarrow$	0.1434	0.1455	0.1350	0.1214	5.3583	2.9729	1.4858	0.7671
CK $\rightarrow$ QN $\uparrow$	0.2067	0.2033	0.2069	0.1919	10.0108	5.1877	2.5876	1.1356
CK $\rightarrow$ QN $\downarrow$	0.1793	0.1699	0.1557	0.1417	5.9538	3.1407	1.5871	0.7791
RN $\rightarrow$ QN $\uparrow$	0.2365	0.2520	0.2871	0.3514	9.9510	5.2307	2.6843	1.2722

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1367	0.1211	0.0781	0.0820
	setup $\downarrow$ $\rightarrow$ CK	0.1562	0.1172	0.0664	0.0156
	hold $\uparrow$ $\rightarrow$ CK	-0.0625	-0.0586	-0.0391	-0.0430
	hold $\downarrow$ $\rightarrow$ CK	-0.0352	-0.0195	0.0000	0.0234
CK	minpwh	0.1564	0.1321	0.1176	0.1030
	minpwl	0.1807	0.1759	0.1467	0.1321
RN	minpwl	0.1516	0.1613	0.1759	0.2050
	recovery	0.1406	0.1211	0.0820	0.0859
	removal	-0.0898	-0.0820	-0.0547	-0.0586



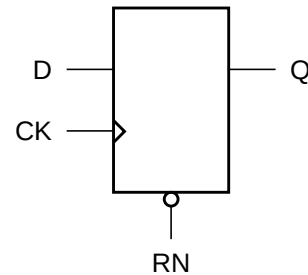
Cell Description

The DFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	D	CK	Q[n+1]
0	x	x	0
1	0		0
1	1		1
1	x		Q[n]

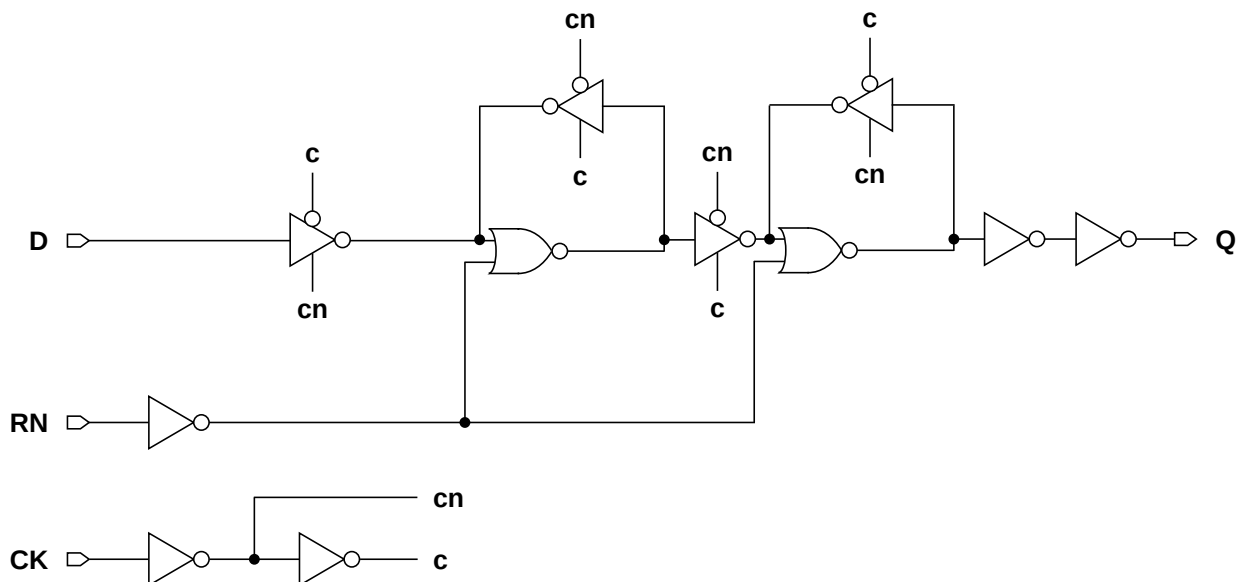
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFRHQX1	3.69	9.66
DFFRHQX2	3.69	10.12
DFFRHQX4	3.69	13.34
DFFRHQX8	3.69	14.72

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
D	0.0137	0.0177	0.0311	0.0319
CK	0.0186	0.0235	0.0354	0.0375
RN	0.0025	0.0035	0.0053	0.0055
Q	0.0081	0.0100	0.0153	0.0274

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0014	0.0017	0.0029	0.0028
CK	0.0032	0.0043	0.0060	0.0061
RN	0.0034	0.0048	0.0073	0.0073

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1614	0.1312	0.1246	0.1414	4.3770	2.2888	1.1429	0.5699
CK $\rightarrow$ Q $\downarrow$	0.1504	0.1317	0.1249	0.1450	2.8729	1.4968	0.7384	0.3816
RN $\rightarrow$ Q $\downarrow$	0.1529	0.1401	0.1378	0.1718	3.0049	1.5685	0.7932	0.4125

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup $\uparrow$ $\rightarrow$ CK	0.1055	0.0742	0.0664	0.0742
	setup $\downarrow$ $\rightarrow$ CK	0.1055	0.0898	0.0781	0.0703
	hold $\uparrow$ $\rightarrow$ CK	-0.0273	-0.0117	-0.0078	-0.0078
	hold $\downarrow$ $\rightarrow$ CK	-0.0586	-0.0469	-0.0352	-0.0273
CK	minpwh	0.0787	0.0593	0.0593	0.0593
	minpwl	0.2099	0.1661	0.1418	0.1516
RN	minpwl	0.1467	0.1321	0.1273	0.1613
	recovery	-0.0469	-0.0352	-0.0352	-0.0352
	removal	0.0859	0.0859	0.1328	0.1211



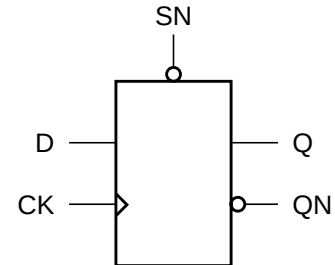
Cell Description

The DFFS cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN).

Functions

SN	D	CK	Q[n+1]	QN[n+1]
0	x	x	1	0
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

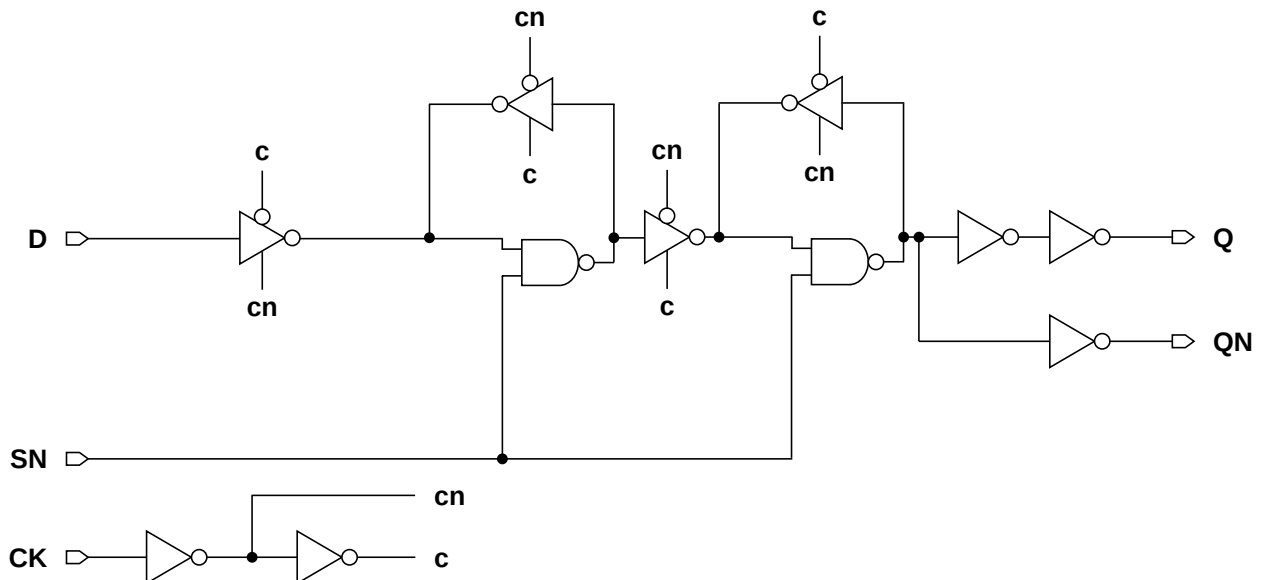
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSXL	3.69	8.28
DFFSX1	3.69	8.28
DFFSX2	3.69	9.66
DFFSX4	3.69	11.96

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0077	0.0090	0.0138	0.0223
CK	0.0101	0.0103	0.0120	0.0160
SN	0.0018	0.0019	0.0020	0.0021
Q	0.0092	0.0121	0.0201	0.0314

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0012	0.0013	0.0022	0.0037
CK	0.0015	0.0015	0.0015	0.0022
SN	0.0025	0.0031	0.0036	0.0064

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2435	0.2363	0.2172	0.1875	9.9051	5.1591	2.3311	1.3023
CK $\rightarrow$ Q $\downarrow$	0.2819	0.2947	0.2961	0.2524	4.8129	2.8441	1.5408	0.7853
SN $\rightarrow$ Q $\uparrow$	0.2116	0.2472	0.3331	0.4887	9.8961	5.1575	2.3393	1.3074
CK $\rightarrow$ QN $\uparrow$	0.2230	0.2122	0.2172	0.1923	10.1485	5.2368	2.3684	1.3145
CK $\rightarrow$ QN $\downarrow$	0.1895	0.1805	0.1699	0.1395	5.9969	3.2137	1.6623	0.8298
SN $\rightarrow$ QN $\downarrow$	0.1607	0.1930	0.2808	0.4037	5.3461	3.1654	1.8866	1.1247

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1016	0.0898	0.0664	0.0625
	setup $\downarrow$ $\rightarrow$ CK	0.0938	0.0859	0.0508	0.0195
	hold $\uparrow$ $\rightarrow$ CK	-0.0234	-0.0234	-0.0234	-0.0273
	hold $\downarrow$ $\rightarrow$ CK	-0.0117	-0.0117	0.0039	0.0156
CK	minpwh	0.1467	0.1370	0.1224	0.0981
	minpwl	0.1564	0.1613	0.1516	0.1273
SN	minpwl	0.1467	0.1807	0.2584	0.4042
	recovery	-0.0547	-0.0781	-0.0938	-0.0977
	removal	0.0820	0.1016	0.1328	0.1562

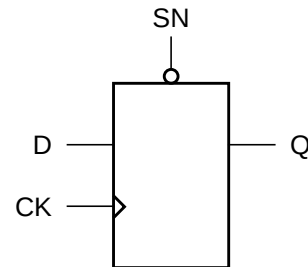


The DFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.

Functions

SN	D	CK	Q[n+1]
0	x	x	1
1	0		0
1	1		1
1	x		Q[n]

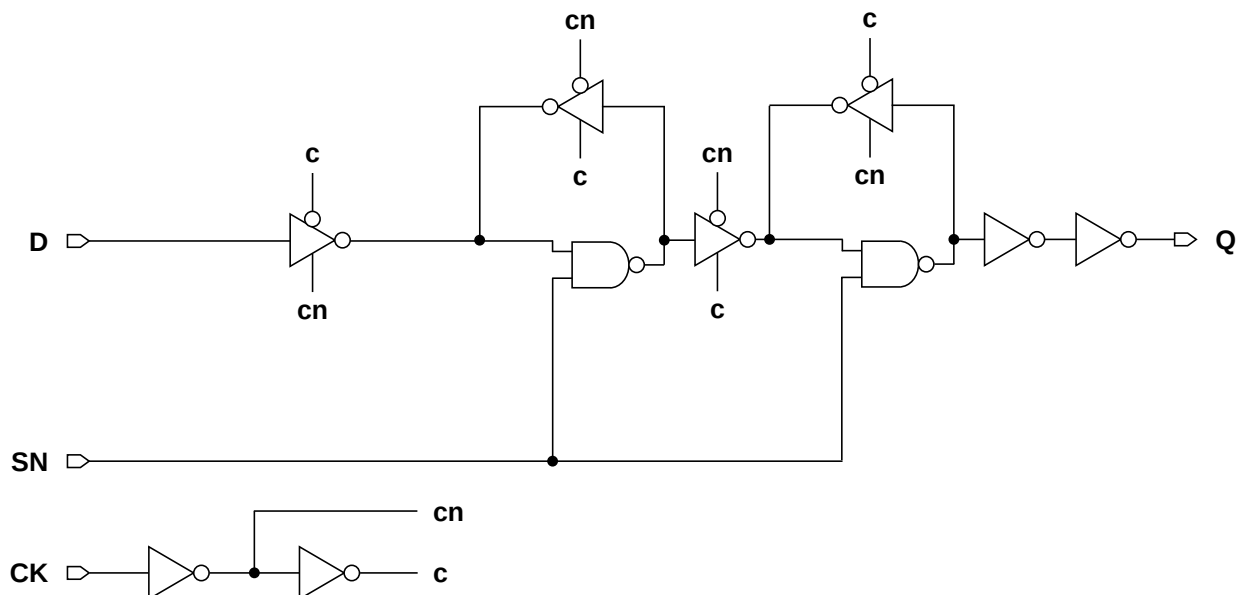
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSHQX1	3.69	10.58
DFFSHQX2	3.69	11.96
DFFSHQX4	3.69	15.64
DFFSHQX8	3.69	16.56

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
D	0.0141	0.0201	0.0332	0.0333
CK	0.0181	0.0257	0.0387	0.0389
SN	0.0059	0.0068	0.0077	0.0079
Q	0.0085	0.0107	0.0161	0.0278

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0013	0.0018	0.0030	0.0030
CK	0.0029	0.0044	0.0073	0.0073
SN	0.0050	0.0061	0.0083	0.0083

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1535	0.1285	0.1147	0.1269	4.3296	2.1520	1.1140	0.5602
CK $\rightarrow$ Q $\downarrow$	0.1646	0.1381	0.1253	0.1435	3.0905	1.4933	0.7390	0.4084
SN $\rightarrow$ Q $\uparrow$	0.1697	0.1639	0.1638	0.1895	4.3359	2.1600	1.1176	0.5642

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup $\uparrow$ $\rightarrow$ CK	0.1172	0.0938	0.0820	0.0859
	setup $\downarrow$ $\rightarrow$ CK	0.1211	0.0977	0.0742	0.0742
	hold $\uparrow$ $\rightarrow$ CK	-0.0430	-0.0273	-0.0195	-0.0156
	hold $\downarrow$ $\rightarrow$ CK	-0.0664	-0.0469	-0.0234	-0.0195
CK	minpwh	0.0738	0.0593	0.0544	0.0544
	minpwl	0.2099	0.1759	0.1613	0.1613
SN	minpwl	0.1370	0.1370	0.1370	0.1613
	recovery	0.0508	0.0430	0.0312	0.0312
	removal	-0.0312	-0.0195	-0.0117	-0.0117



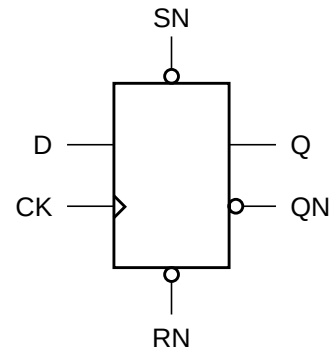
Cell Description

The DFFSR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset.

Functions

RN	SN	D	CK	Q[n+1]	QN[n+1]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0
1	1	0		0	1
1	1	1		1	0
1	1	x		Q[n]	QN[n]

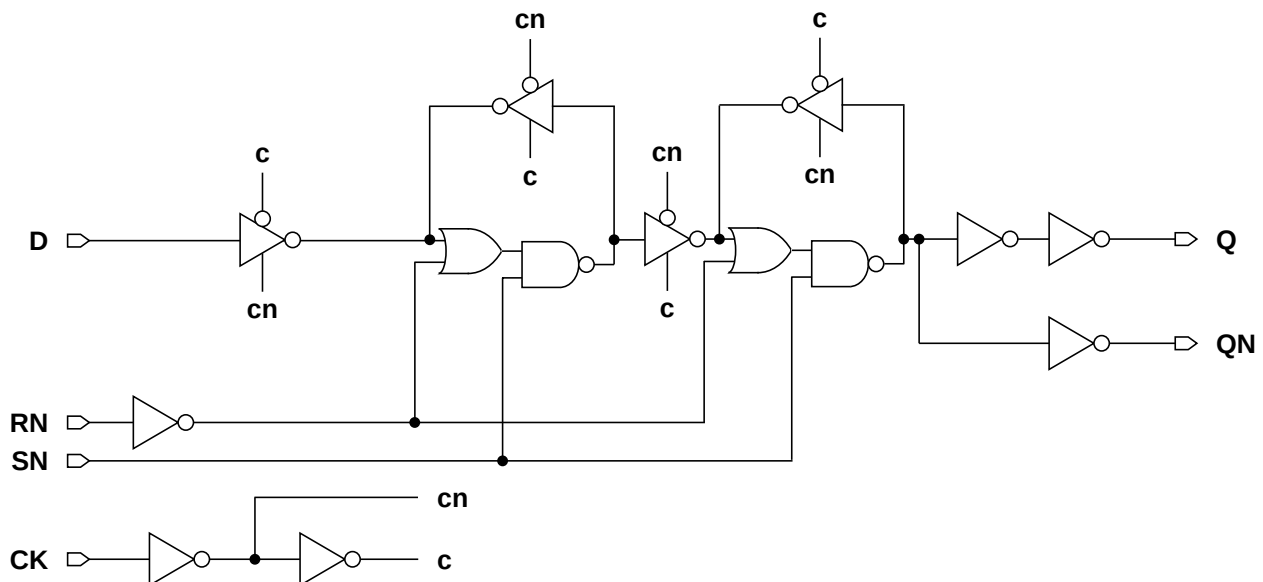
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSRXL	3.69	10.12
DFFSRX1	3.69	10.12
DFFSRX2	3.69	11.50
DFFSRX4	3.69	15.64

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0100	0.0114	0.0161	0.0286
CK	0.0118	0.0121	0.0139	0.0186
SN	0.0017	0.0017	0.0018	0.0025
RN	0.0041	0.0041	0.0043	0.0062
Q	0.0088	0.0113	0.0183	0.0319

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0013	0.0016	0.0027	0.0042
CK	0.0014	0.0014	0.0016	0.0026
SN	0.0025	0.0029	0.0039	0.0067
RN	0.0012	0.0013	0.0014	0.0024

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2753	0.2644	0.2618	0.2397	9.9079	5.1611	2.2658	1.1325
CK $\rightarrow$ Q $\downarrow$	0.2984	0.3150	0.3147	0.2917	4.9534	2.8472	1.5055	0.7463
SN $\rightarrow$ Q $\uparrow$	0.2283	0.2623	0.3518	0.5619	9.8933	5.1570	2.2703	1.1423
SN $\rightarrow$ Q $\downarrow$	0.1973	0.2122	0.2369	0.2916	4.9321	2.8416	1.5068	0.7513
RN $\rightarrow$ Q $\downarrow$	0.2424	0.2762	0.3083	0.3758	4.9253	2.8401	1.5060	0.7512
CK $\rightarrow$ QN $\uparrow$	0.2377	0.2315	0.2190	0.2016	10.1204	5.2369	2.2893	1.1433
CK $\rightarrow$ QN $\downarrow$	0.2192	0.2074	0.1976	0.1686	6.5358	3.3515	1.7518	0.8428
SN $\rightarrow$ QN $\uparrow$	0.1361	0.1275	0.1349	0.1818	10.0892	5.2546	2.3380	1.2107
SN $\rightarrow$ QN $\downarrow$	0.1758	0.2072	0.2832	0.4534	5.6701	3.2406	1.8645	1.1093
RN $\rightarrow$ QN $\uparrow$	0.1810	0.1911	0.2053	0.2640	10.0365	5.2367	2.3348	1.2119

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1641	0.1172	0.1055	0.1055
	setup $\downarrow$ $\rightarrow$ CK	0.1133	0.0781	0.0273	0.0234
	hold $\uparrow$ $\rightarrow$ CK	-0.0391	-0.0312	-0.0273	-0.0312
	hold $\downarrow$ $\rightarrow$ CK	-0.0117	0.0039	0.0234	0.0195
CK	minpwh	0.1613	0.1467	0.1467	0.1321
	minpwl	0.1856	0.1759	0.1564	0.1370
SN	minpwl	0.1661	0.2001	0.2779	0.4722
	recovery	-0.0820	-0.1016	-0.1172	-0.1055
	removal	0.1250	0.1445	0.1758	0.1953
RN	minpwl	0.1467	0.1661	0.1904	0.2536
	recovery	0.1484	0.1016	0.0977	0.0938
	removal	-0.0273	0.0039	0.0195	0.0742



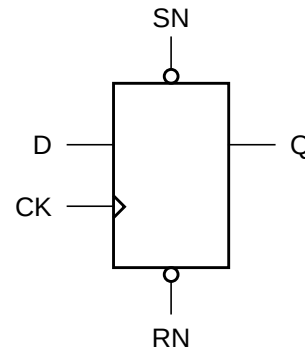
Cell Description

The DFFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	SN	D	CK	Q[n+1]
0	1	x	x	0
1	0	x	x	1
0	0	x	x	1
1	1	0		0
1	1	1		1
1	1	x		Q[n]

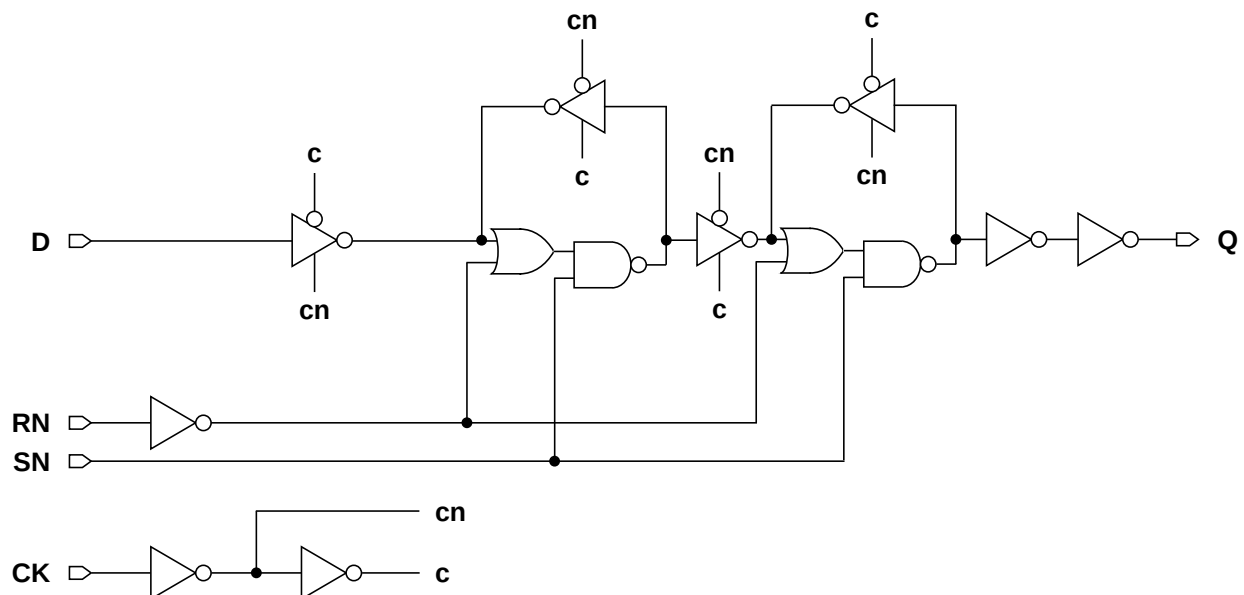
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFSRHQX1	3.69	12.42
DFFSRHQX2	3.69	13.34
DFFSRHQX4	3.69	15.64
DFFSRHQX8	3.69	16.56

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
D	0.0157	0.0204	0.0266	0.0267
CK	0.0181	0.0231	0.0314	0.0315
SN	0.0084	0.0093	0.0114	0.0117
RN	0.0020	0.0026	0.0042	0.0042
Q	0.0085	0.0125	0.0168	0.0297

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0016	0.0020	0.0030	0.0030
CK	0.0027	0.0040	0.0063	0.0064
SN	0.0033	0.0040	0.0052	0.0052
RN	0.0031	0.0039	0.0056	0.0056

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1707	0.1428	0.1403	0.1647	4.4223	2.2135	1.1381	0.5792
CK $\rightarrow$ Q $\downarrow$	0.1802	0.1528	0.1481	0.1795	3.1482	1.5908	0.8119	0.4307
SN $\rightarrow$ Q $\uparrow$	0.2332	0.2600	0.2346	0.2735	4.3545	2.2140	1.1275	0.5707
SN $\rightarrow$ Q $\downarrow$	0.3228	0.2976	0.2298	0.2988	3.4835	1.7852	0.8376	0.4520
RN $\rightarrow$ Q $\downarrow$	0.2760	0.2437	0.1781	0.2451	3.4956	1.7898	0.8391	0.4525

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup $\uparrow$ $\rightarrow$ CK	0.1602	0.1055	0.1055	0.1055
	setup $\downarrow$ $\rightarrow$ CK	0.1172	0.0977	0.0859	0.0859
	hold $\uparrow$ $\rightarrow$ CK	-0.0664	-0.0312	-0.0273	-0.0234
	hold $\downarrow$ $\rightarrow$ CK	-0.0508	-0.0352	-0.0312	-0.0273
CK	minpwh	0.0787	0.0641	0.0641	0.0738
	minpwl	0.2487	0.1807	0.1807	0.1807
SN	minpwl	0.1953	0.2196	0.2001	0.2390
	recovery	0.0469	0.0391	0.0430	0.0430
	removal	-0.0234	-0.0156	-0.0234	-0.0234
RN	minpwl	0.2487	0.2196	0.1564	0.2196
	recovery	-0.0391	-0.0312	-0.0312	-0.0312
	removal	0.0781	0.0820	0.0859	0.0820



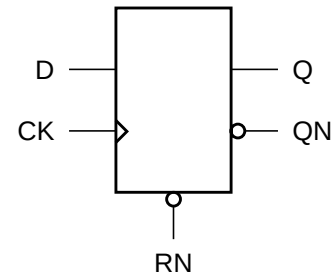
Cell Description

The DFFTR cell is a positive-edge triggered, static D-type flip-flop with synchronous active-low reset (RN).

Functions

RN	D	CK	Q[n+1]	QN[n+1]
0	x		0	1
x	x		Q[n]	QN[n]
1	0		0	1
1	1		1	0

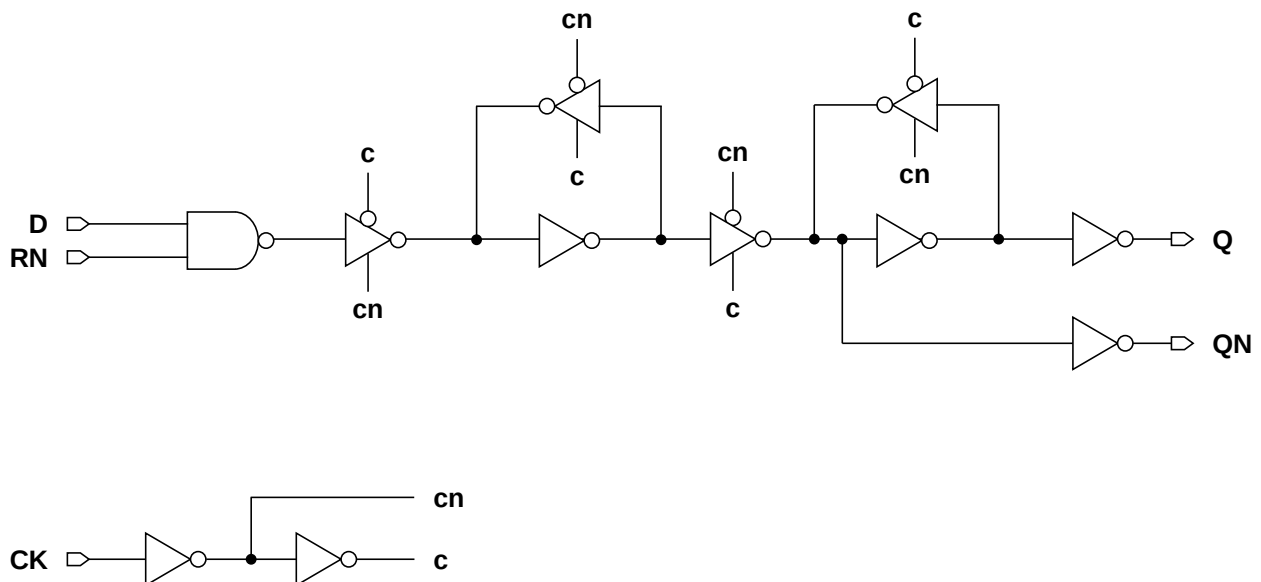
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFTRXL	3.69	7.82
DFFTRX1	3.69	7.82
DFFTRX2	3.69	8.28
DFFTRX4	3.69	11.04

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0089	0.0103	0.0142	0.0225
CK	0.0095	0.0102	0.0113	0.0151
RN	0.0096	0.0112	0.0156	0.0253
Q	0.0094	0.0123	0.0190	0.0324

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0012	0.0015	0.0024	0.0041
CK	0.0014	0.0014	0.0015	0.0023
RN	0.0012	0.0012	0.0019	0.0031

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2411	0.2533	0.2085	0.2014	9.8839	5.1594	2.2577	1.1289
CK $\rightarrow$ Q $\downarrow$	0.2778	0.3091	0.2933	0.2666	4.8599	3.1131	1.4959	0.7404
CK $\rightarrow$ QN $\uparrow$	0.2098	0.2077	0.2001	0.1800	9.9239	5.1952	2.2697	1.1359
CK $\rightarrow$ QN $\downarrow$	0.1777	0.1812	0.1480	0.1384	5.6950	3.2261	1.5409	0.7690

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1367	0.1133	0.0820	0.0742
	setup $\downarrow$ $\rightarrow$ CK	0.1484	0.0977	0.0742	0.0391
	hold $\uparrow$ $\rightarrow$ CK	-0.0625	-0.0547	-0.0430	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	-0.0352	-0.0156	-0.0117	0.0039
CK	minpwh	0.1370	0.1370	0.1030	0.1030
	minpwl	0.1759	0.1807	0.1516	0.1224
RN	setup $\uparrow$ $\rightarrow$ CK	0.1367	0.1133	0.0820	0.0742
	setup $\downarrow$ $\rightarrow$ CK	0.1914	0.1836	0.1602	0.1289
	hold $\uparrow$ $\rightarrow$ CK	-0.0625	-0.0547	-0.0430	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	-0.0508	-0.0547	-0.0508	-0.0352



Cell Description

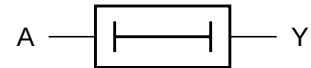
The DLY1 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY1X1	3.69	3.68
DLY1X4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X1	X4
A	0.0163	0.0214

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X4
A	0.0032	0.0033

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	X4	X1	X4
A $\rightarrow$ Y $\uparrow$	0.0917	0.1038	5.1125	1.1226
A $\rightarrow$ Y $\downarrow$	0.1054	0.1338	4.2231	1.3576



Cell Description

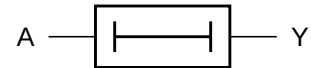
The DLY2 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY2X1	3.69	3.68
DLY2X4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X1	X4
A	0.0184	0.0240

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X4
A	0.0032	0.0033

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	X4	X1	X4
A $\rightarrow$ Y $\uparrow$	0.1622	0.1859	5.1395	1.1386
A $\rightarrow$ Y $\downarrow$	0.1702	0.2236	4.3372	1.4059



Cell Description

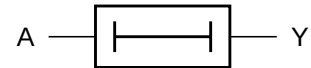
The DLY3 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY3X1	3.69	3.68
DLY3X4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X1	X4
A	0.0214	0.0277

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X4
A	0.0033	0.0033

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	X4	X1	X4
A $\rightarrow$ Y $\uparrow$	0.2524	0.2886	5.1942	1.1639
A $\rightarrow$ Y $\downarrow$	0.2426	0.3210	4.5222	1.4733



Cell Description

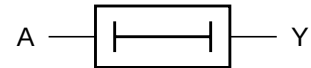
The DLY4 cell provides the logical delay of a single input (A). The output (Y) is represented by the logic equation:

$$Y = A$$

Functions

A	Y
0	0
1	1

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DLY4X1	3.69	3.68
DLY4X4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X1	X4
A	0.0239	0.0321

Pin Capacitance

Pin	Capacitance (pF)	
	X1	X4
A	0.0033	0.0034

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X1	X4	X1	X4
$A \rightarrow Y\uparrow$	0.3603	0.4091	5.2779	1.1969
$A \rightarrow Y\downarrow$	0.3219	0.4261	4.7531	1.5517





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0109	0.0108	0.0146	0.0212
CK	0.0136	0.0133	0.0148	0.0207
E	0.0155	0.0148	0.0190	0.0273
Q	0.0102	0.0130	0.0228	0.0379

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0018	0.0013	0.0017	0.0023
CK	0.0017	0.0022	0.0025	0.0039
E	0.0034	0.0030	0.0032	0.0041

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2331	0.2079	0.1693	0.1517	9.4198	5.8539	2.2648	1.1337
CK $\rightarrow$ Q $\downarrow$	0.1698	0.1422	0.1262	0.1143	5.2032	3.1382	1.5347	0.7585
CK $\rightarrow$ QN $\uparrow$	0.2537	0.2081	0.1823	0.1672	9.4611	5.1627	2.2560	1.1276
CK $\rightarrow$ QN $\downarrow$	0.3319	0.2939	0.2436	0.2117	5.2889	2.8891	1.4668	0.7199

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0898	0.1094	0.1211	0.1133
	setup $\downarrow$ $\rightarrow$ CK	0.2070	0.3438	0.3086	0.2695
	hold $\uparrow$ $\rightarrow$ CK	-0.0664	-0.0820	-0.0898	-0.0820
	hold $\downarrow$ $\rightarrow$ CK	-0.0898	-0.2109	-0.1953	-0.1641
CK	minpwh	0.1176	0.1127	0.0933	0.0835
	minpwl	0.2147	0.1904	0.1661	0.1418
E	setup $\uparrow$ $\rightarrow$ CK	0.2422	0.3711	0.3477	0.3125
	setup $\downarrow$ $\rightarrow$ CK	0.1445	0.2500	0.2227	0.1914
	hold $\uparrow$ $\rightarrow$ CK	-0.0820	-0.0977	-0.1016	-0.0938
	hold $\downarrow$ $\rightarrow$ CK	-0.0820	-0.1367	-0.1562	-0.1641

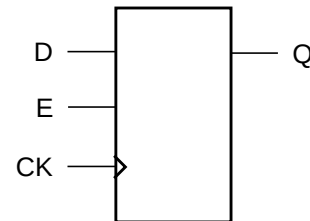


The EDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a synchronous, active-high enable (E). The cell has a single output (Q) and fast clock-to-out path.

Functions

E	D	CK	Q[n+1]
0	x	x	Q[n]
1	0		0
1	1		1
x	x		Q[n]

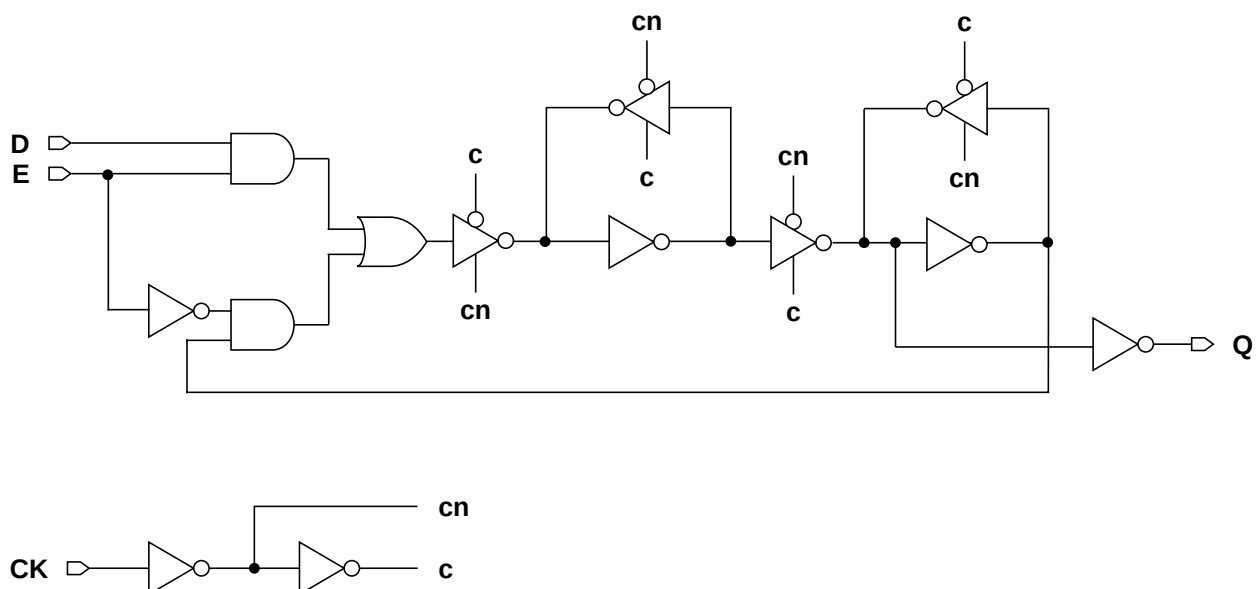
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
EDFFHGX1	3.69	10.12
EDFFHGX2	3.69	11.04
EDFFHGX4	3.69	16.56
EDFFHGX8	3.69	17.94

Functional Schematic





AC Power

Pin	Power (μ W/MHz)			
	X1	X2	X4	X8
D	0.0186	0.0246	0.0410	0.0416
CK	0.0183	0.0229	0.0366	0.0366
E	0.0187	0.0233	0.0400	0.0399
Q	0.0067	0.0098	0.0160	0.0286

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D	0.0019	0.0032	0.0052	0.0051
CK	0.0025	0.0039	0.0069	0.0069
E	0.0031	0.0039	0.0057	0.0056

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1375	0.1175	0.1112	0.1262	4.4415	2.2688	1.1327	0.5707
CK $\rightarrow$ Q $\downarrow$	0.1491	0.1249	0.1249	0.1438	3.2379	1.5234	0.7531	0.3902

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D	setup $\uparrow$ $\rightarrow$ CK	0.1211	0.1094	0.0938	0.0938
	setup $\downarrow$ $\rightarrow$ CK	0.1445	0.1289	0.1016	0.1016
	hold $\uparrow$ $\rightarrow$ CK	-0.0664	-0.0508	-0.0391	-0.0352
	hold $\downarrow$ $\rightarrow$ CK	-0.0938	-0.0781	-0.0547	-0.0508
CK	minpwh	0.0641	0.0544	0.0495	0.0544
	minpwl	0.1904	0.1564	0.1467	0.1467
E	setup $\uparrow$ $\rightarrow$ CK	0.1250	0.1172	0.0977	0.0977
	setup $\downarrow$ $\rightarrow$ CK	0.1992	0.1875	0.1992	0.1953
	hold $\uparrow$ $\rightarrow$ CK	-0.0898	-0.0703	-0.0547	-0.0586
	hold $\downarrow$ $\rightarrow$ CK	-0.1641	-0.1562	-0.1094	-0.1055





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0114	0.0115	0.0148	0.0217
CK	0.0143	0.0141	0.0168	0.0235
E	0.0167	0.0158	0.0200	0.0283
RN	0.0144	0.0136	0.0176	0.0247
Q	0.0099	0.0130	0.0231	0.0377

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0019	0.0013	0.0016	0.0025
CK	0.0014	0.0020	0.0028	0.0044
E	0.0038	0.0032	0.0036	0.0045
RN	0.0020	0.0015	0.0017	0.0024

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2319	0.2098	0.1805	0.1529	9.4091	5.1611	2.2649	1.1341
CK $\rightarrow$ Q $\downarrow$	0.1705	0.1410	0.1307	0.1229	5.1572	3.1566	1.5288	0.7608
CK $\rightarrow$ QN $\uparrow$	0.2503	0.2018	0.1893	0.1768	9.4435	5.1571	2.2578	1.1279
CK $\rightarrow$ QN $\downarrow$	0.3282	0.2892	0.2588	0.2138	5.2613	2.8417	1.4769	0.7211

Timing Constraints at 25°C, 1.2V, Typical Process

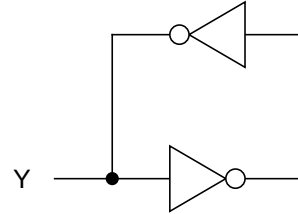
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.1094	0.1484	0.1445	0.1406
	setup $\downarrow$ $\rightarrow$ CK	0.2305	0.3711	0.3242	0.2852
	hold $\uparrow$ $\rightarrow$ CK	-0.0898	-0.1133	-0.1094	-0.1016
	hold $\downarrow$ $\rightarrow$ CK	-0.1250	-0.2578	-0.2109	-0.1836
CK	minpwh	0.1176	0.1127	0.0981	0.0835
	minpwl	0.2487	0.2293	0.2001	0.1661
E	setup $\uparrow$ $\rightarrow$ CK	0.2695	0.4023	0.3594	0.3242
	setup $\downarrow$ $\rightarrow$ CK	0.1797	0.3008	0.2539	0.2227
	hold $\uparrow$ $\rightarrow$ CK	-0.1055	-0.1367	-0.1289	-0.1172
	hold $\downarrow$ $\rightarrow$ CK	-0.1250	-0.1758	-0.1836	-0.1875
RN	setup $\uparrow$ $\rightarrow$ CK	0.1289	0.1680	0.1641	0.1523
	setup $\downarrow$ $\rightarrow$ CK	0.2109	0.3359	0.2969	0.2578
	hold $\uparrow$ $\rightarrow$ CK	-0.1094	-0.1367	-0.1289	-0.1211
	hold $\downarrow$ $\rightarrow$ CK	-0.0703	-0.1992	-0.1680	-0.1445



Cell Description

The HOLD cell holds data at a known value. This cell is often used for holding data on a tri-state bus.

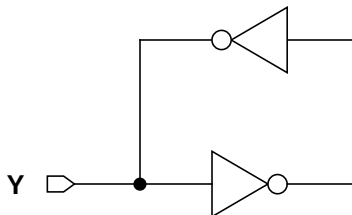
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
HOLDX1	3.69	1.84

Functional Schematic





AC Power

Pin	Power (μ W/MHz)
	X1
Y	0.0036

Pin Capacitance

Pin	Capacitance (pF)
	X1
Y	0.0383

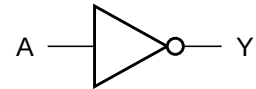
**Cell Description**

The INV cell provides the logical inversion of a single input (A). The output (Y) is represented by the logic equation:

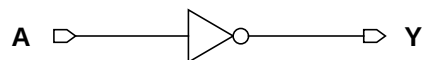
$$Y = \bar{A}$$

Functions

A	Y
0	1
1	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
INVXL	3.69	0.92
INVX1	3.69	0.92
INVX2	3.69	1.38
INVX3	3.69	1.38
INVX4	3.69	1.84
INVX6	3.69	2.30
INVX8	3.69	2.76
INVX12	3.69	3.68
INVX16	3.69	4.60
INVX20	3.69	5.98

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0024	0.0037	0.0069	0.0086	0.0120	0.0186	0.0235	0.0355	0.0475	0.0589

Pin Capacitance

Pin	Capacitance (pF)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0016	0.0025	0.0047	0.0065	0.0092	0.0136	0.0180	0.0265	0.0358	0.0442

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0242	0.0210	0.0186	0.0166	0.0165	0.0173	0.0168	0.0166	0.0165	0.0166
A $\rightarrow$ Y $\downarrow$	0.0150	0.0145	0.0148	0.0137	0.0134	0.0139	0.0135	0.0135	0.0134	0.0135

Description	K_{load} (ns/pF)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	9.3452	5.1168	2.2403	1.5499	1.1202	0.7468	0.5752	0.3779	0.2814	0.2281
A $\rightarrow$ Y $\downarrow$	4.3398	2.5589	1.3691	0.9927	0.6850	0.4566	0.3425	0.2335	0.1722	0.1401



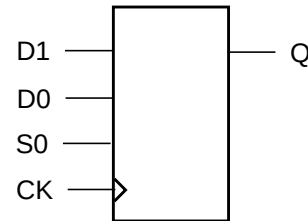
Cell Description

The MDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a 2-to-1 data select control (S0) for the data inputs (D1 and D0). The cell has a single output (Q) and fast clock-to-out path.

Functions

S0	D1	D0	CK	Q[n+1]
0	x	0		0
0	x	1		1
1	0	x		0
1	1	x		1
x	x	x		Q[n]

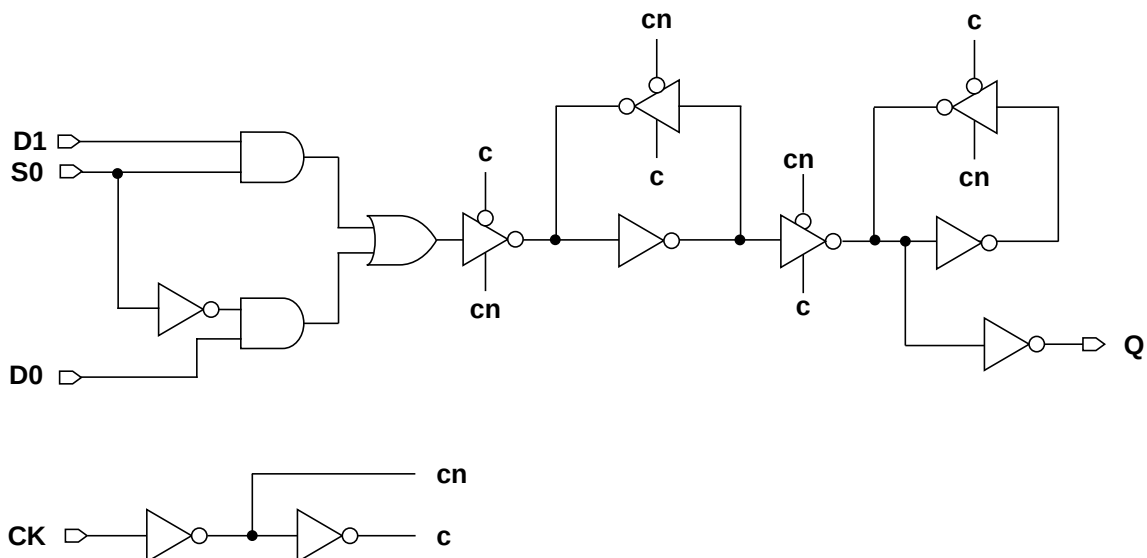
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MDFFHQX1	3.69	10.12
MDFFHQX2	3.69	11.04
MDFFHQX4	3.69	17.48
MDFFHQX8	3.69	18.40

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
D0	0.0174	0.0235	0.0384	0.0386
D1	0.0182	0.0249	0.0435	0.0437
S0	0.0197	0.0269	0.0472	0.0473
CK	0.0283	0.0369	0.0618	0.0617
Q	0.0066	0.0096	0.0142	0.0253

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
D0	0.0020	0.0030	0.0047	0.0047
D1	0.0019	0.0032	0.0052	0.0051
S0	0.0039	0.0061	0.0105	0.0104
CK	0.0026	0.0042	0.0071	0.0071

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1367	0.1154	0.1059	0.1206	4.4356	2.2642	1.1300	0.5684
CK $\rightarrow$ Q $\downarrow$	0.1488	0.1253	0.1206	0.1381	2.9237	1.5045	0.7448	0.3834

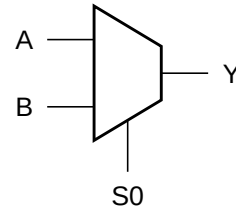
Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
D0	setup $\uparrow \rightarrow$ CK	0.1289	0.1016	0.1016	0.1016
	setup $\downarrow \rightarrow$ CK	0.1445	0.1250	0.0977	0.0977
	hold $\uparrow \rightarrow$ CK	-0.0742	-0.0469	-0.0469	-0.0430
	hold $\downarrow \rightarrow$ CK	-0.0977	-0.0781	-0.0508	-0.0469
D1	setup $\uparrow \rightarrow$ CK	0.1172	0.1094	0.0977	0.1016
	setup $\downarrow \rightarrow$ CK	0.1445	0.1328	0.1094	0.1094
	hold $\uparrow \rightarrow$ CK	-0.0664	-0.0547	-0.0469	-0.0430
	hold $\downarrow \rightarrow$ CK	-0.0977	-0.0820	-0.0625	-0.0586
S0	setup $\uparrow \rightarrow$ CK	0.1250	0.1133	0.0938	0.0938
	setup $\downarrow \rightarrow$ CK	0.1445	0.1094	0.1055	0.1094
	hold $\uparrow \rightarrow$ CK	-0.0508	-0.0352	-0.0312	-0.0273
	hold $\downarrow \rightarrow$ CK	-0.0781	-0.0547	-0.0391	-0.0352
CK	minpwh	0.0690	0.0544	0.0447	0.0495
	minpwl	0.1856	0.1564	0.1467	0.1467

**Cell Description**

The MX2 cell is a 2-to-1 multiplexer. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y). The output (Y) is represented by the logic equation:

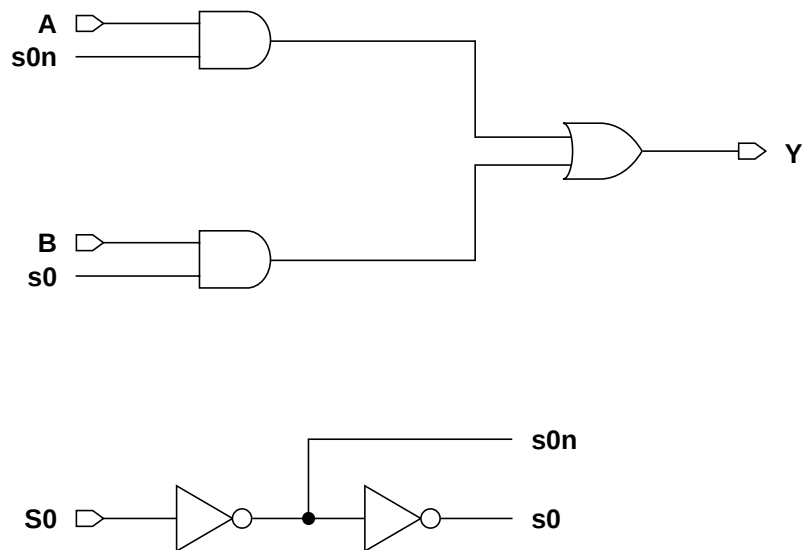
$$Y = (\overline{S0} \bullet A) + (S0 \bullet B)$$

Logic Symbol**Functions**

S0	A	B	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
MX2XL	3.69	3.68
MX2X1	3.69	3.68
MX2X2	3.69	4.14
MX2X4	3.69	5.52
MX2X6	3.69	5.52
MX2X8	3.69	5.98

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
S0	0.0081	0.0112	0.0195	0.0284	0.0345	0.0412
A	0.0062	0.0084	0.0147	0.0212	0.0290	0.0350
B	0.0067	0.0093	0.0166	0.0256	0.0311	0.0384

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
S0	0.0030	0.0039	0.0061	0.0073	0.0076	0.0076
A	0.0015	0.0024	0.0039	0.0047	0.0051	0.0051
B	0.0014	0.0022	0.0037	0.0050	0.0048	0.0048

Delay Tables at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
S0 $\rightarrow$ Y $\uparrow$	0.1179	0.1203	0.1125	0.1242	0.1356	0.1444
S0 $\rightarrow$ Y $\downarrow$	0.1136	0.1021	0.0944	0.1079	0.1228	0.1411
A $\rightarrow$ Y $\uparrow$	0.0930	0.0811	0.0739	0.0787	0.0953	0.1055
A $\rightarrow$ Y $\downarrow$	0.1277	0.1093	0.0946	0.1061	0.1195	0.1337
B $\rightarrow$ Y $\uparrow$	0.0936	0.0770	0.0706	0.0784	0.0894	0.0989
B $\rightarrow$ Y $\downarrow$	0.1331	0.1118	0.0997	0.1180	0.1272	0.1418

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
S0 $\rightarrow$ Y $\uparrow$	9.4183	5.1506	2.2603	1.2597	0.7617	0.5751
S0 $\rightarrow$ Y $\downarrow$	5.3475	2.9227	1.5080	0.7879	0.5325	0.4084
A $\rightarrow$ Y $\uparrow$	9.4226	5.1500	2.2604	1.2608	0.7620	0.5755
A $\rightarrow$ Y $\downarrow$	5.4356	2.9517	1.5073	0.7800	0.5292	0.4096
B $\rightarrow$ Y $\uparrow$	9.4320	5.1509	2.2586	1.2595	0.7606	0.5742
B $\rightarrow$ Y $\downarrow$	5.5162	2.9560	1.5089	0.7885	0.5368	0.4163

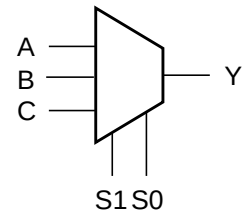
**Cell Description**

The MX3 cell is a 3-to-1 multiplexer. The state of the select inputs (S1, S0) determines which data input (A, B, C) is presented to the output (Y). The output (Y) is represented by the logic equation:

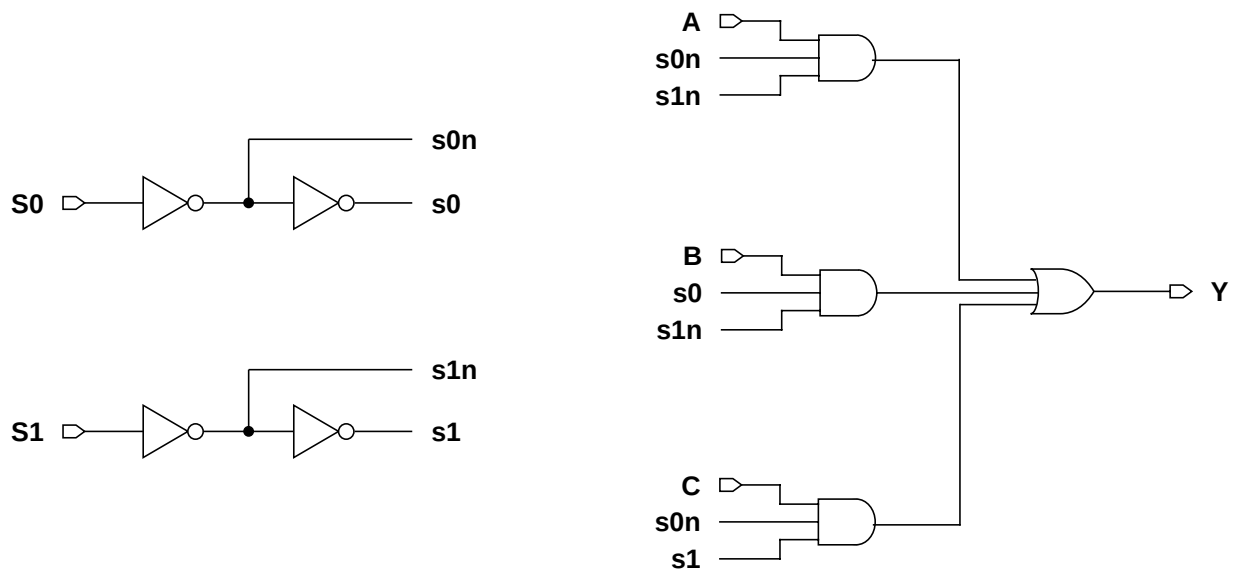
$$Y = (\overline{S0} \cdot \overline{S1} \cdot A) + (S0 \cdot \overline{S1} \cdot B) + (\overline{S0} \cdot S1 \cdot C)$$

Functions

S1	S0	A	B	C	Y
0	0	0	x	x	0
0	0	1	x	x	1
0	1	x	0	x	0
0	1	x	1	x	1
1	x	x	x	0	0
1	x	x	x	1	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
MX3XL	3.69	5.98
MX3X1	3.69	6.90
MX3X2	3.69	7.82
MX3X4	3.69	8.28

Functional Schematic

**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0134	0.0181	0.0290	0.0340
S1	0.0104	0.0120	0.0202	0.0258
A	0.0090	0.0137	0.0224	0.0292
B	0.0098	0.0149	0.0253	0.0322
C	0.0074	0.0107	0.0190	0.0262

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0037	0.0051	0.0083	0.0082
S1	0.0038	0.0043	0.0070	0.0076
A	0.0014	0.0031	0.0047	0.0047
B	0.0013	0.0030	0.0046	0.0046
C	0.0015	0.0021	0.0038	0.0048

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.1828	0.1561	0.1465	0.1594	9.8171	5.1959	2.2870	1.1562
S0 $\rightarrow$ Y $\downarrow$	0.1940	0.1486	0.1403	0.1644	6.6695	3.2300	1.6696	0.8675
S1 $\rightarrow$ Y $\uparrow$	0.1413	0.1188	0.1102	0.1204	9.8064	5.1892	2.2824	1.1548
S1 $\rightarrow$ Y $\downarrow$	0.1088	0.1053	0.0946	0.1097	5.6696	3.0245	1.5453	0.8491
A $\rightarrow$ Y $\uparrow$	0.1610	0.1182	0.1131	0.1272	9.8237	5.1989	2.2873	1.1567
A $\rightarrow$ Y $\downarrow$	0.2149	0.1449	0.1385	0.1598	6.7032	3.2431	1.6724	0.8680
B $\rightarrow$ Y $\uparrow$	0.1587	0.1143	0.1102	0.1239	9.8205	5.1959	2.2860	1.1561
B $\rightarrow$ Y $\downarrow$	0.2172	0.1495	0.1451	0.1662	6.7086	3.2442	1.6747	0.8681
C $\rightarrow$ Y $\uparrow$	0.0993	0.0883	0.0790	0.0806	9.6406	5.1579	2.2620	1.1346
C $\rightarrow$ Y $\downarrow$	0.1428	0.1326	0.1121	0.1127	5.4151	3.0387	1.5493	0.7749



Cell Description

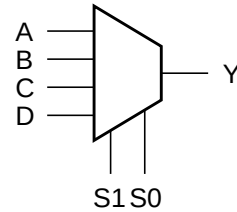
The MX4 cell is a 4-to-1 multiplexer. The state of the select inputs (S1, S0) determines which data input (A, B, C, D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \cdot \overline{S1} \cdot A) + (S0 \cdot \overline{S1} \cdot B) + (\overline{S0} \cdot S1 \cdot C) + (S0 \cdot S1 \cdot D)$$

Functions

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	0
0	0	1	x	x	x	1
0	1	x	0	x	x	0
0	1	x	1	x	x	1
1	0	x	x	0	x	0
1	0	x	x	1	x	1
1	1	x	x	x	0	0
1	1	x	x	x	1	1

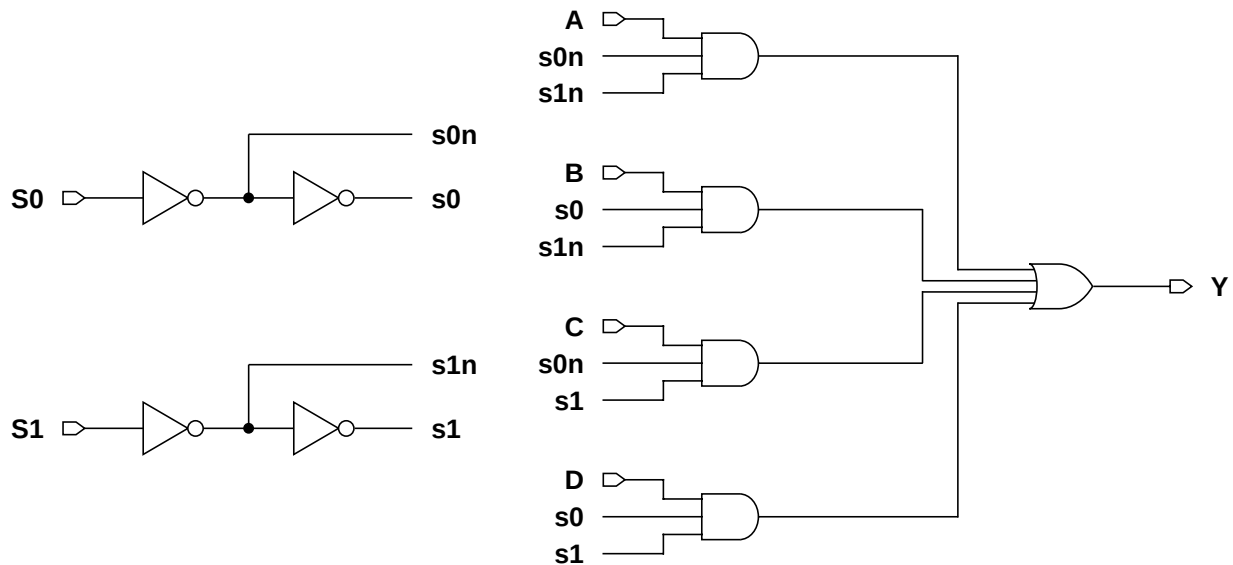
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MX4XL	3.69	8.74
MX4X1	3.69	8.74
MX4X2	3.69	10.12
MX4X4	3.69	10.58

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0131	0.0231	0.0381	0.0428
S1	0.0075	0.0131	0.0213	0.0255
A	0.0083	0.0140	0.0225	0.0293
B	0.0089	0.0152	0.0255	0.0324
C	0.0094	0.0157	0.0262	0.0332
D	0.0103	0.0172	0.0291	0.0358

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0052	0.0093	0.0160	0.0162
S1	0.0028	0.0048	0.0078	0.0078
A	0.0014	0.0030	0.0046	0.0046
B	0.0013	0.0030	0.0045	0.0046
C	0.0013	0.0030	0.0047	0.0047
D	0.0013	0.0030	0.0046	0.0046

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.2063	0.1588	0.1366	0.1500	9.6849	5.2067	2.2879	1.1571
S0 $\rightarrow$ Y $\downarrow$	0.2169	0.1653	0.1592	0.1816	7.0336	3.2867	1.6982	0.8885
S1 $\rightarrow$ Y $\uparrow$	0.1268	0.1207	0.1094	0.1205	9.6482	5.1995	2.2862	1.1563
S1 $\rightarrow$ Y $\downarrow$	0.1146	0.1005	0.0972	0.1174	6.2772	3.2203	1.6722	0.8827
A $\rightarrow$ Y $\uparrow$	0.1533	0.1199	0.1133	0.1266	9.6557	5.2052	2.2884	1.1576
A $\rightarrow$ Y $\downarrow$	0.1954	0.1513	0.1478	0.1683	6.8622	3.2774	1.6958	0.8870
B $\rightarrow$ Y $\uparrow$	0.1528	0.1178	0.1129	0.1260	9.6560	5.2040	2.2882	1.1576
B $\rightarrow$ Y $\downarrow$	0.1993	0.1539	0.1492	0.1692	6.8901	3.2794	1.6858	0.8811
C $\rightarrow$ Y $\uparrow$	0.1615	0.1218	0.1133	0.1259	9.6934	5.2098	2.2908	1.1581
C $\rightarrow$ Y $\downarrow$	0.2084	0.1602	0.1562	0.1772	7.0440	3.2951	1.6997	0.8890
D $\rightarrow$ Y $\uparrow$	0.1620	0.1191	0.1127	0.1253	9.6924	5.2072	2.2912	1.1584
D $\rightarrow$ Y $\downarrow$	0.2150	0.1624	0.1560	0.1765	7.0765	3.2956	1.6875	0.8820



Cell Description

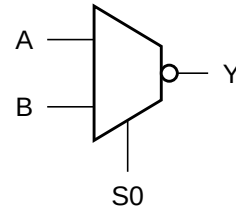
The MXI2 cell is a 2-to-1 multiplexer with inverted output. The state of the select input (S0) determines which data input (A, B) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet A) + (S0 \bullet B)$$

Functions

S0	A	B	Y
0	0	x	1
0	1	x	0
1	x	0	1
1	x	1	0

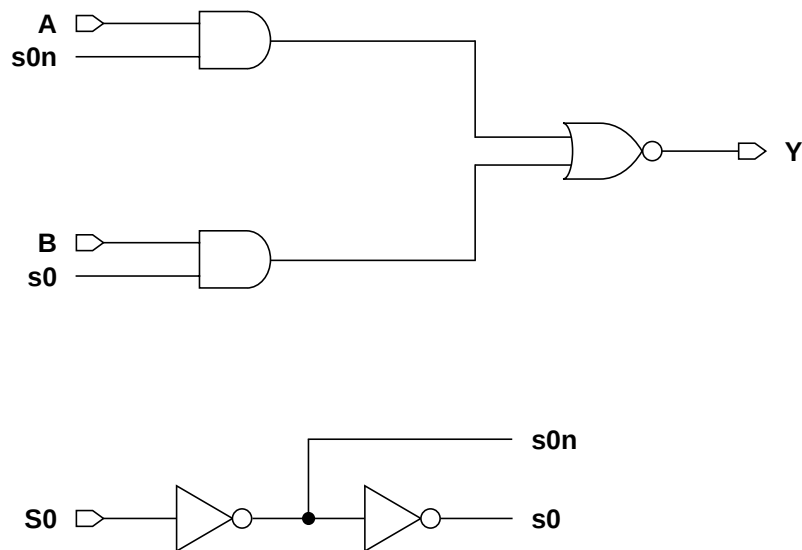
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MXI2XL	3.69	3.22
MXI2X1	3.69	3.22
MXI2X2	3.69	4.60
MXI2X4	3.69	6.90
MXI2X6	3.69	12.88
MXI2X8	3.69	17.02

Functional Schematic





AC Power

Pin	Power (μ W/MHz)					
	XL	X1	X2	X4	X6	X8
S0	0.0061	0.0092	0.0189	0.0319	0.0775	0.1053
A	0.0049	0.0075	0.0120	0.0260	0.0597	0.0788
B	0.0056	0.0091	0.0146	0.0312	0.0725	0.0912

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
S0	0.0031	0.0044	0.0075	0.0145	0.0377	0.0521
A	0.0015	0.0025	0.0045	0.0093	0.0258	0.0343
B	0.0016	0.0025	0.0049	0.0090	0.0261	0.0343

Delay Tables at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
S0 $\rightarrow$ Y $\uparrow$	0.0656	0.0573	0.0528	0.0441	0.0848	0.0816
S0 $\rightarrow$ Y $\downarrow$	0.0654	0.0785	0.0801	0.0645	0.0896	0.0851
A $\rightarrow$ Y $\uparrow$	0.0614	0.0516	0.0443	0.0439	0.0568	0.0565
A $\rightarrow$ Y $\downarrow$	0.0395	0.0373	0.0385	0.0381	0.0360	0.0355
B $\rightarrow$ Y $\uparrow$	0.0625	0.0575	0.0514	0.0475	0.0606	0.0596
B $\rightarrow$ Y $\downarrow$	0.0377	0.0351	0.0350	0.0327	0.0330	0.0328

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
S0 $\rightarrow$ Y $\uparrow$	11.5485	6.3989	3.2075	1.4527	0.8128	0.6095
S0 $\rightarrow$ Y $\downarrow$	6.7113	3.4321	1.9000	0.8792	0.4132	0.3092
A $\rightarrow$ Y $\uparrow$	11.6998	6.3776	3.1508	1.4508	0.8292	0.6217
A $\rightarrow$ Y $\downarrow$	6.8881	3.6339	2.0406	0.9364	0.4167	0.3121
B $\rightarrow$ Y $\uparrow$	11.7034	6.3816	3.3356	1.4512	0.8276	0.6206
B $\rightarrow$ Y $\downarrow$	6.8750	3.6152	2.0527	0.9187	0.4140	0.3104



Cell Description

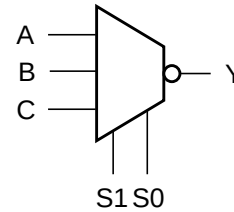
The MXI3 cell is a 3-to-1 multiplexer with inverted output. The state of the select inputs (S1, S0) determines which data input (A, B, C) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (S0 \bullet S1 \bullet \bar{A}) + (\bar{S0} \bullet S1 \bullet \bar{B}) + (\bar{S1} \bullet \bar{C})$$

Functions

S1	S0	A	B	C	Y
0	0	0	x	x	1
0	0	1	x	x	0
0	1	x	0	x	1
0	1	x	1	x	0
1	0	x	x	0	1
1	0	x	x	1	0

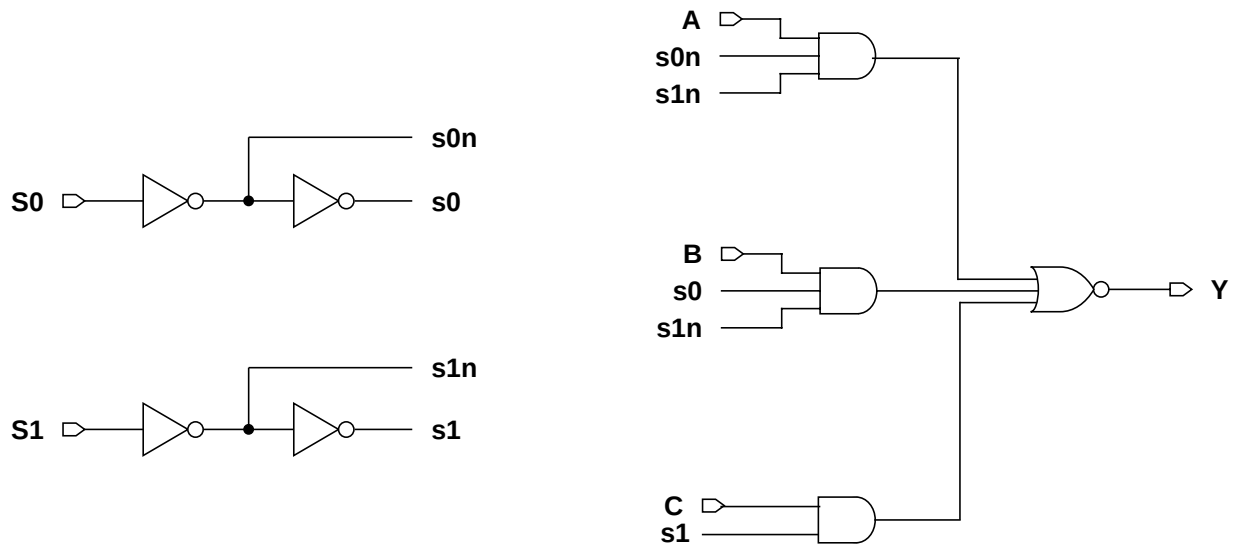
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
MXI3XL	3.69	7.36
MXI3X1	3.69	7.36
MXI3X2	3.69	7.82
MXI3X4	3.69	8.74

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0151	0.0166	0.0274	0.0379
S1	0.0107	0.0120	0.0190	0.0250
A	0.0136	0.0158	0.0250	0.0347
B	0.0140	0.0160	0.0265	0.0365
C	0.0088	0.0106	0.0173	0.0243

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0039	0.0037	0.0051	0.0060
S1	0.0027	0.0028	0.0036	0.0043
A	0.0013	0.0018	0.0029	0.0038
B	0.0014	0.0018	0.0030	0.0037
C	0.0015	0.0016	0.0020	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.2365	0.2148	0.1763	0.1794	9.4631	5.8574	2.2662	1.1380
S0 $\rightarrow$ Y $\downarrow$	0.2554	0.2240	0.2047	0.2227	5.5998	3.2199	1.5178	0.7725
S1 $\rightarrow$ Y $\uparrow$	0.1363	0.1311	0.1123	0.1121	9.4369	5.8467	2.2626	1.1359
S1 $\rightarrow$ Y $\downarrow$	0.1196	0.1080	0.0933	0.1004	5.5925	3.2006	1.5044	0.7754
A $\rightarrow$ Y $\uparrow$	0.2645	0.2196	0.1784	0.1787	9.4631	5.8576	2.2663	1.1381
A $\rightarrow$ Y $\downarrow$	0.2296	0.1916	0.1620	0.1709	5.6003	3.2193	1.5177	0.7724
B $\rightarrow$ Y $\uparrow$	0.2590	0.2174	0.1829	0.1832	9.4617	5.8572	2.2664	1.1380
B $\rightarrow$ Y $\downarrow$	0.2233	0.1862	0.1582	0.1689	5.5987	3.2189	1.5175	0.7723
C $\rightarrow$ Y $\uparrow$	0.1631	0.1582	0.1454	0.1483	9.4374	5.8471	2.2603	1.1344
C $\rightarrow$ Y $\downarrow$	0.1735	0.1512	0.1327	0.1428	5.5960	3.2124	1.5057	0.7760

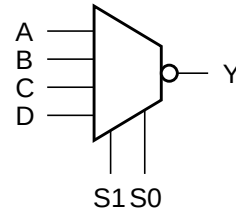


Cell Description

The MXI4 cell is a 4-to-1 multiplexer with inverted output. The state of the select inputs (S1, S0) determines which data input (A, B, C, D) is presented to the output (Y). The output (Y) is represented by the logic equation:

$$Y = (\overline{S0} \bullet \overline{S1} \bullet A) + (S0 \bullet \overline{S1} \bullet B) + (\overline{S0} \bullet S1 \bullet C) + (S0 \bullet S1 \bullet D)$$

Logic Symbol



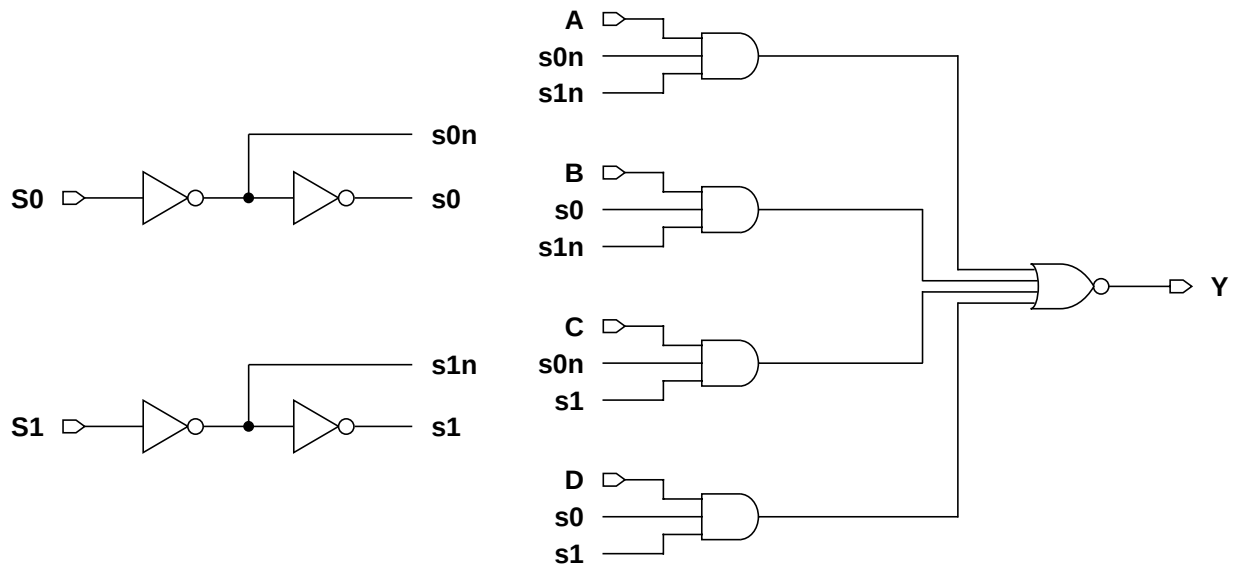
Functions

S1	S0	A	B	C	D	Y
0	0	0	x	x	x	1
0	0	1	x	x	x	0
0	1	x	0	x	x	1
0	1	x	1	x	x	0
1	0	x	x	0	x	1
1	0	x	x	1	x	0
1	1	x	x	x	0	1
1	1	x	x	x	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
MXI4XL	3.69	8.74
MXI4X1	3.69	8.74
MXI4X2	3.69	9.66
MXI4X4	3.69	10.12

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
S0	0.0152	0.0199	0.0329	0.0410
S1	0.0085	0.0117	0.0192	0.0258
A	0.0115	0.0150	0.0258	0.0344
B	0.0116	0.0162	0.0277	0.0362
C	0.0099	0.0135	0.0226	0.0290
D	0.0107	0.0145	0.0239	0.0305

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
S0	0.0048	0.0060	0.0095	0.0107
S1	0.0026	0.0027	0.0036	0.0041
A	0.0014	0.0019	0.0029	0.0037
B	0.0014	0.0018	0.0030	0.0035
C	0.0015	0.0020	0.0033	0.0038
D	0.0016	0.0021	0.0029	0.0035

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
S0 $\rightarrow$ Y $\uparrow$	0.2352	0.2157	0.1835	0.1899	9.4457	5.1626	2.2663	1.2177
S0 $\rightarrow$ Y $\downarrow$	0.2650	0.2388	0.2074	0.2103	5.5148	2.9806	1.5196	0.7802
S1 $\rightarrow$ Y $\uparrow$	0.1266	0.1262	0.1135	0.1143	9.4253	5.1506	2.2618	1.2154
S1 $\rightarrow$ Y $\downarrow$	0.1204	0.1053	0.0948	0.1029	5.4121	2.9519	1.5063	0.7746
A $\rightarrow$ Y $\uparrow$	0.2457	0.2131	0.1867	0.1921	9.4466	5.1632	2.2665	1.2176
A $\rightarrow$ Y $\downarrow$	0.2180	0.1827	0.1669	0.1751	5.5144	2.9803	1.5193	0.7801
B $\rightarrow$ Y $\uparrow$	0.2390	0.2175	0.1910	0.1947	9.4454	5.1629	2.2663	1.2177
B $\rightarrow$ Y $\downarrow$	0.2112	0.1870	0.1663	0.1731	5.5123	2.9810	1.5192	0.7800
C $\rightarrow$ Y $\uparrow$	0.2150	0.1962	0.1680	0.1710	9.4368	5.1575	2.2616	1.2155
C $\rightarrow$ Y $\downarrow$	0.1997	0.1815	0.1574	0.1640	5.5269	2.9795	1.5081	0.7753
D $\rightarrow$ Y $\uparrow$	0.2234	0.2031	0.1688	0.1724	9.4377	5.1577	2.2616	1.2156
D $\rightarrow$ Y $\downarrow$	0.2019	0.1849	0.1540	0.1616	5.5268	2.9797	1.5081	0.7753



Cell Description

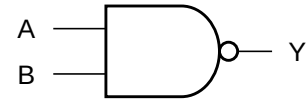
The NAND2 cell provides the logical NAND of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B)}$$

Functions

A	B	Y
0	x	1
x	0	1
1	1	0

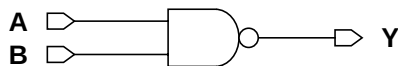
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND2XL	3.69	1.38
NAND2X1	3.69	1.38
NAND2X2	3.69	2.30
NAND2X4	3.69	3.22
NAND2X6	3.69	4.60
NAND2X8	3.69	5.52

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0027	0.0045	0.0084	0.0155	0.0233	0.0294
B	0.0037	0.0059	0.0120	0.0218	0.0332	0.0425

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0019	0.0031	0.0055	0.0105	0.0160	0.0205
B	0.0017	0.0028	0.0059	0.0101	0.0162	0.0199

Delay at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0293	0.0268	0.0226	0.0226	0.0219	0.0223
A $\rightarrow$ Y $\downarrow$	0.0192	0.0188	0.0175	0.0181	0.0178	0.0181
B $\rightarrow$ Y $\uparrow$	0.0366	0.0345	0.0304	0.0294	0.0291	0.0292
B $\rightarrow$ Y $\downarrow$	0.0213	0.0215	0.0211	0.0213	0.0214	0.0215

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.3695	5.1305	2.2460	1.2021	0.7546	0.6010
A $\rightarrow$ Y $\downarrow$	5.3413	3.1353	1.5674	0.8825	0.5602	0.4481
B $\rightarrow$ Y $\uparrow$	9.3367	5.1188	2.2444	1.2007	0.7541	0.6004
B $\rightarrow$ Y $\downarrow$	5.3400	3.1346	1.5688	0.8827	0.5604	0.4482



Cell Description

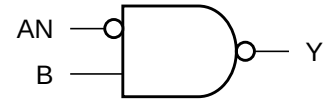
The NAND2B cell provides the logical NAND of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$$Y = \overline{(AN \bullet B)}$$

Functions

AN	B	Y
1	x	1
x	0	1
0	1	0

Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND2BXL	3.69	1.84
NAND2BX1	3.69	1.84
NAND2BX2	3.69	2.76
NAND2BX4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0060	0.0067	0.0113	0.0209
B	0.0039	0.0051	0.0096	0.0177

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0013	0.0015	0.0024	0.0044
B	0.0018	0.0030	0.0059	0.0105

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0725	0.0629	0.0555	0.0568	9.3918	5.1436	2.2322	1.2162
AN $\rightarrow$ Y $\downarrow$	0.0951	0.0914	0.0811	0.0697	5.5635	3.2690	1.6174	0.8449
B $\rightarrow$ Y $\uparrow$	0.0439	0.0334	0.0292	0.0298	9.3523	5.1305	2.2311	1.2151
B $\rightarrow$ Y $\downarrow$	0.0267	0.0230	0.0233	0.0232	5.3900	3.1720	1.5872	0.8359

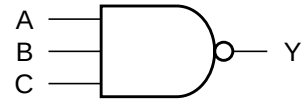


Cell Description

The NAND3 cell provides the logical NAND of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C)}$$

Logic Symbol



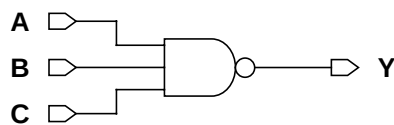
Functions

A	B	C	Y
0	x	x	1
x	0	x	1
x	x	0	1
1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND3XL	3.69	1.84
NAND3X1	3.69	1.84
NAND3X2	3.69	3.22
NAND3X4	3.69	4.60
NAND3X6	3.69	6.90
NAND3X8	3.69	9.20

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0035	0.0056	0.0096	0.0175	0.0258	0.0360
B	0.0046	0.0077	0.0140	0.0245	0.0348	0.0511
C	0.0056	0.0094	0.0179	0.0314	0.0450	0.0624

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0021	0.0033	0.0057	0.0110	0.0162	0.0233
B	0.0020	0.0032	0.0063	0.0110	0.0164	0.0235
C	0.0018	0.0029	0.0067	0.0110	0.0169	0.0238

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0410	0.0361	0.0280	0.0300	0.0263	0.0279
A $\rightarrow$ Y $\downarrow$	0.0287	0.0267	0.0222	0.0245	0.0248	0.0227
B $\rightarrow$ Y $\uparrow$	0.0502	0.0457	0.0383	0.0388	0.0336	0.0367
B $\rightarrow$ Y $\downarrow$	0.0339	0.0327	0.0302	0.0311	0.0318	0.0292
C $\rightarrow$ Y $\uparrow$	0.0585	0.0538	0.0458	0.0463	0.0400	0.0446
C $\rightarrow$ Y $\downarrow$	0.0369	0.0358	0.0338	0.0347	0.0361	0.0334

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.3687	5.1310	2.3032	1.3001	0.7478	0.6070
A $\rightarrow$ Y $\downarrow$	6.5228	3.8156	1.9335	1.1082	0.7770	0.5178
B $\rightarrow$ Y $\uparrow$	9.3444	5.1229	2.3036	1.2991	0.7478	0.6065
B $\rightarrow$ Y $\downarrow$	6.5201	3.8133	1.9340	1.1078	0.7766	0.5178
C $\rightarrow$ Y $\uparrow$	9.3696	5.1392	2.3133	1.3038	0.7514	0.6092
C $\rightarrow$ Y $\downarrow$	6.5196	3.8144	1.9353	1.1083	0.7771	0.5179



Cell Description

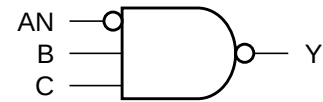
The NAND3B cell provides the logical NAND of one inverted input (AN) and two non-inverted inputs (B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A}N \bullet B \bullet C)}$$

Functions

AN	B	C	Y
1	x	x	1
x	0	x	1
x	x	0	1
0	1	1	0

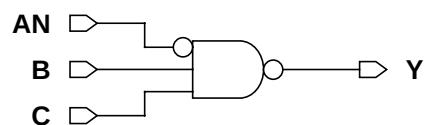
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND3BXL	3.69	2.30
NAND3BX1	3.69	2.30
NAND3BX2	3.69	3.22
NAND3BX4	3.69	5.06

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0078	0.0082	0.0127	0.0260
B	0.0041	0.0054	0.0091	0.0186
C	0.0050	0.0068	0.0125	0.0250

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0015	0.0018	0.0024	0.0045
B	0.0021	0.0029	0.0057	0.0108
C	0.0021	0.0029	0.0060	0.0109

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0822	0.0692	0.0620	0.0626	9.3862	6.0193	2.5511	1.3016
AN $\rightarrow$ Y $\downarrow$	0.1125	0.0910	0.0794	0.0749	6.8580	4.2609	2.2603	1.1158
B $\rightarrow$ Y $\uparrow$	0.0528	0.0472	0.0350	0.0371	9.3641	6.0181	2.5515	1.3015
B $\rightarrow$ Y $\downarrow$	0.0379	0.0348	0.0312	0.0325	6.7049	4.2278	2.2466	1.1119
C $\rightarrow$ Y $\uparrow$	0.0612	0.0568	0.0429	0.0454	9.3835	6.0250	2.5569	1.3042
C $\rightarrow$ Y $\downarrow$	0.0413	0.0390	0.0358	0.0371	6.7107	4.2314	2.2487	1.1122

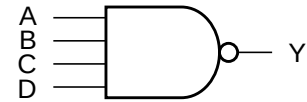


Cell Description

The NAND4 cell provides a logical NAND of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A \bullet B \bullet C \bullet D)}$$

Logic Symbol



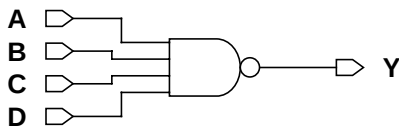
Functions

A	B	C	D	Y
0	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
1	1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4XL	3.69	2.30
NAND4X1	3.69	2.30
NAND4X2	3.69	3.68
NAND4X4	3.69	7.36
NAND4X6	3.69	9.66
NAND4X8	3.69	14.26

Functional Schematic



**AC Power**

Pin	Power (μ W/MHz)					
	XL	X1	X2	X4	X6	X8
A	0.0039	0.0064	0.0101	0.0217	0.0301	0.0440
B	0.0053	0.0088	0.0134	0.0292	0.0415	0.0613
C	0.0064	0.0108	0.0175	0.0382	0.0549	0.0771
D	0.0074	0.0125	0.0211	0.0462	0.0659	0.0926

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0023	0.0034	0.0056	0.0129	0.0188	0.0264
B	0.0022	0.0034	0.0058	0.0132	0.0182	0.0256
C	0.0022	0.0034	0.0062	0.0135	0.0185	0.0258
D	0.0019	0.0032	0.0067	0.0141	0.0189	0.0264

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0434	0.0391	0.0320	0.0321	0.0331	0.0341
A $\rightarrow$ Y $\downarrow$	0.0315	0.0295	0.0263	0.0257	0.0265	0.0267
B $\rightarrow$ Y $\uparrow$	0.0534	0.0503	0.0420	0.0423	0.0435	0.0447
B $\rightarrow$ Y $\downarrow$	0.0391	0.0390	0.0360	0.0352	0.0363	0.0364
C $\rightarrow$ Y $\uparrow$	0.0635	0.0603	0.0506	0.0510	0.0528	0.0542
C $\rightarrow$ Y $\downarrow$	0.0454	0.0458	0.0428	0.0418	0.0434	0.0433
D $\rightarrow$ Y $\uparrow$	0.0715	0.0679	0.0584	0.0590	0.0611	0.0627
D $\rightarrow$ Y $\downarrow$	0.0486	0.0489	0.0475	0.0464	0.0483	0.0481

**Delays at 25°C, 1.2V, Typical Process**

Description	K _{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A → Y↑	9.3789	5.1324	2.5487	1.1726	0.8501	0.6254
A → Y↓	7.6301	4.4603	2.5049	1.1150	0.8123	0.5803
B → Y↑	9.3501	5.1249	2.5469	1.1713	0.8493	0.6247
B → Y↓	7.6215	4.4562	2.5038	1.1143	0.8117	0.5798
C → Y↑	9.3815	5.1443	2.5570	1.1764	0.8530	0.6273
C → Y↓	7.6237	4.4574	2.5044	1.1146	0.8121	0.5801
D → Y↑	9.4442	5.1786	2.5753	1.1857	0.8596	0.6323
D → Y↓	7.6214	4.4570	2.5056	1.1152	0.8124	0.5802



Cell Description

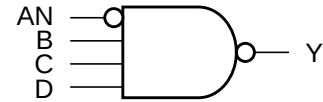
The NAND4B cell provides a logical NAND of one inverted input (AN) and three non-inverted inputs (B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(AN \bullet B \bullet C \bullet D)}$$

Functions

AN	B	C	D	Y
1	x	x	x	1
x	0	x	x	1
x	x	0	x	1
x	x	x	0	1
0	1	1	1	0

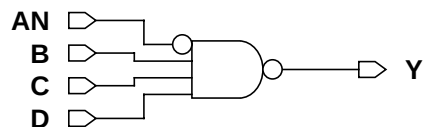
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4BXL	3.69	2.76
NAND4BX1	3.69	2.76
NAND4BX2	3.69	4.14
NAND4BX4	3.69	7.82

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0074	0.0101	0.0146	0.0308
B	0.0043	0.0069	0.0107	0.0245
C	0.0051	0.0085	0.0146	0.0316
D	0.0062	0.0102	0.0182	0.0397

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0013	0.0016	0.0029	0.0048
B	0.0021	0.0034	0.0059	0.0130
C	0.0021	0.0032	0.0063	0.0136
D	0.0020	0.0032	0.0069	0.0143

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0842	0.0804	0.0628	0.0669	9.4128	5.1470	2.5010	1.1742
AN $\rightarrow$ Y $\downarrow$	0.1117	0.1115	0.0824	0.0773	7.7558	4.7461	2.4822	1.1193
B $\rightarrow$ Y $\uparrow$	0.0545	0.0489	0.0405	0.0414	9.3758	5.1369	2.5031	1.1743
B $\rightarrow$ Y $\downarrow$	0.0423	0.0429	0.0379	0.0374	7.6575	4.7074	2.4760	1.1169
C $\rightarrow$ Y $\uparrow$	0.0653	0.0590	0.0497	0.0510	9.3898	5.1455	2.5083	1.1767
C $\rightarrow$ Y $\downarrow$	0.0497	0.0507	0.0453	0.0448	7.6656	4.7094	2.4766	1.1172
D $\rightarrow$ Y $\uparrow$	0.0740	0.0670	0.0582	0.0597	9.4404	5.1730	2.5238	1.1847
D $\rightarrow$ Y $\downarrow$	0.0531	0.0541	0.0502	0.0497	7.6650	4.7096	2.4779	1.1176



Cell Description

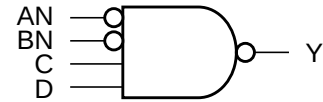
The NAND4BB cell provides a logical NAND of two inverted inputs (AN, BN) and two non-inverted inputs (C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} \bullet \overline{BN} \bullet C \bullet D)}$$

Functions

AN	BN	C	D	Y
1	x	x	x	1
x	1	x	x	1
x	x	0	x	1
x	x	x	0	1
0	0	1	1	0

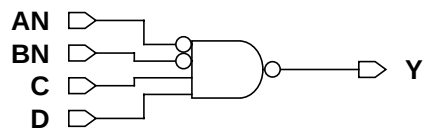
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND4BBXL	3.69	3.22
NAND4BBX1	3.69	3.68
NAND4BBX2	3.69	5.06
NAND4BBX4	3.69	8.74

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0064	0.0089	0.0151	0.0312
BN	0.0074	0.0107	0.0178	0.0380
C	0.0044	0.0072	0.0126	0.0272
D	0.0052	0.0090	0.0158	0.0341

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0013	0.0017	0.0027	0.0048
BN	0.0013	0.0016	0.0028	0.0047
C	0.0022	0.0032	0.0062	0.0135
D	0.0022	0.0032	0.0067	0.0142

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0762	0.0703	0.0654	0.0678	9.4052	5.1420	2.5020	1.1548
AN $\rightarrow$ Y $\downarrow$	0.0994	0.0932	0.0858	0.0784	7.7189	4.4904	2.4805	1.1183
BN $\rightarrow$ Y $\uparrow$	0.0845	0.0844	0.0748	0.0790	9.3924	5.1500	2.5048	1.1557
BN $\rightarrow$ Y $\downarrow$	0.1039	0.1079	0.0965	0.0911	7.7550	4.5301	2.4955	1.1255
C $\rightarrow$ Y $\uparrow$	0.0616	0.0580	0.0491	0.0498	9.4138	5.1655	2.5137	1.1599
C $\rightarrow$ Y $\downarrow$	0.0483	0.0499	0.0469	0.0469	7.6798	4.4914	2.4824	1.1198
D $\rightarrow$ Y $\uparrow$	0.0707	0.0672	0.0573	0.0586	9.4569	5.1911	2.5265	1.1663
D $\rightarrow$ Y $\downarrow$	0.0522	0.0541	0.0518	0.0522	7.6848	4.4927	2.4837	1.1205

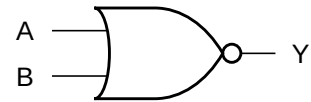


Cell Description

The NOR2 cell provides a logical NOR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B)}$$

Logic Symbol



Functions

A	B	Y
0	0	1
x	1	0
1	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR2XL	3.69	1.38
NOR2X1	3.69	1.38
NOR2X2	3.69	2.30
NOR2X4	3.69	3.22
NOR2X6	3.69	4.14
NOR2X8	3.69	5.52

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0029	0.0047	0.0088	0.0170	0.0241	0.0341
B	0.0036	0.0058	0.0112	0.0222	0.0305	0.0431

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0019	0.0031	0.0056	0.0114	0.0153	0.0219
B	0.0016	0.0027	0.0059	0.0109	0.0153	0.0208

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0385	0.0315	0.0281	0.0273	0.0269	0.0268
A $\rightarrow$ Y $\downarrow$	0.0177	0.0176	0.0172	0.0174	0.0167	0.0172
B $\rightarrow$ Y $\uparrow$	0.0471	0.0398	0.0379	0.0360	0.0361	0.0355
B $\rightarrow$ Y $\downarrow$	0.0203	0.0210	0.0217	0.0216	0.0208	0.0214

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	14.7291	7.2817	3.6378	1.7217	1.2645	0.8759
A $\rightarrow$ Y $\downarrow$	4.3412	2.5616	1.3704	0.6853	0.4778	0.3504
B $\rightarrow$ Y $\uparrow$	14.6832	7.2653	3.6345	1.7192	1.2634	0.8748
B $\rightarrow$ Y $\downarrow$	4.3805	2.5789	1.3799	0.6896	0.4807	0.3525



Cell Description

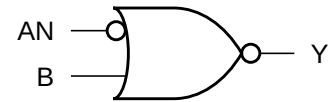
The NOR2B cell provides a logical NOR of one inverted input (AN) and one non-inverted input (B). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A}N + B)}$$

Functions

AN	B	Y
1	0	1
x	1	0
0	x	0

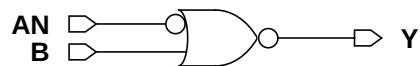
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR2BXL	3.69	1.84
NOR2BX1	3.69	1.84
NOR2BX2	3.69	2.76
NOR2BX4	3.69	3.68

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0051	0.0067	0.0104	0.0197
B	0.0040	0.0059	0.0111	0.0215

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0012	0.0013	0.0027	0.0045
B	0.0017	0.0030	0.0060	0.0108

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0750	0.0671	0.0588	0.0578	14.7398	7.2912	3.6397	1.7593
AN $\rightarrow$ Y $\downarrow$	0.0914	0.0965	0.0861	0.0723	4.9843	2.7530	1.4404	0.7084
B $\rightarrow$ Y $\uparrow$	0.0521	0.0421	0.0393	0.0377	14.6829	7.2713	3.6366	1.7567
B $\rightarrow$ Y $\downarrow$	0.0222	0.0210	0.0213	0.0208	4.7021	2.5821	1.3817	0.6907

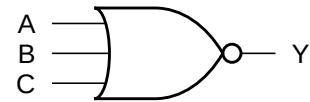


Cell Description

The NOR3 cell provides a logical NOR of three inputs (A, B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C)}$$

Logic Symbol



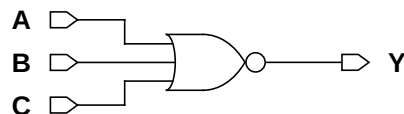
Functions

A	B	C	Y
0	0	0	1
x	x	1	0
x	1	x	0
1	x	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR3XL	3.69	1.84
NOR3X1	3.69	1.84
NOR3X2	3.69	2.76
NOR3X4	3.69	5.52
NOR3X6	3.69	7.36
NOR3X8	3.69	9.20

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0043	0.0073	0.0106	0.0214	0.0325	0.0407
B	0.0052	0.0089	0.0137	0.0279	0.0418	0.0512
C	0.0064	0.0107	0.0180	0.0374	0.0557	0.0673

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0021	0.0032	0.0058	0.0131	0.0198	0.0245
B	0.0020	0.0032	0.0064	0.0136	0.0193	0.0234
C	0.0020	0.0032	0.0065	0.0131	0.0194	0.0232

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0577	0.0495	0.0345	0.0295	0.0319	0.0344
A $\rightarrow$ Y $\downarrow$	0.0227	0.0242	0.0217	0.0212	0.0214	0.0207
B $\rightarrow$ Y $\uparrow$	0.0813	0.0717	0.0568	0.0510	0.0516	0.0539
B $\rightarrow$ Y $\downarrow$	0.0275	0.0308	0.0293	0.0293	0.0283	0.0266
C $\rightarrow$ Y $\uparrow$	0.0905	0.0801	0.0656	0.0597	0.0609	0.0631
C $\rightarrow$ Y $\downarrow$	0.0286	0.0332	0.0323	0.0326	0.0316	0.0294

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	17.7630	9.0830	4.5407	2.0020	1.3378	1.1431
A $\rightarrow$ Y $\downarrow$	4.3426	2.8555	1.6232	0.7915	0.5056	0.3853
B $\rightarrow$ Y $\uparrow$	17.7201	9.0762	4.5381	2.0009	1.3356	1.1412
B $\rightarrow$ Y $\downarrow$	4.3851	2.8738	1.6334	0.7967	0.5086	0.3875
C $\rightarrow$ Y $\uparrow$	17.7129	9.0751	4.5378	2.0005	1.3360	1.1416
C $\rightarrow$ Y $\downarrow$	4.4958	2.9283	1.6654	0.8135	0.5193	0.3951



Cell Description

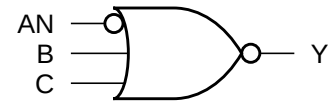
The NOR3B cell provides a logical NOR of one inverted input (AN) and two non-inverted inputs (B, C). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{A}N + B + C)}$$

Functions

AN	B	C	Y
1	0	0	1
x	x	1	0
x	1	x	0
0	x	x	0

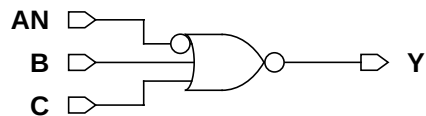
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR3BXL	3.69	2.30
NOR3BX1	3.69	2.76
NOR3BX2	3.69	3.68
NOR3BX4	3.69	6.44

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0057	0.0074	0.0127	0.0240
B	0.0060	0.0082	0.0159	0.0296
C	0.0070	0.0100	0.0198	0.0385

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0014	0.0017	0.0030	0.0055
B	0.0020	0.0034	0.0067	0.0144
C	0.0019	0.0034	0.0072	0.0139

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0970	0.0682	0.0642	0.0601	17.7555	8.0203	4.0067	2.0032
AN $\rightarrow$ Y $\downarrow$	0.0995	0.0912	0.0855	0.0806	4.6017	2.6841	1.4210	0.7086
B $\rightarrow$ Y $\uparrow$	0.0910	0.0604	0.0570	0.0556	17.7161	8.0062	4.0037	2.0027
B $\rightarrow$ Y $\downarrow$	0.0292	0.0269	0.0274	0.0270	4.3855	2.5864	1.3838	0.6922
C $\rightarrow$ Y $\uparrow$	0.1003	0.0687	0.0665	0.0644	17.7076	8.0039	4.0046	2.0023
C $\rightarrow$ Y $\downarrow$	0.0307	0.0292	0.0307	0.0299	4.4883	2.6387	1.4115	0.7061

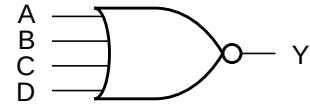


Cell Description

The NOR4 cell provides a logical NOR of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(A + B + C + D)}$$

Logic Symbol



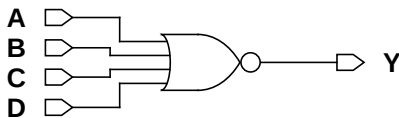
Functions

A	B	C	D	Y
0	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
1	x	x	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4XL	3.69	2.30
NOR4X1	3.69	2.30
NOR4X2	3.69	4.14
NOR4X4	3.69	7.82
NOR4X6	3.69	11.96
NOR4X8	3.69	16.10

Functional Schematic





AC Power

Pin	Power (μ W/MHz)					
	XL	X1	X2	X4	X6	X8
A	0.0055	0.0073	0.0141	0.0260	0.0402	0.0552
B	0.0066	0.0092	0.0172	0.0316	0.0485	0.0660
C	0.0078	0.0112	0.0214	0.0397	0.0605	0.0821
D	0.0090	0.0137	0.0261	0.0486	0.0740	0.0997

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0022	0.0038	0.0065	0.0137	0.0211	0.0288
B	0.0022	0.0036	0.0068	0.0135	0.0211	0.0285
C	0.0022	0.0037	0.0072	0.0140	0.0212	0.0287
D	0.0022	0.0035	0.0076	0.0142	0.0214	0.0293

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0705	0.0417	0.0433	0.0420	0.0439	0.0457
A $\rightarrow$ Y $\downarrow$	0.0266	0.0233	0.0233	0.0240	0.0236	0.0233
B $\rightarrow$ Y $\uparrow$	0.1101	0.0727	0.0774	0.0745	0.0758	0.0774
B $\rightarrow$ Y $\downarrow$	0.0335	0.0301	0.0305	0.0311	0.0303	0.0295
C $\rightarrow$ Y $\uparrow$	0.1349	0.0922	0.0980	0.0950	0.0964	0.0981
C $\rightarrow$ Y $\downarrow$	0.0368	0.0334	0.0339	0.0350	0.0337	0.0327
D $\rightarrow$ Y $\uparrow$	0.1435	0.0999	0.1085	0.1065	0.1077	0.1097
D $\rightarrow$ Y $\downarrow$	0.0365	0.0335	0.0348	0.0363	0.0348	0.0337

**Delays at 25°C, 1.2V, Typical Process**

Description	K _{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A → Y↑	20.4297	9.4141	5.2969	2.7982	1.8659	1.3999
A → Y↓	4.4970	2.5653	1.3965	0.7745	0.4923	0.3516
B → Y↑	20.4102	9.4017	5.2941	2.7950	1.8636	1.3979
B → Y↓	4.5382	2.5840	1.4070	0.7795	0.4955	0.3540
C → Y↑	20.4259	9.4018	5.2941	2.7955	1.8637	1.3980
C → Y↓	4.6520	2.6430	1.4374	0.7951	0.5057	0.3614
D → Y↑	20.4215	9.3991	5.2985	2.7978	1.8653	1.3994
D → Y↓	4.7839	2.7160	1.4747	0.8160	0.5189	0.3709



Cell Description

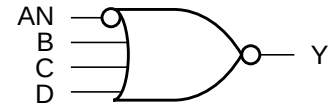
The NOR4B cell provides a logical NOR of one inverted input (AN) and three non-inverted inputs (B, C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(AN + B + C + D)}$$

Functions

AN	B	C	D	Y
1	0	0	0	1
x	x	x	1	0
x	x	1	x	0
x	1	x	x	0
0	x	x	x	0

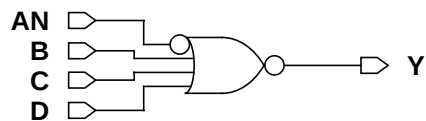
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4BXL	3.69	3.22
NOR4BX1	3.69	3.22
NOR4BX2	3.69	5.06
NOR4BX4	3.69	8.74

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0056	0.0083	0.0162	0.0261
B	0.0059	0.0092	0.0186	0.0326
C	0.0071	0.0112	0.0226	0.0395
D	0.0083	0.0136	0.0275	0.0483

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0013	0.0019	0.0031	0.0049
B	0.0021	0.0036	0.0073	0.0135
C	0.0022	0.0037	0.0077	0.0139
D	0.0021	0.0036	0.0078	0.0142

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.0939	0.0721	0.0752	0.0758	20.4727	9.4125	4.7107	2.8316
AN $\rightarrow$ Y $\downarrow$	0.0998	0.0967	0.0936	0.0851	4.6083	2.6862	1.3409	0.7799
B $\rightarrow$ Y $\uparrow$	0.0994	0.0736	0.0722	0.0757	20.4165	9.4023	4.7010	2.8274
B $\rightarrow$ Y $\downarrow$	0.0299	0.0296	0.0290	0.0300	4.3936	2.5885	1.2950	0.7662
C $\rightarrow$ Y $\uparrow$	0.1235	0.0932	0.0919	0.0967	20.4187	9.4017	4.7007	2.8271
C $\rightarrow$ Y $\downarrow$	0.0329	0.0334	0.0328	0.0341	4.5038	2.6421	1.3216	0.7803
D $\rightarrow$ Y $\uparrow$	0.1316	0.1010	0.1001	0.1075	20.4088	9.4000	4.7010	2.8290
D $\rightarrow$ Y $\downarrow$	0.0322	0.0336	0.0331	0.0353	4.6321	2.7142	1.3580	0.7998



Cell Description

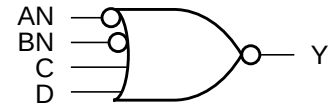
The NOR4BB cell provides a logical NOR of two inverted inputs (AN, BN) and two non-inverted inputs (C, D). The output (Y) is represented by the logic equation:

$$Y = \overline{(\overline{AN} + \overline{BN} + C + D)}$$

Functions

AN	BN	C	D	Y
1	1	0	0	1
x	x	x	1	0
x	x	1	x	0
x	0	x	x	0
0	x	x	x	0

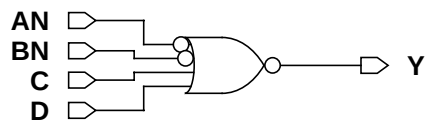
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR4BBXL	3.69	3.68
NOR4BBX1	3.69	3.68
NOR4BBX2	3.69	6.44
NOR4BBX4	3.69	9.66

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
AN	0.0059	0.0083	0.0162	0.0268
BN	0.0069	0.0100	0.0179	0.0329
C	0.0078	0.0119	0.0223	0.0400
D	0.0089	0.0142	0.0259	0.0482

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
AN	0.0014	0.0019	0.0030	0.0048
BN	0.0014	0.0019	0.0031	0.0047
C	0.0022	0.0036	0.0076	0.0140
D	0.0021	0.0036	0.0072	0.0140

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
AN $\rightarrow$ Y $\uparrow$	0.1011	0.0751	0.0826	0.0771	20.4407	9.4066	5.3730	2.8310
AN $\rightarrow$ Y $\downarrow$	0.1033	0.0983	0.0932	0.0858	4.5832	2.6767	1.3440	0.7788
BN $\rightarrow$ Y $\uparrow$	0.1443	0.1100	0.1103	0.1117	20.3449	9.3840	5.3564	2.8294
BN $\rightarrow$ Y $\downarrow$	0.1102	0.1066	0.0965	0.0959	4.5466	2.6578	1.3280	0.7780
C $\rightarrow$ Y $\uparrow$	0.1346	0.0993	0.0997	0.1003	20.4125	9.4022	5.3537	2.8275
C $\rightarrow$ Y $\downarrow$	0.0349	0.0345	0.0305	0.0345	4.5010	2.6441	1.3207	0.7810
D $\rightarrow$ Y $\uparrow$	0.1436	0.1074	0.1080	0.1110	20.4061	9.4007	5.3538	2.8290
D $\rightarrow$ Y $\downarrow$	0.0346	0.0350	0.0308	0.0358	4.6252	2.7132	1.3530	0.7998



Cell Description

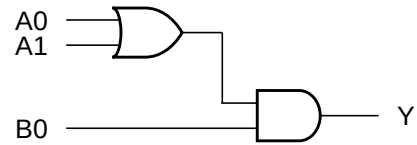
The OA21 cell provides the logical AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

$$Y = (A0 + A1) \bullet B0$$

Functions

A0	A1	B0	Y
x	x	0	0
0	0	x	0
x	1	1	1
1	x	1	1

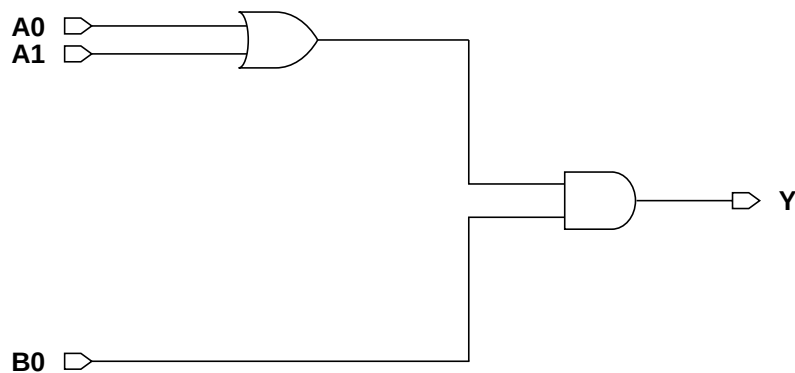
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OA21XL	3.69	2.30
OA21X1	3.69	2.76
OA21X2	3.69	2.76
OA21X4	3.69	3.68

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0058	0.0079	0.0121	0.0201
A1	0.0063	0.0085	0.0132	0.0223
B0	0.0047	0.0060	0.0097	0.0169

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0015	0.0015	0.0026	0.0047
A1	0.0015	0.0014	0.0024	0.0046
B0	0.0015	0.0017	0.0026	0.0042

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0737	0.0845	0.0789	0.0691	9.3978	5.1652	2.5819	1.1317
A0 $\rightarrow$ Y $\downarrow$	0.1440	0.1832	0.1547	0.1179	5.1732	3.1422	1.5837	0.7585
A1 $\rightarrow$ Y $\uparrow$	0.0796	0.0904	0.0861	0.0765	9.4128	5.1743	2.5865	1.1340
A1 $\rightarrow$ Y $\downarrow$	0.1539	0.1924	0.1646	0.1266	5.1735	3.1424	1.5838	0.7587
B0 $\rightarrow$ Y $\uparrow$	0.0708	0.0790	0.0761	0.0690	9.4135	5.1742	2.5866	1.1338
B0 $\rightarrow$ Y $\downarrow$	0.0825	0.1008	0.0949	0.0783	4.8531	2.8969	1.5243	0.7423

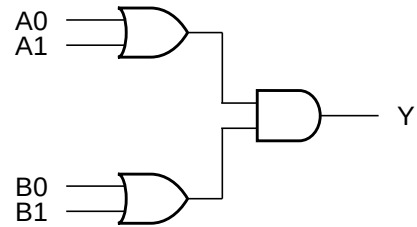
**Cell Description**

The OA22 cell provides the logical AND of two OR groups. The output (Y) is represented by the logic equation:

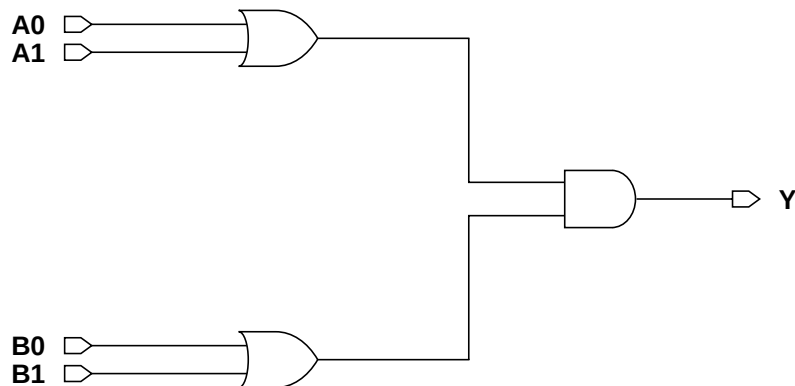
$$Y = (A0 + A1) \bullet (B0 + B1)$$

Functions

A0	A1	B0	B1	Y
x	x	0	0	0
0	0	x	x	0
x	1	x	1	1
x	1	1	x	1
1	x	x	1	1
1	x	1	x	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OA22XL	3.69	2.76
OA22X1	3.69	2.76
OA22X2	3.69	3.22
OA22X4	3.69	4.14

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0053	0.0066	0.0103	0.0168
A1	0.0058	0.0072	0.0115	0.0195
B0	0.0081	0.0095	0.0147	0.0224
B1	0.0087	0.0101	0.0157	0.0251

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0018	0.0019	0.0030	0.0046
A1	0.0017	0.0018	0.0028	0.0045
B0	0.0018	0.0018	0.0028	0.0045
B1	0.0017	0.0017	0.0028	0.0045

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0825	0.0854	0.0767	0.0754	9.4287	5.1824	2.2721	1.2614
A0 $\rightarrow$ Y $\downarrow$	0.1358	0.1594	0.1355	0.1164	5.5519	3.6606	1.6594	0.8303
A1 $\rightarrow$ Y $\uparrow$	0.0887	0.0917	0.0834	0.0824	9.4419	5.1901	2.2764	1.2634
A1 $\rightarrow$ Y $\downarrow$	0.1459	0.1694	0.1452	0.1250	5.5523	3.6609	1.6591	0.8304
B0 $\rightarrow$ Y $\uparrow$	0.0989	0.1014	0.0894	0.0855	9.4340	5.1840	2.2724	1.2615
B0 $\rightarrow$ Y $\downarrow$	0.1957	0.2187	0.1797	0.1466	5.7490	3.7617	1.6811	0.8354
B1 $\rightarrow$ Y $\uparrow$	0.1059	0.1083	0.0968	0.0930	9.4422	5.1915	2.2765	1.2634
B1 $\rightarrow$ Y $\downarrow$	0.2076	0.2300	0.1910	0.1561	5.7479	3.7619	1.6814	0.8354

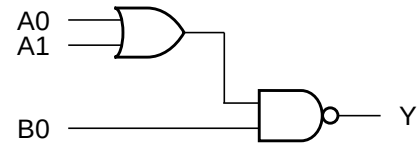
**Cell Description**

The OAI21 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

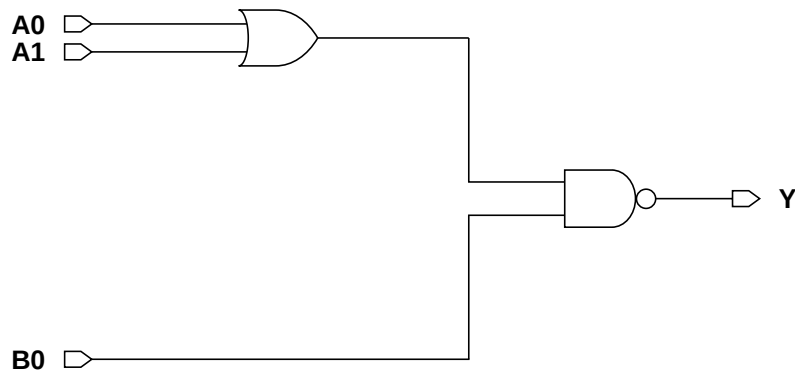
$$Y = \overline{(A0 + A1) \bullet B0}$$

Functions

A0	A1	B0	Y
0	0	x	1
x	x	0	1
x	1	1	0
1	x	1	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI21XL	3.69	1.84
OAI21X1	3.69	2.30
OAI21X2	3.69	3.68
OAI21X4	3.69	5.06

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0058	0.0080	0.0150	0.0287
A1	0.0066	0.0095	0.0181	0.0335
B0	0.0042	0.0061	0.0108	0.0215

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0021	0.0035	0.0064	0.0127
A1	0.0019	0.0032	0.0068	0.0123
B0	0.0018	0.0029	0.0057	0.0107

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0803	0.0574	0.0545	0.0498	14.7487	7.2944	3.6484	1.7260
A0 $\rightarrow$ Y $\downarrow$	0.0315	0.0277	0.0265	0.0266	5.3477	3.1394	1.5703	0.8187
A1 $\rightarrow$ Y $\uparrow$	0.0899	0.0672	0.0647	0.0587	14.7296	7.2869	3.6459	1.7243
A1 $\rightarrow$ Y $\downarrow$	0.0360	0.0330	0.0321	0.0319	5.3838	3.1588	1.5793	0.8146
B0 $\rightarrow$ Y $\uparrow$	0.0348	0.0286	0.0264	0.0249	9.3901	5.1427	2.5725	1.2010
B0 $\rightarrow$ Y $\downarrow$	0.0268	0.0255	0.0241	0.0246	5.3889	3.1621	1.5806	0.8154



Cell Description

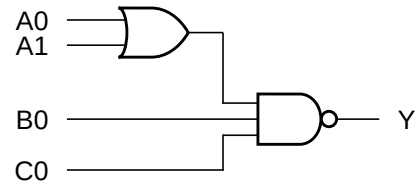
The OAI211 cell provides the logical inverted OR of one OR group and two additional inputs. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet B0 \bullet C0}$$

Functions

A0	A1	B0	C0	Y
0	0	x	x	1
x	x	0	x	1
x	x	x	0	1
x	1	1	1	0
1	x	1	1	0

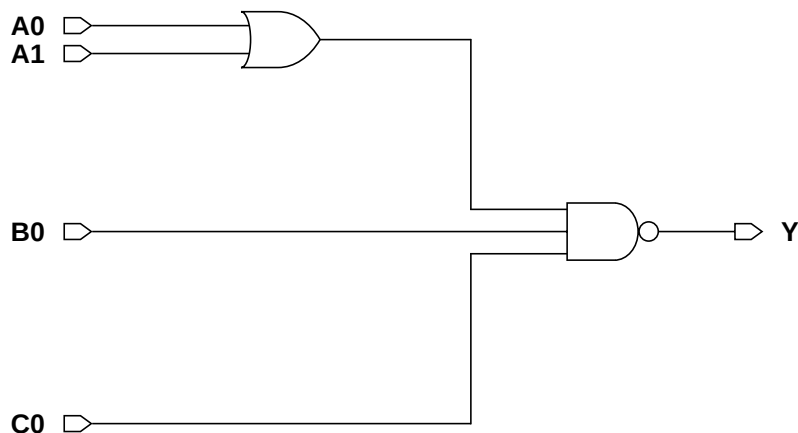
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI211XL	3.69	2.30
OAI211X1	3.69	2.30
OAI211X2	3.69	4.14
OAI211X4	3.69	4.14

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0065	0.0106	0.0208	0.0238
A1	0.0074	0.0121	0.0234	0.0246
B0	0.0041	0.0067	0.0127	0.0199
C0	0.0052	0.0087	0.0167	0.0209

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0022	0.0032	0.0071	0.0023
A1	0.0021	0.0033	0.0060	0.0021
B0	0.0020	0.0032	0.0059	0.0022
C0	0.0019	0.0031	0.0062	0.0021

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0945	0.0790	0.0798	0.2341	14.5113	7.3223	3.6639	1.1266
A0 $\rightarrow$ Y $\downarrow$	0.0404	0.0423	0.0401	0.1463	6.5296	4.0818	1.9318	0.7143
A1 $\rightarrow$ Y $\uparrow$	0.1047	0.0889	0.0880	0.2441	14.5003	7.3187	3.6608	1.1266
A1 $\rightarrow$ Y $\downarrow$	0.0468	0.0469	0.0497	0.1545	6.5663	3.8886	2.0494	0.7146
B0 $\rightarrow$ Y $\uparrow$	0.0380	0.0345	0.0305	0.1426	9.3963	5.1449	2.2523	1.1256
B0 $\rightarrow$ Y $\downarrow$	0.0334	0.0332	0.0337	0.1399	6.5744	3.8933	2.0518	0.7147
C0 $\rightarrow$ Y $\uparrow$	0.0481	0.0446	0.0397	0.1532	9.3784	5.1400	2.2507	1.1259
C0 $\rightarrow$ Y $\downarrow$	0.0393	0.0398	0.0414	0.1449	6.5751	3.8923	2.0513	0.7146

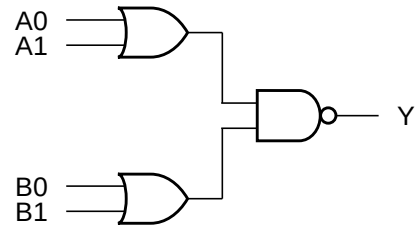
**Cell Description**

The OAI22 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

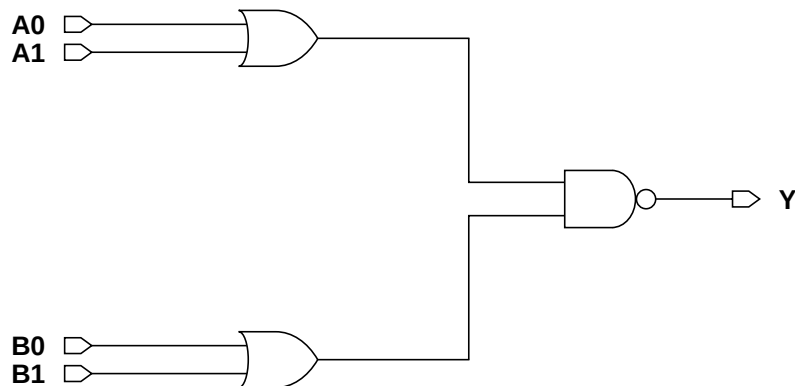
$$Y = \overline{(A0 + A1) \bullet (B0 + B1)}$$

Functions

A0	A1	B0	B1	Y
0	0	x	x	1
x	x	0	0	1
x	1	x	1	0
x	1	1	x	0
1	x	x	1	0
1	x	1	x	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI22XL	3.69	2.30
OAI22X1	3.69	2.76
OAI22X2	3.69	4.60
OAI22X4	3.69	6.44

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0044	0.0072	0.0145	0.0271
A1	0.0052	0.0084	0.0168	0.0325
B0	0.0083	0.0109	0.0220	0.0403
B1	0.0091	0.0124	0.0249	0.0450

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0021	0.0033	0.0065	0.0124
A1	0.0022	0.0033	0.0070	0.0121
B0	0.0021	0.0032	0.0063	0.0122
B1	0.0020	0.0031	0.0067	0.0119

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0497	0.0427	0.0413	0.0398	14.8117	7.3179	3.6602	1.7680
A0 $\rightarrow$ Y $\downarrow$	0.0268	0.0306	0.0286	0.0305	5.3906	3.3664	1.6050	0.8668
A1 $\rightarrow$ Y $\uparrow$	0.0626	0.0530	0.0516	0.0488	14.8200	7.3159	3.6590	1.7662
A1 $\rightarrow$ Y $\downarrow$	0.0331	0.0372	0.0348	0.0368	5.4208	3.3782	1.6115	0.8692
B0 $\rightarrow$ Y $\uparrow$	0.1139	0.0754	0.0746	0.0675	14.7735	7.2901	3.6457	1.7611
B0 $\rightarrow$ Y $\downarrow$	0.0450	0.0409	0.0386	0.0395	5.3945	3.3668	1.6052	0.8667
B1 $\rightarrow$ Y $\uparrow$	0.1242	0.0849	0.0845	0.0764	14.7618	7.2851	3.6439	1.7597
B1 $\rightarrow$ Y $\downarrow$	0.0502	0.0472	0.0446	0.0458	5.4187	3.3782	1.6114	0.8692

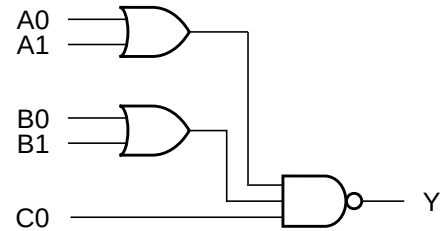
**Cell Description**

The OAI221 cell provides the logical inverted AND of two OR groups and an additional input. The output (Y) is represented by the logic equation:

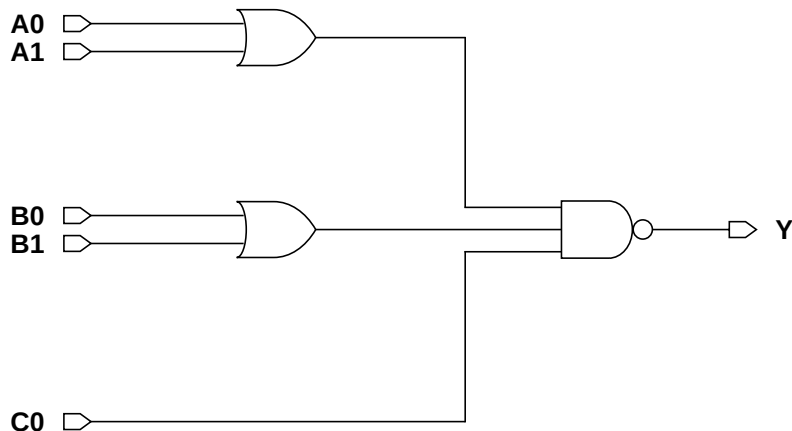
$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet C0}$$

Functions

A0	A1	B0	B1	C0	Y
0	0	x	x	x	1
x	x	0	0	x	1
x	x	x	x	0	1
x	1	x	1	1	0
x	1	1	x	1	0
1	x	x	1	1	0
1	x	1	x	1	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI221XL	3.69	3.22
OAI221X1	3.69	3.68
OAI221X2	3.69	5.06
OAI221X4	3.69	5.06

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0066	0.0111	0.0205	0.0235
A1	0.0074	0.0126	0.0233	0.0243
B0	0.0089	0.0146	0.0272	0.0259
B1	0.0098	0.0160	0.0302	0.0267
C0	0.0050	0.0085	0.0156	0.0217

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0022	0.0035	0.0066	0.0023
A1	0.0022	0.0035	0.0068	0.0023
B0	0.0022	0.0035	0.0067	0.0023
B1	0.0022	0.0033	0.0070	0.0021
C0	0.0020	0.0033	0.0061	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0865	0.0744	0.0688	0.2356	14.5263	7.3261	3.6638	1.1272
A0 $\rightarrow$ Y $\downarrow$	0.0474	0.0481	0.0468	0.1650	6.5778	3.8609	1.9768	0.7199
A1 $\rightarrow$ Y $\uparrow$	0.0971	0.0841	0.0781	0.2464	14.5149	7.3213	3.6614	1.1273
A1 $\rightarrow$ Y $\downarrow$	0.0544	0.0553	0.0543	0.1737	6.6022	3.8530	1.9822	0.7201
B0 $\rightarrow$ Y $\uparrow$	0.1233	0.1033	0.0970	0.2691	14.6005	7.3629	3.6848	1.1274
B0 $\rightarrow$ Y $\downarrow$	0.0568	0.0561	0.0542	0.1737	6.5782	3.8428	1.9764	0.7200
B1 $\rightarrow$ Y $\uparrow$	0.1335	0.1120	0.1067	0.2789	14.5918	7.3584	3.6830	1.1275
B1 $\rightarrow$ Y $\downarrow$	0.0636	0.0631	0.0619	0.1820	6.6022	3.8533	1.9821	0.7202
C0 $\rightarrow$ Y $\uparrow$	0.0409	0.0381	0.0338	0.1530	9.4322	5.1639	2.5812	1.1259
C0 $\rightarrow$ Y $\downarrow$	0.0416	0.0424	0.0398	0.1577	6.6099	3.8573	1.9840	0.7202



Cell Description

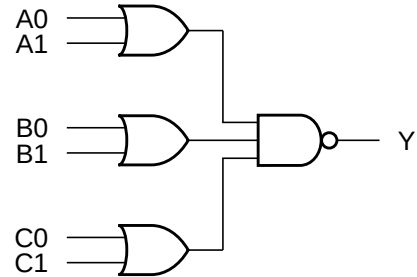
The OAI222 cell provides the logical inverted AND of three OR groups. The output (Y) is represented by the logic equation:

$$Y = \overline{(A0 + A1) \bullet (B0 + B1) \bullet (C0 + C1)}$$

Functions

A0	A1	B0	B1	C0	C1	Y
0	0	x	x	x	x	1
x	x	0	0	x	x	1
x	x	x	x	0	0	1
x	1	x	1	1	x	0
x	1	x	1	x	1	0
x	1	1	x	1	x	0
x	1	1	x	x	1	0
1	x	x	1	1	x	0
1	x	x	1	x	1	0
1	x	1	x	1	x	0
1	x	1	x	x	1	0

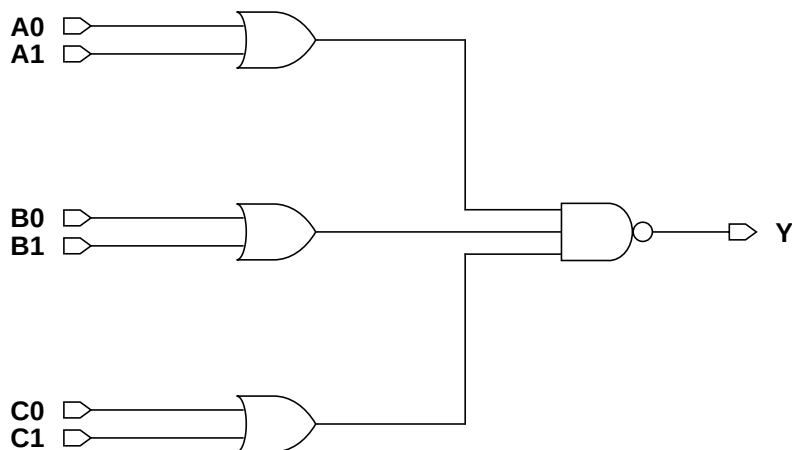
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI222XL	3.69	3.68
OAI222X1	3.69	4.14
OAI222X2	3.69	6.90
OAI222X4	3.69	5.52

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0096	0.0141	0.0285	0.0257
A1	0.0104	0.0155	0.0315	0.0267
B0	0.0118	0.0176	0.0352	0.0279
B1	0.0125	0.0190	0.0381	0.0289
C0	0.0059	0.0096	0.0191	0.0218
C1	0.0066	0.0110	0.0215	0.0228

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0023	0.0034	0.0065	0.0022
A1	0.0021	0.0034	0.0070	0.0023
B0	0.0023	0.0036	0.0065	0.0023
B1	0.0021	0.0034	0.0068	0.0025
C0	0.0023	0.0036	0.0068	0.0026
C1	0.0021	0.0034	0.0073	0.0023

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1233	0.0920	0.0913	0.2711	14.5012	7.3011	3.6517	1.2299
A0 $\rightarrow$ Y $\downarrow$	0.0674	0.0627	0.0635	0.1779	6.6092	3.9094	2.0052	0.7156
A1 $\rightarrow$ Y $\uparrow$	0.1338	0.1018	0.1015	0.2824	14.4904	7.2968	3.6507	1.2299
A1 $\rightarrow$ Y $\downarrow$	0.0746	0.0706	0.0719	0.1869	6.6255	3.9167	2.0089	0.7158
B0 $\rightarrow$ Y $\uparrow$	0.1625	0.1256	0.1233	0.3062	14.5517	7.3369	3.6681	1.2290
B0 $\rightarrow$ Y $\downarrow$	0.0762	0.0715	0.0723	0.1908	6.6085	3.9089	2.0050	0.7160
B1 $\rightarrow$ Y $\uparrow$	0.1718	0.1343	0.1331	0.3169	14.5441	7.3330	3.6674	1.2289
B1 $\rightarrow$ Y $\downarrow$	0.0831	0.0789	0.0806	0.1950	6.6253	3.9167	2.0088	0.7159
C0 $\rightarrow$ Y $\uparrow$	0.0653	0.0543	0.0514	0.2155	14.6301	7.3615	3.6787	1.2298
C0 $\rightarrow$ Y $\downarrow$	0.0434	0.0455	0.0442	0.1558	6.6115	3.9104	2.0052	0.7156
C1 $\rightarrow$ Y $\uparrow$	0.0733	0.0627	0.0618	0.2234	14.5908	7.3497	3.6780	1.2297
C1 $\rightarrow$ Y $\downarrow$	0.0489	0.0525	0.0528	0.1628	6.6268	3.9166	2.0090	0.7157



Cell Description

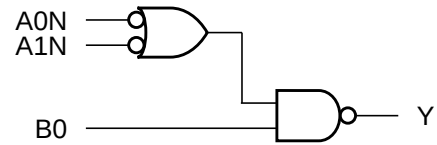
The OAI2BB1 cell provides the logical inverted AND of one OR group of two inverted inputs (A0N, A1N) and an additional non-inverted input (B0). The output (Y) is represented by the logic equation:

$$Y = (\overline{A0N} + \overline{A1N}) \bullet B0$$

Functions

A0N	A1N	B0	Y
1	1	x	1
x	x	0	1
x	0	1	0
0	x	1	0

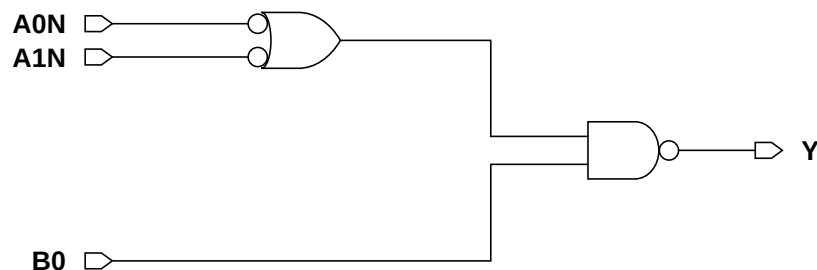
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI2BB1XL	3.69	2.30
OAI2BB1X1	3.69	2.30
OAI2BB1X2	3.69	3.22
OAI2BB1X4	3.69	4.14

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0063	0.0076	0.0132	0.0241
A1N	0.0059	0.0068	0.0119	0.0212
B0	0.0030	0.0050	0.0097	0.0171

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0012	0.0014	0.0023	0.0038
A1N	0.0010	0.0012	0.0018	0.0029
B0	0.0016	0.0025	0.0049	0.0086

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0803	0.0768	0.0713	0.0751	9.4186	5.1595	2.2418	1.1225
A0N $\rightarrow$ Y $\downarrow$	0.1023	0.1000	0.0968	0.0838	5.6202	3.2729	1.6311	0.9505
A1N $\rightarrow$ Y $\uparrow$	0.0775	0.0737	0.0675	0.0716	9.4177	5.1594	2.2419	1.1223
A1N $\rightarrow$ Y $\downarrow$	0.0928	0.0905	0.0865	0.0768	5.5927	3.2598	1.6248	0.9486
B0 $\rightarrow$ Y $\uparrow$	0.0392	0.0342	0.0297	0.0270	9.3476	5.1261	2.2287	1.1130
B0 $\rightarrow$ Y $\downarrow$	0.0242	0.0236	0.0233	0.0242	5.3932	3.1732	1.5876	0.9381



Cell Description

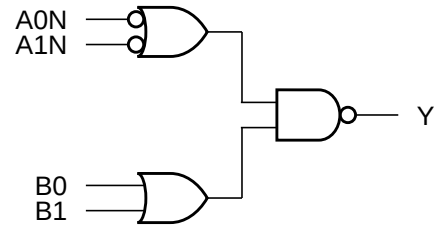
The OAI2BB2 cell provides the logical inverted AND of one OR group of two inverted inputs (A0N, A1N) and one OR group of two non-inverted inputs (B0, B1). The output (Y) is represented by the logic equation:

$$Y = \overline{(A0N + A1N)} \bullet (B0 + B1)$$

Functions

A0N	A1N	B0	B1	Y
1	1	x	x	1
x	x	0	0	1
x	0	x	1	0
x	0	1	x	0
0	x	x	1	0
0	x	1	x	0

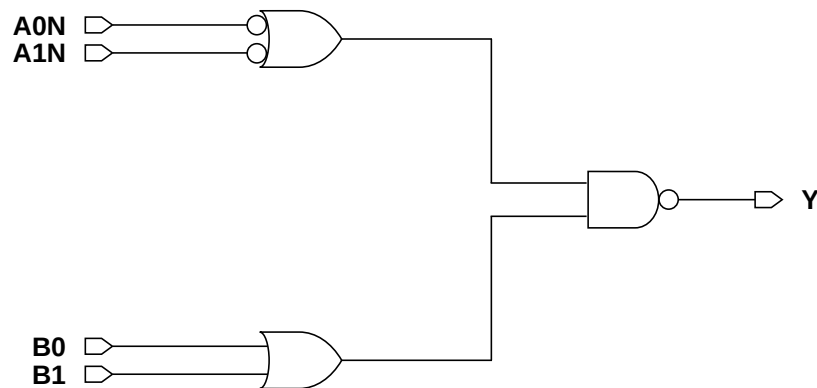
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
OAI2BB2XL	3.69	2.76
OAI2BB2X1	3.69	3.22
OAI2BB2X2	3.69	4.60
OAI2BB2X4	3.69	6.44

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0N	0.0065	0.0089	0.0161	0.0302
A1N	0.0054	0.0072	0.0127	0.0238
B0	0.0042	0.0066	0.0132	0.0228
B1	0.0050	0.0079	0.0162	0.0276

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0N	0.0012	0.0016	0.0026	0.0045
A1N	0.0015	0.0018	0.0028	0.0046
B0	0.0021	0.0032	0.0071	0.0138
B1	0.0020	0.0031	0.0069	0.0129

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0N $\rightarrow$ Y $\uparrow$	0.0759	0.0780	0.0741	0.0816	9.4370	5.0679	2.2620	1.3259
A0N $\rightarrow$ Y $\downarrow$	0.1037	0.1092	0.1063	0.0919	5.5685	3.2524	1.6198	0.8714
A1N $\rightarrow$ Y $\uparrow$	0.0733	0.0753	0.0704	0.0772	9.4094	5.0549	2.2578	1.3221
A1N $\rightarrow$ Y $\downarrow$	0.0873	0.0924	0.0887	0.0763	5.5633	3.3001	1.6421	0.8699
B0 $\rightarrow$ Y $\uparrow$	0.0701	0.0565	0.0583	0.0490	14.7463	7.2961	3.6512	1.7269
B0 $\rightarrow$ Y $\downarrow$	0.0299	0.0302	0.0322	0.0304	5.4035	3.2279	1.6151	0.8625
B1 $\rightarrow$ Y $\uparrow$	0.0799	0.0663	0.0663	0.0569	14.7264	7.2896	3.6461	1.7245
B1 $\rightarrow$ Y $\downarrow$	0.0347	0.0363	0.0361	0.0357	5.4346	3.1932	1.5933	0.8643

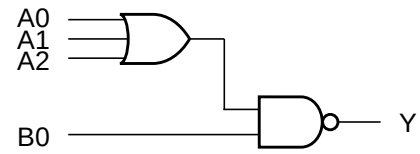
**Cell Description**

The OAI31 cell provides the logical inverted AND of one OR group and an additional input. The output (Y) is represented by the logic equation:

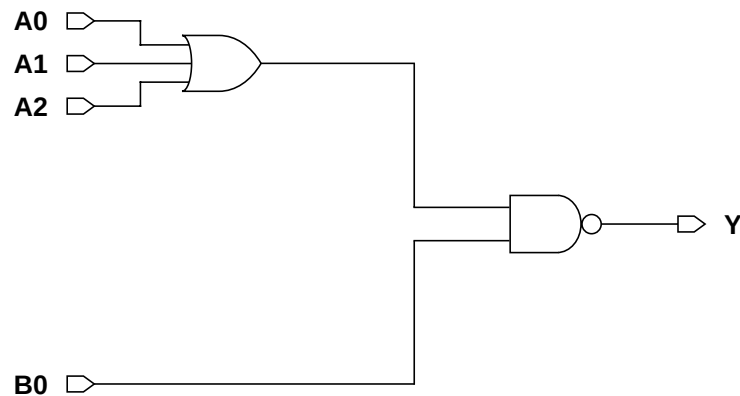
$$Y = \overline{(A0 + A1 + A2) \bullet B0}$$

Functions

A0	A1	A2	B0	Y
0	0	0	x	1
x	x	x	0	1
x	x	1	1	0
x	1	x	1	0
1	x	x	1	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI31XL	3.69	2.30
OAI31X1	3.69	2.76
OAI31X2	3.69	4.60
OAI31X4	3.69	4.60

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0061	0.0089	0.0174	0.0227
A1	0.0071	0.0106	0.0209	0.0233
A2	0.0081	0.0126	0.0248	0.0257
B0	0.0055	0.0082	0.0159	0.0207

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0024	0.0038	0.0071	0.0027
A1	0.0023	0.0037	0.0074	0.0024
A2	0.0022	0.0036	0.0076	0.0023
B0	0.0019	0.0030	0.0058	0.0021

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0924	0.0626	0.0610	0.2228	17.7746	8.0353	4.0211	1.1272
A0 $\rightarrow$ Y $\downarrow$	0.0331	0.0302	0.0300	0.1282	5.3520	3.1455	1.5986	0.7139
A1 $\rightarrow$ Y $\uparrow$	0.1167	0.0834	0.0825	0.2449	17.7495	8.0262	4.0170	1.1273
A1 $\rightarrow$ Y $\downarrow$	0.0399	0.0381	0.0383	0.1371	5.3936	3.1634	1.6077	0.7142
A2 $\rightarrow$ Y $\uparrow$	0.1261	0.0915	0.0914	0.2536	17.7462	8.0255	4.0177	1.1272
A2 $\rightarrow$ Y $\downarrow$	0.0421	0.0413	0.0419	0.1428	5.5215	3.2272	1.6411	0.7146
B0 $\rightarrow$ Y $\uparrow$	0.0378	0.0304	0.0280	0.1421	9.3937	5.1475	2.5745	1.1269
B0 $\rightarrow$ Y $\downarrow$	0.0339	0.0327	0.0314	0.1345	5.5237	3.2295	1.6419	0.7146

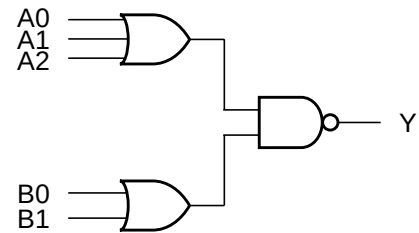
**Cell Description**

The OAI32 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

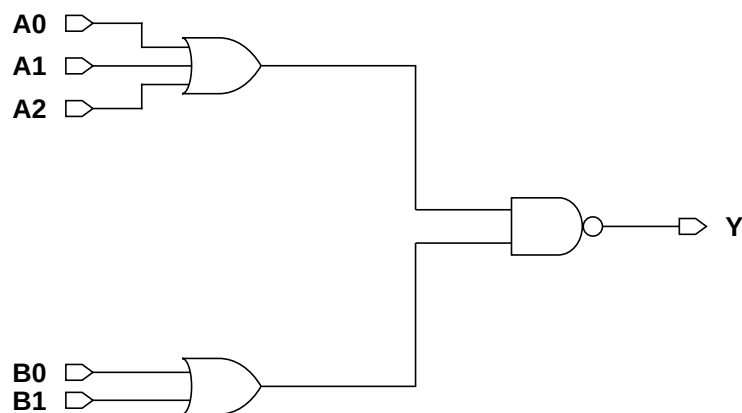
$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1)}$$

Functions

A0	A1	A2	B0	B1	Y
0	0	0	x	x	1
x	x	x	0	0	1
x	x	1	x	1	0
x	x	1	1	x	0
x	1	x	1	x	0
x	1	x	x	1	0
1	x	x	1	x	0
1	x	x	x	1	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI32XL	3.69	3.22
OAI32X1	3.69	3.22
OAI32X2	3.69	5.52
OAI32X4	3.69	5.06

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A0	0.0076	0.0125	0.0252	0.0241
A1	0.0086	0.0142	0.0283	0.0249
A2	0.0096	0.0162	0.0322	0.0273
B0	0.0057	0.0098	0.0199	0.0215
B1	0.0064	0.0112	0.0225	0.0222

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0023	0.0038	0.0080	0.0024
A1	0.0022	0.0037	0.0075	0.0023
A2	0.0021	0.0036	0.0070	0.0023
B0	0.0021	0.0034	0.0071	0.0022
B1	0.0021	0.0033	0.0062	0.0022

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.1162	0.0897	0.0911	0.2524	17.7792	8.0348	4.0170	1.1273
A0 $\rightarrow$ Y $\downarrow$	0.0426	0.0437	0.0460	0.1424	5.3977	3.1640	1.6323	0.7143
A1 $\rightarrow$ Y $\uparrow$	0.1413	0.1102	0.1107	0.2762	17.7573	8.0265	4.0121	1.1273
A1 $\rightarrow$ Y $\downarrow$	0.0500	0.0520	0.0533	0.1520	5.4282	3.1750	1.6116	0.7143
A2 $\rightarrow$ Y $\uparrow$	0.1499	0.1182	0.1187	0.2848	17.7531	8.0251	4.0113	1.1274
A2 $\rightarrow$ Y $\downarrow$	0.0528	0.0560	0.0586	0.1582	5.5348	3.2262	1.6632	0.7148
B0 $\rightarrow$ Y $\uparrow$	0.0542	0.0474	0.0496	0.1940	14.8317	7.3367	3.6671	1.1287
B0 $\rightarrow$ Y $\downarrow$	0.0345	0.0378	0.0400	0.1390	5.5292	3.2253	1.6627	0.7147
B1 $\rightarrow$ Y $\uparrow$	0.0651	0.0570	0.0580	0.2054	14.8174	7.3281	3.6613	1.1287
B1 $\rightarrow$ Y $\downarrow$	0.0406	0.0442	0.0460	0.1467	5.5339	3.2265	1.6630	0.7148

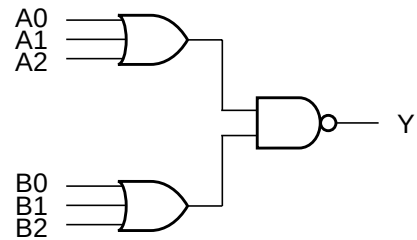
**Cell Description**

The OAI33 cell provides the logical inverted AND of two OR groups. The output (Y) is represented by the logic equation:

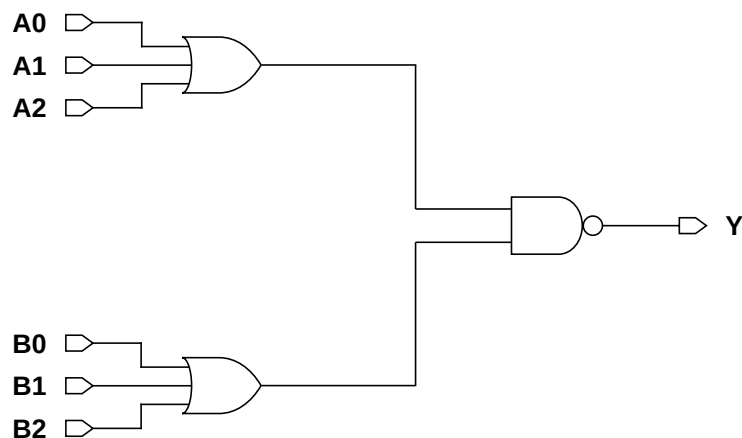
$$Y = \overline{(A0 + A1 + A2) \bullet (B0 + B1 + B2)}$$

Functions

A0	A1	A2	B0	B1	B2	Y
0	0	0	x	x	x	1
x	x	x	0	0	0	1
x	x	1	x	x	1	0
x	x	1	x	1	x	0
x	x	1	1	x	x	0
x	1	x	x	x	1	0
x	1	x	x	1	x	0
x	1	x	1	x	x	0
1	x	x	x	x	1	0
1	x	x	x	1	x	0
1	x	x	1	x	x	0

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
OAI33XL	3.69	3.68
OAI33X1	3.69	3.68
OAI33X2	3.69	6.44
OAI33X4	3.69	5.06

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A0	0.0073	0.0119	0.0235	0.0220
A1	0.0083	0.0136	0.0264	0.0232
A2	0.0094	0.0156	0.0301	0.0254
B0	0.0106	0.0167	0.0332	0.0264
B1	0.0115	0.0184	0.0363	0.0272
B2	0.0127	0.0203	0.0401	0.0297

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A0	0.0024	0.0039	0.0082	0.0024
A1	0.0024	0.0038	0.0077	0.0025
A2	0.0024	0.0038	0.0069	0.0025
B0	0.0023	0.0039	0.0082	0.0024
B1	0.0023	0.0039	0.0077	0.0025
B2	0.0022	0.0036	0.0069	0.0023

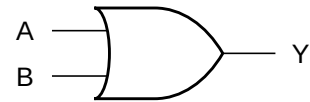
Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A0 $\rightarrow$ Y $\uparrow$	0.0781	0.0561	0.0633	0.2313	17.8965	8.0799	4.2726	1.1308
A0 $\rightarrow$ Y $\downarrow$	0.0404	0.0423	0.0450	0.1507	5.3733	3.2211	1.6474	0.7535
A1 $\rightarrow$ Y $\uparrow$	0.1036	0.0771	0.0824	0.2570	17.8758	8.0704	4.2649	1.1309
A1 $\rightarrow$ Y $\downarrow$	0.0486	0.0516	0.0531	0.1614	5.3807	3.2210	1.6471	0.7535
A2 $\rightarrow$ Y $\uparrow$	0.1154	0.0863	0.0907	0.2668	17.8920	8.0734	4.2643	1.1309
A2 $\rightarrow$ Y $\downarrow$	0.0534	0.0573	0.0582	0.1687	5.4584	3.2576	1.6648	0.7542
B0 $\rightarrow$ Y $\uparrow$	0.1523	0.1089	0.1131	0.3023	17.7590	8.0260	4.0142	1.1286
B0 $\rightarrow$ Y $\downarrow$	0.0544	0.0548	0.0576	0.1667	5.3734	3.2217	1.6469	0.7534
B1 $\rightarrow$ Y $\uparrow$	0.1771	0.1294	0.1320	0.3253	17.7395	8.0181	4.0087	1.1286
B1 $\rightarrow$ Y $\downarrow$	0.0624	0.0640	0.0657	0.1764	5.3801	3.2216	1.6337	0.7537
B2 $\rightarrow$ Y $\uparrow$	0.1858	0.1374	0.1404	0.3342	17.7362	8.0176	4.0089	1.1286
B2 $\rightarrow$ Y $\downarrow$	0.0661	0.0691	0.0718	0.1834	5.4581	3.2585	1.6651	0.7540

**Cell Description**

The OR2 cell provides the logical OR of two inputs (A, B). The output (Y) is represented by the logic equation:

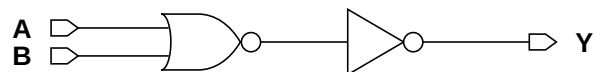
$$Y = (A + B)$$

Logic Symbol**Functions**

A	B	Y
0	0	0
x	1	1
1	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR2XL	3.69	1.84
OR2X1	3.69	1.84
OR2X2	3.69	2.30
OR2X4	3.69	2.76
OR2X6	3.69	4.14
OR2X8	3.69	4.60

Functional Schematic

**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0042	0.0054	0.0086	0.0146	0.0237	0.0293
B	0.0049	0.0061	0.0101	0.0176	0.0273	0.0354

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0015	0.0016	0.0024	0.0040	0.0065	0.0081
B	0.0016	0.0017	0.0026	0.0041	0.0069	0.0086

Delays at 25°C, 1.2V, Typical Process

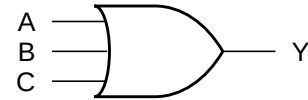
Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0536	0.0576	0.0525	0.0534	0.0499	0.0513
A $\rightarrow$ Y $\downarrow$	0.1054	0.1276	0.1139	0.0938	0.0953	0.0873
B $\rightarrow$ Y $\uparrow$	0.0598	0.0639	0.0586	0.0600	0.0559	0.0578
B $\rightarrow$ Y $\downarrow$	0.1204	0.1426	0.1270	0.1042	0.1061	0.0971

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.3621	5.1327	2.2474	1.2694	0.7492	0.6206
A $\rightarrow$ Y $\downarrow$	5.0115	2.9825	1.6238	0.7500	0.4981	0.3708
B $\rightarrow$ Y $\uparrow$	9.3748	5.1391	2.2505	1.2711	0.7505	0.6214
B $\rightarrow$ Y $\downarrow$	5.0119	2.9835	1.6240	0.7500	0.4982	0.3708

**Cell Description**

The OR3 cell provides the logical OR of three inputs (A, B, C). The output (Y) is represented by the logic equation:

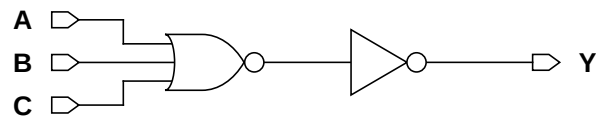
$$Y = (A + B + C)$$

Logic Symbol**Functions**

A	B	C	Y
0	0	0	0
x	x	1	1
x	1	x	1
1	x	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR3XL	3.69	2.76
OR3X1	3.69	2.76
OR3X2	3.69	2.76
OR3X4	3.69	3.68
OR3X6	3.69	4.60
OR3X8	3.69	6.44

Functional Schematic

**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0048	0.0062	0.0100	0.0169	0.0238	0.0333
B	0.0054	0.0069	0.0112	0.0200	0.0272	0.0378
C	0.0063	0.0078	0.0131	0.0235	0.0331	0.0460

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0017	0.0018	0.0029	0.0052	0.0068	0.0102
B	0.0017	0.0018	0.0028	0.0055	0.0071	0.0098
C	0.0016	0.0017	0.0027	0.0059	0.0075	0.0101

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0611	0.0652	0.0592	0.0559	0.0601	0.0554
A $\rightarrow$ Y $\downarrow$	0.1375	0.1637	0.1315	0.1166	0.1106	0.1192
B $\rightarrow$ Y $\uparrow$	0.0673	0.0719	0.0664	0.0643	0.0691	0.0627
B $\rightarrow$ Y $\downarrow$	0.1621	0.1888	0.1533	0.1398	0.1322	0.1406
C $\rightarrow$ Y $\uparrow$	0.0703	0.0753	0.0705	0.0695	0.0753	0.0676
C $\rightarrow$ Y $\downarrow$	0.1711	0.1979	0.1621	0.1503	0.1428	0.1513

Description	K_{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	9.3618	5.1372	2.2497	1.1258	0.8413	0.5628
A $\rightarrow$ Y $\downarrow$	5.5162	3.2624	1.6196	0.8003	0.5274	0.4003
B $\rightarrow$ Y $\uparrow$	9.3744	5.1434	2.2539	1.1280	0.8427	0.5638
B $\rightarrow$ Y $\downarrow$	5.5176	3.2623	1.6197	0.8002	0.5274	0.4004
C $\rightarrow$ Y $\uparrow$	9.4084	5.1607	2.2638	1.1334	0.8462	0.5665
C $\rightarrow$ Y $\downarrow$	5.5167	3.2618	1.6195	0.8004	0.5274	0.4004

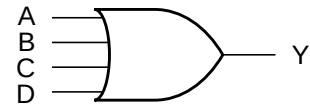


Cell Description

The OR4 cell provides the logical OR of four inputs (A, B, C, D). The output (Y) is represented by the logic equation:

$$Y = (A + B + C + D)$$

Logic Symbol



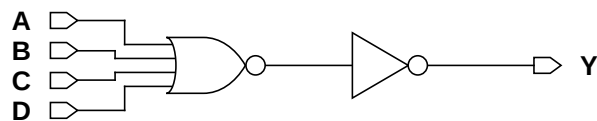
Functions

A	B	C	D	Y
0	0	0	0	0
x	x	x	1	1
x	x	1	x	1
x	1	x	x	1
1	x	x	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
OR4XL	3.69	2.76
OR4X1	3.69	2.76
OR4X2	3.69	3.22
OR4X4	3.69	4.60
OR4X6	3.69	5.98
OR4X8	3.69	7.82

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)					
	XL	X1	X2	X4	X6	X8
A	0.0052	0.0066	0.0098	0.0177	0.0256	0.0332
B	0.0060	0.0074	0.0117	0.0217	0.0308	0.0404
C	0.0069	0.0084	0.0128	0.0241	0.0338	0.0447
D	0.0078	0.0094	0.0152	0.0283	0.0399	0.0526

Pin Capacitance

Pin	Capacitance (pF)					
	XL	X1	X2	X4	X6	X8
A	0.0018	0.0019	0.0031	0.0057	0.0073	0.0103
B	0.0017	0.0018	0.0030	0.0062	0.0077	0.0108
C	0.0020	0.0021	0.0030	0.0064	0.0081	0.0107
D	0.0020	0.0020	0.0030	0.0065	0.0083	0.0108

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)					
	XL	X1	X2	X4	X6	X8
A $\rightarrow$ Y $\uparrow$	0.0698	0.0686	0.0613	0.0585	0.0581	0.0626
A $\rightarrow$ Y $\downarrow$	0.1684	0.1954	0.1453	0.1293	0.1400	0.1326
B $\rightarrow$ Y $\uparrow$	0.0770	0.0759	0.0695	0.0680	0.0659	0.0725
B $\rightarrow$ Y $\downarrow$	0.2080	0.2349	0.1785	0.1643	0.1727	0.1663
C $\rightarrow$ Y $\uparrow$	0.0831	0.0822	0.0749	0.0740	0.0711	0.0788
C $\rightarrow$ Y $\downarrow$	0.2380	0.2656	0.1988	0.1851	0.1925	0.1855
D $\rightarrow$ Y $\uparrow$	0.0852	0.0840	0.0776	0.0771	0.0737	0.0825
D $\rightarrow$ Y $\downarrow$	0.2502	0.2765	0.2096	0.1961	0.2028	0.1959



Delay Table at 25°C, 1.2V, Typical Process

Description	K _{load} (ns/pF)					
	XL	X1	X2	X4	X6	X8
A → Y↑	9.3588	5.1391	2.2511	1.1263	0.7957	0.5888
A → Y↓	6.3547	3.5998	1.7275	0.8530	0.5741	0.4283
B → Y↑	9.3694	5.1457	2.2555	1.1287	0.7970	0.5898
B → Y↓	6.3561	3.5994	1.7274	0.8531	0.5742	0.4283
C → Y↑	9.4063	5.1659	2.2659	1.1344	0.8002	0.5927
C → Y↓	6.3570	3.5997	1.7274	0.8532	0.5743	0.4283
D → Y↑	9.4517	5.1947	2.2825	1.1435	0.8053	0.5970
D → Y↓	6.3530	3.5998	1.7275	0.8532	0.5743	0.4284



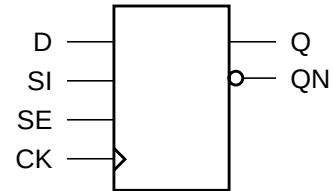
Cell Description

The Sdff cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE).

Functions

D	SI	SE	CK	Q[n+1]	QN[n+1]
1	x	0		1	0
0	x	0		0	1
x	x	x		Q[n]	QN[n]
x	1	1		1	0
x	0	1		0	1

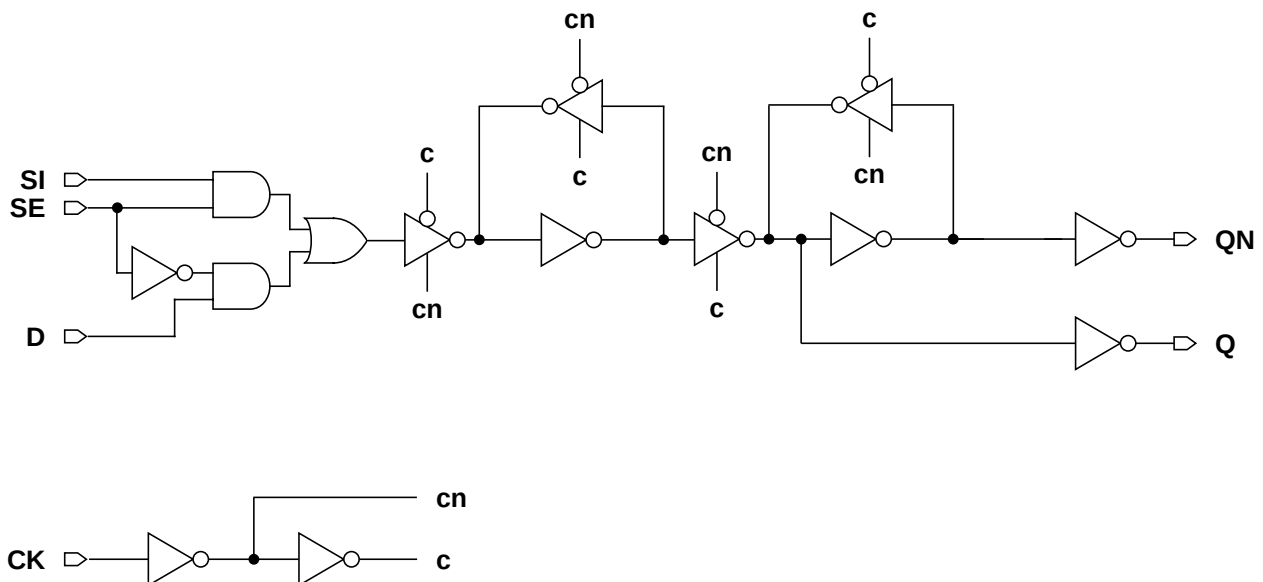
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFXL	3.69	9.66
SDFFX1	3.69	9.66
SDFFX2	3.69	10.58
SDFFX4	3.69	13.34

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0122	0.0127	0.0148	0.0219
SE	0.0144	0.0149	0.0169	0.0241
D	0.0109	0.0114	0.0132	0.0197
CK	0.0120	0.0122	0.0147	0.0190
Q	0.0082	0.0115	0.0198	0.0341

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0014	0.0014	0.0014	0.0014
SE	0.0028	0.0028	0.0030	0.0034
D	0.0011	0.0011	0.0014	0.0021
CK	0.0014	0.0019	0.0023	0.0036

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2180	0.1942	0.1789	0.1494	9.4223	5.1675	2.2686	1.1327
CK $\rightarrow$ Q $\downarrow$	0.1721	0.1439	0.1324	0.1224	5.2217	2.8942	1.5140	0.7568
CK $\rightarrow$ QN $\uparrow$	0.2206	0.1898	0.1777	0.1711	9.3949	5.1439	2.2497	1.1259
CK $\rightarrow$ QN $\downarrow$	0.2761	0.2577	0.2476	0.2075	4.7568	2.7486	1.4499	0.7161

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CK	0.1211	0.1289	0.1406	0.1719
	setup $\downarrow$ $\rightarrow$ CK	0.3047	0.3516	0.3789	0.4414
	hold $\uparrow$ $\rightarrow$ CK	-0.0977	-0.0938	-0.1055	-0.1289
	hold $\downarrow$ $\rightarrow$ CK	-0.1953	-0.2383	-0.2695	-0.3320
SE	setup $\uparrow$ $\rightarrow$ CK	0.3086	0.3594	0.3828	0.4570
	setup $\downarrow$ $\rightarrow$ CK	0.3125	0.3516	0.3047	0.2500
	hold $\uparrow$ $\rightarrow$ CK	-0.0859	-0.0859	-0.0977	-0.1250
	hold $\downarrow$ $\rightarrow$ CK	-0.1367	-0.1367	-0.1367	-0.1289
D	setup $\uparrow$ $\rightarrow$ CK	0.1055	0.1133	0.1133	0.1016
	setup $\downarrow$ $\rightarrow$ CK	0.3086	0.3477	0.3008	0.2461
	hold $\uparrow$ $\rightarrow$ CK	-0.0820	-0.0820	-0.0859	-0.0742
	hold $\downarrow$ $\rightarrow$ CK	-0.2070	-0.2422	-0.1953	-0.1523
CK	minpwh	0.1127	0.1030	0.0933	0.0787
	minpwl	0.2147	0.1856	0.1807	0.1418



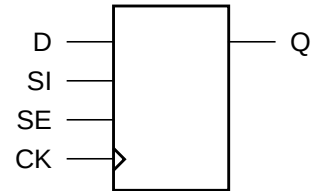
Cell Description

The SDFFQ cell is a positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (Q).

Functions

D	SI	SE	CK	Q[n+1]
1	x	0		1
0	x	0		0
x	x	x		Q[n]
x	1	1		1
x	0	1		0

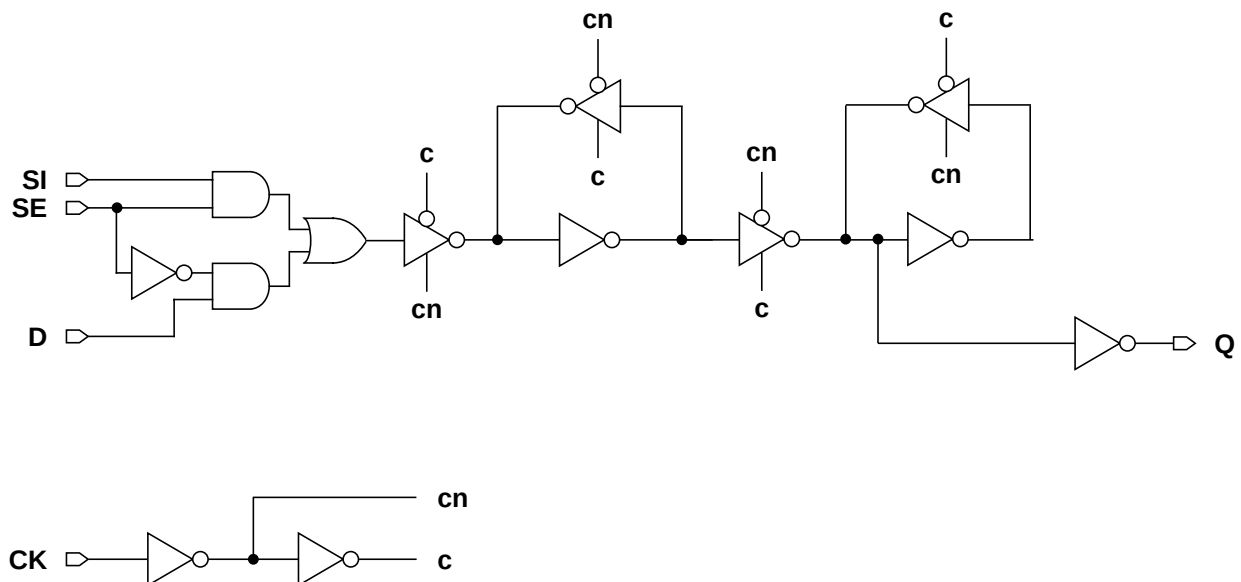
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFQXL	3.69	8.74
SDFFQX1	3.69	8.74
SDFFQX2	3.69	9.20
SDFFQX4	3.69	11.04

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0130	0.0125	0.0148	0.0194
SE	0.0153	0.0147	0.0169	0.0213
D	0.0115	0.0113	0.0133	0.0174
CK	0.0124	0.0124	0.0136	0.0174
Q	0.0058	0.0071	0.0101	0.0162

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0014	0.0014	0.0014	0.0013
SE	0.0032	0.0028	0.0030	0.0033
D	0.0018	0.0011	0.0013	0.0018
CK	0.0014	0.0018	0.0021	0.0030

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2168	0.1971	0.1803	0.1626	9.3940	5.1514	2.2693	1.1327
CK $\rightarrow$ Q $\downarrow$	0.1680	0.1491	0.1276	0.1207	5.2001	3.3121	1.4996	0.7489

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CK	0.1328	0.1289	0.1484	0.1641
	setup $\downarrow$ $\rightarrow$ CK	0.3281	0.3555	0.3867	0.4336
	hold $\uparrow$ $\rightarrow$ CK	-0.1094	-0.0977	-0.1094	-0.1211
	hold $\downarrow$ $\rightarrow$ CK	-0.2188	-0.2344	-0.2695	-0.3164
SE	setup $\uparrow$ $\rightarrow$ CK	0.3398	0.3594	0.3906	0.4453
	setup $\downarrow$ $\rightarrow$ CK	0.2109	0.3789	0.3359	0.2734
	hold $\uparrow$ $\rightarrow$ CK	-0.1016	-0.0859	-0.1016	-0.1172
	hold $\downarrow$ $\rightarrow$ CK	-0.1016	-0.1367	-0.1484	-0.1289
D	setup $\uparrow$ $\rightarrow$ CK	0.0859	0.1133	0.1289	0.1055
	setup $\downarrow$ $\rightarrow$ CK	0.2070	0.3711	0.3281	0.2695
	hold $\uparrow$ $\rightarrow$ CK	-0.0703	-0.0820	-0.0938	-0.0781
	hold $\downarrow$ $\rightarrow$ CK	-0.0977	-0.2578	-0.2109	-0.1680
CK	minpwh	0.1127	0.1030	0.0933	0.0884
	minpwl	0.2001	0.1904	0.1856	0.1516



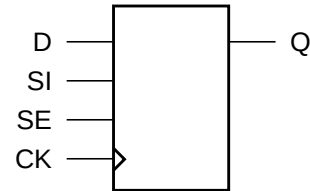
Cell Description

The SDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI) and active-high scan enable (SE). The cell has a single output (Q) and fast clock-to-out path.

Functions

D	SI	SE	CK	Q[n+1]
1	x	0		1
0	x	0		0
x	x	x		Q[n]
x	1	1		1
x	0	1		0

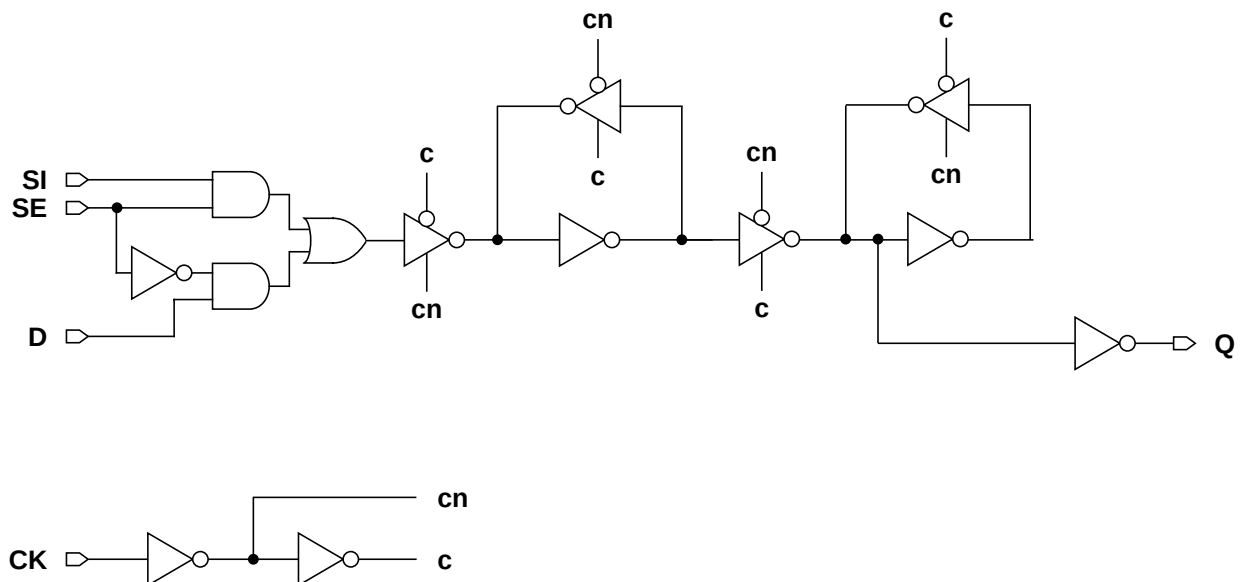
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFHQX1	3.69	11.04
SDFFHQX2	3.69	11.96
SDFFHQX4	3.69	15.18
SDFFHQX8	3.69	16.10

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0151	0.0217	0.0379	0.0370
SE	0.0180	0.0247	0.0390	0.0389
D	0.0170	0.0242	0.0398	0.0398
CK	0.0194	0.0251	0.0378	0.0380
Q	0.0070	0.0104	0.0157	0.0271

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0014	0.0018	0.0030	0.0030
SE	0.0033	0.0040	0.0054	0.0054
D	0.0020	0.0030	0.0049	0.0049
CK	0.0030	0.0042	0.0061	0.0061

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1480	0.1247	0.1133	0.1272	4.2996	2.2245	1.1119	0.5589
CK $\rightarrow$ Q $\downarrow$	0.1522	0.1327	0.1230	0.1417	2.9699	1.4764	0.7309	0.3765

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup $\uparrow$ $\rightarrow$ CK	0.1250	0.1211	0.1094	0.1133
	setup $\downarrow$ $\rightarrow$ CK	0.2266	0.2266	0.1953	0.1953
	hold $\uparrow$ $\rightarrow$ CK	-0.0586	-0.0625	-0.0547	-0.0469
	hold $\downarrow$ $\rightarrow$ CK	-0.1719	-0.1680	-0.1406	-0.1367
SE	setup $\uparrow$ $\rightarrow$ CK	0.2500	0.2539	0.2266	0.2266
	setup $\downarrow$ $\rightarrow$ CK	0.1484	0.1484	0.1406	0.1406
	hold $\uparrow$ $\rightarrow$ CK	-0.0508	-0.0547	-0.0430	-0.0391
	hold $\downarrow$ $\rightarrow$ CK	-0.0742	-0.0547	-0.0391	-0.0352
D	setup $\uparrow$ $\rightarrow$ CK	0.1133	0.0977	0.0859	0.0898
	setup $\downarrow$ $\rightarrow$ CK	0.1445	0.1133	0.0977	0.0938
	hold $\uparrow$ $\rightarrow$ CK	-0.0469	-0.0391	-0.0352	-0.0273
	hold $\downarrow$ $\rightarrow$ CK	-0.0938	-0.0664	-0.0508	-0.0469
CK	minpwh	0.0690	0.0544	0.0544	0.0544
	minpwl	0.2147	0.1807	0.1516	0.1516



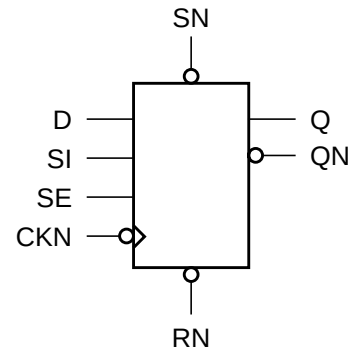
Cell Description

The Sdffnsr cell is a negative-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN).

Functions

RN	SN	D	SI	SE	CKN	Q[n+1]	QN[n+1]
1	1	1	x	0		1	0
1	1	0	x	0		0	1
1	1	x	x	x		Q[n]	QN[n]
1	1	x	1	1		1	0
1	1	x	0	1		0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

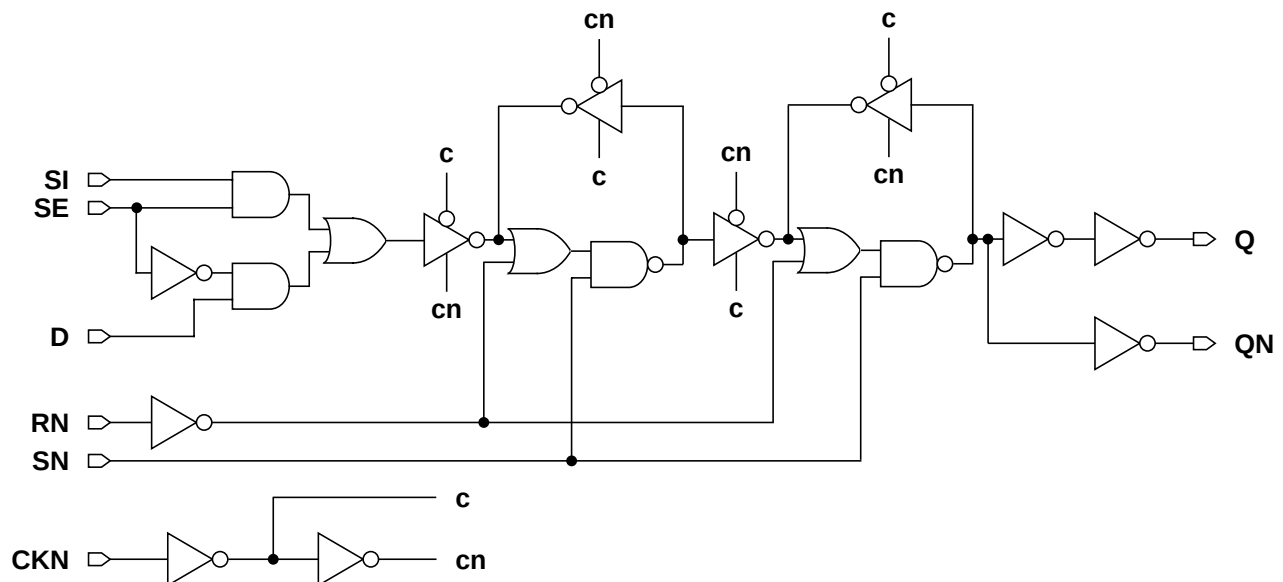
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SdffnsrXL	3.69	11.96
SdffnsrX1	3.69	11.96
SdffnsrX2	3.69	12.88
SdffnsrX4	3.69	17.48

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0136	0.0159	0.0233	0.0366
SE	0.0153	0.0174	0.0234	0.0361
D	0.0130	0.0154	0.0219	0.0343
CKN	0.0097	0.0101	0.0115	0.0158
SN	0.0018	0.0019	0.0019	0.0025
RN	0.0046	0.0045	0.0051	0.0077
Q	0.0096	0.0130	0.0210	0.0358

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0015	0.0015	0.0018	0.0030
SE	0.0023	0.0025	0.0033	0.0050
D	0.0016	0.0019	0.0028	0.0045
CKN	0.0015	0.0016	0.0017	0.0023
SN	0.0026	0.0030	0.0039	0.0066
RN	0.0016	0.0016	0.0015	0.0024

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CKN $\rightarrow$ Q $\uparrow$	0.3145	0.3180	0.3122	0.3103	9.9105	5.1617	2.2642	1.1332
CKN $\rightarrow$ Q $\downarrow$	0.2665	0.2862	0.2830	0.2575	4.9460	2.8453	1.5041	0.7461
SN $\rightarrow$ Q $\uparrow$	0.2216	0.2606	0.3556	0.5687	9.8929	5.1576	2.2714	1.1424
SN $\rightarrow$ Q $\downarrow$	0.1912	0.2108	0.2164	0.2080	4.9302	2.8411	1.5045	0.7467
RN $\rightarrow$ Q $\downarrow$	0.2396	0.2793	0.3039	0.2816	4.9213	2.8385	1.5040	0.7465
CKN $\rightarrow$ QN $\uparrow$	0.2073	0.2036	0.1895	0.1684	10.1246	5.2408	2.2942	1.1444
CKN $\rightarrow$ QN $\downarrow$	0.2612	0.2637	0.2505	0.2354	6.6240	3.3346	1.6505	0.8792
SN $\rightarrow$ QN $\uparrow$	0.1322	0.1270	0.1185	0.1135	10.0611	5.2536	2.3221	1.1601
SN $\rightarrow$ QN $\downarrow$	0.1713	0.2067	0.2859	0.4603	5.5803	3.2182	1.8511	1.1095
RN $\rightarrow$ QN $\uparrow$	0.1804	0.1954	0.2055	0.1874	10.0109	5.2337	2.3153	1.1565

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CKN	0.1211	0.0859	0.0742	0.0859
	setup↓ → CKN	0.3242	0.3594	0.3516	0.3047
	hold↑ → CKN	0.0156	0.0117	0.0078	0.0039
	hold↓ → CKN	-0.1875	-0.2305	-0.2266	-0.1562
SE	setup↑ → CKN	0.3438	0.3789	0.3789	0.3398
	setup↓ → CKN	0.2852	0.2383	0.1680	0.1367
	hold↑ → CKN	0.0273	0.0234	0.0156	0.0117
	hold↓ → CKN	-0.0469	-0.0352	-0.0273	-0.0312
D	setup↑ → CKN	0.1211	0.0625	0.0195	0.0430
	setup↓ → CKN	0.3242	0.2734	0.1992	0.1445
	hold↑ → CKN	0.0117	0.0234	0.0352	0.0273
	hold↓ → CKN	-0.1875	-0.1719	-0.1211	-0.0586
CKN	minpwl	0.2050	0.1953	0.1807	0.1807
	minpwh	0.1418	0.1321	0.1321	0.1078
SN	minpwl	0.1613	0.1953	0.2827	0.4771
	recovery	-0.0234	-0.0430	-0.0625	-0.0469
	removal	0.0742	0.0898	0.1094	0.1094
RN	minpwl	0.1467	0.1710	0.1904	0.1759
	recovery	0.0312	-0.0312	-0.0586	-0.0312
	removal	0.0859	0.1289	0.1953	0.2070



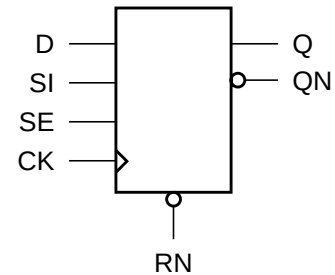
Cell Description

The SDFFR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN).

Functions

RN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	0	1

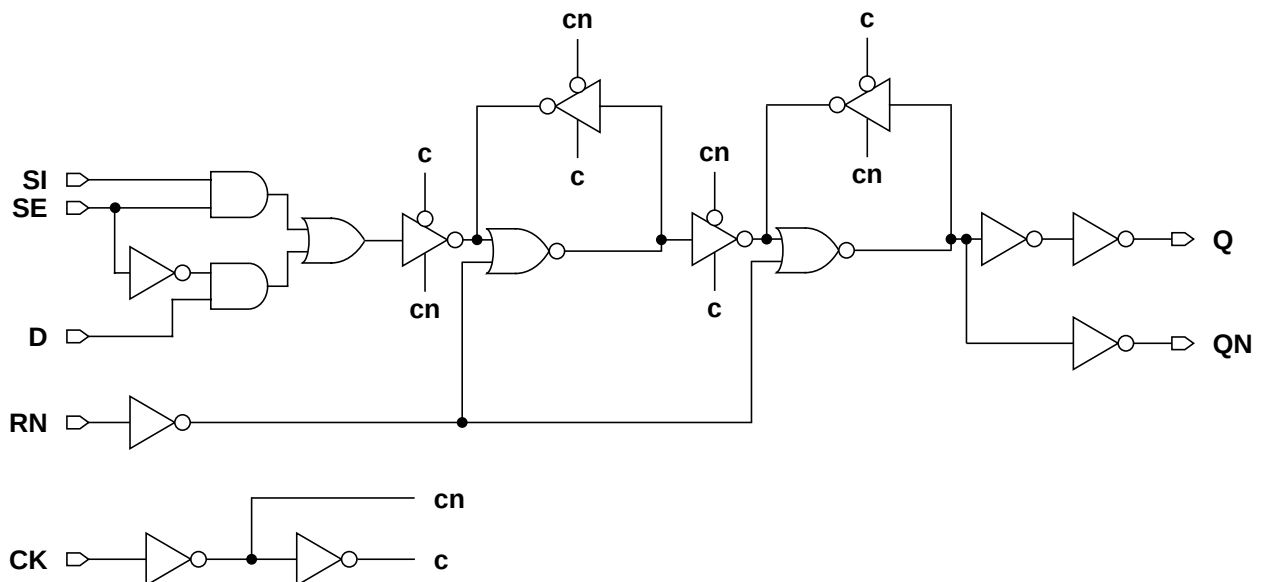
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFRXL	3.69	11.04
SDFFRX1	3.69	11.04
SDFFRX2	3.69	12.42
SDFFRX4	3.69	15.64

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0129	0.0145	0.0191	0.0316
SE	0.0147	0.0161	0.0203	0.0323
D	0.0123	0.0138	0.0185	0.0301
CK	0.0089	0.0091	0.0102	0.0140
RN	0.0019	0.0020	0.0024	0.0034
Q	0.0095	0.0124	0.0198	0.0329

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0011	0.0011	0.0013	0.0022
SE	0.0025	0.0026	0.0032	0.0044
D	0.0012	0.0013	0.0020	0.0031
CK	0.0014	0.0014	0.0016	0.0023
RN	0.0048	0.0051	0.0063	0.0096

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2733	0.2457	0.2237	0.2026	9.9674	5.1817	2.5871	1.1353
CK $\rightarrow$ Q $\downarrow$	0.3047	0.3089	0.3084	0.2778	5.1767	2.9616	1.4657	0.7527
RN $\rightarrow$ Q $\downarrow$	0.1369	0.1445	0.1348	0.1180	5.1887	2.9700	1.4856	0.7638
CK $\rightarrow$ QN $\uparrow$	0.2061	0.2008	0.2016	0.1802	10.0083	5.1875	2.5886	1.1361
CK $\rightarrow$ QN $\downarrow$	0.1810	0.1694	0.1542	0.1376	5.8543	3.1333	1.5856	0.7781
RN $\rightarrow$ QN $\uparrow$	0.2316	0.2505	0.2868	0.3441	9.9532	5.2293	2.6842	1.2703



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.2031	0.1953	0.1758	0.1680
	setup↓ → CK	0.2695	0.2930	0.2734	0.2266
	hold↑ → CK	-0.1133	-0.1172	-0.1094	-0.1016
	hold↓ → CK	-0.1445	-0.1523	-0.1406	-0.1211
SE	setup↑ → CK	0.2812	0.3086	0.2969	0.2578
	setup↓ → CK	0.2422	0.2266	0.1953	0.2188
	hold↑ → CK	-0.1016	-0.1016	-0.0977	-0.0898
	hold↓ → CK	-0.0820	-0.0742	-0.0391	-0.0273
D	setup↑ → CK	0.1836	0.1680	0.1328	0.1289
	setup↓ → CK	0.2188	0.2148	0.1406	0.1055
	hold↑ → CK	-0.1055	-0.1016	-0.0820	-0.0781
	hold↓ → CK	-0.1133	-0.1055	-0.0664	-0.0508
CK	minpwh	0.1564	0.1321	0.1176	0.1030
	minpwl	0.1661	0.1661	0.1467	0.1273
RN	minpwl	0.1467	0.1613	0.1759	0.2001
	recovery	0.1953	0.1797	0.1367	0.1328
	removal	-0.1406	-0.1328	-0.1055	-0.0977



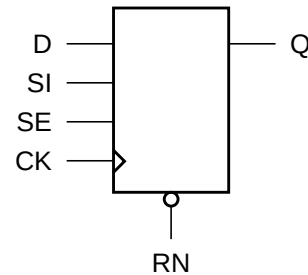
Cell Description

The SDFFRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN). The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	D	SI	SE	CK	Q[n+1]
1	1	x	0		1
1	0	x	0		0
1	x	x	x		Q[n]
1	x	1	1		1
1	x	0	1		0
0	x	x	x	x	0

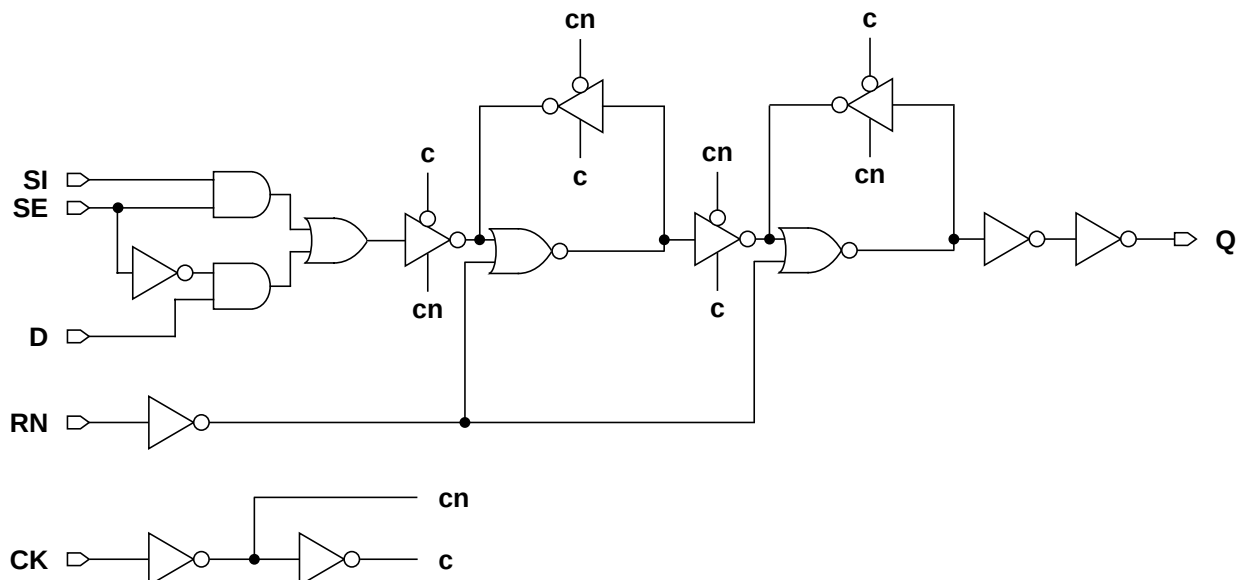
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFRHQX1	3.69	11.96
SDFFRHQX2	3.69	12.42
SDFFRHQX4	3.69	16.56
SDFFRHQX8	3.69	17.48

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0168	0.0238	0.0386	0.0385
SE	0.0191	0.0255	0.0415	0.0415
D	0.0179	0.0252	0.0414	0.0413
CK	0.0204	0.0259	0.0397	0.0397
RN	0.0026	0.0034	0.0056	0.0056
Q	0.0085	0.0110	0.0161	0.0270

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0014	0.0019	0.0031	0.0030
SE	0.0034	0.0041	0.0055	0.0055
D	0.0019	0.0030	0.0049	0.0050
CK	0.0031	0.0044	0.0060	0.0060
RN	0.0035	0.0048	0.0073	0.0073

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1709	0.1402	0.1279	0.1452	4.3777	2.2493	1.1269	0.5699
CK $\rightarrow$ Q $\downarrow$	0.1549	0.1382	0.1304	0.1487	2.8940	1.4907	0.7391	0.3815
RN $\rightarrow$ Q $\downarrow$	0.1547	0.1382	0.1272	0.1716	3.0072	1.5614	0.7803	0.4123



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup↑ → CK	0.1406	0.1172	0.1172	0.1172
	setup↓ → CK	0.2344	0.2227	0.1758	0.1758
	hold↑ → CK	-0.0625	-0.0547	-0.0508	-0.0469
	hold↓ → CK	-0.1719	-0.1719	-0.1250	-0.1211
SE	setup↑ → CK	0.2578	0.2500	0.2109	0.2109
	setup↓ → CK	0.1641	0.1406	0.1523	0.1523
	hold↑ → CK	-0.0547	-0.0469	-0.0430	-0.0352
	hold↓ → CK	-0.0781	-0.0586	-0.0352	-0.0312
D	setup↑ → CK	0.1289	0.0977	0.0938	0.0977
	setup↓ → CK	0.1445	0.1133	0.0938	0.0938
	hold↑ → CK	-0.0469	-0.0391	-0.0352	-0.0312
	hold↓ → CK	-0.0898	-0.0703	-0.0469	-0.0430
CK	minpwh	0.0835	0.0641	0.0544	0.0641
	minpwl	0.2196	0.1710	0.1564	0.1613
RN	minpwl	0.1467	0.1273	0.1224	0.1613
	recovery	-0.0586	-0.0469	-0.0352	-0.0391
	removal	0.0977	0.1055	0.1289	0.1211



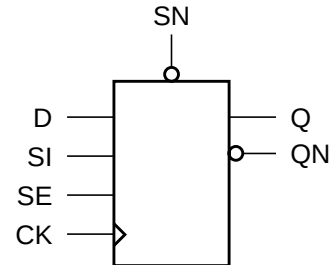
Cell Description

The SDFFS cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN).

Functions

SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	x	0		1	0
1	0	x	0		0	1
1	x	x	x		Q[n]	QN[n]
1	x	1	1		1	0
1	x	0	1		0	1
0	x	x	x	x	1	0

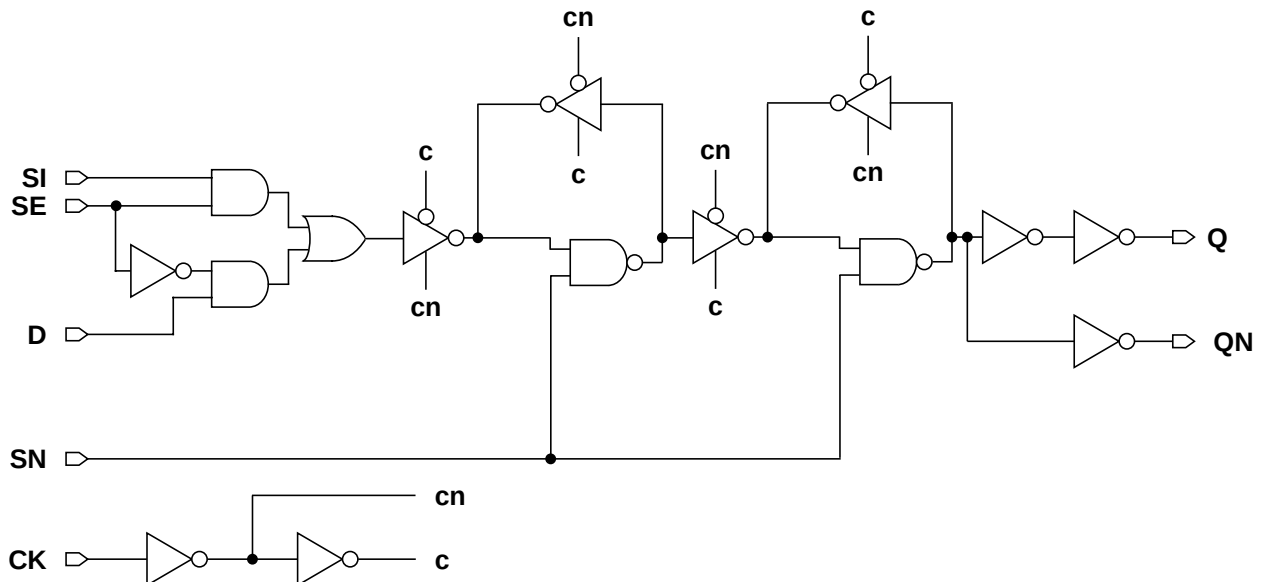
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSXL	3.69	10.58
SDFFSX1	3.69	10.58
SDFFSX2	3.69	11.50
SDFFSX4	3.69	13.34

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0120	0.0126	0.0166	0.0253
SE	0.0136	0.0142	0.0180	0.0255
D	0.0113	0.0119	0.0160	0.0241
CK	0.0100	0.0102	0.0110	0.0126
SN	0.0018	0.0019	0.0020	0.0022
Q	0.0090	0.0116	0.0181	0.0295

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0014	0.0014	0.0015	0.0019
SE	0.0023	0.0023	0.0026	0.0034
D	0.0015	0.0015	0.0020	0.0031
CK	0.0014	0.0014	0.0014	0.0018
SN	0.0025	0.0030	0.0042	0.0064

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2395	0.2774	0.3096	0.3043	9.8982	5.1653	2.2680	1.2701
CK $\rightarrow$ Q $\downarrow$	0.2811	0.2946	0.3069	0.2925	4.9621	2.8432	1.5051	0.8120
SN $\rightarrow$ Q $\uparrow$	0.2057	0.2428	0.3362	0.4818	9.8903	5.1563	2.2667	1.2740
CK $\rightarrow$ QN $\uparrow$	0.2222	0.2137	0.2117	0.2017	10.1312	5.2237	2.2852	1.2777
CK $\rightarrow$ QN $\downarrow$	0.1863	0.2194	0.2354	0.2226	6.0999	3.6145	1.9069	0.9851
SN $\rightarrow$ QN $\downarrow$	0.1557	0.1893	0.2630	0.3793	5.4457	3.1450	1.8238	1.1098



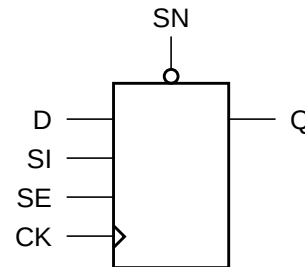
Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1836	0.1875	0.2148	0.1992
	setup↓ → CK	0.2656	0.2578	0.2930	0.2852
	hold↑ → CK	-0.0820	-0.0898	-0.1133	-0.1172
	hold↓ → CK	-0.1758	-0.1797	-0.1953	-0.1797
SE	setup↑ → CK	0.2812	0.2773	0.3203	0.3125
	setup↓ → CK	0.2461	0.2500	0.2344	0.2070
	hold↑ → CK	-0.0703	-0.0781	-0.1055	-0.1094
	hold↓ → CK	-0.1367	-0.1406	-0.0938	-0.0547
D	setup↑ → CK	0.1875	0.1914	0.1719	0.1367
	setup↓ → CK	0.2656	0.2578	0.2031	0.1484
	hold↑ → CK	-0.0859	-0.0938	-0.0898	-0.0820
	hold↓ → CK	-0.1758	-0.1797	-0.1328	-0.0859
CK	minpwh	0.1467	0.1759	0.1953	0.1904
	minpwl	0.1613	0.1710	0.1807	0.1710
SN	minpwl	0.1467	0.1759	0.2633	0.3945
	recovery	-0.0508	-0.0742	-0.1016	-0.1055
	removal	0.0781	0.0977	0.1289	0.1484



Logic Symbol

The SDFFSHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low set (SN). The cell has a single output (Q) and fast clock-to-out path.



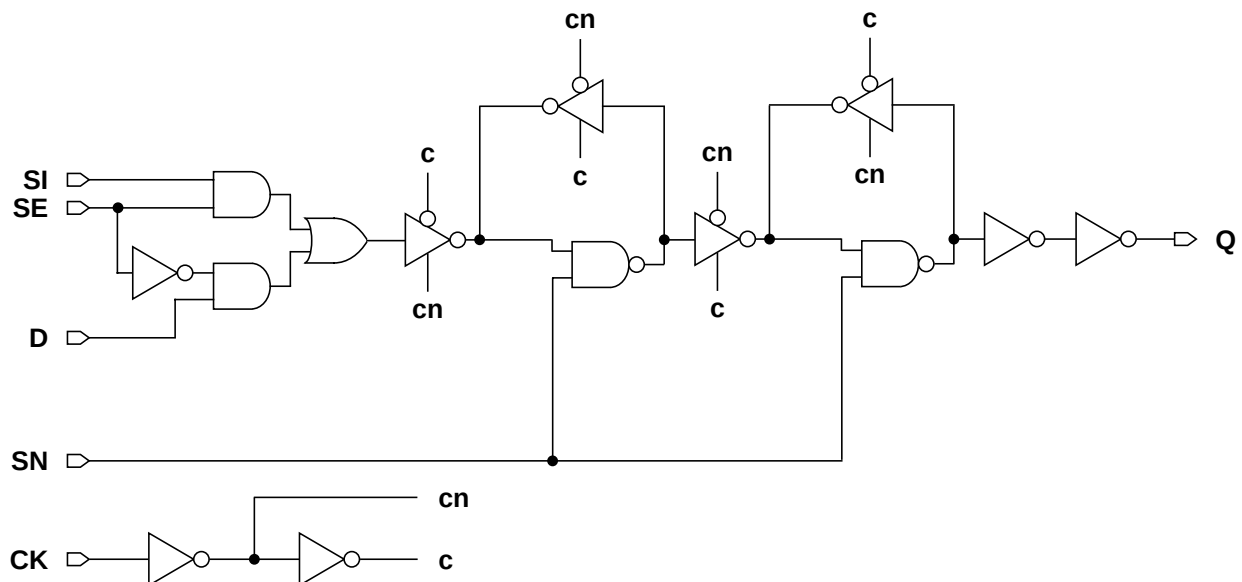
Functions

SN	D	SI	SE	CK	Q[n+1]
1	1	x	0		1
1	0	x	0		0
1	x	x	x		Q[n]
1	x	1	1		1
1	x	0	1		0
0	x	x	x	x	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSHQX1	3.69	12.42
SDFFSHQX2	3.69	13.80
SDFFSHQX4	3.69	18.86
SDFFSHQX8	3.69	19.78

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0173	0.0244	0.0389	0.0392
SE	0.0194	0.0269	0.0421	0.0419
D	0.0187	0.0267	0.0423	0.0417
CK	0.0205	0.0275	0.0426	0.0429
SN	0.0057	0.0070	0.0078	0.0079
Q	0.0083	0.0116	0.0168	0.0277

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0014	0.0016	0.0026	0.0025
SE	0.0036	0.0040	0.0058	0.0058
D	0.0021	0.0032	0.0055	0.0055
CK	0.0031	0.0045	0.0074	0.0074
SN	0.0048	0.0061	0.0083	0.0083

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1621	0.1323	0.1209	0.1333	4.3183	2.1518	1.1139	0.5603
CK $\rightarrow$ Q $\downarrow$	0.1616	0.1408	0.1303	0.1493	2.8866	1.4932	0.7387	0.4084
SN $\rightarrow$ Q $\uparrow$	0.1554	0.1655	0.1630	0.1875	4.3112	2.1600	1.1178	0.5644

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup↑ → CK	0.1562	0.1562	0.1445	0.1445
	setup↓ → CK	0.2617	0.2852	0.2383	0.2383
	hold↑ → CK	-0.0781	-0.0859	-0.0781	-0.0742
	hold↓ → CK	-0.1953	-0.2109	-0.1719	-0.1641
SE	setup↑ → CK	0.2891	0.3164	0.2812	0.2773
	setup↓ → CK	0.1758	0.1602	0.1562	0.1562
	hold↑ → CK	-0.0703	-0.0781	-0.0703	-0.0625
	hold↓ → CK	-0.0859	-0.0508	-0.0273	-0.0234
D	setup↑ → CK	0.1367	0.1133	0.1016	0.1055
	setup↓ → CK	0.1523	0.1172	0.0898	0.0898
	hold↑ → CK	-0.0586	-0.0469	-0.0430	-0.0391
	hold↓ → CK	-0.0938	-0.0625	-0.0391	-0.0352
CK	minpwh	0.0787	0.0593	0.0544	0.0593
	minpwl	0.2196	0.1856	0.1661	0.1710
SN	minpwl	0.1273	0.1370	0.1370	0.1613
	recovery	0.0508	0.0391	0.0273	0.0273
	removal	-0.0273	-0.0195	-0.0117	-0.0117



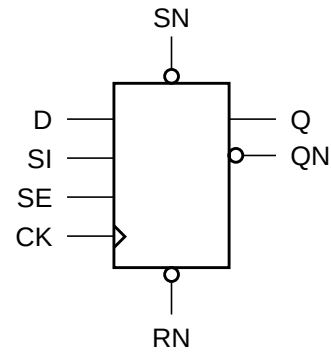
Cell Description

The SDFFSR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN). Set (SN) dominates reset (RN).

Functions

RN	SN	D	SI	SE	CK	Q[n+1]	QN[n+1]
1	1	1	x	0		1	0
1	1	0	x	0		0	1
1	1	x	x	x		Q[n]	QN[n]
1	1	x	1	1		1	0
1	1	x	0	1		0	1
0	1	x	x	x	x	0	1
1	0	x	x	x	x	1	0
0	0	x	x	x	x	1	0

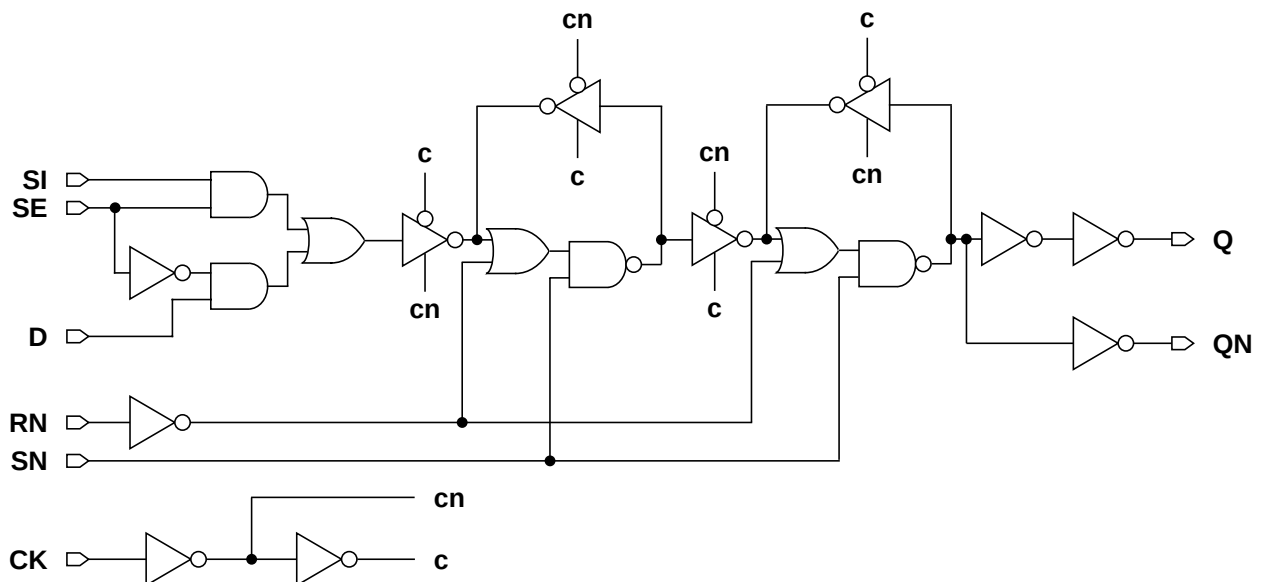
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDFFSRXL	3.69	11.96
SDFFSRX1	3.69	11.96
SDFFSRX2	3.69	12.88
SDFFSRX4	3.69	17.48

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0142	0.0164	0.0231	0.0384
SE	0.0158	0.0178	0.0233	0.0374
D	0.0136	0.0159	0.0218	0.0365
CK	0.0114	0.0118	0.0139	0.0176
SN	0.0017	0.0018	0.0018	0.0024
RN	0.0042	0.0041	0.0051	0.0077
Q	0.0086	0.0112	0.0181	0.0321

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0014	0.0014	0.0018	0.0031
SE	0.0023	0.0025	0.0033	0.0048
D	0.0015	0.0018	0.0028	0.0045
CK	0.0014	0.0014	0.0016	0.0026
SN	0.0025	0.0029	0.0039	0.0066
RN	0.0012	0.0012	0.0015	0.0025

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2699	0.2599	0.2432	0.2411	9.8950	5.1610	2.2645	1.1324
CK $\rightarrow$ Q $\downarrow$	0.2959	0.3130	0.3261	0.2932	4.9185	2.8483	1.5081	0.7459
SN $\rightarrow$ Q $\uparrow$	0.2259	0.2606	0.3569	0.5697	9.8825	5.1569	2.2718	1.1421
SN $\rightarrow$ Q $\downarrow$	0.1934	0.2101	0.2191	0.2075	4.9020	2.8437	1.5083	0.7462
RN $\rightarrow$ Q $\downarrow$	0.2388	0.2731	0.3092	0.2811	4.8922	2.8409	1.5081	0.7461
CK $\rightarrow$ QN $\uparrow$	0.2330	0.2295	0.2297	0.2036	10.1121	5.2306	2.2932	1.1435
CK $\rightarrow$ QN $\downarrow$	0.2106	0.2029	0.1798	0.1704	6.4261	3.3250	1.6352	0.8443
SN $\rightarrow$ QN $\uparrow$	0.1300	0.1251	0.1187	0.1135	10.0898	5.2557	2.3238	1.1602
SN $\rightarrow$ QN $\downarrow$	0.1701	0.2054	0.2872	0.4621	5.6059	3.2168	1.8536	1.1104
RN $\rightarrow$ QN $\uparrow$	0.1753	0.1879	0.2085	0.1873	10.0263	5.2335	2.3165	1.1568



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup $\uparrow$ $\rightarrow$ CK	0.2461	0.2148	0.1836	0.2188
	setup $\downarrow$ $\rightarrow$ CK	0.2578	0.2852	0.2734	0.2461
	hold $\uparrow$ $\rightarrow$ CK	-0.0977	-0.1055	-0.1055	-0.1094
	hold $\downarrow$ $\rightarrow$ CK	-0.1523	-0.1602	-0.1562	-0.1367
SE	setup $\uparrow$ $\rightarrow$ CK	0.2773	0.3086	0.3008	0.2812
	setup $\downarrow$ $\rightarrow$ CK	0.3086	0.2500	0.1953	0.2500
	hold $\uparrow$ $\rightarrow$ CK	-0.0859	-0.0938	-0.0977	-0.1016
	hold $\downarrow$ $\rightarrow$ CK	-0.1094	-0.0781	-0.0430	-0.0156
D	setup $\uparrow$ $\rightarrow$ CK	0.2500	0.1914	0.1289	0.1523
	setup $\downarrow$ $\rightarrow$ CK	0.2578	0.2148	0.1406	0.0938
	hold $\uparrow$ $\rightarrow$ CK	-0.1016	-0.0859	-0.0742	-0.0703
	hold $\downarrow$ $\rightarrow$ CK	-0.1523	-0.1172	-0.0742	-0.0430
CK	minpwh	0.1564	0.1467	0.1321	0.1321
	minpwl	0.1953	0.1856	0.1661	0.1418
SN	minpwl	0.1613	0.1953	0.2827	0.4819
	recovery	-0.0820	-0.1016	-0.1289	-0.1055
	removal	0.1250	0.1445	0.1797	0.1875
RN	minpwl	0.1418	0.1661	0.1904	0.1759
	recovery	0.1602	0.1094	0.0742	0.1016
	removal	-0.0273	0.0039	0.0469	0.0625



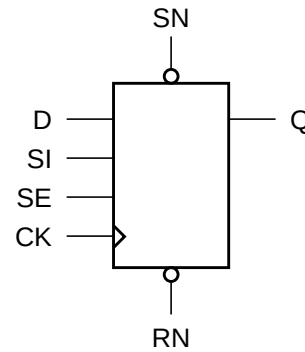
Cell Description

The SDDFSRHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and asynchronous active-low reset (RN) and set (SN), and set dominating reset. The cell has a single output (Q) and fast clock-to-out path.

Functions

RN	SN	D	SI	SE	CK	Q[n+1]
1	1	1	x	0		1
1	1	0	x	0		0
1	1	x	x	x		Q[n]
1	1	x	1	1		1
1	1	x	0	1		0
0	1	x	x	x	x	0
1	0	x	x	x	x	1
0	0	x	x	x	x	1

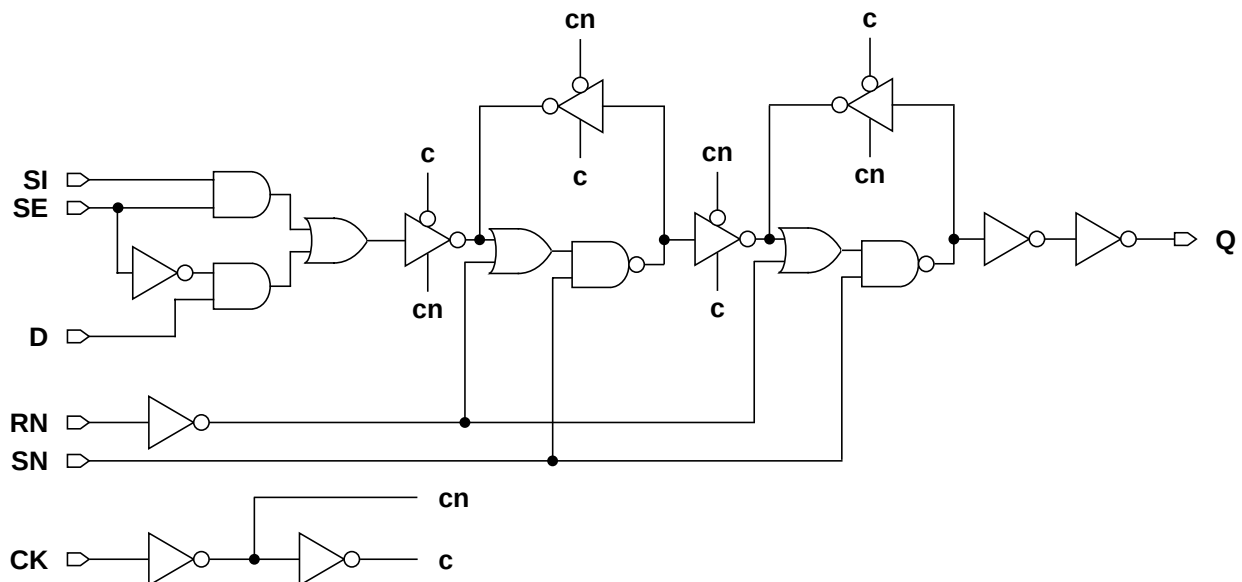
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SDDFSRHQX1	3.69	15.18
SDDFSRHQX2	3.69	16.10
SDDFSRHQX4	3.69	17.94
SDDFSRHQX8	3.69	19.32

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0170	0.0244	0.0314	0.0311
SE	0.0200	0.0262	0.0358	0.0357
D	0.0183	0.0257	0.0335	0.0340
CK	0.0199	0.0237	0.0331	0.0331
SN	0.0081	0.0094	0.0111	0.0113
RN	0.0020	0.0026	0.0042	0.0042
Q	0.0092	0.0121	0.0175	0.0306

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0015	0.0020	0.0031	0.0031
SE	0.0041	0.0046	0.0061	0.0061
D	0.0020	0.0031	0.0043	0.0043
CK	0.0029	0.0038	0.0066	0.0066
SN	0.0030	0.0036	0.0050	0.0050
RN	0.0026	0.0039	0.0055	0.0055

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1772	0.1484	0.1441	0.1696	4.4162	2.1981	1.1379	0.5792
CK $\rightarrow$ Q $\downarrow$	0.1823	0.1598	0.1511	0.1840	3.1385	1.6004	0.8107	0.4313
SN $\rightarrow$ Q $\uparrow$	0.2338	0.2715	0.2289	0.2672	4.3548	2.2003	1.1271	0.5707
SN $\rightarrow$ Q $\downarrow$	0.3199	0.3050	0.2251	0.2963	3.4779	1.8028	0.8345	0.4525
RN $\rightarrow$ Q $\downarrow$	0.2708	0.2498	0.1761	0.2464	3.4903	1.8074	0.8360	0.4530

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup↑ → CK	0.1680	0.1484	0.1445	0.1445
	setup↓ → CK	0.2383	0.2227	0.1914	0.1875
	hold↑ → CK	-0.0820	-0.0742	-0.0625	-0.0586
	hold↓ → CK	-0.1641	-0.1484	-0.1211	-0.1172
SE	setup↑ → CK	0.2656	0.2539	0.2305	0.2266
	setup↓ → CK	0.1914	0.1797	0.1953	0.1953
	hold↑ → CK	-0.0703	-0.0625	-0.0508	-0.0469
	hold↓ → CK	-0.0586	-0.0352	-0.0430	-0.0430
D	setup↑ → CK	0.1445	0.1211	0.1250	0.1289
	setup↓ → CK	0.1367	0.1055	0.1133	0.1094
	hold↑ → CK	-0.0625	-0.0469	-0.0469	-0.0430
	hold↓ → CK	-0.0703	-0.0430	-0.0508	-0.0469
CK	minpwh	0.0787	0.0690	0.0641	0.0738
	minpwl	0.2390	0.1856	0.1856	0.1904
SN	minpwl	0.1953	0.2342	0.1953	0.2342
	recovery	0.0391	0.0352	0.0430	0.0430
	removal	-0.0195	-0.0117	-0.0195	-0.0234
RN	minpwl	0.2439	0.2244	0.1564	0.2196
	recovery	-0.0469	-0.0352	-0.0352	-0.0352
	removal	0.0859	0.0859	0.0898	0.0859



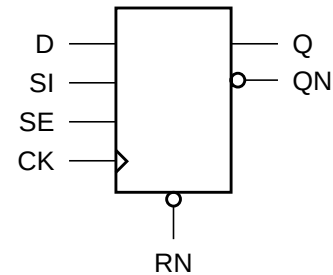
Cell Description

The Sdfftr cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-low reset (RN). Scan enable (SE) dominates reset (RN).

Functions

RN	D	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	0	1		0	1
x	x	1	1		1	0
0	x	x	0		0	1
1	0	x	0		0	1
1	1	x	0		1	0
x	x	x	x		Q[n]	QN[n]

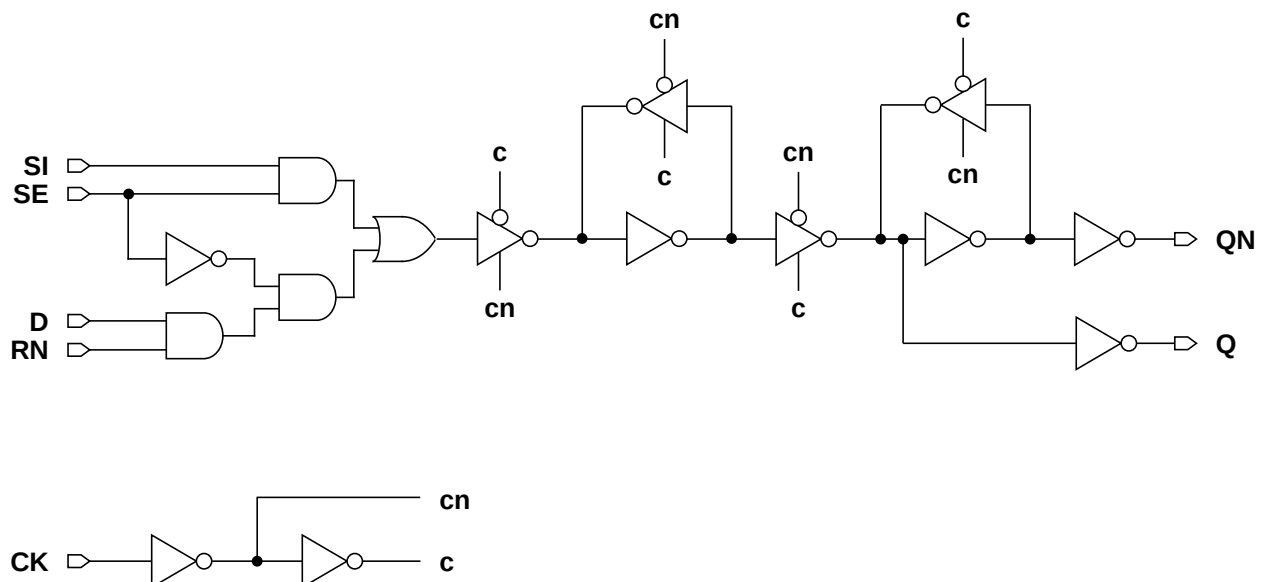
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SdfftrXL	3.69	10.12
SdfftrX1	3.69	10.12
SdfftrX2	3.69	10.58
SdfftrX4	3.69	13.34

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0120	0.0137	0.0180	0.0308
SE	0.0146	0.0161	0.0210	0.0335
D	0.0116	0.0133	0.0180	0.0297
CK	0.0088	0.0090	0.0102	0.0137
RN	0.0126	0.0148	0.0209	0.0357
Q	0.0090	0.0114	0.0181	0.0317

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0015	0.0015	0.0018	0.0031
SE	0.0040	0.0042	0.0053	0.0079
D	0.0014	0.0016	0.0021	0.0032
CK	0.0015	0.0015	0.0017	0.0023
RN	0.0020	0.0021	0.0028	0.0048

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2368	0.2298	0.2160	0.2090	9.9098	5.1571	2.2587	1.1291
CK $\rightarrow$ Q $\downarrow$	0.2580	0.2716	0.2792	0.2691	4.8000	2.8345	1.4957	0.7419
CK $\rightarrow$ QN $\uparrow$	0.1950	0.1881	0.1854	0.1806	10.0268	5.1876	2.2707	1.1358
CK $\rightarrow$ QN $\downarrow$	0.1772	0.1713	0.1535	0.1442	5.7668	3.1433	1.5810	0.7824

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1445	0.1523	0.1445	0.1445
	setup↓ → CK	0.2422	0.2656	0.2383	0.2227
	hold↑ → CK	-0.0859	-0.0898	-0.0898	-0.0938
	hold↓ → CK	-0.1406	-0.1484	-0.1289	-0.1211
SE	setup↑ → CK	0.2539	0.2812	0.2617	0.2617
	setup↓ → CK	0.2227	0.2109	0.2031	0.2422
	hold↑ → CK	-0.0742	-0.0820	-0.0820	-0.0820
	hold↓ → CK	-0.0820	-0.0703	-0.0352	-0.0273
D	setup↑ → CK	0.1680	0.1562	0.1328	0.1406
	setup↓ → CK	0.2031	0.1875	0.1211	0.1016
	hold↑ → CK	-0.1055	-0.0977	-0.0859	-0.0938
	hold↓ → CK	-0.1133	-0.1016	-0.0625	-0.0508
CK	minpwh	0.1370	0.1273	0.1127	0.1127
	minpwl	0.1467	0.1467	0.1321	0.1176
RN	setup↑ → CK	0.1758	0.1602	0.1367	0.1445
	setup↓ → CK	0.2617	0.3008	0.2852	0.2734
	hold↑ → CK	-0.1094	-0.1016	-0.0898	-0.0938
	hold↓ → CK	-0.1445	-0.1641	-0.1484	-0.1406



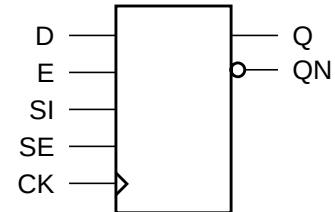
Cell Description

The SEDFF cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E).

Functions

D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	1	1		1	0
x	x	0	1		0	1
x	0	x	0		Q[n]	QN[n]
0	1	x	0		0	1
1	1	x	0		1	0
x	x	x	x		Q[n]	QN[n]

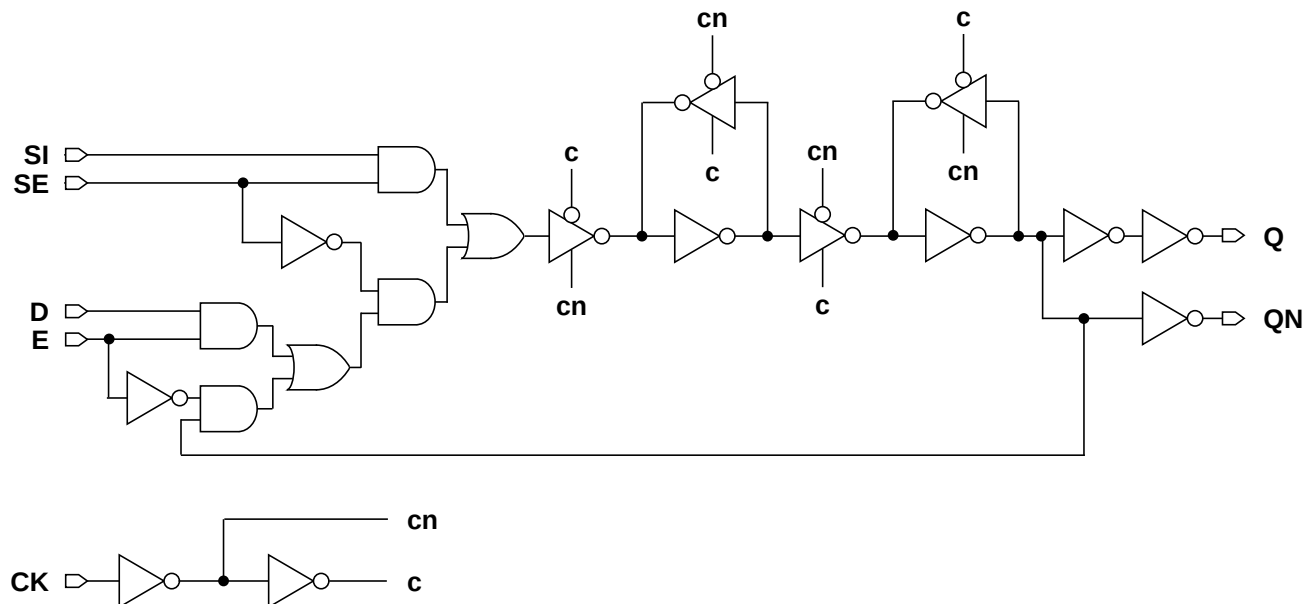
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFXL	3.69	11.50
SEDFFX1	3.69	11.50
SEDFFX2	3.69	13.80
SEDFFX4	3.69	16.10

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0143	0.0152	0.0190	0.0276
SE	0.0172	0.0183	0.0219	0.0306
D	0.0120	0.0132	0.0160	0.0228
CK	0.0205	0.0226	0.0267	0.0385
E	0.0173	0.0185	0.0219	0.0300
Q	0.0106	0.0136	0.0221	0.0378

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0013	0.0012	0.0015	0.0022
SE	0.0026	0.0026	0.0030	0.0036
D	0.0013	0.0012	0.0016	0.0024
CK	0.0014	0.0019	0.0025	0.0039
E	0.0031	0.0031	0.0036	0.0042

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2287	0.2061	0.1688	0.1502	9.4430	5.6191	2.2641	1.1337
CK $\rightarrow$ Q $\downarrow$	0.1707	0.1355	0.1283	0.1145	5.2728	3.1104	1.5414	0.7587
CK $\rightarrow$ QN $\uparrow$	0.2504	0.1944	0.1847	0.1668	9.4402	5.1525	2.2562	1.1276
CK $\rightarrow$ QN $\downarrow$	0.3211	0.2813	0.2430	0.2106	5.1383	2.8112	1.4669	0.7203



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.1523	0.1562	0.1523	0.1406
	setup↓ → CK	0.3203	0.4023	0.3398	0.3125
	hold↑ → CK	-0.1367	-0.1406	-0.1406	-0.1328
	hold↓ → CK	-0.3008	-0.3984	-0.3242	-0.2969
SE	setup↑ → CK	0.4258	0.5312	0.4414	0.4062
	setup↓ → CK	0.4648	0.5430	0.4414	0.3789
	hold↑ → CK	-0.1094	-0.1133	-0.1133	-0.1055
	hold↓ → CK	-0.2148	-0.2188	-0.2188	-0.2188
D	setup↑ → CK	0.1641	0.1719	0.1680	0.1641
	setup↓ → CK	0.4180	0.4961	0.3984	0.3398
	hold↑ → CK	-0.1328	-0.1367	-0.1289	-0.1211
	hold↓ → CK	-0.3008	-0.3516	-0.2812	-0.2266
CK	minpwh	0.1176	0.1127	0.0933	0.0835
	minpwl	0.2487	0.2147	0.1904	0.1710
E	setup↑ → CK	0.4609	0.5391	0.4492	0.3906
	setup↓ → CK	0.3477	0.4141	0.3281	0.2695
	hold↑ → CK	-0.1445	-0.1523	-0.1406	-0.1367
	hold↓ → CK	-0.2031	-0.2109	-0.2109	-0.2188



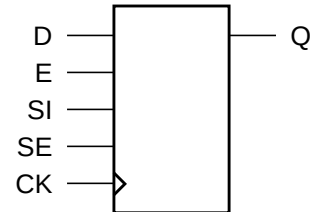
Cell Description

The SEDFFHQ cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), and synchronous active-high enable (E). The cell has a single output (Q) and fast clock-to-output path.

Functions

D	E	SI	SE	CK	Q[n+1]
x	x	1	1		1
x	x	0	1		0
x	0	x	0		Q[n]
0	1	x	0		0
1	1	x	0		1
x	x	x	x		Q[n]

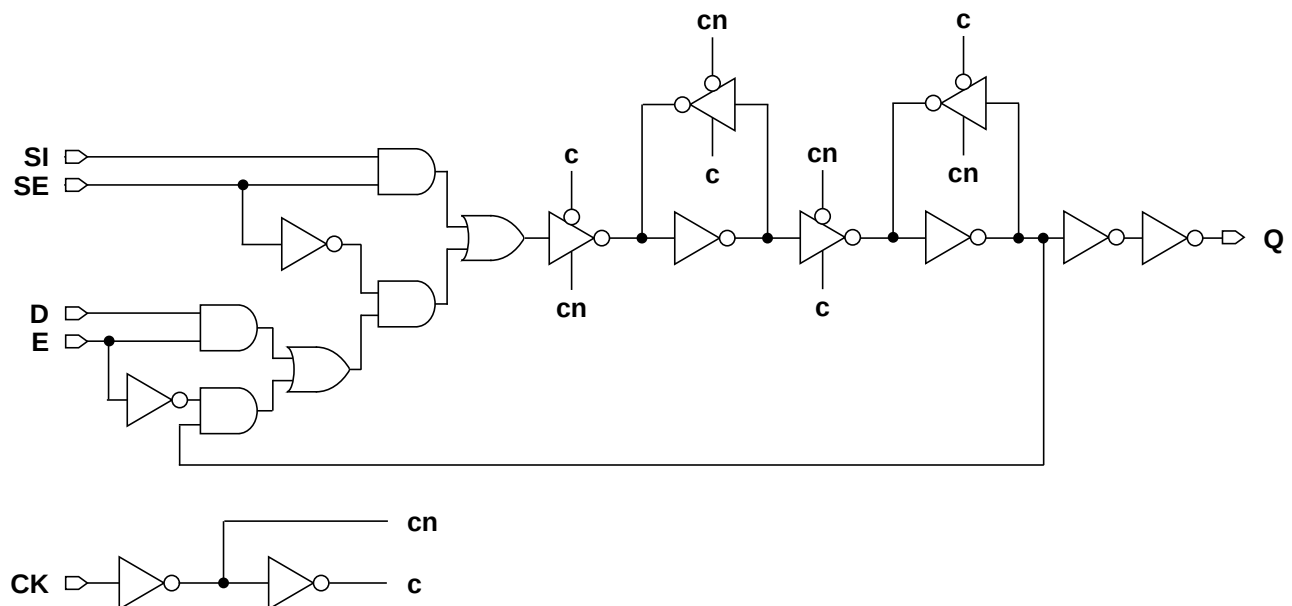
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFHQX1	3.69	14.72
SEDFFHQX2	3.69	16.10
SEDFFHQX4	3.69	20.70
SEDFFHQX8	3.69	21.62

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0157	0.0214	0.0380	0.0380
SE	0.0242	0.0307	0.0529	0.0529
D	0.0165	0.0229	0.0411	0.0412
CK	0.0275	0.0369	0.0608	0.0607
E	0.0250	0.0311	0.0477	0.0477
Q	0.0073	0.0103	0.0162	0.0264

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0013	0.0017	0.0027	0.0027
SE	0.0038	0.0040	0.0045	0.0045
D	0.0018	0.0027	0.0051	0.0051
CK	0.0030	0.0039	0.0068	0.0068
E	0.0026	0.0028	0.0031	0.0031

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1400	0.1187	0.1111	0.1252	4.3009	2.2700	1.1332	0.5702
CK $\rightarrow$ Q $\downarrow$	0.1530	0.1246	0.1246	0.1422	2.8581	1.5245	0.7534	0.3882



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup↑ → CK	0.1641	0.1445	0.1445	0.1445
	setup↓ → CK	0.2734	0.2695	0.2266	0.2266
	hold↑ → CK	-0.1055	-0.0820	-0.0859	-0.0820
	hold↓ → CK	-0.2188	-0.2109	-0.1719	-0.1680
SE	setup↑ → CK	0.2773	0.2891	0.2500	0.2539
	setup↓ → CK	0.2656	0.2539	0.2578	0.2578
	hold↑ → CK	-0.1367	-0.1289	-0.1406	-0.1328
	hold↓ → CK	-0.2031	-0.2070	-0.2227	-0.2188
D	setup↑ → CK	0.1328	0.1094	0.1016	0.1016
	setup↓ → CK	0.1484	0.1250	0.0977	0.1016
	hold↑ → CK	-0.0781	-0.0547	-0.0469	-0.0430
	hold↓ → CK	-0.1016	-0.0742	-0.0547	-0.0508
CK	minpwh	0.0690	0.0544	0.0495	0.0544
	minpwl	0.1856	0.1564	0.1467	0.1467
E	setup↑ → CK	0.2070	0.1875	0.1914	0.1914
	setup↓ → CK	0.3320	0.3594	0.3555	0.3555
	hold↑ → CK	-0.1797	-0.1602	-0.1367	-0.1367
	hold↓ → CK	-0.1992	-0.1992	-0.2109	-0.2070



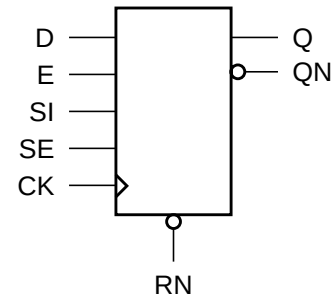
Cell Description

The SEDFFTR cell is a positive-edge triggered, static D-type flip-flop with scan input (SI), active-high scan enable (SE), synchronous active-high enable (E) and synchronous active low reset (RN). Scan enable (SE) dominates reset (RN) and enable (E).

Functions

RN	D	E	SI	SE	CK	Q[n+1]	QN[n+1]
x	x	x	0	1		0	1
x	x	x	1	1		1	0
1	x	0	x	0		Q[n]	QN[n]
0	x	x	x	0		0	1
1	1	1	x	0		1	0
1	0	1	x	0		0	1
x	x	x	x	x		Q[n]	QN[n]

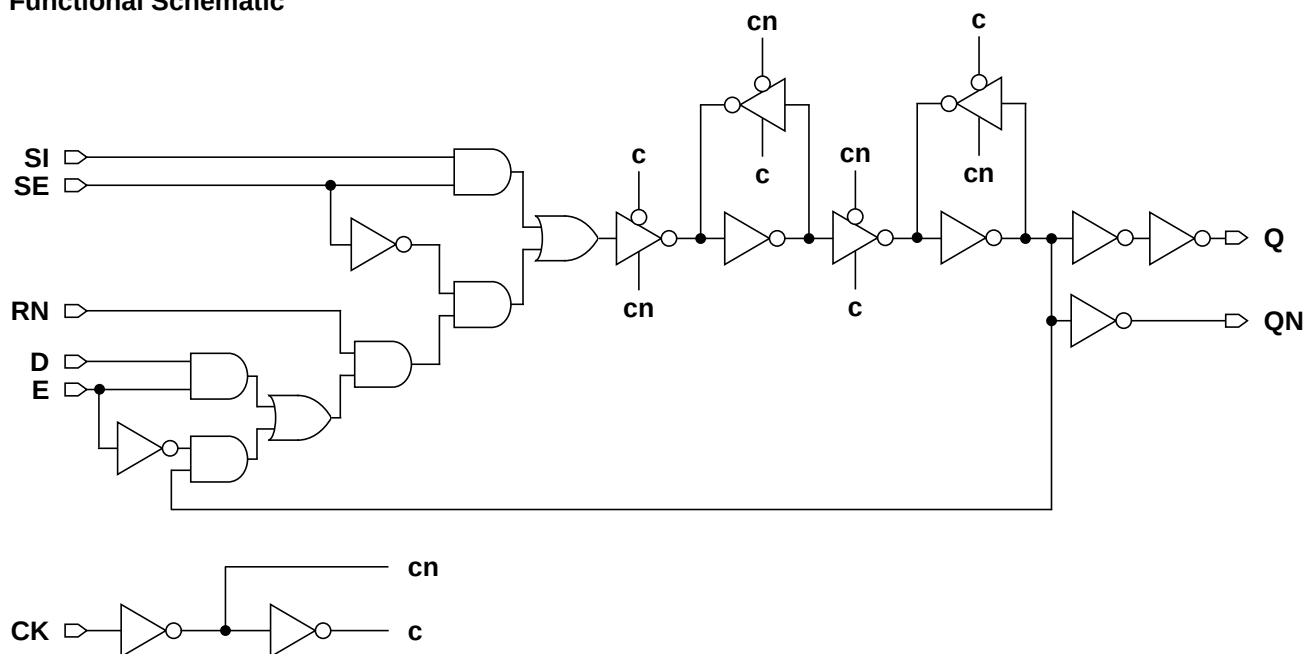
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
SEDFFFTRXL	3.69	15.64
SEDFFFTRX1	3.69	16.10
SEDFFFTRX2	3.69	18.40
SEDFFFTRX4	3.69	19.32

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
SI	0.0171	0.0211	0.0266	0.0352
SE	0.0237	0.0271	0.0325	0.0419
D	0.0229	0.0275	0.0329	0.0417
CK	0.0199	0.0262	0.0349	0.0483
E	0.0278	0.0312	0.0361	0.0451
RN	0.0153	0.0197	0.0248	0.0337
Q	0.0106	0.0166	0.0241	0.0378

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
SI	0.0015	0.0015	0.0014	0.0014
SE	0.0025	0.0024	0.0024	0.0024
D	0.0014	0.0020	0.0018	0.0018
CK	0.0014	0.0023	0.0028	0.0041
E	0.0015	0.0015	0.0014	0.0014
RN	0.0018	0.0017	0.0017	0.0018

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.2921	0.1805	0.1648	0.1464	9.4559	5.1372	2.2515	1.1254
CK $\rightarrow$ Q $\downarrow$	0.3177	0.2177	0.1993	0.1869	5.4533	2.7362	1.4389	0.7205
CK $\rightarrow$ QN $\uparrow$	0.2004	0.1558	0.1396	0.1267	9.3956	5.1520	2.2588	1.1309
CK $\rightarrow$ QN $\downarrow$	0.1912	0.1264	0.1172	0.1048	5.6052	2.8317	1.4906	0.7376



Timing Constraints at 25°C, 1.2V, Typical Process

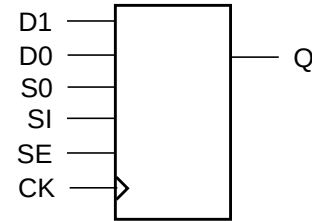
Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
SI	setup↑ → CK	0.2656	0.3086	0.3398	0.3828
	setup↓ → CK	0.2031	0.2266	0.2344	0.2500
	hold↑ → CK	-0.1484	-0.1953	-0.2148	-0.2617
	hold↓ → CK	-0.1875	-0.1953	-0.1992	-0.2109
SE	setup↑ → CK	0.3281	0.3750	0.4023	0.4453
	setup↓ → CK	0.3555	0.3711	0.4023	0.4453
	hold↑ → CK	-0.2031	-0.2227	-0.2266	-0.2344
	hold↓ → CK	-0.1953	-0.2305	-0.2266	-0.2344
D	setup↑ → CK	0.3555	0.3555	0.3867	0.4219
	setup↓ → CK	0.3164	0.2773	0.2852	0.2930
	hold↑ → CK	-0.2227	-0.2383	-0.2578	-0.3047
	hold↓ → CK	-0.2930	-0.2422	-0.2461	-0.2539
CK	minpwh	0.1516	0.0884	0.0835	0.0738
	minpwl	0.2244	0.2342	0.2439	0.2779
E	setup↑ → CK	0.4141	0.4297	0.4570	0.5000
	setup↓ → CK	0.3242	0.3164	0.3320	0.3750
	hold↑ → CK	-0.2969	-0.2852	-0.2891	-0.3008
	hold↓ → CK	-0.1094	-0.1250	-0.1602	-0.2734
RN	setup↑ → CK	0.2539	0.2852	0.3164	0.3594
	setup↓ → CK	0.1992	0.2266	0.2344	0.2461
	hold↑ → CK	-0.1250	-0.1641	-0.1836	-0.2305
	hold↓ → CK	-0.1797	-0.1914	-0.1992	-0.2070



Cell Description

The SMDFFHQ cell is a high-speed, positive-edge triggered, static D-type flip-flop with a 2-to-1 data select control (S0) for the data inputs (D1 and D0), scan input (SI), and active-high scan enable (SE). The cell has a single output (Q) and fast clock-to-out path.

Logic Symbol



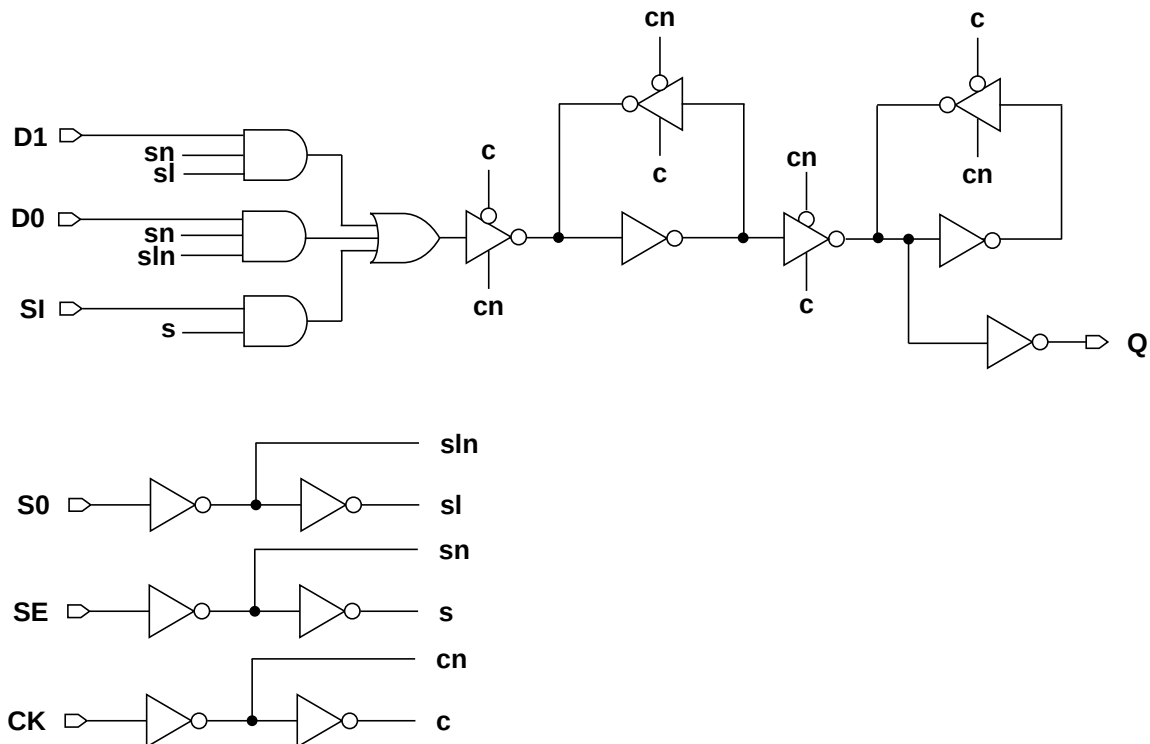
Functions

SE	SI	S0	D1	D0	CK	Q[n+1]
0	x	0	x	0		0
0	x	0	x	1		1
0	x	1	0	x		0
0	x	1	1	x		1
1	0	x	x	x		0
1	1	x	x	x		1
x	x	x	x	x		Q[n]

Cell Size

Drive Strength	Height (μm)	Width (μm)
SMDFFHQX1	3.69	14.26
SMDFFHQX2	3.69	15.64
SMDFFHQX4	3.69	21.16
SMDFFHQX8	3.69	22.54

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	X1	X2	X4	X8
SI	0.0167	0.0224	0.0398	0.0394
SE	0.0245	0.0314	0.0553	0.0545
D0	0.0179	0.0240	0.0413	0.0417
D1	0.0172	0.0234	0.0423	0.0414
S0	0.0264	0.0322	0.0498	0.0497
CK	0.0282	0.0371	0.0634	0.0630
Q	0.0062	0.0089	0.0142	0.0233

Pin Capacitance

Pin	Capacitance (pF)			
	X1	X2	X4	X8
SI	0.0014	0.0017	0.0027	0.0027
SE	0.0037	0.0039	0.0046	0.0045
D0	0.0020	0.0030	0.0048	0.0048
D1	0.0019	0.0027	0.0050	0.0050
S0	0.0027	0.0027	0.0032	0.0030
CK	0.0026	0.0042	0.0071	0.0071

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	X1	X2	X4	X8	X1	X2	X4	X8
CK $\rightarrow$ Q $\uparrow$	0.1352	0.1150	0.1062	0.1205	4.4357	2.2644	1.1302	0.5685
CK $\rightarrow$ Q $\downarrow$	0.1467	0.1248	0.1209	0.1377	2.9214	1.5051	0.7454	0.3835



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		X1	X2	X4	X8
SI	setup $\uparrow$ → CK	0.1641	0.1484	0.1562	0.1562
	setup $\downarrow$ → CK	0.2852	0.2852	0.2500	0.2461
	hold $\uparrow$ → CK	-0.1055	-0.0898	-0.0977	-0.0938
	hold $\downarrow$ → CK	-0.2344	-0.2266	-0.1914	-0.1836
SE	setup $\uparrow$ → CK	0.3008	0.3047	0.2812	0.2773
	setup $\downarrow$ → CK	0.2891	0.2656	0.3047	0.3047
	hold $\uparrow$ → CK	-0.1172	-0.1094	-0.1289	-0.1211
	hold $\downarrow$ → CK	-0.1797	-0.1641	-0.1953	-0.1758
D0	setup $\uparrow$ → CK	0.1328	0.1055	0.1133	0.1133
	setup $\downarrow$ → CK	0.1523	0.1289	0.1016	0.1016
	hold $\uparrow$ → CK	-0.0781	-0.0508	-0.0586	-0.0508
	hold $\downarrow$ → CK	-0.1055	-0.0820	-0.0547	-0.0508
D1	setup $\uparrow$ → CK	0.1250	0.1094	0.1094	0.1094
	setup $\downarrow$ → CK	0.1484	0.1289	0.1094	0.1094
	hold $\uparrow$ → CK	-0.0703	-0.0586	-0.0547	-0.0469
	hold $\downarrow$ → CK	-0.1016	-0.0820	-0.0625	-0.0547
S0	setup $\uparrow$ → CK	0.2031	0.1914	0.2070	0.2031
	setup $\downarrow$ → CK	0.2422	0.2344	0.2422	0.2266
	hold $\uparrow$ → CK	-0.1445	-0.1367	-0.1250	-0.1211
	hold $\downarrow$ → CK	-0.1719	-0.1562	-0.1875	-0.1680
CK	minpwh	0.0641	0.0544	0.0447	0.0495
	minpwl	0.1856	0.1564	0.1467	0.1467

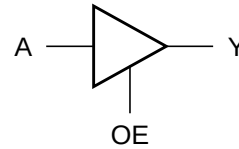


Cell Description

The TBUF cell provides the logical buffer of a single input (A) with an active-high output enable (OE). When the enable is high, the output (Y) is represented by the logic equation:

$$Y = A$$

Logic Symbol



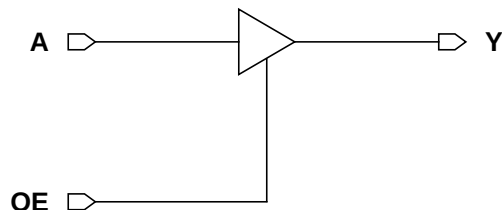
Functions

OE	A	Y
0	x	Z
1	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
TBUFXL	3.69	2.76
TBUF1	3.69	3.22
TBUF2	3.69	3.22
TBUF3	3.69	3.68
TBUF4	3.69	4.14
TBUF6	3.69	4.60
TBUF8	3.69	5.52
TBUF12	3.69	6.90
TBUF16	3.69	7.82
TBUF20	3.69	9.66

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0066	0.0075	0.0107	0.0133	0.0170	0.0249	0.0322	0.0453	0.0566	0.0681
OE	0.0041	0.0049	0.0075	0.0095	0.0125	0.0188	0.0245	0.0339	0.0430	0.0512

Pin Capacitance

Pin	Capacitance (pF)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A	0.0019	0.0019	0.0027	0.0033	0.0039	0.0063	0.0083	0.0109	0.0138	0.0168
OE	0.0022	0.0023	0.0025	0.0026	0.0029	0.0039	0.0047	0.0068	0.0086	0.0102
Y	0.0014	0.0019	0.0033	0.0037	0.0049	0.0080	0.0096	0.0139	0.0174	0.0224

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	0.0969	0.1006	0.0938	0.0892	0.0853	0.0886	0.0821	0.0772	0.0736	0.0772
A $\rightarrow$ Y $\downarrow$	0.1441	0.1698	0.1356	0.1210	0.1140	0.1127	0.1084	0.1021	0.0959	0.1043
OE $\rightarrow$ Y $\uparrow$	0.0654	0.0661	0.0699	0.0676	0.0657	0.0669	0.0657	0.0613	0.0572	0.0627
OE $\rightarrow$ Y $\downarrow$	0.0882	0.0979	0.0938	0.0919	0.0932	0.0937	0.0901	0.0900	0.0858	0.0912

Description	K_{load} (ns/pF)									
	XL	X1	X2	X3	X4	X6	X8	X12	X16	X20
A $\rightarrow$ Y $\uparrow$	9.4406	5.1904	2.2803	1.5787	1.1405	0.7951	0.5888	0.4029	0.3320	0.2542
A $\rightarrow$ Y $\downarrow$	6.8123	3.3439	1.6627	1.0676	0.7984	0.5376	0.3955	0.2661	0.2147	0.1679
OE $\rightarrow$ Y $\uparrow$	9.4167	5.1788	2.2822	1.5781	1.1399	0.7946	0.5887	0.4028	0.3319	0.2542
OE $\rightarrow$ Y $\downarrow$	6.7460	3.2896	1.6462	1.0592	0.7939	0.5352	0.3937	0.2651	0.2142	0.1673

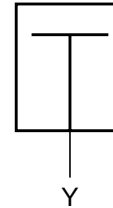
**Cell Description**

The TIEHI cell drives the output (Y) to a logic high. The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

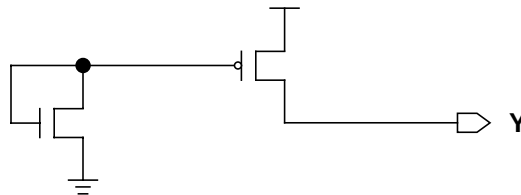
$$Y = 1$$

Function

Y
1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
TIEHI	3.69	0.92

Functional Schematic



Cell Description

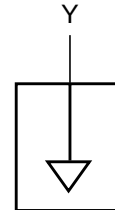
The TIELO cell drives the output (Y) to a logic low. The output is driven through diffusion and not tied directly to the power rail to provide some ESD protection. The output (Y) is represented by the logic equation:

$$Y = 0$$

Function

Y
0

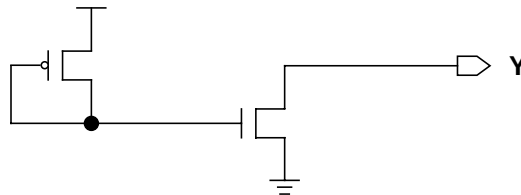
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TIELO	3.69	0.92

Functional Schematic

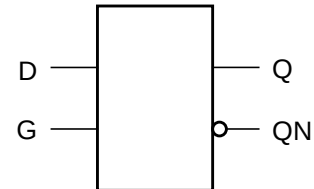


**Cell Description**

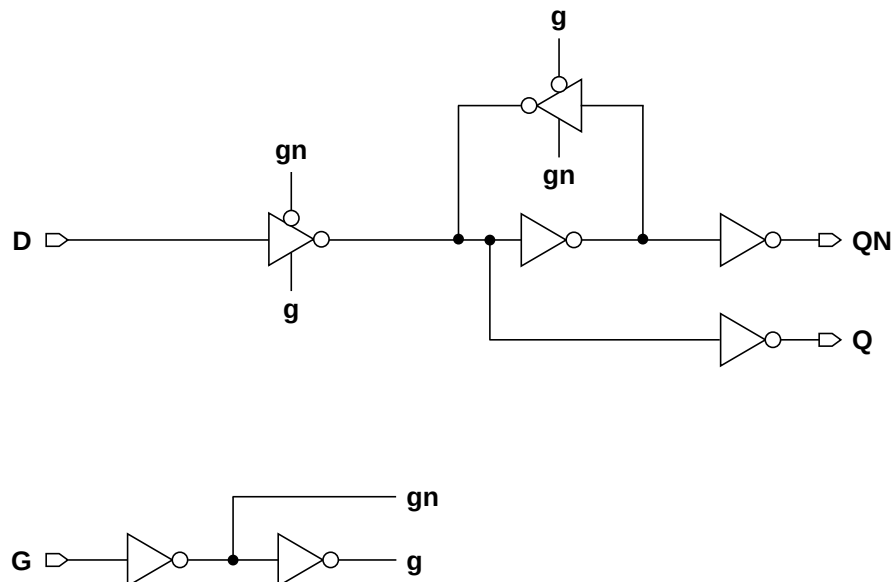
The TLAT cell is an active-high D-type transparent latch. When the enable (G) is high, data is transferred to the outputs (Q, QN).

Functions

G	D	Q[n+1]	QN[n+1]
1	0	0	1
1	1	1	0
0	x	Q[n]	QN[n]

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
TLATXL	3.69	4.60
TLATX1	3.69	4.60
TLATX2	3.69	5.52
TLATX4	3.69	7.36

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0010	0.0021	0.0034	0.0062
G	0.0070	0.0077	0.0089	0.0132
Q	0.0094	0.0115	0.0199	0.0330

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0014	0.0028	0.0043	0.0078
G	0.0013	0.0016	0.0020	0.0029

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.0959	0.0755	0.0722	0.0695	9.5428	5.1659	2.2651	1.1350
D $\rightarrow$ Q $\downarrow$	0.1804	0.1235	0.1164	0.1077	5.9453	2.9073	1.5172	0.7539
G $\rightarrow$ Q $\uparrow$	0.1987	0.1869	0.1695	0.1462	9.5366	5.1648	2.2646	1.1344
G $\rightarrow$ Q $\downarrow$	0.1898	0.1290	0.1282	0.1167	5.9265	2.9036	1.5208	0.7550
D $\rightarrow$ QN $\uparrow$	0.2421	0.1770	0.1739	0.1711	9.4259	5.1517	2.2570	1.1296
D $\rightarrow$ QN $\downarrow$	0.1648	0.1578	0.1626	0.1504	4.8351	3.0498	1.4968	0.7367
G $\rightarrow$ QN $\uparrow$	0.2527	0.1841	0.1866	0.1807	9.4262	5.1522	2.2574	1.1296
G $\rightarrow$ QN $\downarrow$	0.2682	0.2696	0.2602	0.2271	4.8361	3.0506	1.4971	0.7370

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow \rightarrow$ G	-0.0078	-0.0273	-0.0273	-0.0156
	setup $\downarrow \rightarrow$ G	0.1367	0.0781	0.0703	0.0703
	hold $\uparrow \rightarrow$ G	0.0156	0.0352	0.0352	0.0234
	hold $\downarrow \rightarrow$ G	-0.1328	-0.0703	-0.0625	-0.0664
G	minpwh	0.1030	0.0981	0.0884	0.0738



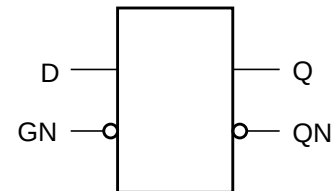
Cell Description

The TLATN cell is an active-low D-type transparent latch. When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Functions

GN	D	Q[n+1]	QN[n+1]
0	0	0	1
0	1	1	0
1	x	Q[n]	QN[n]

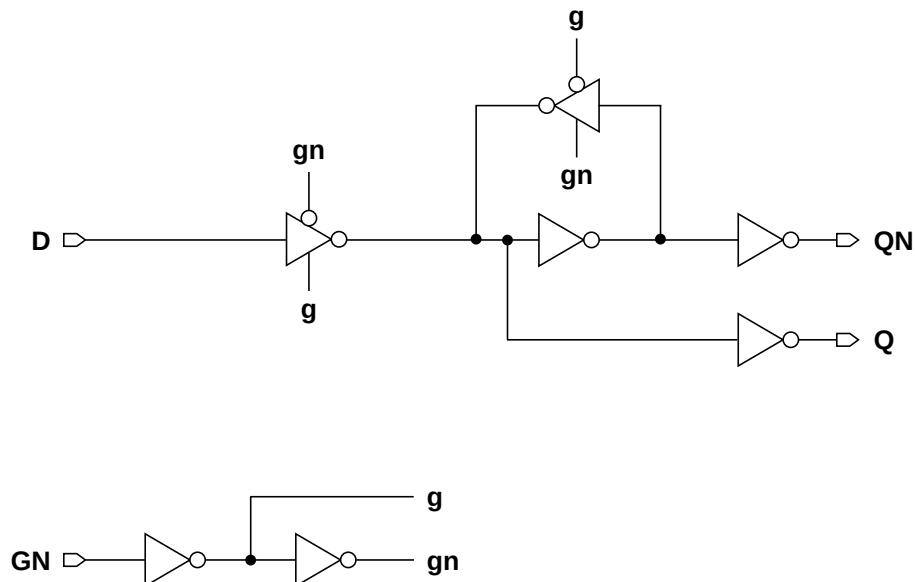
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNXL	3.69	4.60
TLATNX1	3.69	4.60
TLATNX2	3.69	5.52
TLATNX4	3.69	7.36

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0011	0.0023	0.0035	0.0062
GN	0.0071	0.0083	0.0113	0.0174
Q	0.0094	0.0120	0.0195	0.0329

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0013	0.0026	0.0043	0.0079
GN	0.0013	0.0015	0.0018	0.0029

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.0949	0.0781	0.0729	0.0682	9.5109	5.1680	2.2682	1.1344
D $\rightarrow$ Q $\downarrow$	0.1782	0.1204	0.1134	0.1086	5.8859	2.8895	1.5109	0.7527
GN $\rightarrow$ Q $\uparrow$	0.1719	0.1335	0.1389	0.1104	9.5183	5.1756	2.2747	1.1369
GN $\rightarrow$ Q $\downarrow$	0.2769	0.2005	0.2048	0.1776	5.8809	2.8886	1.5102	0.7524
D $\rightarrow$ QN $\uparrow$	0.2394	0.1748	0.1664	0.1697	9.4125	5.1499	2.2540	1.1291
D $\rightarrow$ QN $\downarrow$	0.1632	0.1629	0.1629	0.1463	4.8314	3.1776	1.4917	0.7353
GN $\rightarrow$ QN $\uparrow$	0.3388	0.2555	0.2580	0.2388	9.4120	5.1506	2.2543	1.1293
GN $\rightarrow$ QN $\downarrow$	0.2410	0.2196	0.2305	0.1899	4.8344	3.1788	1.4922	0.7355

Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ GN	0.0625	0.0508	0.0469	0.0430
	setup $\downarrow$ $\rightarrow$ GN	0.1055	0.0430	0.0273	0.0312
	hold $\uparrow$ $\rightarrow$ GN	-0.0586	-0.0469	-0.0430	-0.0391
	hold $\downarrow$ $\rightarrow$ GN	-0.0898	-0.0312	-0.0156	-0.0234
GN	minpwl	0.1467	0.1127	0.1224	0.0933



Cell Description

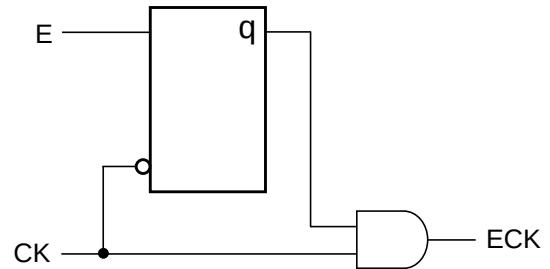
The TLATNCA cell is a positive-edge triggered clock-gating latch. The positive-edge clock (CK) is qualified by the latched enable signal (E) to create the gated positive-edge clock ECK.

Functions

CK	E	q[n+1]	ECK[n+1]
1	x	q[n]	q[n]
0	0	0	0
0	1	1	0

Note: q is an internal node, and is not accessible.

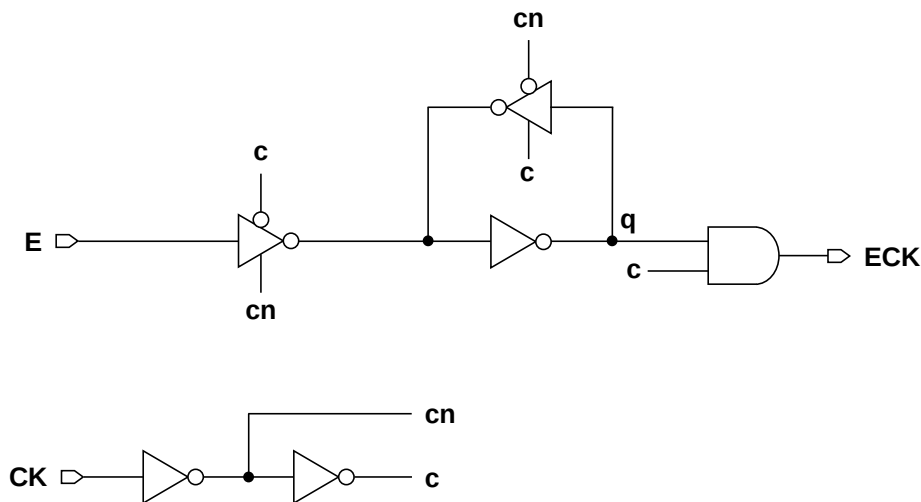
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNCAX2	3.69	5.06
TLATNCAX3	3.69	5.98
TLATNCAX4	3.69	6.90
TLATNCAX6	3.69	8.28
TLATNCAX8	3.69	13.80
TLATNCAX12	3.69	12.88
TLATNCAX16	3.69	16.10
TLATNCAX20	3.69	18.40

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0145	0.0166	0.0212	0.0279	0.0486	0.0146	0.0172	0.0210
CK	0.0140	0.0170	0.0214	0.0282	0.0537	0.0142	0.0175	0.0219
ECK	0.0065	0.0091	0.0121	0.0189	0.0416	0.0750	0.0991	0.1176

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0037	0.0044	0.0067	0.0089	0.0173	0.0036	0.0042	0.0062
CK	0.0032	0.0041	0.0049	0.0073	0.0139	0.0030	0.0039	0.0048

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	0.0782	0.0774	0.0758	0.0729	0.0705	0.2261	0.2257	0.2249
CK $\rightarrow$ ECK $\downarrow$	0.1021	0.1004	0.1022	0.0838	0.0850	0.2046	0.1896	0.1900

Delays at 25°C, 1.2V, Typical Process

Description	K_{load} (ns/pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	4.5077	3.1403	2.2708	1.5209	0.7592	0.2108	0.1605	0.1398
CK $\rightarrow$ ECK $\downarrow$	2.4724	1.7264	1.3964	0.8444	0.4918	0.2232	0.1764	0.1596



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)							
		X2	X3	X4	X6	X8	X12	X16	X20
E	setup $\uparrow$ → CK	0.0391	0.0391	0.0312	0.0352	0.0312	0.0391	0.0391	0.0312
	setup $\downarrow$ → CK	0.0078	0.0039	0.0039	0.0078	0.0039	0.0078	0.0078	0.0078
	hold $\uparrow$ → CK	-0.0352	-0.0352	-0.0273	-0.0312	-0.0273	-0.0352	-0.0352	-0.0273
	hold $\downarrow$ → CK	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0039	0.0000	-0.0039
CK	minpwl	0.1030	0.1030	0.0933	0.0835	0.0787	0.1030	0.0981	0.0835



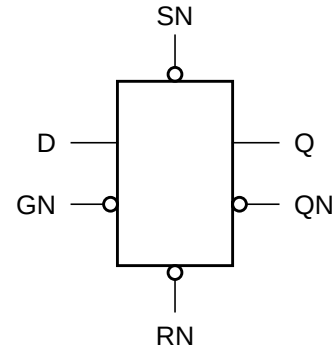
Cell Description

The TLATNSR cell is an active-low D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (GN) is low, data is transferred to the outputs (Q, QN).

Functions

RN	SN	GN	D	Q[n+1]	QN[n+1]
1	1	0	0	0	1
1	1	0	1	1	0
1	1	1	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

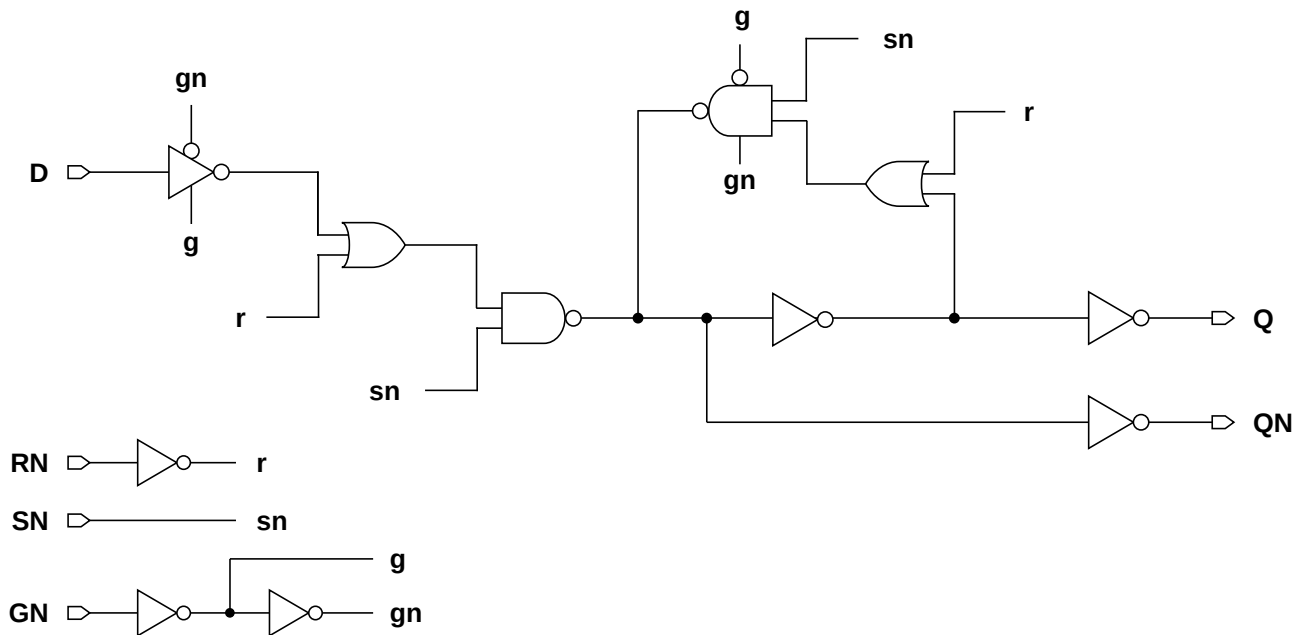
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNSRXL	3.69	6.90
TLATNSRX1	3.69	6.90
TLATNSRX2	3.69	7.82
TLATNSRX4	3.69	11.50

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0014	0.0018	0.0030	0.0054
GN	0.0081	0.0101	0.0132	0.0217
SN	0.0052	0.0063	0.0085	0.0135
RN	0.0017	0.0021	0.0034	0.0060
Q	0.0140	0.0162	0.0253	0.0432

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0018	0.0024	0.0038	0.0072
GN	0.0013	0.0017	0.0021	0.0026
SN	0.0011	0.0017	0.0025	0.0044
RN	0.0024	0.0029	0.0042	0.0075

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1655	0.1500	0.1451	0.1331	9.7229	6.3620	2.3500	1.1720
D $\rightarrow$ Q $\downarrow$	0.2962	0.2819	0.2232	0.2309	7.1879	4.2300	1.8481	0.9390
GN $\rightarrow$ Q $\uparrow$	0.2330	0.2060	0.1924	0.1736	9.7182	6.3636	2.3521	1.1739
GN $\rightarrow$ Q $\downarrow$	0.3691	0.3471	0.2932	0.2887	7.1818	4.2279	1.8486	0.9390
SN $\rightarrow$ Q $\uparrow$	0.2215	0.1879	0.1707	0.1517	9.4662	6.2635	2.3012	1.1516
SN $\rightarrow$ Q $\downarrow$	0.3300	0.3047	0.2424	0.2478	7.1073	4.1884	1.8323	0.9301
RN $\rightarrow$ Q $\uparrow$	0.1610	0.1450	0.1405	0.1293	9.7233	6.3624	2.3499	1.1721
RN $\rightarrow$ Q $\downarrow$	0.2268	0.2061	0.1705	0.1670	6.9408	4.1901	1.8808	0.9364
D $\rightarrow$ QN $\uparrow$	0.3813	0.3562	0.2950	0.3077	9.4340	6.2446	2.2656	1.1327
D $\rightarrow$ QN $\downarrow$	0.2510	0.2406	0.2468	0.2188	5.0031	3.2281	1.5053	0.7371
GN $\rightarrow$ QN $\uparrow$	0.4551	0.4222	0.3656	0.3658	9.4383	6.2458	2.2666	1.1329
GN $\rightarrow$ QN $\downarrow$	0.3200	0.2983	0.2953	0.2602	5.0158	3.2329	1.5069	0.7379
SN $\rightarrow$ QN $\uparrow$	0.4142	0.3784	0.3133	0.3233	9.4359	6.2451	2.2658	1.1327
SN $\rightarrow$ QN $\downarrow$	0.3042	0.2754	0.2677	0.2340	4.9598	3.2147	1.5004	0.7352
RN $\rightarrow$ QN $\uparrow$	0.3097	0.2786	0.2419	0.2406	9.4156	6.2457	2.2674	1.1332
RN $\rightarrow$ QN $\downarrow$	0.2468	0.2361	0.2425	0.2151	5.0078	3.2295	1.5061	0.7375

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ → GN	0.1445	0.1250	0.1289	0.1055
	setup $\downarrow$ → GN	0.2344	0.1992	0.1367	0.1445
	hold $\uparrow$ → GN	-0.1328	-0.1172	-0.1211	-0.1016
	hold $\downarrow$ → GN	-0.1914	-0.1602	-0.1094	-0.1250
GN	minpwl	0.2099	0.1856	0.1759	0.1516
SN	minpwl	0.1856	0.1516	0.1370	0.1127
	recovery	0.2617	0.2148	0.1484	0.1562
	removal	-0.2617	-0.2148	-0.1484	-0.1562
RN	minpwl	0.2342	0.1953	0.1516	0.1370
	recovery	0.1367	0.1211	0.1211	0.1016
	removal	-0.1328	-0.1172	-0.1172	-0.0977



Cell Description

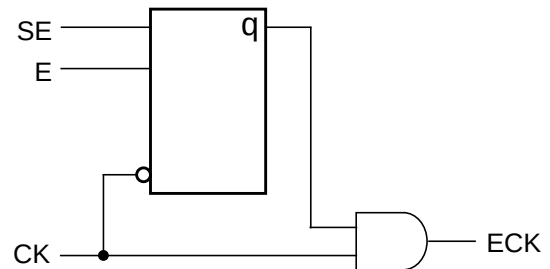
The TLATNTSCA cell is a positive-edge triggered clock-gating latch. The positive-edge clock (CK) is qualified by the latched enable signals (SE) and (E) to create the gated positive-edge clock ECK.

Functions

CK	SE	E	q[n+1]	ECK[n+1]
1	x	x	q[n]	q[n]
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0

Note: q is an internal node, and is not accessible.

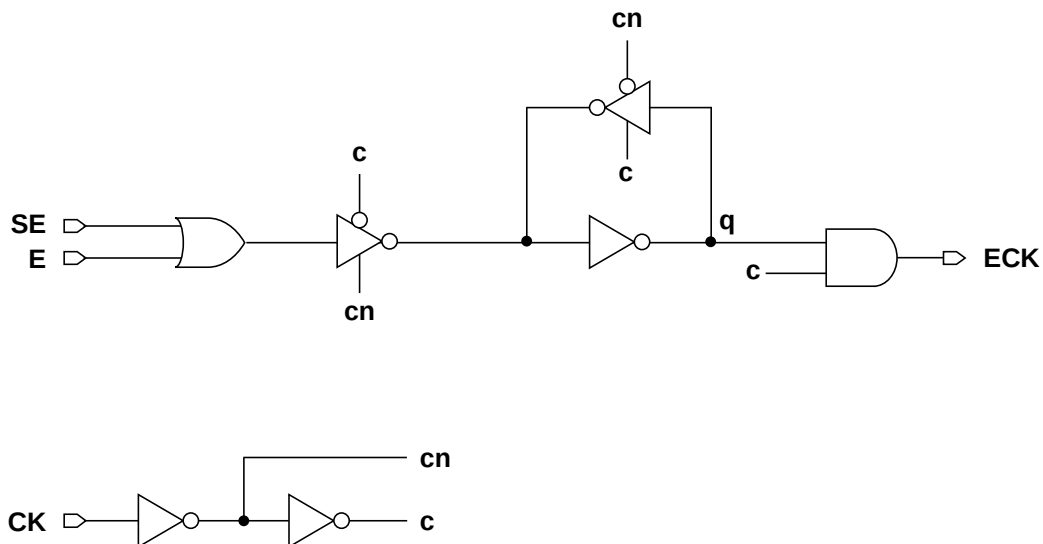
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATNTSCAX2	3.69	6.44
TLATNTSCAX3	3.69	7.36
TLATNTSCAX4	3.69	8.74
TLATNTSCAX6	3.69	11.04
TLATNTSCAX8	3.69	17.48
TLATNTSCAX12	3.69	14.26
TLATNTSCAX16	3.69	17.02
TLATNTSCAX20	3.69	20.24

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0207	0.0232	0.0281	0.0396	0.0681	0.0209	0.0232	0.0282
SE	0.0213	0.0240	0.0286	0.0406	0.0685	0.0215	0.0237	0.0288
CK	0.0147	0.0183	0.0220	0.0316	0.0550	0.0149	0.0180	0.0221
ECK	0.0057	0.0081	0.0090	0.0128	0.0480	0.0786	0.1041	0.1243

Pin Capacitance

Pin	Capacitance (pF)							
	X2	X3	X4	X6	X8	X12	X16	X20
E	0.0016	0.0016	0.0017	0.0023	0.0037	0.0016	0.0016	0.0017
SE	0.0014	0.0015	0.0016	0.0021	0.0034	0.0014	0.0014	0.0015
CK	0.0033	0.0042	0.0049	0.0073	0.0136	0.0033	0.0042	0.0047

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	0.0800	0.0783	0.0763	0.0749	0.0696	0.2275	0.2198	0.2245
CK $\rightarrow$ ECK $\downarrow$	0.1042	0.1043	0.1031	0.0836	0.0917	0.2075	0.2032	0.1952

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)							
	X2	X3	X4	X6	X8	X12	X16	X20
CK $\rightarrow$ ECK $\uparrow$	4.5145	3.1406	2.5676	1.6932	0.6951	0.2108	0.1541	0.1372
CK $\rightarrow$ ECK $\downarrow$	2.4751	1.7746	1.4448	0.8458	0.5075	0.2232	0.1676	0.1479



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)							
		X2	X3	X4	X6	X8	X12	X16	X20
E	setup $\uparrow$ → CK	0.1445	0.1445	0.1367	0.1250	0.1172	0.1445	0.1445	0.1484
	setup $\downarrow$ → CK	0.1484	0.1523	0.1641	0.1602	0.1172	0.1484	0.1562	0.1562
	hold $\uparrow$ → CK	-0.1406	-0.1406	-0.1328	-0.1211	-0.1133	-0.1406	-0.1406	-0.1445
	hold $\downarrow$ → CK	-0.1445	-0.1484	-0.1602	-0.1523	-0.1133	-0.1445	-0.1484	-0.1523
SE	setup $\uparrow$ → CK	0.1445	0.1484	0.1406	0.1289	0.1211	0.1484	0.1445	0.1523
	setup $\downarrow$ → CK	0.1562	0.1602	0.1719	0.1680	0.1250	0.1562	0.1641	0.1641
	hold $\uparrow$ → CK	-0.1445	-0.1445	-0.1367	-0.1250	-0.1172	-0.1445	-0.1406	-0.1484
	hold $\downarrow$ → CK	-0.1484	-0.1523	-0.1680	-0.1602	-0.1211	-0.1484	-0.1562	-0.1562
CK	minpwl	0.1078	0.1078	0.0933	0.0787	0.0835	0.1078	0.1030	0.0933



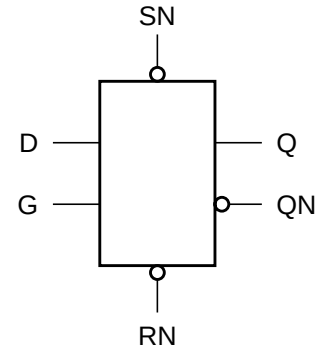
Cell Description

The TLATSR cell is an active-high D-type transparent latch with asynchronous active-low set (SN) and reset (RN), and set dominating reset. When the enable (G) is high, data is transferred to the outputs (Q, QN).

Functions

RN	SN	G	D	Q[n+1]	QN[n+1]
1	1	1	0	0	1
1	1	1	1	1	0
1	1	0	x	Q[n]	QN[n]
0	1	x	x	0	1
1	0	x	x	1	0
0	0	x	x	1	0

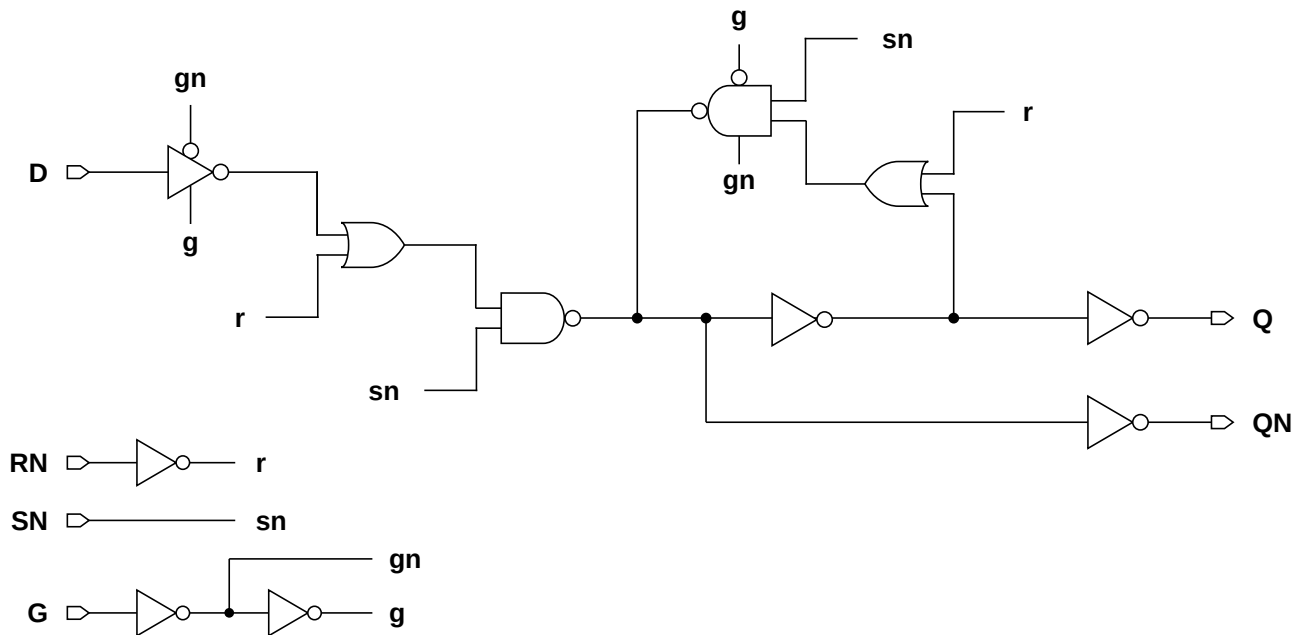
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
TLATSRXL	3.69	6.90
TLATSRX1	3.69	6.90
TLATSRX2	3.69	7.82
TLATSRX4	3.69	11.50

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
D	0.0014	0.0018	0.0030	0.0054
G	0.0089	0.0092	0.0109	0.0182
SN	0.0052	0.0063	0.0085	0.0135
RN	0.0016	0.0021	0.0033	0.0061
Q	0.0137	0.0154	0.0248	0.0427

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0018	0.0024	0.0038	0.0072
G	0.0012	0.0015	0.0019	0.0027
SN	0.0011	0.0017	0.0025	0.0045
RN	0.0024	0.0029	0.0042	0.0076

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
D $\rightarrow$ Q $\uparrow$	0.1658	0.1523	0.1469	0.1341	9.7225	6.3654	2.3529	1.1722
D $\rightarrow$ Q $\downarrow$	0.2976	0.2846	0.2246	0.2294	7.1968	4.2396	1.8548	0.9385
G $\rightarrow$ Q $\uparrow$	0.2672	0.2517	0.2318	0.2196	9.7175	6.3620	2.3514	1.1714
G $\rightarrow$ Q $\downarrow$	0.2947	0.2708	0.2216	0.2152	7.1928	4.2374	1.8559	0.9384
SN $\rightarrow$ Q $\uparrow$	0.2222	0.1887	0.1706	0.1522	9.4657	6.2662	2.3030	1.1524
SN $\rightarrow$ Q $\downarrow$	0.3314	0.3068	0.2430	0.2463	7.1165	4.1985	1.8390	0.9294
RN $\rightarrow$ Q $\uparrow$	0.1611	0.1470	0.1421	0.1304	9.7230	6.3655	2.3528	1.1722
RN $\rightarrow$ Q $\downarrow$	0.2276	0.2084	0.1718	0.1645	6.9529	4.1976	1.8868	0.9320
D $\rightarrow$ QN $\uparrow$	0.3832	0.3584	0.2948	0.3065	9.4353	6.2456	2.2651	1.1327
D $\rightarrow$ QN $\downarrow$	0.2517	0.2426	0.2472	0.2200	5.0086	3.2294	1.5000	0.7374
G $\rightarrow$ QN $\uparrow$	0.3824	0.3467	0.2927	0.2926	9.4390	6.2474	2.2660	1.1330
G $\rightarrow$ QN $\downarrow$	0.3541	0.3428	0.3328	0.3058	5.0212	3.2340	1.5015	0.7380
SN $\rightarrow$ QN $\uparrow$	0.4162	0.3801	0.3124	0.3220	9.4365	6.2459	2.2653	1.1327
SN $\rightarrow$ QN $\downarrow$	0.3052	0.2759	0.2662	0.2348	4.9665	3.2158	1.4950	0.7355
RN $\rightarrow$ QN $\uparrow$	0.3111	0.2805	0.2416	0.2381	9.4158	6.2464	2.2668	1.1330
RN $\rightarrow$ QN $\downarrow$	0.2475	0.2377	0.2425	0.2164	5.0134	3.2312	1.5009	0.7377



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ G	0.0586	0.0508	0.0352	0.0156
	setup $\downarrow$ $\rightarrow$ G	0.2500	0.2305	0.1719	0.1875
	hold $\uparrow$ $\rightarrow$ G	-0.0352	-0.0273	-0.0234	-0.0117
	hold $\downarrow$ $\rightarrow$ G	-0.2383	-0.2227	-0.1602	-0.1758
G	minpwh	0.1661	0.1564	0.1321	0.1176
SN	minpwl	0.1856	0.1516	0.1370	0.1127
	recovery	0.2812	0.2539	0.1875	0.2031
	removal	-0.2773	-0.2500	-0.1836	-0.1992
RN	minpwl	0.2342	0.2001	0.1516	0.1321
	recovery	0.0469	0.0391	0.0273	0.0117
	removal	-0.0469	-0.0391	-0.0273	-0.0117

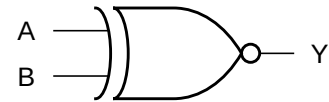


Cell Description

The XNOR2 cell provides a logical EXCLUSIVE NOR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = (A \bullet B) + (\bar{A} \bullet \bar{B})$$

Logic Symbol



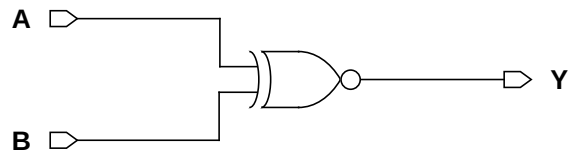
Functions

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
XNOR2XL	3.69	3.22
XNOR2X1	3.69	3.22
XNOR2X2	3.69	5.52
XNOR2X4	3.69	8.28

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0058	0.0115	0.0203	0.0354
B	0.0078	0.0131	0.0290	0.0537

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0024	0.0035	0.0052	0.0084
B	0.0017	0.0032	0.0059	0.0107

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0511	0.0503	0.0496	0.0432	11.6397	5.1241	2.5243	1.2529
A $\rightarrow$ Y $\downarrow$	0.0634	0.0783	0.0703	0.0580	6.7306	2.7914	1.4010	0.7601
B $\rightarrow$ Y $\uparrow$	0.0860	0.0732	0.0803	0.0765	11.7322	5.0645	2.5320	1.2337
B $\rightarrow$ Y $\downarrow$	0.0940	0.0767	0.0819	0.0801	6.9613	3.0068	1.5035	0.7987

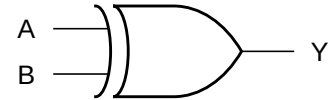


Cell Description

The XOR2 cell provides a logical EXCLUSIVE OR of two inputs (A, B). The output (Y) is represented by the logic equation:

$$Y = (A \bullet \bar{B}) + (\bar{A} \bullet B)$$

Logic Symbol



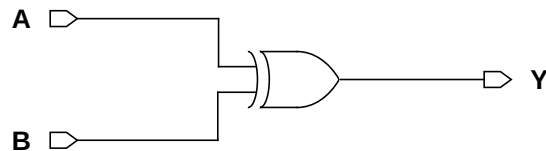
Functions

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
XOR2XL	3.69	3.22
XOR2X1	3.69	3.22
XOR2X2	3.69	5.52
XOR2X4	3.69	8.28

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0057	0.0108	0.0204	0.0350
B	0.0082	0.0150	0.0309	0.0565

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0023	0.0033	0.0051	0.0085
B	0.0017	0.0031	0.0059	0.0115

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.0551	0.0494	0.0503	0.0421	11.6193	5.1124	2.5248	1.2491
A $\rightarrow$ Y $\downarrow$	0.0629	0.0730	0.0709	0.0566	6.7134	2.7952	1.3974	0.7185
B $\rightarrow$ Y $\uparrow$	0.0848	0.0731	0.0752	0.0684	11.7533	5.0564	2.5297	1.2735
B $\rightarrow$ Y $\downarrow$	0.0944	0.0821	0.0843	0.0794	6.9730	3.0293	1.5140	0.7735

Synthesis Optimized Arithmetic Cells



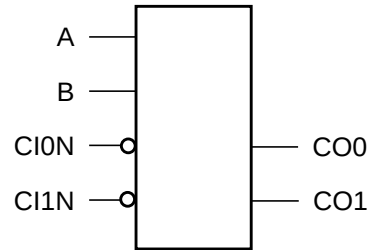
Cell Description

The ACCSHCIN cell provides a carry-select adder-carry generation function with active-low carry inputs. The function produces the carry-outs (CO0, CO1) of the operands (A,B) with active-low carry-ins (CI0N, CI1N). The outputs (CO0, CO1) are represented by the logic equations:

$$CO0 = (A \bullet B) + (A \bullet \overline{CI0N}) + (B \bullet \overline{CI0N})$$

$$CO1 = (A \bullet B) + (A \bullet \overline{CI1N}) + (B \bullet \overline{CI1N})$$

Logic Symbol



Functions

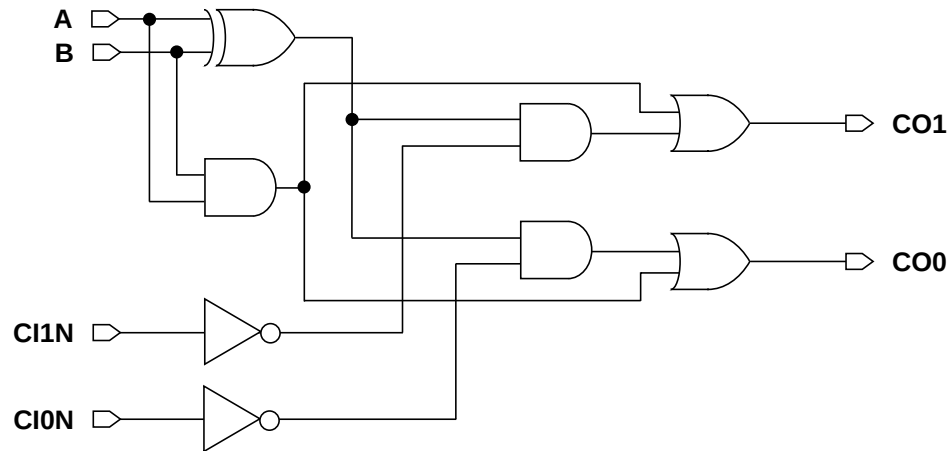
A	B	CI0N	CI1N	CO0	CO1
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	1
0	1	0	1	1	0
0	1	1	0	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
ACCSHCINX2	3.69	14.72
ACCSHCINX4	3.69	17.48



Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.0630	0.0715
B	0.0562	0.0676
CI0N	0.0132	0.0238
CI1N	0.0155	0.0254

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0047	0.0047
B	0.0130	0.0130
CI0N	0.0048	0.0090
CI1N	0.0048	0.0091



Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A → CO0↑	0.1526	0.1778	3.1109	3.0884
A → CO0↓	0.1828	0.2257	2.0031	1.9158
B → CO0↑	0.1160	0.1380	2.9240	3.0135
B → CO0↓	0.1310	0.1659	1.9225	1.8981
CI0N → CO0↑	0.0460	0.0396	2.9432	1.4928
CI0N → CO0↓	0.0353	0.0322	1.9080	0.9704
A → CO1↑	0.1731	0.2045	3.2596	3.1579
A → CO1↓	0.2021	0.2380	2.0301	1.9342
B → CO1↑	0.1015	0.1326	3.2054	3.1298
B → CO1↓	0.1308	0.1721	2.0074	1.9148
CI1N → CO1↑	0.0529	0.0423	2.9467	1.4697
CI1N → CO1↓	0.0407	0.0347	1.9096	0.9584



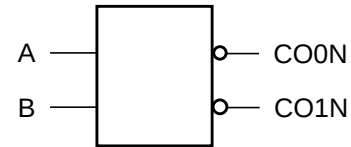
Cell Description

The ACCSIHCON cell provides a carry-select adder-carry generation function for the first stage of a carry-select adder block (i.e., there are no carry-inputs). The function produces active-low carry-outs (CO0N, CO1N) of the operands (A,B). The outputs (CO0N, CO1N) are represented by the logic equations:

$$CO0N = \overline{A \bullet B}$$

$$CO1N = \overline{A + B}$$

Logic Symbol



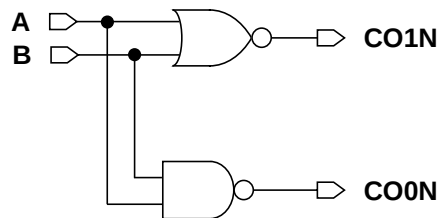
Functions

A	B	CO0N	CO1N
0	0	1	1
0	1	1	0
1	0	1	0
1	1	0	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
ACCSIHCONX2	3.69	2.76
ACCSIHCONX4	3.69	4.60

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0108	0.0200
B	0.0116	0.0228

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0070	0.0136
B	0.0079	0.0156

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ CO0N $\uparrow$	0.0330	0.0302	3.1010	1.5505
A $\rightarrow$ CO0N $\downarrow$	0.0232	0.0214	2.1917	1.0957
B $\rightarrow$ CO0N $\uparrow$	0.0252	0.0224	3.1028	1.5513
B $\rightarrow$ CO0N $\downarrow$	0.0195	0.0176	2.1912	1.0954
A $\rightarrow$ CO1N $\uparrow$	0.0280	0.0262	4.3745	2.1873
A $\rightarrow$ CO1N $\downarrow$	0.0176	0.0168	1.7340	0.8670
B $\rightarrow$ CO1N $\uparrow$	0.0366	0.0355	4.3701	2.1859
B $\rightarrow$ CO1N $\downarrow$	0.0217	0.0213	1.7423	0.8714



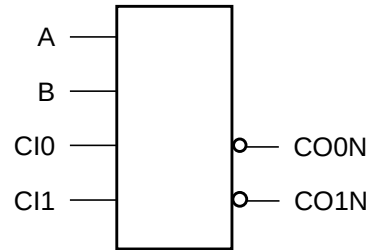
Cell Description

The ACCSHCON cell provides a carry-select adder-carry generation function that produces active-low carry-outs (CO0N, CO1N) of the operands (A,B) with carry-ins (CI0, CI1). The outputs (CO0N, CO1N) are represented by the logic equations:

$$CO0N = \overline{(A \bullet B) + (A \bullet CI0) + (B \bullet CI0)}$$

$$CO1N = \overline{(A \bullet B) + (A \bullet CI1) + (B \bullet CI1)}$$

Logic Symbol



Functions

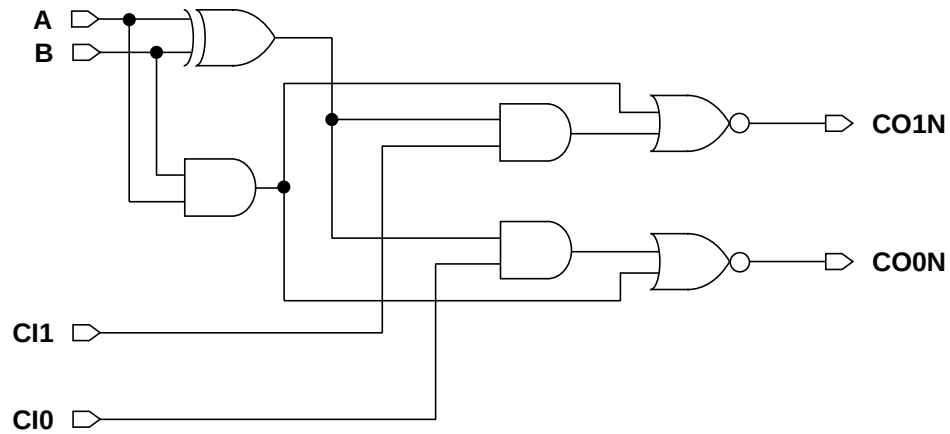
A	B	CI0	CI1	CO0N	CO1N
0	0	0	0	1	1
0	0	0	1	1	1
0	0	1	0	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	0
0	1	1	0	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
ACCSHCONX2	3.69	11.96
ACCSHCONX4	3.69	14.72



Functional Schematic



AC Power

Pin	Power (μ W/MHz)	
	X2	X4
A	0.0581	0.0689
B	0.0538	0.0638
CI0	0.0130	0.0240
CI1	0.0156	0.0251

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0093	0.0093
B	0.0142	0.0142
CI0	0.0047	0.0093
CI1	0.0048	0.0091



Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
A → CO0N↑	0.1827	0.2187	2.7830	2.9237
A → CO0N↓	0.1681	0.2109	1.9983	1.9727
B → CO0N↑	0.1515	0.1893	2.8246	2.9237
B → CO0N↓	0.1305	0.1720	1.9988	1.9730
CI0 → CO0N↑	0.0467	0.0417	2.9694	1.4870
CI0 → CO0N↓	0.0351	0.0319	1.9091	0.9398
A → CO1N↑	0.1809	0.2176	2.8578	2.7934
A → CO1N↓	0.1665	0.2054	2.0079	1.9125
B → CO1N↑	0.1521	0.1881	2.7809	2.7923
B → CO1N↓	0.1285	0.1661	2.0064	1.9109
CI1 → CO1N↑	0.0537	0.0428	2.9734	1.4873
CI1 → CO1N↓	0.0403	0.0339	1.9096	0.9548



Cell Description

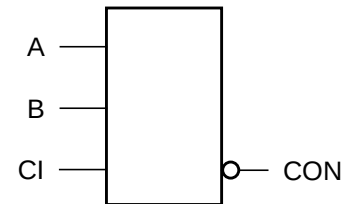
The ACHCON cell is a full adder carry-generator that provides the arithmetic active-low carry-out (CON) of two operands (A, B) with carry-in (CI). The output (CON) is represented by the logic equation:

$$CON = \overline{(A \bullet B) + (A \bullet CI) + (B \bullet CI)}$$

Functions

A	B	CI	CON
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

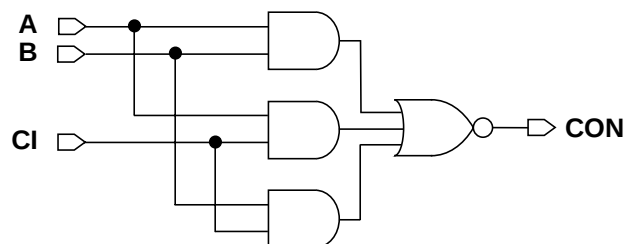
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
ACHCONX2	3.69	7.82
ACHCONX4	3.69	10.12

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0427	0.0528
B	0.0406	0.0477
CI	0.0123	0.0220

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0046	0.0046
B	0.0145	0.0157
CI	0.0048	0.0087

Delays at TypTemp°C, TypVoltV, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ CON $\uparrow$	0.1481	0.1710	2.9927	2.8505
A $\rightarrow$ CON $\downarrow$	0.1393	0.1593	1.8646	1.8305
B $\rightarrow$ CON $\uparrow$	0.0872	0.1049	2.9756	2.9042
B $\rightarrow$ CON $\downarrow$	0.0822	0.0983	1.8975	1.8738
CI $\rightarrow$ CON $\uparrow$	0.0459	0.0391	3.1892	1.6061
CI $\rightarrow$ CON $\downarrow$	0.0341	0.0308	1.9329	0.9762



Cell Description

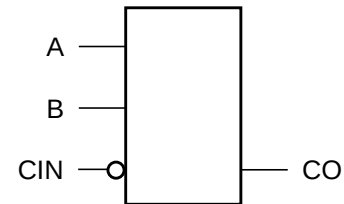
The ACHCIN cell is a full adder carry-generator that provides the arithmetic carry-out (CO) of two operands (A, B) with active low carry-in (CIN). The output (CO) is represented by the logic equation:

$$CO = (A \bullet B) + (A \bullet \overline{CIN}) + (B \bullet \overline{CIN})$$

Functions

A	B	CIN	CO
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

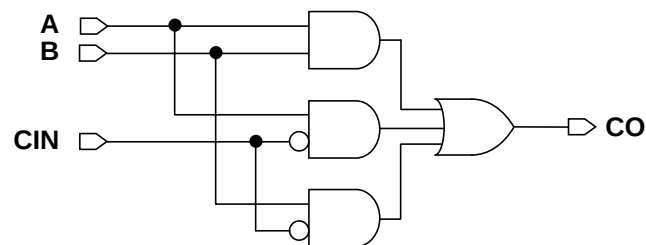
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
ACHCINX2	3.69	7.82
ACHCINX4	3.69	10.12

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0338	0.0414
B	0.0384	0.0450
CIN	0.0123	0.0220

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0046	0.0046
B	0.0106	0.0116
CIN	0.0048	0.0087

Delays at TypTemp°C, TypVoltV, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ CO $\uparrow$	0.1124	0.1361	3.3091	2.9838
A $\rightarrow$ CO $\downarrow$	0.1234	0.1519	2.0102	1.9231
B $\rightarrow$ CO $\uparrow$	0.0916	0.1085	3.0894	2.9355
B $\rightarrow$ CO $\downarrow$	0.0978	0.1146	1.9322	1.9059
CIN $\rightarrow$ CO $\uparrow$	0.0458	0.0391	3.1888	1.6061
CIN $\rightarrow$ CO $\downarrow$	0.0339	0.0308	1.9331	0.9762

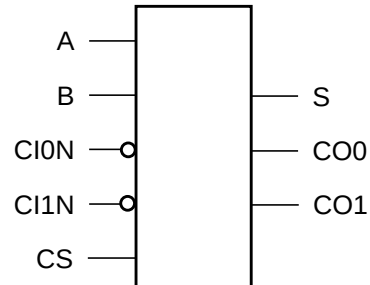
**Cell Description**

The AFCSHCIN cell provides a carry-select adder function that produces the arithmetic sum (S) and carry-outs (CO0, CO1) of the operands (A,B) with active-low carry-ins (CI0N, CI1N). The three outputs (S, CO0, CO1) are represented by the logic equations:

$$S = CS \bullet (A \oplus B \oplus \overline{CI1N}) + \overline{CS} \bullet (A \oplus B \oplus \overline{CI0N})$$

$$CO0 = (A \bullet B) + (A \bullet \overline{CI0N}) + (B \bullet \overline{CI0N})$$

$$CO1 = (A \bullet B) + (A \bullet \overline{CI1N}) + (B \bullet \overline{CI1N})$$

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
AFCSHCINX2	3.69	20.70
AFCSHCINX4	3.69	27.14

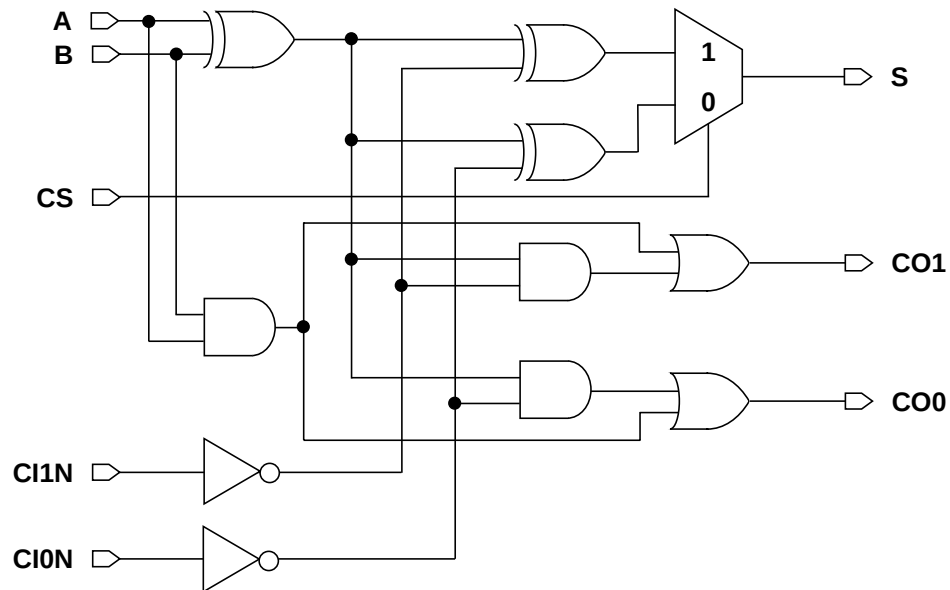
Functions

A	B	CI0N	CI1N	CS	S	CO0	CO1
0	0	0	0	0	1	0	0
0	0	0	0	1	1	0	0
0	0	0	1	0	1	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	0	0	0
0	0	1	0	1	1	0	0
0	0	1	1	0	0	0	0
0	0	1	1	1	0	0	0
0	1	0	0	0	0	1	1
0	1	0	0	1	0	1	1
0	1	0	1	0	0	1	0
0	1	0	1	1	1	1	0
0	1	1	0	0	1	0	1
0	1	1	0	1	0	0	1
0	1	1	1	0	1	0	0
0	1	1	1	1	1	0	0

A	B	CI0N	CI1N	CS	S	CO0	CO1
1	0	0	0	0	0	1	1
1	0	0	0	1	0	1	1
1	0	0	1	0	0	1	0
1	0	0	1	1	1	1	0
1	0	1	0	0	1	0	1
1	0	1	0	1	0	0	1
1	0	1	1	0	1	0	0
1	0	1	1	1	1	0	0
1	1	0	0	0	1	1	1
1	1	0	0	1	1	1	1
1	1	0	1	0	1	1	1
1	1	0	1	1	0	1	1
1	1	1	0	0	0	1	1
1	1	1	0	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1	1	0	1	1



Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
CS	0.0200	0.0207
A	0.0979	0.1137
B	0.0861	0.1031
CI0N	0.0411	0.0542
CI1N	0.0436	0.0517

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0037	0.0038
A	0.0049	0.0048
B	0.0107	0.0107
CI0N	0.0072	0.0097
CI1N	0.0079	0.0096



Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
CS → S↑	0.1164	0.1214	2.2261	2.2440
CS → S↓	0.0948	0.1022	1.4724	1.4209
A → S↑	0.3466	0.3829	2.2297	2.2452
A → S↓	0.3117	0.3457	1.4823	1.4251
B → S↑	0.2750	0.3052	2.2295	2.2450
B → S↓	0.2691	0.3164	1.4823	1.4249
CI0N → S↑	0.2033	0.2294	2.2299	2.2453
CI0N → S↓	0.1901	0.2369	1.4794	1.4239
CI1N → S↑	0.1890	0.1982	2.2266	2.2435
CI1N → S↓	0.1832	0.2182	1.4744	1.4214
A → CO0↑	0.1547	0.1974	4.0792	3.0458
A → CO0↓	0.1846	0.2467	2.4322	1.8747
B → CO0↑	0.1159	0.1649	3.8164	2.9617
B → CO0↓	0.1316	0.1866	2.3560	1.8465
CI0N → CO0↑	0.0614	0.0502	3.8469	1.6901
CI0N → CO0↓	0.0408	0.0368	2.3098	1.0219
A → CO1↑	0.1584	0.2051	3.6524	3.0274
A → CO1↓	0.1943	0.2527	2.2840	1.9529
B → CO1↑	0.1112	0.1526	3.5856	2.9718
B → CO1↓	0.1369	0.1881	2.3471	1.9300
CI1N → CO1↑	0.0533	0.0475	3.5221	1.6471
CI1N → CO1↓	0.0396	0.0362	2.2933	1.0251

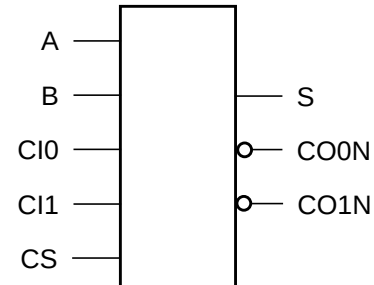
**Cell Description**

The AFCSHCON cell provides a carry-select adder function that produces the arithmetic sum (S) and active-low carry-outs (CO0N, CO1N) of two operands (A,B) with carry-ins (CI0, CI1). The three outputs (S, CO0N, CO1N) are represented by the logic equations:

$$S = CS \bullet (A \oplus B \oplus CI1) + \overline{CS} \bullet (A \oplus B \oplus CI0)$$

$$CO0N = \overline{(A \bullet B) + (A \bullet CI0) + (B \bullet CI0)}$$

$$CO1N = \overline{(A \bullet B) + (A \bullet CI1) + (B \bullet CI1)}$$

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
AFCSHCONX2	3.69	23.92
AFCSHCONX4	3.69	27.60

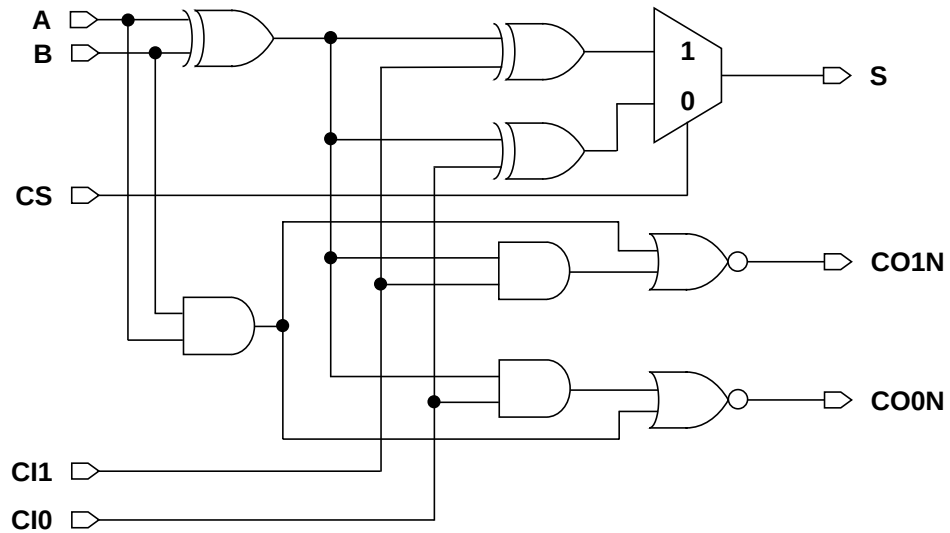
Functions

A	B	CI0	CI1	CS	S	CO0N	CO1N
0	0	0	0	0	0	1	1
0	0	0	0	1	0	1	1
0	0	0	1	0	0	1	1
0	0	0	1	1	1	1	1
0	0	1	0	0	1	1	1
0	0	1	0	1	0	1	1
0	0	1	1	0	1	1	1
0	0	1	1	1	1	1	1
0	1	0	0	0	1	1	1
0	1	0	0	1	1	1	1
0	1	0	1	0	1	1	0
0	1	0	1	1	0	1	0
0	1	1	0	0	0	0	1
0	1	1	0	1	1	0	1
0	1	1	1	0	0	0	0
0	1	1	1	1	0	0	0

A	B	CI0	CI1	CS	S	CO0N	CO1N
1	0	0	0	0	1	1	1
1	0	0	0	1	1	1	1
1	0	0	1	0	1	1	0
1	0	0	1	1	0	1	0
1	0	1	0	0	0	0	1
1	0	1	0	1	1	0	1
1	0	1	1	0	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	0	0	0	0
1	1	0	0	1	0	0	0
1	1	0	1	0	0	0	0
1	1	0	1	1	1	0	0
1	1	1	0	0	1	0	0
1	1	1	0	1	0	0	0
1	1	1	1	0	1	0	0
1	1	1	1	1	1	0	0



Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
CS	0.0206	0.0206
A	0.1062	0.1224
B	0.1018	0.1161
CI0	0.0489	0.0645
CI1	0.0436	0.0580

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0065	0.0065
A	0.0089	0.0089
B	0.0108	0.0108
CI0	0.0082	0.0161
CI1	0.0084	0.0159



Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
CS → S↑	0.1198	0.1208	2.2632	2.2441
CS → S↓	0.0971	0.1018	1.5055	1.4209
A → S↑	0.3862	0.4219	2.2675	2.2453
A → S↓	0.3474	0.3801	1.5150	1.4258
B → S↑	0.3471	0.3820	2.2674	2.2454
B → S↓	0.3082	0.3402	1.5163	1.4258
CI0 → S↑	0.2141	0.2124	2.2673	2.2453
CI0 → S↓	0.2115	0.2028	1.5133	1.4245
CI1 → S↑	0.1901	0.1837	2.2629	2.2435
CI1 → S↓	0.1921	0.1831	1.5065	1.4213
A → CO0N↑	0.2223	0.2639	2.8956	2.8761
A → CO0N↓	0.2170	0.2703	2.1849	2.0023
B → CO0N↑	0.1911	0.2381	3.1612	2.8720
B → CO0N↓	0.1781	0.2302	2.1822	2.0000
CI0 → CO0N↑	0.0575	0.0454	3.3472	1.6559
CI0 → CO0N↓	0.0423	0.0349	2.0536	1.0189
A → CO1N↑	0.2183	0.2635	3.2694	2.8331
A → CO1N↓	0.2168	0.2689	2.0918	1.9174
B → CO1N↑	0.1964	0.2378	2.8399	2.8307
B → CO1N↓	0.1780	0.2291	2.0892	1.9162
CI1 → CO1N↑	0.0562	0.0442	3.3531	1.6490
CI1 → CO1N↓	0.0412	0.0340	2.0729	1.0220



Cell Description

The AFCSIHCON cell provides a carry-select adder function for the initial stage of carry-select adder block. The function produces the arithmetic sum (S) and active-low carry-outs (CO0N, CO1N) of two operands (A,B). The three outputs (S, CO0N, CO1N) are represented by the logic equations:

$$S = A \oplus B \oplus CS$$

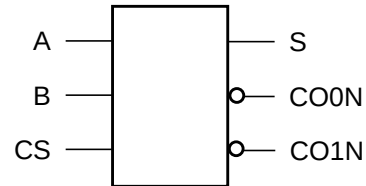
$$CO0N = \overline{A \bullet B}$$

$$CO1N = \overline{A + B}$$

Functions

A	B	CS	S	CO0N	CO1N
0	0	0	0	1	1
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	1	0	0

Logic Symbol

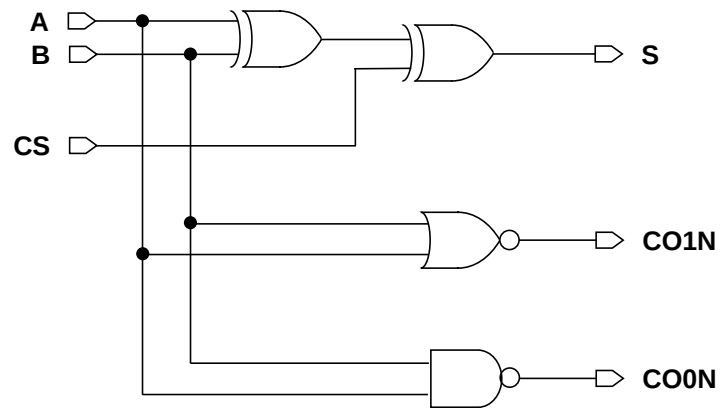


Cell Size

Drive Strength	Height (μm)	Width (μm)
AFCSIHCONX2	3.69	9.20
AFCSIHCONX4	3.69	11.04



Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0449	0.0540
B	0.0418	0.0535
CS	0.0189	0.0189

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0112	0.0184
B	0.0142	0.0218
CS	0.0061	0.0061



Delays at 25°C, 1.2V, Typical Process

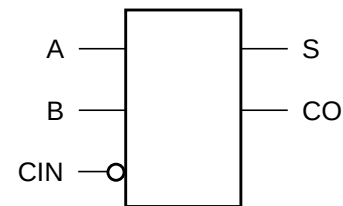
Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
A → S↑	0.2169	0.2184	2.2301	2.2087
A → S↓	0.2401	0.2552	1.8708	1.5896
B → S↑	0.1983	0.2032	2.3183	2.2424
B → S↓	0.2252	0.2383	1.8383	1.5819
CS → S↑	0.1106	0.1125	2.2243	2.2068
CS → S↓	0.1179	0.1312	1.8112	1.5653
A → CO0N↑	0.0247	0.0225	3.1037	1.5515
A → CO0N↓	0.0192	0.0179	2.1917	1.1199
B → CO0N↑	0.0319	0.0297	3.1006	1.5504
B → CO0N↓	0.0225	0.0214	2.1911	1.1197
A → CO1N↑	0.0283	0.0286	4.3744	2.1875
A → CO1N↓	0.0176	0.0178	1.7342	0.8671
B → CO1N↑	0.0379	0.0379	4.3722	2.1860
B → CO1N↓	0.0224	0.0224	1.7426	0.8713

**Cell Description**

The AFHCIN cell is a full adder that provides the arithmetic sum (S) and carry-out (CO) of two operands (A, B) with active-low carry-in (CIN). The outputs (S, CO) are represented by the logic equations:

$$S = A \oplus B \oplus \overline{CIN}$$

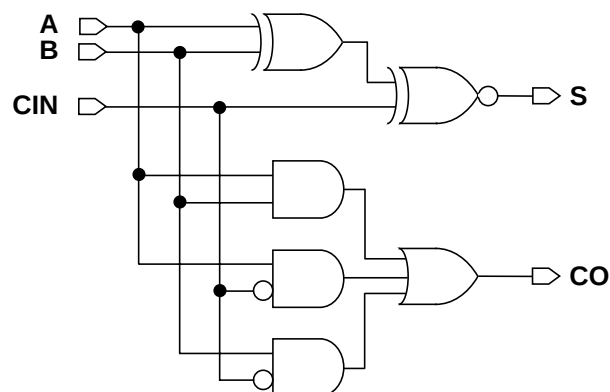
$$CO = (A \bullet B) + (A \bullet \overline{CIN}) + (B \bullet \overline{CIN})$$

Logic Symbol**Functions**

A	B	CIN	S	CO
0	0	0	1	0
0	0	1	0	0
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
AFHCINX2	3.69	13.34
AFHCINX4	3.69	15.18

Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0579	0.0615
B	0.0586	0.0636
CIN	0.0411	0.0592

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0048	0.0048
B	0.0117	0.0117
CIN	0.0093	0.0179

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.2114	0.2290	2.5782	2.5599
A $\rightarrow$ S $\downarrow$	0.2395	0.2594	1.5104	1.4262
B $\rightarrow$ S $\uparrow$	0.1662	0.1870	2.5822	2.5618
B $\rightarrow$ S $\downarrow$	0.1744	0.1916	1.5085	1.4252
CIN $\rightarrow$ S $\uparrow$	0.1151	0.1022	2.5734	2.5568
CIN $\rightarrow$ S $\downarrow$	0.1240	0.1229	1.5145	1.4268
A $\rightarrow$ CO $\uparrow$	0.1398	0.1599	3.4724	3.3917
A $\rightarrow$ CO $\downarrow$	0.1625	0.1942	2.1459	2.0251
B $\rightarrow$ CO $\uparrow$	0.1089	0.1261	3.3616	3.3399
B $\rightarrow$ CO $\downarrow$	0.1277	0.1543	2.0651	1.9987
CIN $\rightarrow$ CO $\uparrow$	0.0482	0.0397	3.0890	1.4868
CIN $\rightarrow$ CO $\downarrow$	0.0394	0.0344	1.9864	0.9720



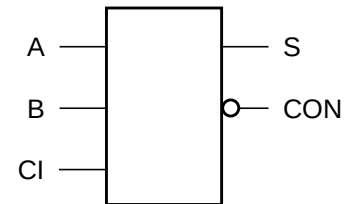
Cell Description

The AFHCON cell is a full adder that provides the arithmetic sum (S) and active-low carry-out (CON) of two operands (A, B) with carry-in (CI). The outputs (S, CON) are represented by the logic equations:

$$S = A \oplus B \oplus CI$$

$$CON = \overline{(A \bullet B) + (A \bullet CI) + (B \bullet CI)}$$

Logic Symbol



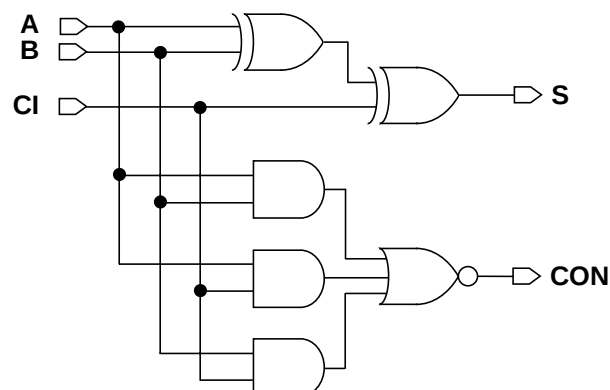
Functions

A	B	CI	S	CON
0	0	0	0	1
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AFHCONX2	3.69	11.50
AFHCONX4	3.69	13.80

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0608	0.0679
B	0.0573	0.0623
CI	0.0383	0.0548

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0048	0.0048
B	0.0159	0.0159
CI	0.0090	0.0179

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.2119	0.2240	2.2669	2.2446
A $\rightarrow$ S $\downarrow$	0.2301	0.2444	1.5265	1.4255
B $\rightarrow$ S $\uparrow$	0.1368	0.1428	2.2673	2.2448
B $\rightarrow$ S $\downarrow$	0.1571	0.1678	1.5241	1.4243
CI $\rightarrow$ S $\uparrow$	0.1116	0.0993	2.2631	2.2425
CI $\rightarrow$ S $\downarrow$	0.1274	0.1247	1.5299	1.4258
A $\rightarrow$ CON $\uparrow$	0.1791	0.1998	3.3279	3.1106
A $\rightarrow$ CON $\downarrow$	0.1598	0.1825	2.0945	1.8801
B $\rightarrow$ CON $\uparrow$	0.1056	0.1251	3.4354	3.1754
B $\rightarrow$ CON $\downarrow$	0.0933	0.1136	2.1524	1.9235
CI $\rightarrow$ CON $\uparrow$	0.0428	0.0388	3.0581	1.5283
CI $\rightarrow$ CON $\downarrow$	0.0331	0.0317	1.9765	0.9968



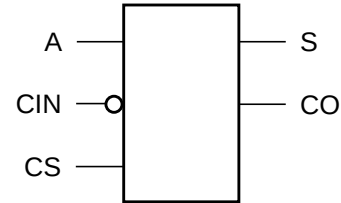
Cell Description

The AHCSHCIN cell provides a carry-select half-adder function that produces the arithmetic sum (S) and carry-out (CO) of a single operand (A) with active-low carry-in (CIN). The outputs (S, CO) are represented by the following equations:

$$S = CS \bullet (A \oplus \overline{CIN}) + \overline{CS} \bullet (A)$$

$$CO = A \bullet \overline{CIN}$$

Logic Symbol



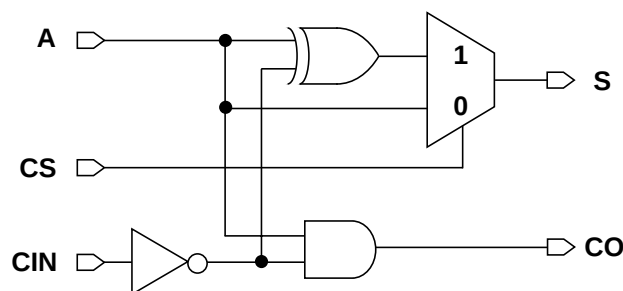
Functions

A	CIN	CS	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AHCSHCINX2	3.69	7.36
AHCSHCINX4	3.69	8.74

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
CS	0.0187	0.0179
A	0.0392	0.0457
CIN	0.0303	0.0363

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0061	0.0060
A	0.0050	0.0049
CIN	0.0118	0.0161

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
CS $\rightarrow$ S $\uparrow$	0.1080	0.1096	2.2248	2.2442
CS $\rightarrow$ S $\downarrow$	0.1120	0.1209	1.6903	1.5132
A $\rightarrow$ S $\uparrow$	0.1865	0.2046	2.2680	2.2593
A $\rightarrow$ S $\downarrow$	0.2015	0.2214	1.7066	1.5208
CIN $\rightarrow$ S $\uparrow$	0.1616	0.1651	2.2672	2.2591
CIN $\rightarrow$ S $\downarrow$	0.1640	0.1763	1.7032	1.5193
A $\rightarrow$ CO $\uparrow$	0.0821	0.0883	4.3854	2.1958
A $\rightarrow$ CO $\downarrow$	0.1076	0.1193	1.8781	0.9481
CIN $\rightarrow$ CO $\uparrow$	0.0287	0.0262	4.3788	2.1887
CIN $\rightarrow$ CO $\downarrow$	0.0180	0.0170	1.7342	0.8668



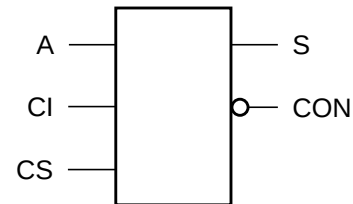
Cell Description

The AHCSHCIN cell provides a carry-select half-adder function that produces the arithmetic sum (S) and active-low carry-out (CON) of a single operand (A) with carry-in (CI). The outputs (S, CON) are represented by the following equations:

$$S = CS \bullet (A \oplus CI) + \overline{CS} \bullet (A)$$

$$CON = \overline{A} \bullet CI$$

Logic Symbol



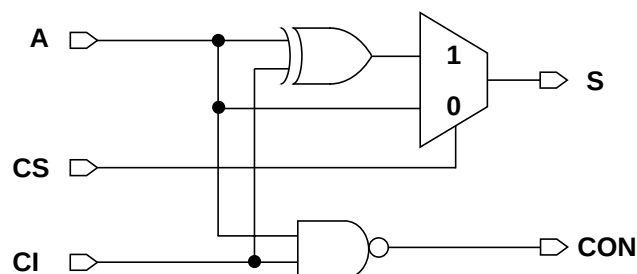
Functions

A	CI	CS	S	CON
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	0
1	1	1	0	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
AHCSHCONX2	3.69	7.36
AHCSHCONX4	3.69	8.28

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
CS	0.0181	0.0187
A	0.0398	0.0480
CI	0.0307	0.0363

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
CS	0.0061	0.0061
A	0.0086	0.0131
CI	0.0102	0.0128

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
CS $\rightarrow$ S $\uparrow$	0.1084	0.1095	2.2595	2.2429
CS $\rightarrow$ S $\downarrow$	0.1114	0.1212	1.6973	1.5132
A $\rightarrow$ S $\uparrow$	0.1667	0.1687	2.3001	2.2573
A $\rightarrow$ S $\downarrow$	0.1978	0.2075	1.7137	1.5196
CI $\rightarrow$ S $\uparrow$	0.1425	0.1443	2.2880	2.2529
CI $\rightarrow$ S $\downarrow$	0.1611	0.1710	1.7134	1.5196
A $\rightarrow$ CON $\uparrow$	0.0313	0.0304	3.0998	1.5504
A $\rightarrow$ CON $\downarrow$	0.0220	0.0215	2.1909	1.0956
CI $\rightarrow$ CON $\uparrow$	0.0246	0.0230	3.1037	1.5517
CI $\rightarrow$ CON $\downarrow$	0.0191	0.0180	2.1918	1.0958



Cell Description

The AHHCIN cell is a half adder that provides the arithmetic sum (S) and carry-out (CO) of the input operand (A) with an active-low carry-in (CIN). The outputs (S, CO) are represented by the logic equations:

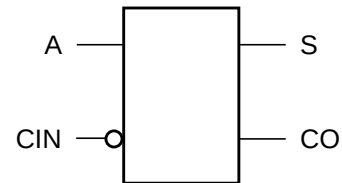
$$S = A \oplus \overline{CIN}$$

$$CO = A \bullet \overline{CIN}$$

Functions

A	CIN	S	CO
0	0	1	0
0	1	0	0
1	0	0	1
1	1	1	0

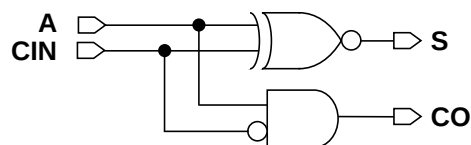
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AHHCINX2	3.69	5.98
AHHCINX4	3.69	7.82

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0327	0.0439
CIN	0.0274	0.0391

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0076	0.0088
CIN	0.0129	0.0173

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.0765	0.0771	3.0197	3.0137
A $\rightarrow$ S $\downarrow$	0.0767	0.0770	1.9399	1.9343
CIN $\rightarrow$ S $\uparrow$	0.0474	0.0480	2.9610	2.9569
CIN $\rightarrow$ S $\downarrow$	0.0697	0.0678	1.7856	1.8355
A $\rightarrow$ CO $\uparrow$	0.0570	0.0567	3.5359	1.6715
A $\rightarrow$ CO $\downarrow$	0.0842	0.0715	1.4399	0.7079
CIN $\rightarrow$ CO $\uparrow$	0.0381	0.0374	3.5334	1.6686
CIN $\rightarrow$ CO $\downarrow$	0.0211	0.0214	1.3827	0.6910



Cell Description

The AHHCON cell is a half adder that provides the arithmetic sum (S) and active-low carry-out (CON) of the input operand (A) with carry-in (CI). The outputs (S, CON) are represented by the logic equations:

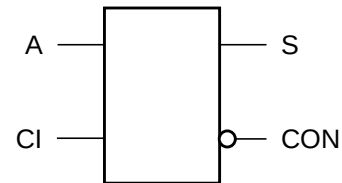
$$S = A \oplus CI$$

$$CON = \overline{A \bullet CI}$$

Functions

A	CI	S	CON
0	0	0	1
0	1	1	1
1	0	1	1
1	1	0	0

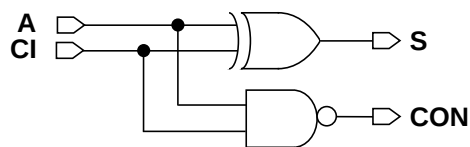
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
AHHCONX2	3.69	5.98
AHHCONX4	3.69	7.36

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0356	0.0472
CI	0.0211	0.0291

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0104	0.0155
CI	0.0138	0.0196

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.0743	0.0749	3.0172	3.0121
A $\rightarrow$ S $\downarrow$	0.0823	0.0833	1.9506	1.9383
CI $\rightarrow$ S $\uparrow$	0.0515	0.0533	3.0507	3.0168
CI $\rightarrow$ S $\downarrow$	0.0741	0.0722	1.8248	1.8715
A $\rightarrow$ CON $\uparrow$	0.0299	0.0292	2.2444	1.1216
A $\rightarrow$ CON $\downarrow$	0.0208	0.0208	1.5688	0.8090
CI $\rightarrow$ CON $\uparrow$	0.0222	0.0224	2.2460	1.1231
CI $\rightarrow$ CON $\downarrow$	0.0172	0.0177	1.5680	0.8090

**Cell Description**

The booth encoder block (BENC) cell performs a 2-bit multiplier recoding per a modified Booth's algorithm. Each BENC cell examines 3 bits of the multiplier (M0, M1, M2) and generates the appropriate control signals to adjust the multiplicand for subsequent partial product reduction. The outputs (S, A, X2) are represented by the logic equations:

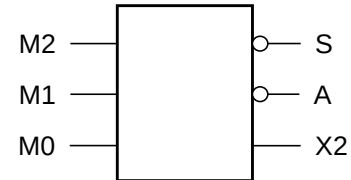
$$A = M2 + (\overline{M0} \bullet \overline{M1})$$

$$S = \overline{M2} + (M0 \bullet M1)$$

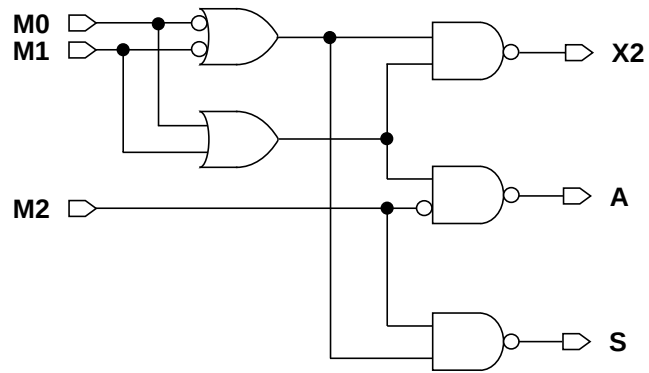
$$X2 = \overline{M1} \oplus \overline{M0}$$

Functions

M2	M1	M0	X2	A	S
0	0	0	1	1	1
0	0	1	0	0	1
0	1	0	0	0	1
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	1	0
1	1	1	1	1	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
BENCX1	3.69	14.72
BENCX2	3.69	19.32
BENCX4	3.69	34.50

**Functional Schematic****AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)		
	X1	X2	X4
M2	0.0241	0.0415	0.0787
M1	0.0475	0.0859	0.1690
M0	0.0539	0.0963	0.1868

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
M2	0.0049	0.0065	0.0112
M1	0.0071	0.0119	0.0231
M0	0.0067	0.0115	0.0211

**Delays at 25°C, 1.2V, Typical Process**

Description	Intrinsic Delay (ns)			K _{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
M2 → A↑	0.1835	0.1872	0.1752	1.1825	0.5797	0.2894
M2 → A↓	0.1706	0.1792	0.1753	0.8748	0.4185	0.2092
M1 → A↑	0.2106	0.1743	0.1674	1.1875	0.5797	0.2898
M1 → A↓	0.1399	0.1330	0.1304	0.8778	0.4201	0.2099
M0 → A↑	0.2028	0.1675	0.1590	1.1885	0.5795	0.2896
M0 → A↓	0.1331	0.1261	0.1222	0.8774	0.4195	0.2097
M2 → S↑	0.1581	0.1458	0.1330	1.1832	0.5780	0.2891
M2 → S↓	0.1092	0.1004	0.0972	0.8753	0.4202	0.2093
M1 → S↑	0.2572	0.2117	0.2051	1.1892	0.5782	0.2896
M1 → S↓	0.2405	0.2147	0.2035	0.8790	0.4219	0.2099
M0 → S↑	0.2418	0.1903	0.1863	1.1893	0.5783	0.2895
M0 → S↓	0.1966	0.1707	0.1611	0.8783	0.4213	0.2096
M1 → X2↑	0.1647	0.1447	0.1417	1.1870	0.5782	0.2894
M1 → X2↓	0.1968	0.1742	0.1659	0.9560	0.4633	0.2266
M0 → X2↑	0.2059	0.1708	0.1683	1.1865	0.5782	0.2896
M0 → X2↓	0.2002	0.1781	0.1677	0.9552	0.4633	0.2267

**Cell Description**

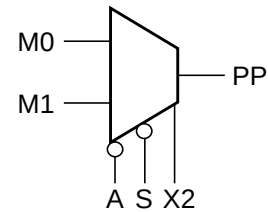
The BMX cell performs the shifting and 2's complement inversion of the multiplicand bits (M1 and M0) based on the recode control signals (X2, A, and S) from the BENC cell. The partial product output (PP) is represented by the logic equation:

$$PP = X2 \bullet ((M0 \bullet \bar{A}) + (\bar{M0} \bullet \bar{S})) + \bar{X2} \bullet ((M1 \bullet \bar{A}) + (\bar{M1} \bullet \bar{S}))$$

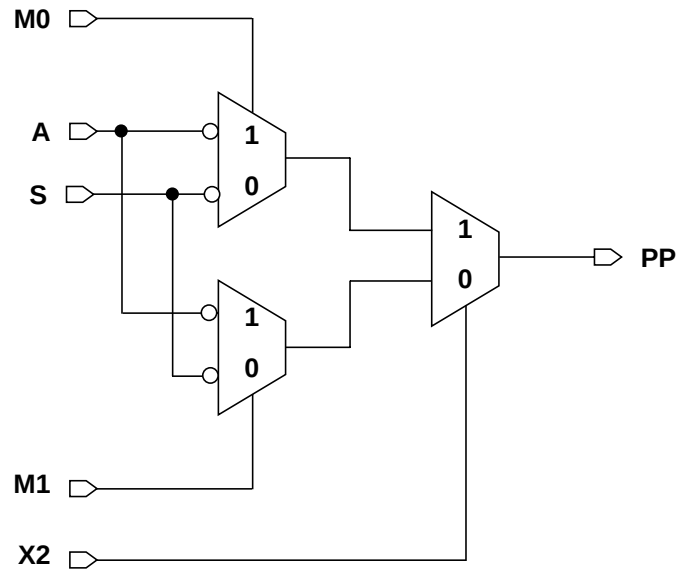
Functions¹

X2	A	S	M0	M1	PP
0	0	0	x	x	x
0	0	1	x	0	0
0	0	1	x	1	1
0	1	0	x	0	1
0	1	0	x	1	0
0	1	1	x	x	0
1	0	0	x	x	x
1	0	1	0	x	0
1	0	1	1	x	1
1	1	0	0	x	1
1	1	0	1	x	0
1	1	1	x	x	0

¹ Shaded areas represent illegal conditions.

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
BMXX2	3.69	11.04
BMXX4	3.69	17.02

**Functional Schematic****AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
X2	0.0191	0.0343
M0	0.0298	0.0528
A	0.0320	0.0600
S	0.0409	0.0750
M1	0.0246	0.0474

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
X2	0.0035	0.0059
M0	0.0059	0.0101
A	0.0057	0.0106
S	0.0059	0.0107
M1	0.0049	0.0090

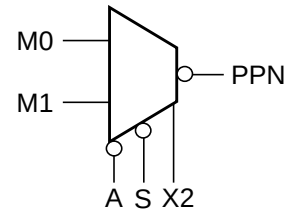
**Delays at 25°C, 1.2V, Typical Process**

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
X2 → PP↑	0.1111	0.1042	2.2598	1.1330
X2 → PP↓	0.0930	0.0950	1.5079	0.7643
M0 → PP↑	0.1602	0.1519	2.2620	1.1333
M0 → PP↓	0.2061	0.1984	1.5110	0.7653
A → PP↑	0.1652	0.1533	2.2617	1.1331
A → PP↓	0.1555	0.1494	1.5110	0.7650
S → PP↑	0.1777	0.1642	2.2635	1.1350
S → PP↓	0.1683	0.1626	1.5117	0.7656
M1 → PP↑	0.1487	0.1490	2.2620	1.1343
M1 → PP↓	0.1835	0.1873	1.5038	0.7660

**Cell Description**

The BMXI cell performs the shifting and 2's complement inversion of the multiplicand bits (M1 and M0) based on the recode control signals (X2, A, and S) from the BENC cell. The inverted partial product output (PPN) is represented by the logic equation:

$$PPN = X2 \bullet ((M0 \bullet \bar{A}) + (\bar{M0} \bullet \bar{S})) + \bar{X2} \bullet ((M1 \bullet \bar{A}) + (\bar{M1} \bullet \bar{S}))$$

Logic Symbol**Functions¹**

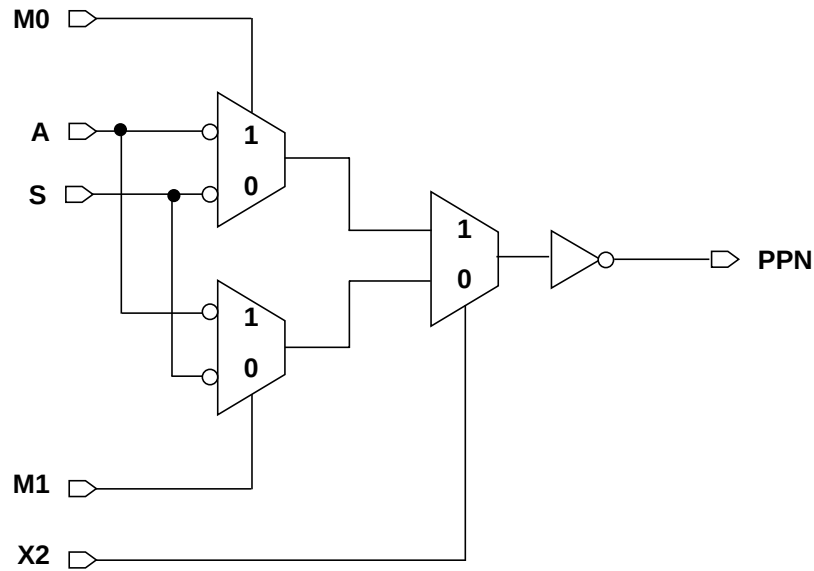
X2	A	S	M0	M1	PPN
0	0	0	x	x	x
0	0	1	x	0	1
0	0	1	x	1	0
0	1	0	x	0	0
0	1	0	x	1	1
0	1	1	x	x	1
1	0	0	x	x	x
1	0	1	0	x	1
1	0	1	1	x	0
1	1	0	0	x	0
1	1	0	1	x	1
1	1	1	x	x	1

¹ Shaded areas represent illegal conditions.

Drive Strength	Height (μm)	Width (μm)
BMXIX2	3.69	10.12
BMXIX4	3.69	14.72



Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
X2	0.0162	0.0300
M0	0.0203	0.0374
A	0.0274	0.0500
S	0.0306	0.0571
M1	0.0197	0.0377

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
X2	0.0049	0.0088
M0	0.0059	0.0101
A	0.0057	0.0106
S	0.0058	0.0108
M1	0.0049	0.0090



Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K _{load} (ns/pF)	
	X2	X4	X2	X4
X2 → PPN↑	0.1221	0.1198	2.2845	1.1457
X2 → PPN↓	0.1132	0.1136	1.6691	0.8400
M0 → PPN↑	0.1602	0.1546	2.2891	1.1474
M0 → PPN↓	0.1595	0.1520	1.6861	0.8452
A → PPN↑	0.1109	0.1082	2.2862	1.1464
A → PPN↓	0.1623	0.1539	1.6869	0.8461
S → PPN↑	0.1266	0.1232	2.2891	1.1478
S → PPN↓	0.1707	0.1597	1.7081	0.8538
M1 → PPN↑	0.1518	0.1506	2.2882	1.1473
M1 → PPN↓	0.1461	0.1400	1.6811	0.8443



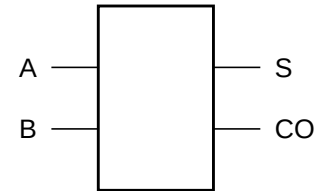
Cell Description

The CMPR22 cell provides the arithmetic sum (S) and carry out (CO) of two operands (A, B). The two outputs (S, CO) are represented by the logic equations:

$$S = (\bar{A} \bullet B) + (A \bullet \bar{B})$$

$$CO = A \bullet B$$

Logic Symbol



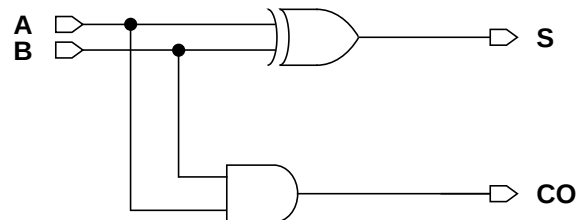
Functions

A	B	S	CO
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
CMPR22X2	3.69	8.74
CMPR22X4	3.69	13.80

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0634	0.1201
B	0.0339	0.0620

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0145	0.0273
B	0.0086	0.0159

Delays (25°C, 1.2V, Typical Process)

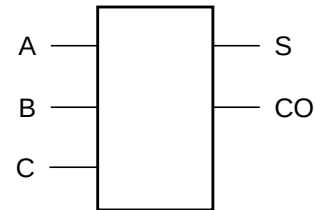
Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.0641	0.0644	1.5071	0.7395
A $\rightarrow$ S $\downarrow$	0.0709	0.0690	0.9933	0.4825
B $\rightarrow$ S $\uparrow$	0.0417	0.0410	1.5267	0.7409
B $\rightarrow$ S $\downarrow$	0.0605	0.0592	0.8954	0.4330
A $\rightarrow$ CO $\uparrow$	0.0689	0.0641	2.2626	1.1310
A $\rightarrow$ CO $\downarrow$	0.1007	0.0864	1.4759	0.7245
B $\rightarrow$ CO $\uparrow$	0.0673	0.0615	2.2622	1.1308
B $\rightarrow$ CO $\downarrow$	0.0939	0.0787	1.4708	0.7222

**Cell Description**

The CMPR32 cell takes in 3 bits of the partial product (A, B, C) and compresses them into 2-bits of partial product (S, CO). The two outputs (S, CO) are represented by the logic equations:

$$S = A \oplus B \oplus C$$

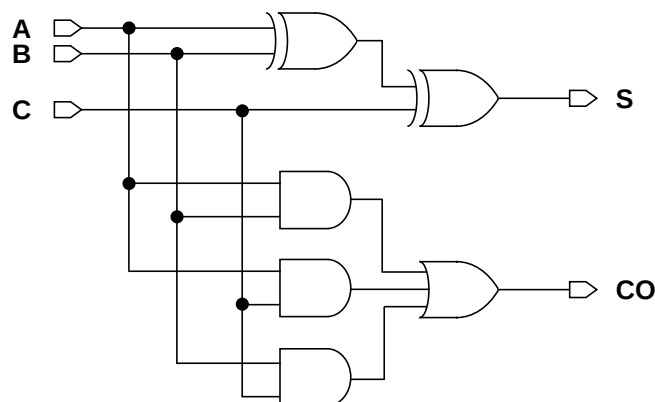
$$CO = (A \bullet B) + (A \bullet C) + (B \bullet C)$$

Logic Symbol**Functions**

A	B	C	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
CMPR32X2	3.69	9.66
CMPR32X4	3.69	10.58

Functional Schematic

**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)	
	X2	X4
A	0.0396	0.0564
B	0.0516	0.0663
C	0.0246	0.0410

Pin Capacitance

Pin	Capacitance (pF)	
	X2	X4
A	0.0047	0.0047
B	0.0049	0.0049
C	0.0045	0.0045

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)		K_{load} (ns/pF)	
	X2	X4	X2	X4
A $\rightarrow$ S $\uparrow$	0.1882	0.2259	2.3203	1.1831
A $\rightarrow$ S $\downarrow$	0.2592	0.3083	1.8026	0.9923
B $\rightarrow$ S $\uparrow$	0.2178	0.2457	2.3291	1.1902
B $\rightarrow$ S $\downarrow$	0.2908	0.3405	1.8026	0.9923
C $\rightarrow$ S $\uparrow$	0.1667	0.2131	2.3214	1.1922
C $\rightarrow$ S $\downarrow$	0.1469	0.1944	1.8430	1.0257
A $\rightarrow$ CO $\uparrow$	0.2383	0.2844	2.2719	1.1552
A $\rightarrow$ CO $\downarrow$	0.2375	0.2816	1.6934	0.9426
B $\rightarrow$ CO $\uparrow$	0.2695	0.3161	2.2716	1.1551
B $\rightarrow$ CO $\downarrow$	0.2517	0.2879	1.5535	0.8433
C $\rightarrow$ CO $\uparrow$	0.1333	0.1795	2.3046	1.1796
C $\rightarrow$ CO $\downarrow$	0.1776	0.2224	1.7372	0.9641

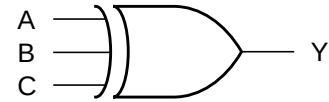


Cell Description

The XOR3 cell provides a logical EXCLUSIVE OR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$$Y = A \oplus B \oplus C$$

Logic Symbol



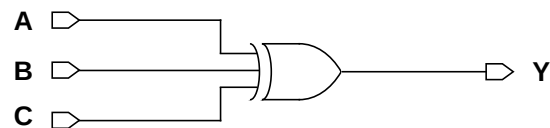
Functions

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Cell Size

Drive Strength	Height (μm)	Width (μm)
XOR3XL	3.69	7.82
XOR3X1	3.69	7.82
XOR3X2	3.69	8.28
XOR3X4	3.69	11.50

Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)			
	XL	X1	X2	X4
A	0.0208	0.0279	0.0443	0.0662
B	0.0177	0.0233	0.0370	0.0569
C	0.0087	0.0108	0.0184	0.0329

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0019	0.0029	0.0048	0.0048
B	0.0056	0.0069	0.0102	0.0136
C	0.0039	0.0039	0.0063	0.0114

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.3067	0.2598	0.2134	0.2151	9.5710	5.1653	2.3065	1.1516
A $\rightarrow$ Y $\downarrow$	0.3020	0.2848	0.2302	0.2346	6.3667	3.6825	1.6890	0.9344
B $\rightarrow$ Y $\uparrow$	0.1799	0.1709	0.1437	0.1595	9.5715	5.1550	2.3061	1.1766
B $\rightarrow$ Y $\downarrow$	0.2080	0.2029	0.1696	0.1947	6.1736	3.5561	1.6846	0.9332
C $\rightarrow$ Y $\uparrow$	0.1147	0.1307	0.1218	0.1213	9.5400	5.1494	2.3022	1.1737
C $\rightarrow$ Y $\downarrow$	0.1076	0.1156	0.1081	0.1236	5.2613	3.4563	1.6641	0.9173

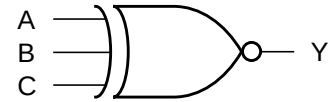


Cell Description

The XNOR3 cell provides a logical EXCLUSIVE NOR of three inputs (A,B,C). The output (Y) is represented by the following equation:

$$Y = \overline{A \oplus B \oplus C}$$

Logic Symbol



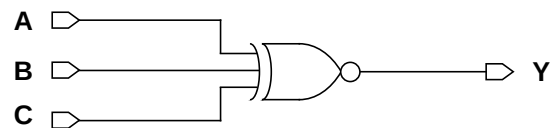
Functions

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
XNOR3XL	3.69	7.82
XNOR3X1	3.69	7.82
XNOR3X2	3.69	8.28
XNOR3X4	3.69	11.50

Functional Schematic



**AC Power**

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
A	0.0214	0.0276	0.0436	0.0682
B	0.0173	0.0228	0.0367	0.0575
C	0.0087	0.0109	0.0184	0.0324

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0019	0.0029	0.0048	0.0048
B	0.0056	0.0069	0.0102	0.0136
C	0.0027	0.0025	0.0036	0.0062

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
A $\rightarrow$ Y $\uparrow$	0.3077	0.2581	0.2161	0.2162	9.5709	5.1570	2.3073	1.1519
A $\rightarrow$ Y $\downarrow$	0.3038	0.2878	0.2315	0.2359	6.3611	3.6678	1.6942	0.9371
B $\rightarrow$ Y $\uparrow$	0.1789	0.1716	0.1458	0.1589	9.5705	5.1554	2.3074	1.1770
B $\rightarrow$ Y $\downarrow$	0.2099	0.2007	0.1682	0.1982	6.1665	3.5589	1.6936	0.9318
C $\rightarrow$ Y $\uparrow$	0.1146	0.1309	0.1218	0.1226	9.5153	5.1380	2.3015	1.1756
C $\rightarrow$ Y $\downarrow$	0.1127	0.1174	0.1089	0.1288	5.2531	3.4077	1.6700	0.9261

**Cell Description**

The CMPR42 cell takes in 4 bits of the partial product (A, B, C, D) and compresses them into 2-bits of partial product (S, CO). The cell requires an intermediate carry-in input (ICI) from the n-1 compressor and an intermediate carry-out output (CO) to the n+1 compressor. The CMPR42 cell also contains an internal sum (IS). The internal sum (IS), carry-in output (ICO), and the two outputs (S, CO) are represented by the logic equations:

$$IS = A \oplus B \oplus C$$

$$ICO = (A \bullet B) + (A \bullet C) + (B \bullet C)$$

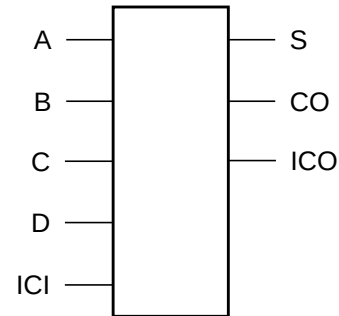
$$S = IS \oplus D \oplus ICI$$

$$CO = (IS \bullet D) + (IS \bullet ICI) + (D \bullet ICI)$$

Functions

A	B	C	IS	ICO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

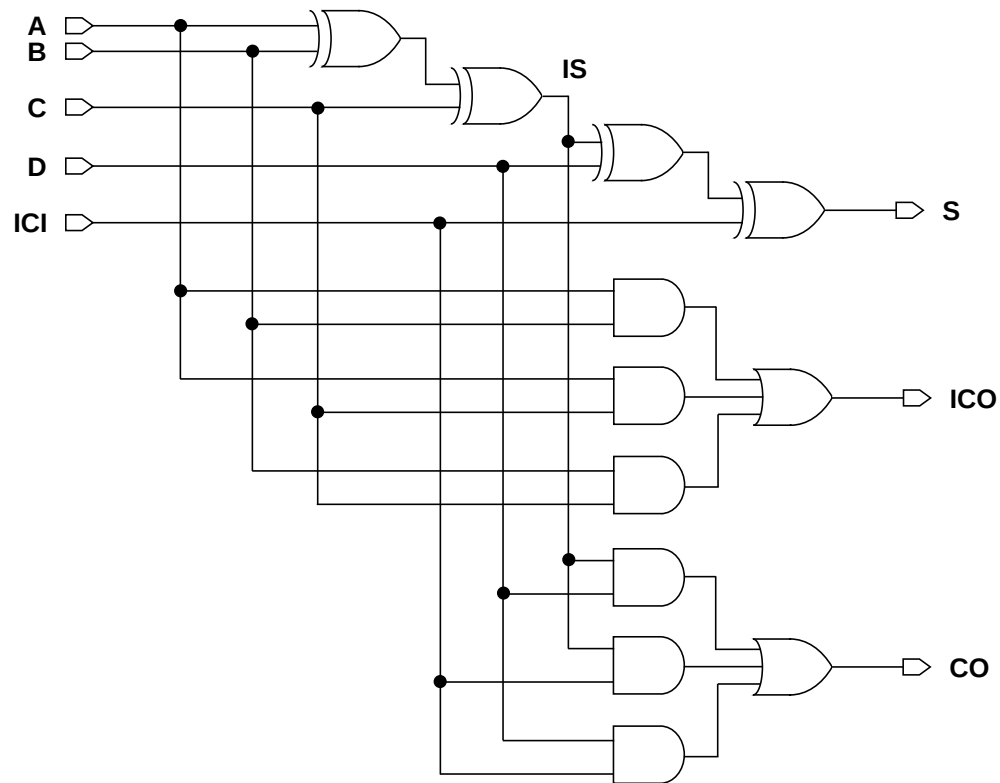
IS	D	ICI	S	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
CMPR42X1	3.69	16.56
CMPR42X2	3.69	18.40
CMPR42X4	3.69	26.22



Functional Schematic



**AC Power**

Pin	Power ($\mu\text{W}/\text{MHz}$)		
	X1	X2	X4
A	0.0552	0.0854	0.1313
B	0.0531	0.0818	0.1260
C	0.0492	0.0768	0.1173
D	0.0410	0.0623	0.0998
ICI	0.0207	0.0344	0.0609

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
A	0.0072	0.0112	0.0173
B	0.0078	0.0119	0.0192
C	0.0062	0.0086	0.0111
D	0.0047	0.0063	0.0065
ICI	0.0029	0.0046	0.0047



Delays at 25°C, 1.2V, Typical Process

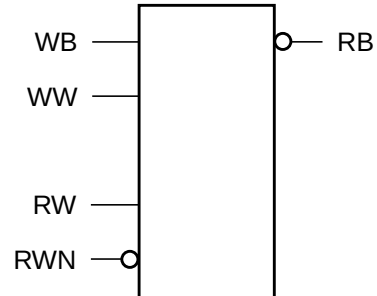
Description	Intrinsic Delay (ns)			K _{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
A → S↑	0.4188	0.3943	0.4614	5.1857	2.2757	1.1432
A → S↓	0.5076	0.4613	0.5244	2.9385	1.5092	0.7577
B → S↑	0.3707	0.3401	0.4074	5.1494	2.2757	1.1432
B → S↓	0.4481	0.4071	0.4704	2.9389	1.5094	0.7577
C → S↑	0.3160	0.3151	0.3683	5.1845	2.2757	1.1436
C → S↓	0.4054	0.3828	0.4364	2.9386	1.5094	0.7577
D → S↑	0.2943	0.2679	0.3094	5.1492	2.2582	1.1471
D → S↓	0.3424	0.3299	0.3929	2.9392	1.5090	0.7566
ICI → S↑	0.1617	0.1343	0.1802	5.1700	2.2656	1.1361
ICI → S↓	0.1869	0.1617	0.1880	2.9492	1.5114	0.7589
A → ICO↑	0.0821	0.0765	0.0746	5.1689	2.2679	1.1360
A → ICO↓	0.1565	0.1284	0.1261	2.9669	1.5104	0.7615
B → ICO↑	0.0827	0.0778	0.0765	5.1757	2.2695	1.1365
B → ICO↓	0.1499	0.1244	0.1229	2.9667	1.5107	0.7618
C → ICO↑	0.0725	0.0687	0.0679	5.1588	2.2629	1.1335
C → ICO↓	0.1259	0.1075	0.1084	2.9732	1.5239	0.7686
A → CO↑	0.4003	0.3762	0.4462	5.1646	2.2681	1.1512
A → CO↓	0.4704	0.4273	0.5154	2.8815	1.4790	0.7934
B → CO↑	0.3673	0.3262	0.3966	5.1344	2.2678	1.1512
B → CO↓	0.4181	0.3773	0.4657	2.8818	1.4792	0.7934
C → CO↑	0.3107	0.2994	0.3682	5.1365	2.2686	1.1512
C → CO↓	0.3359	0.3122	0.4031	2.8821	1.4793	0.7934
D → CO↑	0.2819	0.2580	0.2840	5.1364	2.2569	1.1396
D → CO↓	0.2500	0.2388	0.2936	2.7722	1.4765	0.7961
ICI → CO↑	0.0875	0.0787	0.1102	5.1694	2.2648	1.1504
ICI → CO↓	0.1267	0.1054	0.1503	3.0050	1.5096	0.8290

**Cell Description**

The RF1R1W register file cell is an active-high D-type transparent latch with an active-high tri-state output. The output (RB) is inverted.

Functions for Write Operations

WW	WB	Q[n+1]
0	0	0
0	1	Q[n]
0	0	Q[n]
0	1	Q[n]
1	0	0
1	1	1
1	0	Q[n]
1	1	1

Logic Symbol**Cell Size**

Drive Strength	Height (μm)	Width (μm)
RF1R1WX1	3.69	3.68

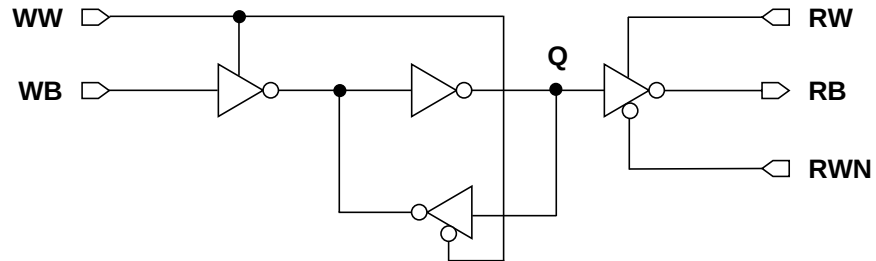
Functions for Read Operations¹

RW	RWN	Q	RB
0	0	0	1
0	0	1	Hi-Z
0	1	0	Hi-Z
0	1	1	Hi-Z
1	0	0	1
1	0	1	0
1	1	0	Hi-Z
1	1	1	0

¹ Shaded areas represent operations that are legal only during RW/RWN transitions.



Functional Schematic



AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)
	X1
WW	0.0073
WB	0.0079
RW	0.0011
RB	0.0047

Pin Capacitance

Pin	Capacitance (pF)
	X1
WW	0.0023
WB	0.0012
RW	0.0011
RWN	0.0005
RB	0.0037

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)	K_{load} (ns/pF)
	X1	X1
WW $\rightarrow$ RB $\uparrow$	0.2371	10.0860
WW $\rightarrow$ RB $\downarrow$	0.1483	4.4963
WB $\rightarrow$ RB $\uparrow$	0.2198	10.0860
WB $\rightarrow$ RB $\downarrow$	0.1608	4.4963
RW $\rightarrow$ RB $\uparrow$	0.0364	10.0578
RW $\rightarrow$ RB $\downarrow$	0.0161	4.4374

**Timing Constraints at 25°C, 1.2V, Typical Process**

Pin	Requirement	Interval (ns)
		X1
WW	minpwh	0.0690
WB	setup $\uparrow$ $\rightarrow$ WW	0.0859
	setup $\downarrow$ $\rightarrow$ WW	0.1094
	hold $\uparrow$ $\rightarrow$ WW	-0.0703
	hold $\downarrow$ $\rightarrow$ WW	-0.1016



Cell Description

The RF2R1W register file cell is an active-high D-type transparent latch with two independently controlled, active-high tri-state outputs. The cell has two read ports and one write port. The outputs (R1B, R2B) are inverted.

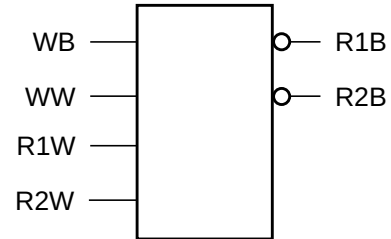
Functions for Write Operations

WW	WB	Q[n+1]
0	0	Q[n]
0	1	Q[n]
1	0	0
1	1	1

Functions for Read Operations

R1W/ R2W	Q	R1B/ R2B
0	0	Hi-Z
0	1	Hi-Z
1	0	1
1	1	0

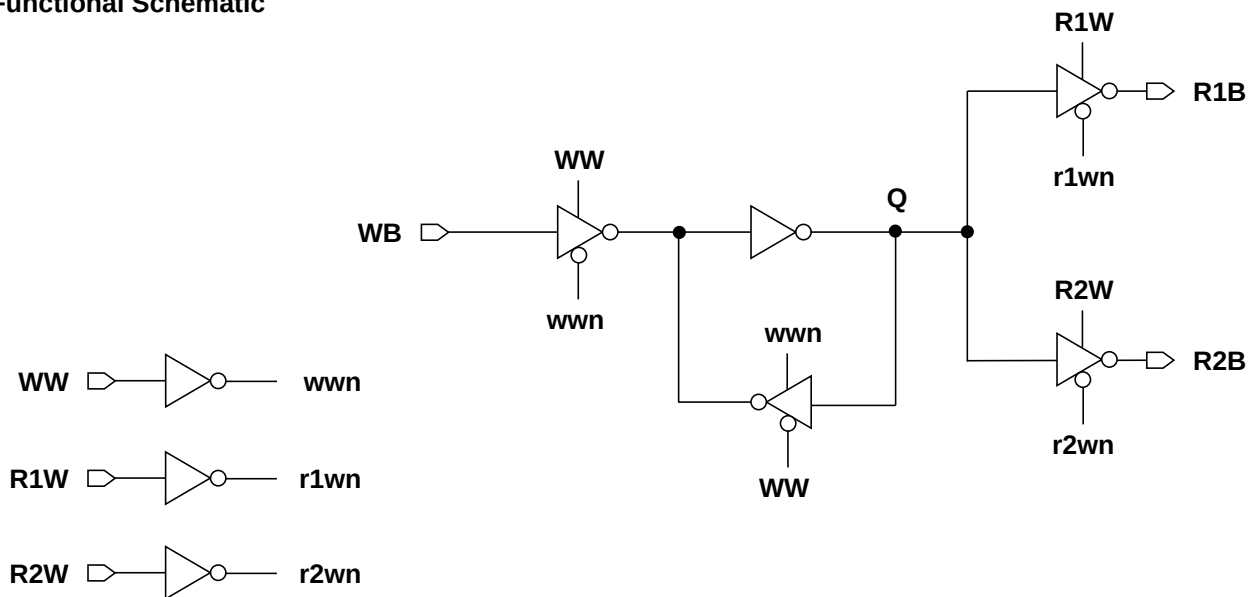
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RF2R1WX1	3.69	6.90

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)
	X1
WB	0.0214
WW	0.0059
R1W	0.0040
R2W	0.0037
R1B	0.0329

Pin Capacitance

Pin	Capacitance (pF)
	X1
WB	0.0030
WW	0.0031
R1W	0.0027
R2W	0.0034
R1B	0.0032
R2B	0.0032

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)	K_{load} (ns/pF)
	X1	X1
WB $\rightarrow$ R1B $\uparrow$	0.1933	4.4768
WB $\rightarrow$ R1B $\downarrow$	0.1349	2.3201
WW $\rightarrow$ R1B $\uparrow$	0.2183	4.4768
WW $\rightarrow$ R1B $\downarrow$	0.1209	2.3204
R1W $\rightarrow$ R1B $\uparrow$	0.0533	4.4436
R1W $\rightarrow$ R1B $\downarrow$	0.0139	2.2481
WB $\rightarrow$ R2B $\uparrow$	0.1923	4.4736
WB $\rightarrow$ R2B $\downarrow$	0.1350	2.3703
WW $\rightarrow$ R2B $\uparrow$	0.2173	4.4736
WW $\rightarrow$ R2B $\downarrow$	0.1209	2.3704
R2W $\rightarrow$ R2B $\uparrow$	0.0526	4.4432
R2W $\rightarrow$ R2B $\downarrow$	0.0141	2.3002



Timing Constraints at 25°C, 1.2V, Typical Process

Pin	Requirement	Interval (ns)
		X1
WB	setup $\uparrow$ $\rightarrow$ WW	0.0742
	setup $\downarrow$ $\rightarrow$ WW	0.0742
	hold $\uparrow$ $\rightarrow$ WW	-0.0586
	hold $\downarrow$ $\rightarrow$ WW	-0.0586
WW	minpwh	0.0544



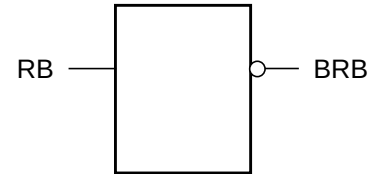
Cell Description

The RFRD output buffer has a “keeper” function that holds the input and output ports at the present level when the input (RB) is in a state of high-impedance.

Functions

RB	BRB
0	1
1	0
Hi-Z	Keep

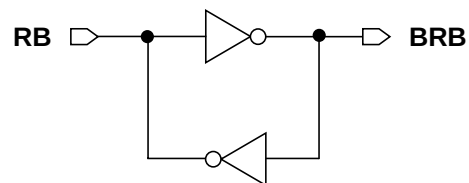
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
RFRDX1	3.69	2.30
RFRDX2	3.69	2.30
RFRDX4	3.69	2.76

Functional Schematic





AC Power

Pin	Power ($\mu\text{W}/\text{MHz}$)		
	X1	X2	X4
RB	0.0074	0.0102	0.0159

Pin Capacitance

Pin	Capacitance (pF)		
	X1	X2	X4
RB	0.0416	0.0588	0.0621

Delays at 25°C, 1.2V, Typical Process

Description	Intrinsic Delay (ns)			K_{load} (ns/pF)		
	X1	X2	X4	X1	X2	X4
RB $\rightarrow$ BRB $\uparrow$	0.0353	0.0247	0.0196	5.1357	2.2447	1.1215
RB $\rightarrow$ BRB $\downarrow$	0.0227	0.0190	0.0156	2.5618	1.3700	0.6851